॥ श्री गणेश प्रसन्न ॥

# Crazy 2

**Papers 1–4 combined · Sections 4–9**

All four answer books merged section by section; overlapping answers are printed once with every fact kept, tagged P1–P4 by where they were asked.

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# Section 4 · Paediatric Urology 1

## Long answer questions (LAQ)

### LAQ 1 Primary VUR: VUJ embryology/anatomy, pathophysiology, management (incl. bilateral, adult grade 3), reimplantation techniques and STING

P1 · P3 · P4 **merged from 6**

*Asked as:* P1 LAQ "Primary (congenital) VUR: VUJ embryology, pathophysiology, natural history and management" (Describe embryological development of VU Junction and outline management of primary VUR?, Summer 2020 Paper I Q2; B/l congenital VUR - Pathophysiology and natural history, Practice Set 1 Paper I Q3f SAQ) · P3 LAQ "Management of bilateral VUR: medical treatment, surgical techniques, selection and long-term outcome" (Describe medical management of VUR. Describe treatment of an 8-yr male child with b/l grade IV VUR, Summer 2015 Paper III Q2; Discuss mgt. of b/l VUR, Practice Set 2 Paper III Q2; Discuss treatment of b/l VUR. Describe surgical techniques and criteria for their selection, Practice Set 2 Paper III Q3d SAQ; Discuss the management of B/l VUR and discuss its long term outcome, Practice Set 6 Paper III Q1) · P4 LAQ "VUR: etiopathogenesis, recent trends in management and STING technique" (Describe etiopathogenesis of VUR and recent trends in management, Practice Set 2 Paper IV Q2; STING technique for VUR, Summer 2020 Paper IV Q3c and Practice Set 5 Paper IV Q3e SAQ) · P3 LAQ "Development and clinical anatomy of Vesico-Ureteric junction and management of grade 3 VUR in adult?" · P3 SAQ "Endoscopic correction of VUR (STING procedure): indications and complications" (Endoscopic correction of VUR : indications and complications, Summer 2019 Paper III Q3e; Sting procedure, Winter 2024 Paper III Q3f) · P3 SAQ "Paquin and Gil-Vernet techniques of ureteric reimplantation"

*"B/l congenital VUR" is read as bilateral primary (congenital) VUR (vesicoureteral reflux), often detected antenatally. STING is taken as subureteric Teflon injection, the endoscopic treatment of VUR.*

#### Definition and epidemiology

- **VUR**: retrograde flow of urine from the bladder into the upper tract.
- **Primary**: congenitally deficient UVJ (ureterovesical junction) flap valve with a short intravesical tunnel and a normal bladder. **Secondary**: high bladder pressure (PUV (posterior urethral valves), neurogenic bladder, BBD (bladder and bowel dysfunction)).
- Found in about **30% of children with UTI (urinary tract infection)**, about 30% of siblings and about 17% of children imaged for hydronephrosis. The younger the child, the likelier VUR.
- Infant boys predominate (higher voiding pressures, more UTI if uncircumcised); overall girls are affected twice as often. Less common in Black children.
- **Bilateral VUR is a high-risk feature:** it resolves less and scars more.
- **Infection, not sterile reflux, injures the kidney.** Infected urine plus intrarenal reflux causes scars, hypertension and CKD (chronic kidney disease); some high-grade "scars" are congenital dysplasia. So management targets infection risk, not the reflux itself.

## Embryology of the VUJ (vesicoureteric junction)

1. **Ureteric bud (weeks 4–6)** arises from the caudal mesonephric (Wolffian) duct: ureter and duct form a "Y", and the common nephric (excretory) duct is the stem.
2. **Reciprocal induction:** the bud enters the metanephric blastema, branches into the collecting system and induces nephrons.
3. **Weeks 5–6:** the urorectal septum divides the cloaca into urogenital sinus (UGS) and anorectal canal.
4. **Absorption:** the common nephric duct is taken into the UGS (current view: removed by **apoptosis**), so ureter and duct open separately.
5. **Rotation (Mackie and Stephens):** the ureteric orifice moves **cranially and laterally** to the bladder base; the duct stays caudal as the ejaculatory duct, ending in the prostatic urethra (Gartner duct in females).
6. **Trigone:** classically ureteral; lineage tracing shows it is mostly bladder muscle. The **Chwalla membrane** over the orifice ruptures; its persistence explains ureterocele.
7. **Maturation:** ureteric muscle joins the detrusor as **Waldeyer's sheath**; the tunnel lengthens with growth, so VUR resolves with age.

| Bud origin | Orifice | Result |
|---|---|---|
| Normal | Trigone corner | Tunnel length to diameter about **5:1** (Paquin); refluxing junctions about 1.4:1 |
| Too low (early budding, early contact with UGS) | High, **lateral** | Short tunnel, so **primary VUR**; abnormal bud–metanephros contact gives **dysplasia** |
| Too high (late budding, late contact) | Caudal (distal), **medial** | **Ectopic ureter** (bladder neck, urethra, Wolffian structures), often obstructed |
| Two buds | Duplex | **Weigert–Meyer rule:** upper pole caudal-medial (ectopia, ureterocele); lower pole lateral (reflux) |

- **Concept:** bud position fixes both orifice and kidney, so this single embryological fault (lateral orifice, short tunnel, dysplasia) explains why high-grade VUR and dysplasia coexist.

## Clinical anatomy of the VUJ

- **Course:** juxtavesical ureter in **Waldeyer's sheath**, then an oblique intramural segment through the detrusor hiatus, then the submucosal tunnel, then the orifice (caudal and medial to the entry).
- **Intramural length 1.2–2.5 cm in adults** (0.5–0.8 cm in neonates). The **narrowest part** of the ureter (about 3–4 mm): the commonest stone impaction site.
- **Trigone:** superficial trigone (ureteric longitudinal muscle, running to the verumontanum); deep trigone (Waldeyer's sheath plus detrusor, to the bladder neck). Its base is the interureteric ridge. Waldeyer's sheath anchors the intramural ureter to the deep trigone.
- **Orifice (Lyon):** position A (medial trigone) to C (lateral); shape cone (normal), stadium, horseshoe, **golf-hole**. The more lateral and open, the more reflux.
- **Relations:** the vas (male) or uterine artery (female) crosses the ureter. **Pelvic plexus nerves lie dorsomedial to the VUJ**; wide extravesical dissection causes voiding dysfunction.
- **Clinical:** Hutch diverticulum at the hiatus; "fish-hook" ureter with BPH (benign prostatic hyperplasia).

## Normal antireflux mechanism

- Balance of three factors: **ureteric peristalsis**, **UVJ anatomy** and **bladder dynamics** (low bladder pressure).
- **Passive flap valve:** the submucosal (intramural) ureter is compressed against firm detrusor backing as the bladder fills. This needs adequate tunnel length and fixation.
- **Paquin:** tunnel length to diameter about **5:1** normally versus 1.4:1 in refluxing units. Orifice shape also matters.
- **Active:** ureteric peristalsis opens the tunnel for each bolus, then it closes.
- **Bladder pressure:** high storage or voiding pressure overwhelms a normal valve.

## Etiopathogenesis

**1. Embryology (Mackie and Stephens):** early budding gives a high, lateral orifice with a short tunnel (VUR) and dysplasia; late budding gives a distal, medial, ectopic and often obstructed ureter (see above).

**2. Primary versus secondary**

| Primary | Secondary |
|---|---|
| Congenitally deficient UVJ: short tunnel, weak detrusor backing, lateral orifice | Normal UVJ overwhelmed by pressure: PUV, neurogenic bladder (VUR in up to 30% of spina bifida), dysfunctional voiding; or UVJ anomaly (ureterocele, duplication, paraureteral diverticulum) |

- They overlap: "primary" VUR often has a treatable secondary component.

**3. Bladder and bowel dysfunction (BBD):** high voiding pressures and incomplete emptying distort the UVJ. Detrusor overactivity is the commonest urodynamic finding. Infant boys have high voiding pressures that settle with maturation.

**4. Genetics:** sibling prevalence about 30%, highest under 3 years; identical twins almost fully concordant; polygenic.

**5. Infection:** cystitis lowers the reflux threshold; *E. coli* endotoxin causes ureteric atony.

## Pathogenesis of renal damage

| | Reflux-associated dysplasia | Acquired scar |
|---|---|---|
| Cause | Abnormal bud (congenital) | Infected urine plus VUR |
| Typical patient | Boy, grade IV–V, antenatal hydronephrosis | Girl, recurrent febrile UTI, BBD |
| Preventable | No | Yes |

- **Intrarenal reflux** through flat compound papillae at the poles needs only 2 mmHg in neonates, so scars are polar.
- **Sterile reflux rarely scars** unless pressures are high (Ransley and Risdon). Most scars follow early infections ("big bang", now debated).
- Sequelae: hypertension in **34–38%** of scarred children; CKD; poor growth; pre-eclampsia.
- **Also know:** massive bilateral VUR gives the megacystis–megaureter association.

## International grading (International Reflux Study, 1981; VCUG (voiding cystourethrogram))

| Grade | Findings |
|---|---|
| I | Non-dilated ureter only |

| Grade | Findings |
|---|---|
| II | Ureter, pelvis and calyces reached; no dilatation, sharp fornices |
| III | Mild–moderate dilatation of ureter, pelvis and calyces; minimal forniceal blunting |
| IV | Moderate dilatation and tortuosity; fornices obliterated but **papillary impressions kept** |
| V | Gross dilatation and tortuosity; **papillary impressions lost**; intrarenal reflux |

- **Also know:** grading varies between observers; UDR (ureteral diameter ratio) predicts resolution better.

### Evaluation (children)

- History of febrile UTIs and BBD; urine culture; BP (blood pressure), growth, spine, urine protein, creatinine.
- **RBUS (renal and bladder ultrasound)**: a normal scan does not exclude VUR.
- **VCUG** (gold standard): grade and urethra (excludes PUV). **Nuclear cystogram** (lower dose) for follow-up; **ceVUS (contrast-enhanced voiding urosonography)** avoids radiation.
- **DMSA (dimercaptosuccinic acid) scan**: scars and split function; a defect persisting 6 months is a scar.
- **UDR** predicts resolution and breakthrough UTI better than grade.

### Natural history

- Resolution comes from **elongation of the intravesical ureter** with growth.
- Resolution falls with grade and age: grade II 63–85%, grade III about 50%, **grades IV–V and bilateral grade III rarely** (International Reflux Study). Younger age resolves better; infants with grade IV–V may still resolve fairly well.
- **Bilaterality and older age delay resolution:** persistence is highest in bilateral grade IV (AUA (American Urological Association) 1997: about 10% resolve in 5 years), and in bilateral grade III at 6–10 years.
- **Consequences of bilateral damage:** adult hypertension in 34–38% of scarred children; CKD, with reflux nephropathy a leading paediatric transplant diagnosis; poor growth; pre-eclampsia (worse with bilateral scars). Correction cannot reverse dysplasia.

### Risk stratification (Hidas)

| Low risk | High risk |
|---|---|
| Male, >5 years | Female, <1 year (infant) |
| Unilateral, grade I–II | **Bilateral**, grade III–V |
| No BBD, no scars | BBD, scars, multiple febrile UTIs |

- Two-year breakthrough UTI on CAP (continuous antibiotic prophylaxis): 8.6% low, 26% intermediate, 62.5% high risk.

### Recent trends in management

1. **Treat infection risk, not reflux:** risk stratification by sex, age, grade, laterality, BBD, scars and presentation (Hidas model).
2. **Fewer VCUGs:** ultrasound first (AAP (American Academy of Pediatrics) 2011, NICE (National Institute for Health and Care Excellence)); or "top-down" DMSA first. Contrast-enhanced voiding urosonography avoids radiation.

3. **Objective measures:** UDR predicts resolution and breakthrough UTI better than grade; machine-learning models.
4. **Selective CAP:** RIVUR halved recurrent UTI but did not reduce scarring; NNT (number needed to treat) 5 in high-risk and 18 in low-risk children. The Swedish Reflux Trial showed benefit in girls with dilating VUR.
5. **BBD first:** timed voiding, constipation treatment, anticholinergics. Topical steroid for phimosis in boys.
6. **Endoscopic injection:** Dx/HA (dextranomer/hyaluronic acid); HIT and double HIT (hydrodistension implantation technique) give about 90% resolution. Vantris works better but can obstruct late.
7. **RALUR-EV (robot-assisted extravesical reimplantation):** less pain and shorter stay; results approaching the open benchmark of **95.9%**.
8. **Less follow-up imaging:** VCUG may be omitted after surgery for low-grade VUR if ultrasound is normal.

## Medical management

- **Aim:** prevent febrile UTI and scars while VUR resolves.
- **Treat BBD first:** timed voiding, hydration, constipation treatment, anticholinergics if needed. BBD raises breakthrough UTI and lowers resolution and endoscopic success.
- **CAP:** one low dose at night, a quarter of the treatment dose.

| Agent | Age | Dose | Note |
|---|---|---|---|
| Trimethoprim or amoxicillin | <2 months (young infants) | — | Avoid sulfonamide (neonatal jaundice) |
| TMP-SMX (trimethoprim-sulfamethoxazole) | >2 months | 2 mg/kg TMP | Avoid under 2 months (kernicterus); used in RIVUR |
| Nitrofurantoin | >2 months | 1–2 mg/kg | Avoid in G6PD (glucose-6-phosphate dehydrogenase) deficiency; rare lung, liver toxicity |

- **RIVUR (Randomized Intervention for Children with Vesicoureteral Reflux, 2014)**, 607 children: recurrent UTI fell from **23.6% to 12.9%** on TMP-SMX; **scarring unchanged**; most benefit with BBD and in high risk.
- **Swedish Reflux Trial** (grade III–IV, age 1–2 years): febrile UTI in girls 19% on CAP, 23% after endoscopy, 57% on surveillance; **no difference in boys**.
- **IRSC (International Reflux Study in Children) and Birmingham trials:** surgery equalled medical care for (new) scarring. **PRIVENT:** 6% absolute UTI reduction.
- **Circumcision** or topical steroid for phimosis: about 4 boys with VUR circumcised prevent one UTI (NNT about 4).
- After stopping CAP, 8–10% have a febrile UTI.
- Low-risk children: observation without CAP.
- **Guidelines:** AUA 2010/2017 recommends CAP for VUR with BBD and treating BBD first; EAU/ESPU (European Association of Urology/European Society for Paediatric Urology) give high-risk children CAP and correct early after breakthrough infection.

## Indications for correction

- Breakthrough febrile UTI on CAP despite good bladder and bowel habits.

- New scars or falling function.
- Persistent high-grade (IV–V) bilateral VUR (girls traditionally corrected before puberty; benefit unproven).
- Non-adherence or intolerance to CAP; parental choice.
- **Controversy:** asymptomatic bilateral high-grade VUR in an older boy with normal kidneys can be observed.

## Case: 8-year-old boy with bilateral grade IV VUR

- Exclude PUV, neurogenic bladder and BBD; DMSA, creatinine, BP.
- This rarely resolves at 8 years, but older boys have low UTI risk, so treating an **asymptomatic** boy is **controversial**.
- **No UTI, no scars:** observation or CAP with BBD therapy. **Breakthrough UTI, scars or falling function:** correct surgically.
- **Choice here: bilateral Cohen.** `Exclude PUV/neurogenic bladder, treat BBD -> febrile UTI, scars or renal loss? Yes -> bilateral Cohen (or Dx/HA) No -> observe or CAP; correct if UTI/scar`

## Endoscopic correction: STING procedure

### Definition and principle

- **STING (subureteric Teflon injection):** endoscopic injection of a **bulking agent** beneath the ureteric orifice. The mound supports and lengthens the intramural tunnel, restoring the **flap valve**.
- First done with Teflon by Matouschek (1981); popularised by **O'Donnell and Puri, 1986**. **Dx/HA (Deflux)** is now standard.

### Bulking agents

| Agent | Particle / property | Problem |
|---|---|---|
| Teflon (PTFE (polytetrafluoroethylene) paste) | 90% <40 µm, permanent | **Migration**; never FDA (US Food and Drug Administration) approved |
| **Dx/HA (Deflux)** | Microspheres **80–250 µm**; gel resorbed | Volume loss about 23%; late recurrence; FDA approved 2001; standard agent |
| Polyacrylate (Vantris) | 300 µm, permanent | Works better but late ureteric obstruction |

- **Also know:** Macroplastique.

### Indications

- Persistent primary VUR (grades II–IV) with **breakthrough febrile UTI** on prophylaxis, **new scars**, poor adherence or parental preference.
- Older girls; residual VUR after reimplantation; duplex or secondary VUR (lower success).
- **Relative contraindications:** grade V, **UVJ obstruction**, untreated BBD, large diverticulum.

### Technique

1. Antibiotic at induction; offset-lens cystoscope; rigid 3.7–5 Fr needle; bladder partly filled.
2. **STING:** needle enters **2–3 mm below the orifice at 6 o'clock**, bevel up, advanced **4–5 mm** submucosally.
3. Inject slowly; a mound should appear within **0.1–0.2 mL**, else reposition.

4. Aim for a **"volcano" mound** with a slit-like orifice. Do not cauterise bleeding (the implant extrudes). Day case.

| Technique | Needle entry | Result |
|---|---|---|
| STING | 2–3 mm below the orifice, at 6 o'clock | Baseline |
| **HIT** (hydrodistension implantation technique) | Inside the hydrodistended orifice | Better than STING, all grades |
| **Double HIT** | Two intraureteric sites | About 90% resolution |

### Results

- Low-grade VUR about 90% after one injection.
- **Elder 2006 meta-analysis** (one injection): grade I–II about 79%, III 72%, **IV 63%**, V 51%; repeat injections raise success.
- **Swedish Reflux Trial:** 20% recurrence at 2 years after initial success.
- Less effective in adults.

### Complications

- **Failure or recurrence** (commonest): repeat injection once or twice, then reimplant (reimplantation is not harder after Dx/HA).
- **New contralateral VUR** after unilateral injection.
- **Ureteric obstruction:** early or **late-onset**, so long-term ultrasound.
- Febrile UTI despite a negative VCUG.
- **Calcified mound** (over half) mimics a ureteric stone. Teflon migration.

### Follow-up

- Antibiotics 3 months; ultrasound at 2–6 weeks; ultrasound with or without VCUG at 3 months.

## Ureteric reimplantation: principles and techniques

- **Principles:** exclude secondary VUR; gentle handling; tension-free; straight course; tunnel **5:1** (Paquin's rule).
- Open success **95.9%** overall (Elder 1997; AUA meta-analysis benchmark).

| Technique | Method / tunnel | Strength / weakness |
|---|---|---|
| **Cohen** (intravesical) | Cross-trigonal | Commonest; success over 98%; one cystotomy for both sides; later ureteroscopy or retrograde access hard |
| Politano–Leadbetter (intravesical) | New superomedial hiatus above the old one | Long tunnel; blind hiatus risks bowel injury |
| Glenn–Anderson (intravesical) | Orifice advanced towards the bladder neck | Least kinking; short tunnel |
| **Lich–Gregoir** (extravesical) | Detrusorotomy, ureter laid in | No haematuria or spasm; no cystotomy; **retention up to 10% after bilateral repair** (pelvic plexus injury); stay close to the ureter |

| Technique | Method / tunnel | Strength / weakness |
|---|---|---|
| **RALUR / RALUR-EV** (robot-assisted laparoscopic ureteral reimplantation, extravesical) | Y-shaped detrusorotomy | Less pain, shorter stay; results approaching open |
| **Paquin** | Combined extra- and intravesical, suprahiatal | Long tunnel; for dilated ureters |
| **Gil-Vernet** | Intravesical trigonoplasty, no tunnel | Low-grade bilateral VUR |

### Paquin technique

- **Combined extravesical–intravesical** reimplant described by Paquin (1959), with the 5:1 rule.
- Steps: 1. Ureter mobilised **extravesically** and divided at the bladder. 2. Bladder opened; a **new hiatus** created higher on the posterior wall (**suprahiatal**). 3. Long **submucosal tunnel** from the new hiatus down to the original orifice site, length about 5 × diameter. 4. Ureter drawn through, spatulated, sutured to mucosa and anchored to detrusor. 5. Old hiatus closed.
- **Advantages:** long tunnel under direct vision; good for **dilated ureters and megaureter** (can combine with tapering).
- **Disadvantages:** kinking or obstruction at the new hiatus.

### Gil-Vernet trigonoplasty

- **Intravesical, with no tunnelling and no ureteric mobilisation.**
- Steps: 1. Bladder opened. 2. **Transverse mucosal incision** across the trigone between the two orifices. 3. Intramural ureters with their muscle sheath **advanced medially** by one non-absorbable or absorbable stitch across the midline. 4. Mucosa closed.
- **Effect:** lengthens and medialises the intravesical ureters; corrects both sides at once.
- **Indications:** low to moderate grade (I–III), usually bilateral VUR, with non-dilated ureters.
- **Advantages:** quick, simple, low obstruction risk; easy later retrograde access.
- **Disadvantages:** unsuitable for high grade, dilated ureters or paraureteral diverticulum.

| Feature | Paquin | Gil-Vernet |
|---|---|---|
| Approach | Extra- + intravesical | Intravesical |
| Ureter divided | Yes | No |
| Tunnel | New long submucosal | None (advancement) |
| Best for | High grade, dilated | Low grade, bilateral |

## Criteria for selecting the technique

| Situation | Preferred option |
|---|---|
| Grades II–IV, parents want least invasive | Dx/HA injection (lower success at higher grade) |
| **Bilateral VUR, small or trabeculated bladder** | **Cohen** (both sides through one cystotomy) |
| Grade V, dilated ureter | Intravesical reimplant with or without tapering (Politano–Leadbetter or Cohen) |

| Situation | Preferred option |
|---|---|
| Unilateral, older child | Lich–Gregoir or RALUR-EV |
| Bilateral extravesical | Nerve-sparing dissection; warn about retention |
| Duplex system | Common sheath reimplant |
| Failed injection | Reimplant (not harder after Dx/HA) |
| Neurogenic or PUV bladder (secondary VUR) | Treat the bladder first (anticholinergics, CIC (clean intermittent catheterisation)) |

## Complications and follow-up after reimplantation

- Ultrasound at 6–12 weeks.
- Complications: transient obstruction; persistent reflux (usually resolves by 1 year); contralateral VUR (about 1 in 10, usually resolves); leak; retention after bilateral extravesical repair.
- Obstruction early (oedema): stent or nephrostomy; late (ischaemia, kink): redo.
- Lifelong BP and renal checks if scarred; scarred women risk pre-eclampsia.

## Long-term outcome

| Outcome | Facts |
|---|---|
| **Renal scarring** | About 15% get a febrile UTI within 2 years; about 15% of those scar. Risk highest under 1 year. Much "scarring" in boys with high grade is **congenital dysplasia**, not preventable by any treatment |
| **Hypertension** | Reflux nephropathy may be the commonest cause of childhood hypertension after essential hypertension (renin–angiotensin). **34–38%** of scarred children are hypertensive as adults. Correcting VUR does not lower BP; nephrectomy of a small scarred kidney helps selected cases |
| **CKD and ESKD (end-stage kidney disease)** | Now rare: RRT (renal replacement therapy) from reflux nephropathy fell from 1.9 to 0.6 per million (USRDS (United States Renal Data System), 1996–2014). Bilateral scars cause hyperfiltration, proteinuria, concentrating defect, tubular acidosis |
| **Pregnancy** | VUR in late pregnancy: pyelonephritis about one-third versus under 5%. Scars: about **3× hypertension, 8× pre-eclampsia**; bilateral worse. Impaired function: fetal loss. Reimplanted women still get UTIs |
| Growth; offspring | Catch-up growth after infection control; about 43% of screened offspring have VUR |

- **Concept:** in adult life **the scar, not the reflux, carries the risk**, so correction does not remove it. Scarred patients need lifelong BP, proteinuria and creatinine checks.

## Grade 3 VUR in an adult

- **Grade 3:** mild–moderate dilatation of ureter, pelvis and calyces, minimal forniceal blunting.
- **Adults rarely lose function from reflux alone**, unless high bladder pressure, stasis and UTI coexist. The **scar**, not the reflux, carries the risk. Spontaneous resolution is unlikely.
- **Causes:** persistent primary VUR; **secondary**: BOO (bladder outlet obstruction), neurogenic bladder, resection near the orifice, ureterocele incision, paraureteral diverticulum, transplant ureter, TB (tuberculosis).

- **Presentation:** recurrent febrile UTI, pyelonephritis (especially in pregnancy), loin pain on voiding, hypertension, proteinuria, CKD.
- **Evaluation:** VCUG (grade, bladder, diverticulum, outlet); ultrasound and **DMSA** for scars and split function; MAG3 (mercaptoacetyltriglycine) renogram if obstruction is suspected; **UDS (urodynamics)**: compliance, voiding pressure, residual; cystoscopy; blood pressure, creatinine, urine protein.
- **Treat secondary causes first** (relieve BOO; CIC and antimuscarinic for neurogenic bladder), then reassess.

| Situation (primary grade 3) | Management |
|---|---|
| Incidental; no UTI, no scars | Observe: timed voiding, bowel care, prompt UTI treatment |
| Recurrent febrile UTI, new scars, falling function | **Correct the reflux** |
| Woman planning pregnancy with scars or prior pyelonephritis | Consider correction before pregnancy |
| Poorly functioning scarred kidney with infection or hypertension | Nephrectomy |

- Low-dose prophylaxis selectively; adult evidence is limited.
- **Correction:** endoscopic Dx/HA injection (day case, repeatable, but less effective in adults); or **ureteroneocystostomy** (open, laparoscopic or robotic). Lich–Gregoir/RALUR-EV suits adults (no cystotomy; stay close to the ureter to protect nerves); Politano–Leadbetter (long tunnel); Cohen (over 98%; later retrograde access hard).
- **Psoas hitch** if length is needed; tapering seldom needed in grade 3.
- Follow-up: ultrasound at 6–12 weeks; lifelong blood pressure and renal checks if scarred.

### Overall management algorithm

```
Treat BBD -> low risk: observe | high risk (bilateral): CAP Breakthrough UTI / new scar / persistent IV-V -> Dx/HA injection or reimplant (Cohen) Scars -> lifelong BP and proteinuria checks
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**Diagrams to draw:**

- "Y" of ureteric bud and mesonephric duct with orifice migration (early-budding Y diagram).
- Long oblique tunnel versus short lateral refluxing tunnel; oblique intramural ureter in Waldeyer's sheath with submucosal tunnel and detrusor backing.
- Grades I–V.
- Cohen cross-trigonal reimplant; Gil-Vernet transverse trigonal incision and medial advancement stitch.
- Needle sites: STING below the orifice, HIT and double HIT inside the tunnel, with volcano mound.

**Key points to remember:**

- Low (early) bud → high lateral orifice → short tunnel → VUR plus dysplasia; high (late) bud → ectopia.
- One embryological fault: lateral orifice, short tunnel, dysplasia.
- Tunnel length to diameter 5:1 normal versus 1.4:1 refluxing (Paquin); adult intramural ureter 1.2–2.5 cm.
- Dysplasia (boys, congenital) versus scar (girls, infection); sterile reflux rarely scars.
- Grade V loses papillary impressions; grade IV–V and bilateral grade III rarely resolve; bilateral grade IV persists most; bilateral VUR is high risk.
- RIVUR: CAP halves recurrent UTI, not scarring.
- Treat BBD and give CAP first; correct for breakthrough febrile UTI, new scars or persistent grade IV–V.
- Trend: risk-stratified care, fewer VCUGs, BBD first, endoscopic or robotic correction.
- STING: O'Donnell and Puri 1986; needle 2–3 mm below orifice at 6 o'clock.
- Dx/HA 80–250 µm; HIT better than STING; double HIT about 90%; Elder 2006: grade IV 63% after one injection.
- Watch for late obstruction; calcified mound mimics a stone.
- Open reimplantation 95.9%; Cohen cross-trigonal (over 98%) suits bilateral VUR through one cystotomy.
- Bilateral extravesical (Lich–Gregoir) repair risks retention up to 10%.
- Paquin: tunnel 5:1; combined, suprahiatal reimplant. Gil-Vernet: trigonoplasty, medial advancement, no tunnel; for low-grade bilateral VUR.
- 34–38% of scarred children become hypertensive; correction does not lower BP; scars bring 8× pre-eclampsia risk.
- In adults the scar, not the reflux, carries the risk; exclude secondary causes first; correct for recurrent pyelonephritis, new scars or pregnancy plans.

## LAQ 2 Testicular descent and undescended testis: evaluation, treatment algorithm and orchidopexy methods

P1 · P2 · P3 **merged from 4**

*Asked as:* P1 LAQ "Testicular development, gonadal descent, abnormalities of descent and bilateral undescended testis" (Write a note on the embryological descent of the gonads, explaining the various phases of descent. Outline the evaluation and management of bilateral undescended testis in a 3 year old child, Winter 2024 Paper I Q2; Describe developmental embryology of testes and its physiological descent. Mention various abnormalities associated with testicular descent, Practice Set 14 Paper I Q1) · P3 SAQ "Maldescended (undescended) testes and treatment algorithm" (Treatment algorithm of undescended testis, Summer 2018 Paper III Q3a; Maldescended testes, Practice Set 5 Paper III Q3b) · P2 SAQ "Undescended testis" · P1 SAQ "Methods of orchidopexy"

*Maldescent is taken as any testis not in a stable scrotal position: undescended, ectopic, retractile and acquired.*

### Development of the testis

- **Indifferent gonad:** the gonadal (germinal) ridge forms at **4 weeks** on the medial urogenital ridge. It has four components: coelomic epithelium, underlying mesenchyme, adjacent mesonephros and **PGCs (primordial germ cells)**.
- PGCs arise in the **yolk sac wall in week 3** and migrate along the dorsal mesentery of the hindgut to the ridge (about 6 weeks).
- **SRY (sex-determining region Y)** on the short arm of Y, partly activated by SF1 (steroidogenic factor 1), switches on **SOX9** in pre-Sertoli cells. Sertoli cells then appear (about 40 days). Without SRY no Sertoli cells form and the gonad becomes an ovary.
- Sertoli cells enclose germ cells as **testicular cords (about 6 weeks)**, which canalise into seminiferous tubules.
- **Rete testis** forms with the cords and joins **5–12 mesonephric tubules**, which become efferent ductules. Full continuity (tubules, rete, efferent ductules, epididymis) is established by the **third month**.
- The testis turns round and a thick **tunica albuginea** forms. Testes become spherical after 7–8 weeks; ovaries stay oblong.

| Hormone | Cell | Action | Male remnants |
|---|---|---|---|
| **AMH (anti-Müllerian hormone)** | Sertoli | Müllerian regression, weeks 8–10 | Appendix testis, prostatic utricle |
| **Testosterone** | Leydig (from 8 weeks; driven by placental hCG (human chorionic gonadotrophin), later fetal LH (luteinising hormone)) | Wolffian duct to epididymis, vas, seminal vesicle (weeks 8–12) | Appendix epididymis, paradidymis |
| **DHT (dihydrotestosterone)** | Formed locally by 5α-reductase type 2 | External genitalia, prostate, scrotum (fused by 10–12 weeks) | – |

| Hormone | Cell | Action | Male remnants |
|---|---|---|---|
| **INSL3 (insulin-like peptide 3)** | Leydig | Gubernacular swelling (via RXFP2 receptor) | – |

### Gonadal descent: starting point

- At **7–8 weeks** the testis lies near the kidney (upper lumbar), held by the **cranial suspensory ligament (CSL)** above and the **gubernaculum** below. The gubernaculum appears at 7 weeks, attaches to the **epididymis** and runs to the future inguinal canal.
- **Female:** the CSL persists; the gubernaculum becomes the ovarian and round ligaments; the ovary descends only into the pelvis.

### Phases of testicular descent (classic two-phase model)

| | Transabdominal phase | Inguinoscrotal phase |
|---|---|---|
| Timing | Weeks 10–15 | From the 7th month; inguinal passage rarely before 22 weeks; most testes scrotal after 27 weeks |
| Events | **CSL regresses**; **gubernaculum swells** (outgrowth); its cord shortens, holding the testis at the internal ring while the kidney and abdomen grow away (the kidney moves cranially) | Gubernaculum migrates **3–5 cm** to the scrotum; **processus vaginalis** forms within it; the bulb then regresses; the processus closes |
| Control | CSL regression by **androgen**; swelling by **INSL3** from Leydig cells, via its receptor RXFP2 (occurs even in androgen resistance) | **Androgen**, mainly indirectly via the **genitofemoral nerve (GFN)** releasing **CGRP (calcitonin gene-related peptide)**; intra-abdominal pressure |
| Failure gives | Intra-abdominal testis | Canalicular or ectopic testis |

- **Modern view:** the "transabdominal descent" is largely relative, not true migration. Barteczko and Jacob describe **five phases** (5 weeks to 28 weeks). The gubernaculum **swells to make room** (outgrowth), then **shrinks to let the testis pass** (regression). Testosterone (peak 14–16 weeks) and INSL3 (15–17 weeks) act together on gubernacular and cremaster growth (myogenesis).
- Spontaneous postnatal descent occurs mainly in the first 3–6 months.
- **Also know:** AMH defects (persistent Müllerian duct syndrome) cause cryptorchidism; Insl3-knockout mice are cryptorchid; the scrotum forms by 10–12 weeks under DHT.

| Failed step | Result |
|---|---|
| CSL retained | Intra-abdominal testis |
| Abnormal or ectopic gubernacular attachment | Canalicular testis; ectopic (superficial inguinal pouch, perineal, femoral, prepenile) |
| Processus not closed | Hernia, hydrocele (patent in 87% of unilateral cases) |
| Fibrous remnant of processus / processus not fully involuted | Acquired (ascending) testis; acquired cryptorchidism |

## Abnormalities of descent: definitions and classification

| Type | Description |
|---|---|
| **Undescended (cryptorchid)** | Arrested on the normal path: abdominal, "peeping" at the ring, canalicular, high scrotal/prescrotal |
| **Ectopic** | Off the normal path: **superficial inguinal pouch** (commonest), perineal, femoral, prepubic, penile, contralateral scrotum (transverse, crossed) |
| **Retractile** | Scrotal testis pulled up by the cremaster; stays down after sustained traction; review yearly (risk of ascent); some become acquired |
| **Acquired (ascending)** | Documented scrotal earlier, ascends later; diagnosed at 8–11 years; fibrous processus remnant |
| Recurrent, secondary, iatrogenic | Re-ascent after spontaneous descent; after hernia repair or orchidopexy |
| **Vanishing** vs agenesis | Lost after formation, in utero or perinatally (torsion; monorchia, anorchia) vs never formed |

- Clinically: **palpable** (canalicular, high scrotal, ectopic) or **non-palpable** (intra-abdominal, "peeping", vanished/absent).

## Epidemiology and aetiology

- 1–4% of term boys (P2: 2–4% at birth, 1–2% at 3 months); up to 45% of preterm boys. Most descend by 6 months.
- 75–80% palpable; 60–70% unilateral.
- Descent needs Leydig hormones: **INSL3** (transabdominal phase) and **androgen** (inguinoscrotal phase), with the genitofemoral nerve/CGRP guiding the gubernaculum.
- Risk factors: prematurity, low birth weight, syndromes (prune belly, cerebral palsy).

## Associated anomalies and sequelae

- **Testicular: TTE (transverse testicular ectopia)**, polyorchidism, splenogonadal fusion; epididymal non-fusion.
- **PMDS (persistent Müllerian duct syndrome):** mutation of AMH or its receptor (AMHR2); bilateral cryptorchidism or TTE.
- **Genital and DSD (differences of sex development):** hypospadias (12–24%); bilateral non-palpable testes need karyotype and 17-OHP (17-hydroxyprogesterone) to exclude **CAH (congenital adrenal hyperplasia)**.
- **Syndromes:** prune belly (all), Klinefelter, androgen insensitivity, hypogonadotrophic hypogonadism, cerebral palsy, gastroschisis, omphalocele, myelomeningocele, PUV (posterior urethral valves).
- **Sequelae:** subfertility (worse if bilateral), **TGCT (testicular germ cell tumour) 2–5-fold**, torsion, hernia, trauma.

## Evaluation

- **History:** prematurity, testes once scrotal (acquired), hernia, family history, salt-wasting illness.
- **Examination:** warm room; supine and frog-leg, then cross-legged or squatting. Sweep ("milk") from the anterior superior iliac spine down the canal to the scrotum to find the lowest position; check the superficial inguinal pouch.

- Sustained traction fatigues the cremaster: a **retractile** testis stays in the scrotum after gentle traction.
- Note penis size, meatus (hypospadias), scrotal development, syndromic features. Contralateral hypertrophy suggests a vanished testis.
- **No ultrasound or MRI (magnetic resonance imaging)**: inaccurate (ultrasound sensitivity about 45% for non-palpable testes); they never change the operation or remove the need for surgery, and delay surgery.

### Bilateral undescended testis at 3 years

- This is **late presentation**; surgery is advised from 6 months, preferably by 12 months and by 18 months at the latest.
- **Bilateral palpable:** no investigations needed; plan orchidopexy, both sides usually in one sitting.
- **Bilateral non-palpable (or with hypospadias, small phallus or micropenis): treat as DSD until proven otherwise**, with urgent work-up:
- **Karyotype**; electrolytes and **17-OHP** to exclude **CAH** in a virilised 46,XX child.
- **Hormones:** LH, FSH (follicle-stimulating hormone), testosterone, **AMH and inhibin B**. Measurable AMH or inhibin B confirms testicular tissue; high gonadotrophins with absent AMH suggest **anorchia**. The **hCG stimulation test** (testosterone rise) is now less used.
- Pelvic ultrasound for Müllerian structures.
- **Both testes very high or of doubtful viability:** stage the sides, the second at 6 months, to avoid bilateral atrophy.
- **Biopsy** only if DSD or dysgenesis is suspected. **Bilateral anorchia:** testosterone at puberty, prostheses later.
- Adjuvant LHRH (buserelin) in bilateral cases is experimental.

### Management principles (AUA (American Urological Association) 2014; EAU (European Association of Urology) similar)

- **Observe to 6 months** (corrected age) for spontaneous descent; refer if still undescended.
- **Orchidopexy from 6 months, preferably by 12 months, and by 18 months at the latest.** Reason: germ-cell and Leydig-cell loss starts in infancy; early surgery restores testicular growth and lowers cancer risk. **Kollin RCT (randomised controlled trial):** surgery at 9 months gave better testicular growth than at 3 years.
- **No hormonal therapy** (hCG or LHRH (luteinising hormone-releasing hormone)) to induce descent: descent in about 20% only; hCG may increase germ-cell apoptosis.
- **Goals of surgery:** tension-free **dependent scrotal** testis with intact blood supply; ligate the patent processus.

### Treatment algorithm

```
Boy with empty scrotum ├─ Bilateral non-palpable ± abnormal phallus → DSD work-up (karyotype, hormones, electrolytes) └─ Otherwise → re-examine at 6 months corrected age Still undescended → refer → examination under anaesthesia (EUA) ├─ Retractile → yearly review (risk of ascent) ├─ Palpable (undescended or ectopic) → inguinal or scrotal (Bianchi) orchidopexy └─ Non-palpable → diagnostic laparoscopy ├─ Testis seen → laparoscopic orchidopexy │ vessels short (testis not within 2–4 cm of ring) → Fowler–Stephens │ (clip testicular
```

vessels; testis lives on vasal collaterals; │ one-stage, or two-stage with second stage at 6 months) or Shehata traction ├─ Vessels + vas enter ring → inguinal or scrotal exploration (excise nubbin) └─ Blind-ending vessels → stop (vanished testis)

- **Scrotal nubbin with contralateral hypertrophy** may start with scrotal exploration.
- **Postpubertal intra-abdominal testis with a normal contralateral testis: orchidectomy** (poor function, cancer risk); offer a prosthesis.

### Methods of orchidopexy: palpable testis

- **Inguinal orchidopexy** (steps): 1. Groin incision. 2. Open external oblique (spare the ilioinguinal nerve). 3. Free the cord; divide the gubernaculum. 4. Separate and **ligate the sac (patent processus) at the internal ring**. 5. Retroperitoneal mobilisation; divide lateral bands. 6. **Prentiss manoeuvre** if needed (medial transposition under the inferior epigastric vessels) for extra length. 7. **Subdartos pouch.** Transalbugineal sutures are unnecessary.
- **Trans-scrotal (Bianchi) orchidopexy** for low testes: high scrotal incision; sac ligated through it; inguinal conversion 3.5–4.4%; atrophy 0.3–0.6%; less pain.

### Methods of orchidopexy: non-palpable testis

Examination under anaesthesia, then laparoscopy.

| Method | Principle | Success |
|---|---|---|
| **Primary laparoscopic** | Mobilise vessels, preserve vasal collaterals, new hiatus medial to epigastric vessels | >90% (97%); primary orchidopexy, when possible, is most successful |
| **Open transabdominal (Jones and Bagley)** | Preperitoneal high mobilisation | 95% |
| **Fowler–Stephens** (one- or two-stage) | Divide testicular vessels high; testis survives on **vasal and gubernacular collaterals**; second stage after 6 months | 60–97% (one-stage 63%, two-stage 68%) |
| **Shehata traction** | Fix testis near the opposite anterior superior iliac spine; vessels lengthen; orchidopexy at 12 weeks | 84–85%; atrophy 10% |
| **Microvascular autotransplantation** | Testicular vessels to inferior epigastric vessels | 96% |
| **Orchidectomy** | Postpubertal, dysgenetic or atrophic testis | — |

- Fowler–Stephens is for testes **not within 2–4 cm of the internal ring**.
- **Also know:** staged (two-stage) inguinal orchidopexy; Dessanti PTFE (polytetrafluoroethylene) sleeve traction; gubernaculum-sparing technique; redo approaches (Redman lateral, Cartwright fascial patch).

### Complications of orchidopexy

- **Atrophy** (highest after Fowler–Stephens), retraction or re-ascent, vas or vessel injury, haematoma, infection; torsion very rare after subdartos fixation.

### Follow-up and prognosis

- Check size and position at 6 months. Offer a prosthesis for an atrophic testis.

- **Fertility:** paternity about 80% (four-fifths) after unilateral and one-third after bilateral; worse when bilateral (oligospermia 42%, azoospermia 31% in one large review).
- **Cancer:** 2–5-fold TGCT risk (P1: about 3-fold), higher with late surgery; orchidopexy after 10–11 years carries relative risk 5.8 (Walsh).
- Counsel on fertility and cancer risk; teach **self-examination** after puberty.

**Diagrams to draw:**

- Sagittal fetus showing CSL, gubernaculum, processus vaginalis and both phases of descent.
- Line of descent with ectopic sites.
- Laparoscopic findings at the internal ring: testis, vessels entering the ring, blind-ending vessels.
- Fowler–Stephens collateral supply: testicular artery clipped, deferential and cremasteric arteries intact.

> **Key points to remember:**
>
> - SRY → SOX9 → Sertoli cells (AMH); Leydig cells → testosterone and INSL3.
> - Transabdominal: CSL regression plus INSL3-driven gubernacular swelling.
> - Inguinoscrotal: androgen via genitofemoral nerve (GFN) and CGRP; most testes scrotal after 27 weeks.
> - Ectopic = off the path; commonest site superficial inguinal pouch.
> - Operate from 6 months, preferably by 12 months, and by 18 months at the latest.
> - No imaging, no hormones.
> - Non-palpable: EUA then diagnostic laparoscopy; laparoscopy decides.
> - Bilateral non-palpable: DSD work-up first (karyotype, 17-OHP, AMH) to exclude CAH/DSD.
> - Subdartos pouch; Prentiss manoeuvre gives extra length.
> - Fowler–Stephens relies on vasal collaterals.

## LAQ 3 Retrocaval ureter and IVC anomalous development

P1 · P3 **merged from 2**

*Asked as:* P1 LAQ "IVC anomalous development and retrocaval ureter" · P3 SAQ "Retrocaval ureter"

### Introduction

- The **IVC (inferior vena cava)** forms between the 6th and 8th weeks from three pairs of fetal veins by anastomosis, regression and persistence. The final IVC is **right-sided**, so long left veins must cross the aorta to reach it.
- Most anomalies are silent CT (computed tomography) findings. They **bleed at surgery, mimic lymph nodes, trap the ureter and alter IVC filter placement**.

### Embryology of the IVC

#### Three paired venous systems

- Fetal veins (posterior cardinal, supracardinal, subcardinal) form a venous collar around each ureter and ascending kidney: posterior cardinal and supracardinal veins dorsally, subcardinal veins ventrally.

| Vein | Position | Adult derivatives |
|---|---|---|
| **Posterior cardinal** (first) | Dorsolateral | Iliac confluence; mostly regresses |
| **Subcardinal** | Ventromedial | Right: **suprarenal IVC**; intersubcardinal anastomosis: **left renal vein**; left: adrenal and gonadal veins |
| **Supracardinal** (last) | Dorsomedial | Right: **infrarenal IVC**; thoracic parts: **azygos and hemiazygos** |

#### Segments of the definitive IVC (top to bottom)

1. **Hepatic**: right vitelline vein.
2. **Suprarenal**: right subcardinal vein (via the subcardinal–hepatic anastomosis).
3. **Renal**: subcardinal–supracardinal anastomosis.
4. **Infrarenal**: right supracardinal vein.
5. **Iliac confluence**: posterior cardinal veins.

#### Renal collar

- A venous ring encircles the aorta at renal level. The **ventral (pre-aortic) limb** persists as the left renal vein; the **dorsal (retro-aortic) limb** regresses.
- Normally the right **supracardinal** vein, which lies **dorsal** to the ureter, forms the infrarenal IVC, so the normal ureter lies in front of the IVC.

`Left-sided channels regress -> right-sided persist -> single right IVC + pre-aortic left renal vein`

**Diagram to draw:** three stages: paired cardinal veins, anastomoses with renal collar, definitive IVC.

## Anomalies of IVC development

| Anomaly | Embryological error | Key features |
|---|---|---|
| **Retrocaval ureter** ("preureteral vena cava") | Infrarenal IVC forms from a vein **ventral** to the ureter | Right only; 1 in 1500 autopsies; male 3:1; presents 3rd–4th decade. **Type I** low loop, fishhook, obstructs; **type II** high loop, sickle, mild |
| **Circumaortic left renal vein** | Dorsal limb persists with ventral | Commonest (2–16%); posterior limb runs lower into lumbar IVC |
| **Retroaortic left renal vein** | Ventral limb regresses, dorsal persists | Less common; may be compressed ("posterior nutcracker") |
| **Left IVC** | Left supracardinal persists, right regresses | Crosses to the right at renal level, in front of the aorta |
| **Duplicated IVC** | Both supracardinals persist | Left IVC joins left renal vein; double right cava can trap the ureter |
| **Azygos continuation** | Subcardinal–hepatic anastomosis fails | Large azygos; hepatic veins drain directly to right atrium; polysplenia, heterotaxy |
| **Absent infrarenal IVC** | Supracardinal and posterior cardinal failure | Deep vein thrombosis in the young |

## Retrocaval ureter: definition

- The **right ureter** passes **behind the IVC**, then crosses in front of it from medial to lateral to resume a normal course.
- Better termed "**preureteral vena cava**": the anomaly is venous; the ureter is trapped.
- In P3 the question was also asked as "Retrocecal ureter"; this is taken as a typo for retrocaval (circumcaval) ureter.

## Retrocaval ureter: embryology

- If a vein **ventral** to the ureter persists as the infrarenal IVC, the ureter is trapped behind it.
- Campbell names a persistent **lumbar subcardinal** vein (the lumbar part of the ventral subcardinal vein; Chapter 52) or a **persistent posterior cardinal** vein (Chapter 93, and most texts); both mean that a ventral vein forms the IVC.
- Variant: double right IVC, with the ureter between the limbs.

## Retrocaval ureter: incidence and presentation

- About **1 in 1500** autopsies; **male 3:1**; right side only.
- Presents in the **third–fourth decade** with right flank pain, UTI (urinary tract infection), stones or incidental hydronephrosis.
- Associations: horseshoe kidney, left renal anomalies; bilateral cases occur in situs inversus.

## Retrocaval ureter: classification (Bateson and Atkinson)

| | Type I (low loop), commoner | Type II (high loop) |
|---|---|---|
| Crossing | Lower, at L3–L4 | Higher, at or near renal pelvis level |
| Shape | **J or "fishhook" / reverse-J**, S-curve | Smooth horizontal **sickle** curve |
| Obstruction | Yes, moderate–severe, at the iliopsoas edge | Mild or none (pelvis also crosses) |

## Retrocaval ureter: investigations

- **CT urography (computed tomography urography): investigation of choice**, showing ureter and cava together; MRI (magnetic resonance imaging) is the alternative.
- IVU (intravenous urography): fishhook; ureter not seen beyond the J hook.
- Retrograde pyelography: **S-curve** or fishhook at L3–L4.
- **Diuretic renogram:** grades the degree of obstruction and gives split function.
- Pitfall: mistaken for **PUJ (pelviureteric junction) obstruction**, so pyeloplasty fails.

## Retrocaval ureter: management

- **Non-obstructed (usually type II): observe.**
- **Obstructed: dismembered ureteroureterostomy** (or ureteropyelostomy): divide the ureter, transpose it **anterior to the IVC**, and do a spatulated, tension-free anastomosis over a **double-J stent**. Excise or bypass the **aperistaltic retrocaval segment**; respect and preserve the ureteric blood supply.
- **Laparoscopic or robotic** repair (transperitoneal or retroperitoneal) is now preferred: equal success and faster recovery.
- Nephrectomy if the kidney is non-functioning.
- A pyeloplasty that misses the diagnosis fails.
- **Also know:** retroiliac ureter (L5–S1, with vasal anomalies); mimics of extrinsic ureteric crossing: spermatic / gonadal or lumbar vein, psoas (iliopsoas) tendon.

**Diagram to draw:** right ureter hooking behind the IVC with a fishhook pelvis.

## Clinical implications of IVC anomalies

### 1. Diagnosis

- Venous-phase CT or MRI. A left or duplicated IVC mimics **para-aortic lymphadenopathy**; a large azygos mimics a mediastinal mass, risking needless biopsy.
- Retrocaval ureter: S-shaped or fishhook ureter at L3–L4 on retrograde pyelography; CT urography confirms; diuretic renography grades obstruction.

### 2. RPLND (retroperitoneal lymph node dissection)

- Preoperative CT must look for circumaortic or retroaortic veins and retrocaval ureter. A left or duplicated IVC lies in the left template; a retroaortic limb tears when lumbar vessels are divided.

### 3. Radical nephrectomy

- The hidden posterior limb of a circumaortic vein bleeds when the artery is approached from behind: ligate both limbs. Caval thrombus in a duplicated IVC needs rerouting of the left IVC.

### 4. Living donor nephrectomy

- The left kidney is preferred for its long vein. A circumaortic or retroaortic vein shortens it and adds bleeding risk; CT angiography is mandatory, and the right kidney may be chosen.

### 5. Retrocaval ureter repair

- As above: observe non-obstructed type II; for obstruction, divide, transpose in front of the IVC and re-anastomose over a stent, excising the aperistaltic segment.

### 6. Interventional and vascular

- **IVC filter:** with duplication, emboli bypass a single filter, so filter both cavae or place the filter suprarenally.
- Aortic surgery risks retroaortic vein injury. A retroaortic vein can cause haematuria or left varicocele.
- **Also know:** variable left adrenal vein entry (relevant to adrenal vein sampling); right gonadal vein drains into the right renal vein in up to 10%.

**Key points to remember:**

- Infrarenal IVC = right supracardinal; suprarenal = right subcardinal; hepatic = vitelline.
- Circumaortic left renal vein is commonest (2–16%): ligate both limbs.
- Retrocaval ureter: venous anomaly (persistent subcardinal vein); right side only; 1 in 1500; male 3:1; transpose anterior to IVC.
- Type I fishhook obstructs; type II sickle often does not.
- CT urography; anterior transposition and ureteroureterostomy.
- Duplicated IVC: filter both cavae; do not mistake it for nodes.

## LAQ 4 Hypospadias: present concepts, single-stage vs staged repairs, and hypospadias fistula

P1 · P4 **merged from 2**

*Asked as:* P4 LAQ "Hypospadias: present concepts in management, and hypospadias fistula" · P1 SAQ "Hypospadias repairs – single-stage versus staged techniques"

### Introduction

- Read "present concepts" as the present (current) concepts in the management of hypospadias.
- Hypospadias is an arrest of the androgen-dependent "closing zipper" of urethral fold fusion (weeks 8–16). Triad: ventral ectopic meatus, ventral curvature, dorsal hooded prepuce.
- 1 in 150–300 male births; distal 70–85%, proximal 10–25%.
- **Goals:** a straight erection, a urethra to the tip (standing voiding, semen delivery) and acceptable appearance, with the fewest operations and a result lasting into adulthood.
- The choice of repair depends on **curvature after degloving**, plate quality and glans size, not on meatal position alone.

### Aetiology and associations

- **Aetiology:** causes converge on deficient androgen action (placental insufficiency, assisted reproduction, genes, endocrine disruptors).
- Associations: hernia or hydrocele 9–16%, cryptorchidism about 7% (10% in proximal), enlarged utricle.

### Evaluation

- **DSD (disorder of sex development) work-up** if hypospadias plus undescended testis (especially non-palpable), perineal meatus or ambiguity: karyotype, ultrasound for Müllerian remnants, testosterone, LH (luteinising hormone), FSH (follicle-stimulating hormone), AMH (anti-Müllerian hormone).
- No routine VCUG (voiding cystourethrogram) or renal ultrasound.

### Classification: severity is judged in theatre, not by the meatus

- A "distal" meatus may hide absent spongiosum and severe curvature, so classify **after degloving**.

| GMS (glans–meatus–shaft) score | Glans | Meatus | Shaft curvature |
|---|---|---|---|
| 1 | Good size, deep groove | Glanular | None |
| 2 | Adequate, grooved | Coronal | <30° |
| 3 | Small; narrow or flat plate | Mid or distal shaft | 30–60° |
| 4 | Very small; flat plate | Proximal shaft, penoscrotal | >60° |

- Total 3–12; a higher score predicts more complications.
- **POST (plate objective scoring tool):** a low ratio of neomeatus length to glans fusion length means a poor plate, which favours an inlay graft over plain TIP (tubularised incised

plate).

### Timing and preparation

- Repair at **6–18 months** (AAP (American Academy of Pediatrics): 6–12 months), before genital awareness at about 18 months. One short anaesthetic appears low-risk.
- A straight penis that voids standing may gain little from surgery: counsel honestly.
- **Preoperative androgen** if glans width <14–15 mm: testosterone 2 mg/kg IM (intramuscular) at 5 and 2 weeks before surgery (about 4 mm glans gain). Evidence mixed; two RCTs (randomised controlled trials) showed fewer complications.

### Curvature decides the operation

```
Deglove + excise dysplastic dartos + artificial erection ├─ straight → urethroplasty (single-stage) ├─ residual <30° → dorsal midline plication (12 o'clock nerve-free zone), single-stage └─ >30° → divide plate → still >30° → ventral corporotomy ± graft → staged repair
```

- Recurrence 28% after plication versus 9% after grafting. Under-correction causes repeated surgery, so correct fully first time.

### Choice of technique

| Situation | Current choice |
|---|---|
| Glanular meatus, no curvature | Advancement: MAGPI (meatal advancement and glanuloplasty), MIV (M inverted V) glansplasty; selectively |
| Distal or midshaft, good plate | **TIP (Snodgrass)**: >90% of distal, about 80% of midshaft repairs; Thiersch–Duplay; GAP (glans approximation procedure) for a deep groove |
| Narrow or shallow plate | TIP with **dorsal inlay graft**; onlay island flap; Mathieu (SLAM modification) |
| Proximal, curvature >30° | **Two-stage**: STAG (staged tubularised autograft) or Bracka graft; Byars flaps; distal-first (Ulaanbaatar) |
| Proximal, curvature corrected, good tissue | One-stage onlay or TPIF (transverse preputial island flap); proximal TIP abandoned |

### Single-stage techniques

| Group | Examples | Indication |
|---|---|---|
| Advancement | **MAGPI** (Duckett 1981); MIV | Glanular meatus, no curvature |
| Tubularisation | **TIP** (Snodgrass 1994); Thiersch–Duplay; GAP | Distal–midshaft; TIP in >90% of distal |
| Flap | Mathieu (flip-flap); **OIF** (onlay island flap) | Thin plate |
| Proximal one-stage | TPIF (Duckett tube); proximal OIF | Curvature corrected; good tissue |
| Graft | Dorsal buccal inlay | Narrow plate |

- Pros: one anaesthetic.
- Proximal TIP is abandoned: devascularised plate, stricture, recurrent curvature.

### Staged techniques

- Now the approach of choice for most proximal hypospadias.
- **Rationale (why stage):** a urethroplasty laid over a fresh corporal graft fails more often; stage 1 builds a supple, vascular bed. Stage 2 after **≥6 months**.

| Type | Stage 1 | Stage 2 (≥6 months) |
|---|---|---|
| **Bracka / STAG** (staged tubularised autograft) | Correct curvature, divide plate, quilt inner preputial or buccal graft | Tubularise, glansplasty, tunica vaginalis cover |
| **Byars flaps** | Split dorsal prepuce, transpose ventrally | Tubularise |
| Distal-first (Ulaanbaatar) | Curvature and distal urethra | Join proximal urethra |

- Repeat the artificial erection before stage 2.

## Single-stage versus staged: comparison of outcomes

| Measure | Single-stage | Staged |
|---|---|---|
| Distal success | 83–95% | – |
| Proximal complications | About 60% (long TIP, mixed) | STAG about 23% (lowest); Byars 43–81% |
| Recurrent curvature | Higher after plication | 0–10% |
| Penile length, perception | Lower | Better |
| Extra procedures | Fewer planned | 25–75% need more |

## Operative principles: protect blood supply

- Magnification, tension-free closure, offset suture lines, **vascularised barrier layer** (dartos, tunica vaginalis), rapidly absorbed fine sutures, no hair-bearing skin, haemostasis.
- Postoperative: stent 3–21 days by complexity, compressive dressing, oxybutynin; antibiotic prophylaxis debated.

## Complications and outcomes (standardised)

- Definitions: glans dehiscence (fusion <2 mm); meatal stenosis (symptoms plus <8 Fr before puberty, <12 Fr after); recurrent curvature >30°.
- TIP meta-analysis (Pfistermuller 2015): complications 10.6%, fistula 5.7%. Proximal complications 23–68%; STAG about 23% (lowest); Byars 43–81%.
- Fistula is the commonest complication. A long, rigid, spongiosum-free neourethra raises resistance (Poiseuille's law), causing fistula and diverticulum.
- **Also know:** BXO (balanitis xerotica obliterans): excise and stage with buccal graft; redo repairs use buccal (lip, inside glans) mucosa, Cecil scrotal parasitisation, tissue expansion.

## Hypospadias fistula: definition and incidence

- A UCF (urethrocutaneous fistula) is a urethral leak anywhere below the meatus: the **commonest complication** of hypospadias repair, just under 10% at short-term follow-up.
- After TIP: 5.7% (**Pfistermuller 2015**); after proximal repair 3–45%. Usually at the original meatus or coronal margin.

## Hypospadias fistula: aetiology

- **Distal obstruction** (meatal stenosis, stricture): raises voiding pressure and turbulence; the **key correctable cause**.
- **Ischaemia**, oedema, infection, haematoma, tension.
- **Technical:** overlapping suture lines, poorly inverted epithelium, slowly absorbed sutures, **no barrier layer**.
- **Patient:** proximal meatus, glans <14 mm, redo surgery. A long, rigid neourethra raises resistance (Poiseuille's law).

### Hypospadias fistula: prevention

- **Vascularised barrier** (dartos; **tunica vaginalis** in proximal or redo), offset suture lines, stent in complex repairs.

### Hypospadias fistula: evaluation

- Split stream / multiple streams, usually within 2 years.
- Under anaesthesia: **bougie calibration** for distal narrowing, **urethroscopy** if irregular, and a **stress test** (compress proximally and distally, inject saline or methylene blue) for hidden tracts.

### Hypospadias fistula: management

- Rarely closes spontaneously. **Wait 6–12 months**; **correct distal obstruction at the same sitting**.

| Fistula | Repair |
|---|---|
| Small, shaft | Excise the epithelialised tract, inverting subepithelial closure with fine absorbable suture, multiple offset layers, **dartos barrier** |
| Large or thin tissue | Trap-door or island skin flap plus second layer |
| Coronal margin (glans dehiscence) | Implies distal narrowing or poor tissue: **redo urethroplasty and glansplasty** |
| With diverticulum | Excise, or reuse the redundant wall as a flap |

- Recurrence about **20%**; reduce with interposed vascularised layers (dartos, tunica vaginalis, Cecil modification when tissue is scarce).
- Catheter: none for simple repairs, 7–14 days for complex.
- **Also know:** preputial fistula does not involve the urethra; congenital urethral fistula is closed the same way; follow through puberty.

```
UCF → wait 6–12 months → calibrate, urethroscopy, stress test ├─ distal obstruction → correct at same sitting ├─ small → excise + layered closure + barrier flap └─ large/coronal → flap repair or redo urethroplasty
```

### Follow-up and patient-reported outcomes

- About three-quarters of complications appear by 3 years; stricture and curvature appear late. Follow proximal cases **through puberty**.
- Tools: HOSE (Hypospadias Objective Scoring Evaluation, surgeon), PPS (Penile Perception Score), HOPE (Hypospadias Objective Penile Evaluation, photo panels). Decision aids reduce parental regret.

### Recent advances

- Registries, AI (artificial intelligence) video review, genomic risk profiling, a quality-of-life measure, engineered tissue.

**Diagrams to draw:** curvature algorithm; STAG stage 1 graft and stage 2 tubularisation; fistula repair showing neourethra, dartos barrier and skin with offset suture lines.

**Key points to remember:**

- Classify after degloving (GMS); curvature, not meatus, decides one versus two stages: curvature >30° after degloving means stage.
- TIP for distal; staged graft repair (STAG/Bracka) for severe proximal; STAG has the lowest proximal complications.
- Stage 2 after ≥6 months.
- Hypospadias plus undescended testis: DSD work-up.
- Follow through puberty; use patient-reported outcomes.
- Fistula: commonest complication, just under 10%; TIP 5.7%.
- Exclude distal obstruction; wait 6–12 months.
- Layered closure with barrier flap; recurrence about 20%.

## LAQ 5 Wilms tumour, including bilateral Wilms tumour

P2 · P3 **merged from 2**

*Asked as:* P2 LAQ "Wilms tumour: classification, staging and management" · P3 SAQ "Bilateral Wilms tumour: treatment strategies"

### Introduction

- Read as "Wilms tumour" (nephroblastoma); "Wilm's bladder" in one P2 wording is taken as a typo.
- WT (Wilms tumour) is an embryonal tumour of metanephric blastema. It forms 95% of kidney cancers under 15 years; median age 3.5 years; over 80% present before 5 years.

### Genetics, syndromes and precursors

- **Genetics:** mostly sporadic; familial 1–2%. Genes: **WT1 (11p13)**, IGF2/11p15 (commonest change, about 70%), WTX, CTNNB1; **TP53 in anaplasia (in most anaplastic tumours)**.

| Syndrome | WT risk |
|---|---|
| **WAGR** (Wilms, aniridia, genitourinary anomalies, intellectual disability / retardation; WT1 deletion) | 50% |
| **Denys–Drash** (46,XY DSD [disorder of sex development], mesangial sclerosis; WT1 point mutation) | 50% |
| **BWS (Beckwith–Wiedemann syndrome)**, hemihypertrophy (11p15) | 5–10% |

- **Screening:** ultrasound every 3–4 months until 7 years when risk is >1%.
- **Nephrogenic rests** (precursors) occur in over one-third of WT kidneys and over 90% of bilateral cases.

### Classification: histology

| Group | Features | Significance |
|---|---|---|
| **Favourable histology** | **Triphasic:** blastema, epithelium (tubules), stroma; pseudocapsule | Survival >90% |
| **Anaplasia** (unfavourable) | Nuclei ≥3× neighbours (larger), hyperchromasia, abnormal mitoses; **focal** or **diffuse** (diffuse worse than focal) | Chemoresistant; poor even at stage I |

- **SIOP (International Society of Paediatric Oncology) post-chemotherapy risk groups:**

| Risk | Histology |
|---|---|
| Low | Completely necrotic |
| Intermediate | All others (regressive, mixed, epithelial, stromal, focal anaplasia) |
| High | **Blastemal-predominant**; diffuse anaplasia |

- Residual blastema predicts relapse; stromal and epithelial tumours shrink little but are cured by excision.

### Staging

#### COG (Children's Oncology Group): surgical-pathological, before chemotherapy

| Stage | Description |
|---|---|
| I | Confined to kidney, completely resected, capsule intact, no renal sinus vessel invasion |
| II | Beyond kidney (renal sinus vessels, perirenal fat, vein thrombus) but completely resected |
| III | Residual abdominal disease: **positive nodes, any spill, prior biopsy**, positive margin, peritoneal implants |
| IV | Haematogenous metastases (lung commonest, liver, bone, brain) |
| V | Bilateral at diagnosis (each side staged separately) |

- **SIOP staging** uses the same five stages but is applied **after preoperative chemotherapy**. Necrotic, chemotherapy-induced change in the sinus or perirenal fat does not upstage.
- **Biology:** combined **1p and 16q LOH (loss of heterozygosity)** predicts relapse; 1q gain is the next stratifier.

## Clinical features

- Abdominal mass; haematuria about 20%; hypertension about 25% (renin).
- Look for aniridia, hemihypertrophy and varicocele. Varicocele or hepatomegaly suggests IVC (inferior vena cava) thrombus.
- Acquired von Willebrand disease in up to 8%.

## Investigations

- Ultrasound with Doppler (IVC), then contrast CT (computed tomography) or MRI (magnetic resonance imaging) abdomen, then **chest CT** (lung is the commonest metastatic site; do it before anaesthesia).
- Blood count, renal and liver function, coagulation; syndrome review.

## Management: two philosophies

| | COG (North America) | SIOP (Europe) |
|---|---|---|
| Sequence | **Upfront transperitoneal radical nephrectomy**, then chemotherapy | **Preoperative VCR + AMD** (4 weeks; add DOX for 6 weeks if metastatic), no biopsy, then nephrectomy |
| Advantage | Untreated histology and biology for risk assignment | Tumour shrinks; spill almost zero |
| Risk basis | Stage + histology + LOH | Post-chemotherapy histology + stage |

(VCR = vincristine; AMD = actinomycin D (dactinomycin); DOX = doxorubicin; RT = radiotherapy; NSS = nephron-sparing surgery.)

## Surgical principles (radical nephrectomy)

- Transperitoneal approach; palpate renal vein and IVC before ligation; spare a tumour-free adrenal.
- **Avoid spill** (it makes the tumour stage III).
- **Sample at least 5 lymph nodes** (hilar, paracaval, para-aortic, interaortocaval); this is essential, formal dissection is not. Omitted sampling is the commonest protocol violation.
- IVC thrombus at or above the hepatic veins: preoperative chemotherapy to avoid bypass.

## Postoperative therapy (COG, favourable histology)

| Group | Treatment |
|---|---|
| Very low risk (<2 years, <550 g, stage I) | Nephrectomy alone (AREN0532) |
| Stage I–II | VCR + AMD |
| Stage III | VCR + AMD + DOX + flank RT |
| Stage IV | Three drugs + whole-lung RT; lung RT omitted if lung nodules clear by week 6 (AREN0533) |
| Anaplasia or 1p/16q LOH | Intensified regimens (regimen M adds cyclophosphamide + etoposide) |

## Bilateral WT (stage V): principles

- **Synchronous bilateral WT** is **5–7%**: **COG stage V**. Each kidney is staged separately.
- Linked to WAGR, Denys–Drash (about 50% risk) and BWS; **nephrogenic rests in over 90%**.
- Aim: **cure while preserving renal tissue (nephrons)**. Renal failure is about 15% at 15 years; ESRD (end-stage renal disease) mainly follows bilateral nephrectomy.
- **No upfront biopsy and no upfront nephrectomy.**

## Bilateral WT: COG approach (AREN0534)

```
VCR + AMD + DOX × 6 weeks (no biopsy) → imaging at week 6 ├─ resectable / NSS feasible → bilateral NSS └─ not → 6 more weeks chemotherapy → surgery by week 12 poor response → bilateral OPEN biopsy → further therapy by each kidney's stage and histology ± flank RT
```

- **Why stop at 12 weeks:** in NWTS-4 (National Wilms Tumor Study 4), prolonged chemotherapy in poor responders delayed surgery and some tumours became anaplastic. A non-shrinking tumour is often differentiated and is cured by excision.
- **Why an open, bilateral biopsy:** anaplasia is discordant between kidneys in about 4 of 5.

## Bilateral WT: surgery

- **Bilateral NSS** (partial nephrectomy or enucleation); frozen section; lymph node sampling; avoid spill.
- Compressed normal parenchyma is often underestimated, so try NSS in every case. Expert centres achieve bilateral NSS in over 90%.
- Enucleation: positive margins 23–36%, good outcome with flank radiotherapy.
- Unresectable both sides: bilateral nephrectomy and dialysis are rare and only for refractory tumour; transplant after **1–2 years** tumour-free.

## Bilateral WT: adjuvant therapy and SIOP approach

- Adjuvant therapy is guided by the worst histology and margins: anaplasia or positive margin means intensified chemotherapy and flank radiotherapy.
- **SIOP:** same principle (no biopsy, NSS by week 12) but **two drugs, VCR + AMD**, if non-metastatic.

## Unilateral nephron-sparing surgery

- For syndromic children and solitary kidney; SIOP UMBRELLA criteria are volume <300 mL, a substantial remnant and negative nodes.

## Prognosis and outcomes

- Favourable histology: survival >90%.
- **AREN0534 (bilateral):** surgery by week 12 in 84%; bilateral NSS 39%; **4-year EFS (event-free survival) 82.1%, OS (overall survival) 94.9%**, versus 56% and 80.8% on NWTS-5.
- Salvage survival after relapse: 70–80% after VCR/AMD only; about 10% for anaplastic or blastemal relapse.

## Late effects and follow-up

- **Late effects:** cardiotoxicity (DOX), second cancers (RT), infertility, renal failure (bilateral).
- **Follow-up (bilateral):** lifelong blood pressure, proteinuria, creatinine (renal function) and imaging, as late relapse occurs.

**Diagrams to draw:** COG staging diagram showing capsule, renal sinus, nodes and spill; bilateral tumours with planned partial nephrectomies; AREN0534 timeline (weeks 0, 6, 12).

> **Key points to remember:**
>
> - WT1 at 11p13 (WAGR, Denys–Drash); BWS at 11p15.
> - Anaplasia (≥3× nuclei, hyperchromasia, abnormal mitoses) is unfavourable; TP53.
> - Any spill, biopsy or positive node = stage III; sample ≥5 nodes.
> - COG operates first; SIOP gives chemotherapy (VCR + AMD) first.
> - Bilateral: stage V, 5–7%; no upfront nephrectomy or biopsy; three drugs for 6 weeks (reassess at 6 weeks), surgery by week 12, NSS; SIOP uses two drugs.
> - AREN0534: 4-year OS about 95%; bilateral NSS to avoid ESRD.

# LAQ 6 Proximal (perineoscrotal) hypospadias: penile embryology and management including Koyanagi repair

P2 · P3 **merged from 2**

*Asked as:* P3 LAQ "Embryology of the penis and management of proximal hypospadias (including Koyanagi repair)" · P2 SAQ "Perineo-scrotal hypospadias"

## Embryology of the penis

### Indifferent stage (up to 8–9 weeks)

- Cloacal folds form the **genital tubercle**; after the cloaca divides (7 weeks), **urethral folds** flank the urogenital membrane, with **labioscrotal swellings** laterally. Both sexes look identical until about 9 weeks.

### Hormonal control

- SRY (sex-determining region Y) drives the testis, whose **Leydig cells (about 8 weeks)** secrete testosterone.
- **Type 2 5α-reductase** in the tubercle converts it to **DHT (dihydrotestosterone)**, which acts on androgen receptors. **DHT at 8–12 weeks** is the key mediator; the critical window is weeks 8–16. Without androgen action the default is female.
- First sign: longer anogenital distance, then tubercle elongation (sonic hedgehog-driven).

| Primordium (germ layer) | Male | Female |
|---|---|---|
| Genital tubercle (mesoderm: corpora; ectoderm: skin, prepuce) | Glans, corpora | Clitoris |
| Urethral folds | Fuse to form penile urethra, median raphe | Labia minora |
| Labioscrotal swellings | Scrotum | Labia majora |
| Urethral plate (endoderm) | Whole urethra, including glanular | Vestibule |

### Urethral formation: the "double zipper"

1. **Opening zipper** (both sexes): the urethral plate canalises proximal to distal into a urethral groove.
2. **Closing zipper** (androgen-dependent, male only): the folds fuse in the ventral midline, forming the tubular urethra and raphe. - The **glanular urethra** forms by direct canalisation of the endodermal plate; Glenister's ectodermal ingrowth theory (1954) is outdated. - Timetable: coronal sulcus 11–12 weeks; shaft fused by 16 weeks; complete by about 18 weeks; fetal ventral curvature gone by 20 weeks. - **Prepuce** fuses dorsally first; ventral fusion awaits urethral closure, hence the **dorsal hood** in hypospadias. - Dorsal nerves run at 11 and 1 o'clock, leaving a **12 o'clock nerve-free zone** (safe plication site). - **Hypospadias = arrest of the closing zipper**: the meatus stops where closure stopped, and unfused dysplastic ventral tissue tethers the shaft into curvature.

**Diagram to draw:** double-zipper urethral formation.

## Definition of proximal (perineoscrotal) hypospadias

- Meatus on the proximal shaft, at the penoscrotal junction, scrotum or perineum (10–25% of hypospadias).

- Perineoscrotal hypospadias is a severe form, with severe ventral curvature, a short penis, bifid scrotum and often **penoscrotal transposition**.
- Cause: failed androgen-dependent urethral fold fusion (8–16 weeks).

### Evaluation

- Severity is judged **after degloving**: meatus, plate and curvature (GMS (glans–meatus–shaft) score).
- Examine stretched penile length, glans width, penoscrotal transposition, bifid scrotum and **both testes**.
- **DSD (disorder of sex development) work-up** if hypospadias plus an undescended testis (especially non-palpable), perineal meatus or ambiguity: karyotype, ultrasound for Müllerian remnants, testosterone, LH (luteinising hormone), FSH (follicle-stimulating hormone), AMH (anti-Müllerian hormone).
- DSD risk is 64% in proximal versus 7% in distal cases. A 46,XY karyotype can still hide 5α-reductase deficiency or androgen insensitivity.
- Prostatic utricle is commoner; cystoscopy in severe cases. No routine VCUG (voiding cystourethrogram) or renal ultrasound.

### Preparation

- **Timing:** 6–18 months (AAP (American Academy of Pediatrics): 6–12 months).
- **Preoperative androgen** if glans width <14–15 mm: e.g. testosterone 2 mg/kg IM (intramuscular) at 5 and 2 weeks before surgery (about 4 mm glans gain). Side effects are transient; evidence is mixed, but two RCTs (randomised controlled trials) showed fewer complications.

### Curvature correction (decides one- versus two-stage)

```
Deglove + excise dysplastic dartos + artificial erection ├─ straight → urethroplasty ├─ residual <30° → dorsal midline plication (12 o'clock) └─ >30° → divide urethral plate └─ still >30° → ventral corporotomy ± graft → staged repair
```

| | **Dorsal plication (Nesbit, Baskin)** | **Ventral lengthening** |
|---|---|---|
| Method | Shortens the convex dorsal side | 1–3 transverse corporotomies, or one full-thickness 9-to-3 o'clock corporotomy with dermis, tunica vaginalis or SIS (small intestinal submucosa) graft |
| Use | Mild (≤15–30°) | Moderate–severe |
| Recurrence | 28% | 9% |

**Diagram to draw:** curvature algorithm.

### Urethroplasty options

| **Approach** | **Technique** | **Comment** |
|---|---|---|
| One-stage **proximal TIP (tubularised incised plate)** | Plate kept, incised and tubularised | Recurrent curvature, stricture: largely abandoned for severe cases |
| One-stage **onlay island flap** | Inner preputial flap on preserved plate | Diverticulum, fistula; higher complications |

| Approach | Technique | Comment |
|---|---|---|
| One-stage **TPIF (transverse preputial island flap, Duckett tube)** | Tube after plate division | High complications |
| One-stage **Koyanagi** | Meatus-based parameatal foreskin flap (see below) | No proximal circular anastomosis |
| **Two-stage Bracka / STAG (staged tubularised autograft)** | Stage 1: correct curvature, divide plate, quilt inner-prepuce or **buccal graft**. Stage 2 (≥6 months): tubularise, glansplasty, barrier layer | **Preferred for most proximal cases**; STAG lowest complications (about 23%) |
| **Two-stage Byars** | Dorsal preputial flaps moved ventrally, tubularised later | Supple flap bed; complications 43–81% |

- **Why stage?** A urethroplasty laid over a fresh corporal graft fails more often; staging gives a supple, vascular bed.
- Principles: magnification, tension-free closure, offset suture lines, **vascularised barrier layer** (dartos, tunica vaginalis), fine absorbable sutures, no hair-bearing skin. Stent, compressive dressing, oxybutynin; **scrotoplasty** for transposition.

**Diagram to draw:** two-stage graft repair: stage 1 graft bed, stage 2 tubularisation.

## Koyanagi repair: definition and principle

- **One-stage repair of proximal hypospadias** with a **meatus-based parameatal foreskin flap** (Koyanagi). Skin strips round the meatus continue into the inner prepuce on both sides; they are swung ventrally and tubularised.
- **Principle:** the neourethra stays attached to the native meatus, so there is **no circular proximal anastomosis**, and the urethral plate can be divided to correct severe chordee.

## Koyanagi repair: indications and contraindications

- **Indications:** proximal hypospadias (penoscrotal, scrotal, perineal) with **severe curvature** needing plate transection, and an **ample, unoperated prepuce**.
- **Contraindications:** prior circumcision or failed repair; small glans (give preoperative testosterone); curvature needing a ventral corporal graft (stage instead).

## Koyanagi repair: procedure

`Stay suture; artificial erection → incise round the meatus, along both sides of the plate to the corona and round the inner prepuce (meatus-based "wings") → deglove; divide the plate; excise dysplastic dartos → residual curvature <30° → dorsal midline plication → split the dorsal prepuce; bring both wings ventrally on dartos pedicles → join inner edges, then outer edges, over a catheter → tube → glanuloplasty, meatus at the tip → dartos barrier layer; skin cover ± scrotoplasty → stent 7–10 days; compressive dressing`

- Modified and "neo-modified" versions add flap fixation and barrier layers to reduce fistula and diverticulum.

**Diagram to draw (Koyanagi):** meatus-based foreskin wings split dorsally and joined ventrally into a tube.

## Koyanagi repair: complications

| Complication | Cause |
|---|---|
| **Fistula** (commonest) | Long suture line; repair ≥6 months later |
| Meatal stenosis, stricture | Distal flap ischaemia |
| **Diverticulum** | No spongiosum; flap not fixed to the shaft |
| Glans dehiscence | Tension |
| Residual or recurrent curvature | Plication alone in severe cases |
| Flap necrosis, infection, poor cosmesis | Pedicle injury |

- Because proximal repairs overall have **23–68%** complications, many now prefer **two-stage repair** (STAG, Bracka).

## Complications and outcomes of proximal repair

- Complications **23–68%**; 25–75% need further surgery after planned staging.
- **Fistula** is the commonest (3–45%; repair ≥6 months later; exclude distal obstruction).
- Others: glans dehiscence, meatal stenosis (<8 Fr before puberty), stricture, **diverticulum** (no spongiosum), **recurrent curvature >30°**, BXO (balanitis xerotica obliterans), hair-bearing urethra.
- A long, rigid neourethra raises resistance (Poiseuille's law), causing fistula and diverticulum.

## Follow-up

- Follow through puberty; patient-reported outcomes (PPS (Penile Perception Score), HOPE (Hypospadias Objective Penile Evaluation) score).

> **Key points to remember:**
> - DHT at 8–12 weeks; opening then closing zipper; 12 o'clock nerve-free zone.
> - Hypospadias (including perineoscrotal) plus undescended testis: DSD work-up.
> - <30° plication; >30° divide plate ± ventral graft and stage.
> - STAG/Bracka staged repair preferred; fistula commonest complication.
> - Koyanagi: one stage; meatus-based parameatal foreskin flap; plate divided; no proximal circular anastomosis.
> - Koyanagi: fistula commonest; diverticulum, meatal stenosis.

# LAQ 7 Urological evaluation and management of pediatric kidney transplant patient.

P1

**Type:** LAQ (25 marks)
**Source:** 2022-Summer Paper 1, Q2

## Introduction

- Children needing KRT (kidney replacement therapy) often have CAKUT (congenital anomalies of the kidney and urinary tract). The urologist must ensure that the bladder that destroyed the native kidneys does not destroy the graft.
- **Five goals:** unobstructed drainage; a capacious, low-pressure reservoir with social continence; volitional emptying; no infection; the fewest operations.

## Causes of ESKD (end-stage kidney disease) in children

- Commonest age at transplant 13–17 years.
- **Younger:** dysplasia, hypoplasia, aplasia and obstructive uropathy, mainly PUV (posterior urethral valves). **Older:** FSGS (focal segmental glomerulosclerosis) and neuropathic bladder.

## Pretransplant evaluation

- **Every child:** urological history, renal ultrasound, uroflow with PVR (post-void residual) if toilet-trained.
- **VCUG (voiding cystourethrogram)** only with neuropathic bladder, PUV, VUR (vesicoureteral reflux), previous lower-tract surgery, recurrent febrile UTI (urinary tract infection) or unexplained hydronephrosis.
- **Urodynamics** for neuropathic bladder, PUV or large ureterocele, refractory voiding dysfunction, high-grade hydronephrosis or recurrent UTI. They measure capacity, compliance, detrusor activity, emptying and sphincter function, and are repeated.

```
History positive → ultrasound + VCUG → urodynamics Poor capacity/compliance → anticholinergic, CIC, cycling → still poor → augment Poor emptying → CIC ± catheterisable stoma Adequate → clear for transplant History negative → ultrasound + PVR → normal → clear
```

- An anuric child with no urological history needs no further tests.

## Pretransplant management

### 1. Bladder optimisation (drugs → CIC → cycling → augmentation)

- Commonest problem: low capacity, detrusor overactivity, poor compliance. High pressure harms the kidney even without reflux.
- **Target:** age-expected capacity at detrusor pressure **<40 cm $H_2O$**.
- Antimuscarinics (oxybutynin titrated; solifenacin), mirabegron, intradetrusor botulinum toxin; α-blockers aid emptying. Better storage may worsen emptying, so add **CIC (clean intermittent catheterisation)**. Prove volitional emptying by repeated low PVRs.
- A family's adherence to CIC predicts adherence to immunosuppression.

### 2. Valve bladder versus neuropathic bladder

- PUV bladder moves from overactivity to **myogenic failure**: postpone augmentation. Neuropathic bladder may deteriorate under high post-transplant urine output: reconstruct

earlier.

3. **Defunctionalised bladder** (anuria or diversion)

- **Bladder cycling:** catheter instillation of increasing saline volumes, with drugs, until age-expected capacity; confirm by urodynamics. **Never use the graft to test the bladder.**

4. **Augmentation**

- Indication: cannot reach **>75% expected capacity at <40 cm $H_2O$** with CIC every 3–4 hours despite maximal therapy. It does not clearly raise transplant complications.

| Segment | Point |
|---|---|
| Dilated ureter (ureterocystoplasty) | Explore first: urothelium, no mucus or acidosis |
| Ileum, sigmoid | Most used; mucus, stones, infection, malignancy; lifelong follow-up |
| Stomach | Haematuria–dysuria syndrome in anuric patients; rarely used |

- Continent stoma (Mitrofanoff) if the urethra is sensate or difficult; site it medial to the planned incision. Allow **6–12 weeks** healing before transplant.

5. **Infection, nephrectomy and dialysis**

- Treat stasis (double voiding, CIC); distinguish colonisation from UTI; consider circumcision in boys with valves.
- **Native nephrectomy** (default is to keep): uncontrolled hypertension, nephrotic syndrome, massive polyuria, recurrent pyelonephritis, large stones, malignancy risk, very large kidneys, massive VUR. Retroperitoneal route keeps PD (peritoneal dialysis).
- Keep a normal native ureter for later ureteroureterostomy. Intraperitoneal surgery on PD needs temporary haemodialysis.

## At transplantation

- Retroperitoneal Gibson incision; children **<15 kg** may need aorta–caval anastomosis, sometimes transperitoneal.
- **Ureter:** extravesical non-refluxing ureteroneocystostomy (3–4 cm detrusorotomy) preferred; Barry tunnel; ureteroureterostomy to a normal native ureter.
- **Stent:** selective; double-J removed at 4–6 weeks, or early removal at 5–7 days.

## Post-transplant complications

| Complication | Rate | Management |
|---|---|---|
| Ureteric obstruction | 6% | Usually distal ischaemic stenosis; >half within 100 days. Stent or nephrostomy; balloon dilatation; open repair (native ureter, bladder flap) |
| Urine leak | 3% | Early; drain creatinine; catheter, nephrostomy, stent, then reimplantation |
| Graft VUR | 2% (significant) | Febrile UTI → reimplant; asymptomatic → observe; injection works in one-third |
| Stones | 2% | Treat if >1 cm and growing, obstructing or infected |

- **Rising creatinine with hydronephrosis:** catheter drainage 1–2 weeks (bladder cause?) → biopsy if rejection is suspected → stent or nephrostomy. Obstruction and rejection can coexist.

- Also: lymphocele, **PTLD (post-transplant lymphoproliferative disorder)**, graft thrombosis (a leading cause of early graft loss), late infection reflecting bladder dysfunction.

### Follow-up

- Lifelong ultrasound, PVR and bladder surveillance; recurrent UTIs lower graft GFR (glomerular filtration rate).

**Diagram to draw:** extravesical ureteroneocystostomy with detrusor flaps closed over the ureter.

> **Key points to remember:**
>
> - Bladder target: age-expected capacity at <40 cm $H_2O$; augment if <75% despite CIC.
> - PUV bladder heads to myogenic failure; neuropathic bladder needs earlier reconstruction.
> - Never test a bladder with a graft; cycle the defunctionalised bladder.
> - Stenosis 6%, leak 3%, stones 2%, significant reflux 2%.

## LAQ 8 Embryology and management of ectopic ureter.

P3

**Type:** LAQ (25 marks)
**Source:** 2018-Summer Paper 3, Q1

### Definition

- An **ectopic ureter** is any ureter, single or duplex, that does **not open on the trigone**. True ectopia means an orifice **at or distal to the bladder neck**.
- Commoner in girls. In girls most ectopic ureters drain the **upper pole of a duplex kidney**; in boys most are **single-system**.

### Embryology

- **Week 4–6:** the **ureteric bud** arises from the **mesonephric (wolffian) duct** and induces the metanephric blastema to form the kidney.
- The duct segment distal to the bud is absorbed into the **urogenital sinus**; the ureteric orifice then moves cranially and laterally, the wolffian orifice caudally to the verumontanum.
- **Mackie–Stephens bud theory:**

| Bud position on wolffian duct | Absorption | Orifice | Result |
|---|---|---|---|
| Too caudal | Early | High and lateral, short tunnel | VUR (vesicoureteral reflux) |
| Too cranial | Late or never | Low and medial (bladder neck, urethra) or stays on wolffian duct | **Ectopic ureter**, often obstructed |

- A bud from the wrong place meets poor blastema, so **renal dysplasia increases with the degree of ectopia**.
- **Duplex system:** the upper-pole bud is more cranial, absorbed later, and ends **lower and more medial** (**Weigert–Meyer law**): upper pole obstructs; lower pole refluxes.

### Sites of ectopia (the key sex difference)

| | Female | Male |
|---|---|---|
| Sites (commonest first) | Urethra, vestibule, vagina; rarely cervix, uterus, rectum (via **Gartner duct**) | Posterior urethra, seminal vesicle; also vas, ejaculatory duct |
| Relation to sphincter | Can open **below** the external sphincter | Always **above** the external sphincter |
| Continence | **Continuous dribbling with normal voiding** | **Never incontinent** |
| Typical presentation | Constant dampness, perineal pus, UTI (urinary tract infection) | **Epididymo-orchitis**, UTI, flank pain, hydronephrosis |

- **Zinner syndrome:** ectopic ureter into the seminal vesicle with ipsilateral renal agenesis or dysplasia and a seminal vesicle cyst.

### Clinical features

- **Prenatal hydronephrosis**: most are detected this way.

- **Girl:** wet day and night despite normal voids; she is never dry for 30–60 minutes. Often misdiagnosed as voiding dysfunction.
- **Infection:** low-grade fever with spikes; urine culture may be negative because the infected moiety does not drain into the bladder.
- **Boy:** recurrent epididymitis in a young boy demands an upper-tract ultrasound.
- **Also know:** pain then perineal discharge; bilateral single-system ectopia with small bladder and wide bladder neck.

## Evaluation

| Test | What it shows |
|---|---|
| **Ultrasound** (first) | Dilated upper pole and ureter behind the bladder; parenchyma. Bladder view separates it from ureterocele |
| **VCUG** (voiding cystourethrogram) | Lower-pole or contralateral VUR ("drooping lily"), reflux into the ectopic ureter |
| **DMSA** (dimercaptosuccinic acid) | Gold standard for moiety function ("some" versus "none"); MAG3 (mercaptoacetyltriglycine) if drainage needed |
| **MR (magnetic resonance) urography** | Non-dilated upper pole, small dysplastic kidney, complex anatomy |
| **Examination / cystovaginoscopy** | Perineal orifice between urethra and vagina, cannulated for contrast |

- **Also know:** clear urine leaking despite methylene blue in the bladder suggests an ectopic ureter; CT for a small unseen kidney.

## Management

**Goals:** preserve renal function; remove infection, obstruction and reflux; achieve continence; with least morbidity. Start antibiotic prophylaxis in the neonate with significant dilation.

`Neonate with sepsis or massive dilation → end cutaneous ureterostomy (or nephrostomy); DMSA later DUPLEX system Upper pole non-functioning → upper-pole heminephroureterectomy Upper pole worth keeping → ureteroureterostomy (UU) / pyeloureterostomy + lower-pole VUR → common sheath reimplant (or lower-pole reimplant + distal UU) SINGLE system Functioning → ureteral reimplant (± tapering) Non-functioning → nephroureterectomy (laparoscopic) Bilateral → bladder neck reconstruction ± augmentation`

- **Upper-pole heminephroureterectomy** (open, laparoscopic or robotic): upper-pole ureter as a handle, passed behind the hilum; ligate polar vessels; excise the pole; close over fat. Dissect on the ureter wall to protect the **lower-pole blood supply**. The stump is rarely a problem.
- **Ureteroureterostomy (UU):** end-to-side upper into lower pole ureter, distal (inguinal incision) or proximal (preferred with big size mismatch). **Stent the recipient lower-pole ureter** cystoscopically for identification. Now used even for non-functioning poles; feared hypertension has not been shown.
- **Common sheath reimplantation:** do not separate the two ureters (they share blood supply).

### Complications

- Loss of lower-pole function (most serious); urinoma (up to 20% after minimally invasive partial nephrectomy, mostly harmless); stump infection; new reflux; persistent incontinence.

### Recent advances

- Robotic UU and heminephrectomy; single-site surgery; non-dismembered side-to-side ureterovesicostomy in infants.

**Diagram to draw:** duplex kidney showing Weigert–Meyer orifices, and female versus male ectopic sites relative to the external sphincter.

> **Key points to remember:**
> - Ectopia = orifice at or below the bladder neck; bud too cranial on the wolffian duct.
> - Girls: continuous dribbling with normal voiding; boys: never incontinent, epididymitis.
> - DMSA decides keep (UU/reimplant) versus remove (heminephroureterectomy).
> - Protect the lower-pole blood supply; stent the recipient ureter for UU.

## Short answer questions (SAQ)

## SAQ 1 TIP (Snodgrass) repair for hypospadias

P1 · P3 **merged from 2**

*Asked as:* P3 SAQ "Snodgrass (TIP) repair for hypospadias, including extended indications" · P1 SAQ "TIP repair in hypospadias"

### Introduction

- "Shodgrass" in one P3 wording is taken as the Snodgrass TIP repair, and "distal hypos." as distal hypospadias.
- **TIP (tubularised incised plate) urethroplasty** was described by Snodgrass in 1994, built on Rich's hinging incision (1989).
- Designed for distal hypospadias, it is now used for **over 90% of distal and about 80% of midshaft repairs**.
- Distal hypospadias forms 70–85% of cases.

### Principle

- A **deep midline relaxing incision** of the urethral plate widens a narrow plate so that it can be tubularised **without tension**.
- The incised raw area **heals by re-epithelialisation** without a graft, giving a **vertical slit-like meatus**.

### Indications, prerequisites and assessment

- Distal and midshaft hypospadias, and redo cases with a healthy plate, when curvature is mild (**<30°** after degloving) and corrected by degloving or dorsal plication **without dividing the plate**.
- Supple, well-vascularised plate of adequate width and depth, and an adequate glans.
- **Glans width <14 mm** raises complications (glans width ≥14 mm is needed): give preoperative testosterone to enlarge it.

- **POST (plate objective scoring tool):** a low ratio (AB/BC ratio) of neomeatus length to glans fusion length means a poor plate, which favours a dorsal inlay graft.
- Timing: 6–12 months of age.
- Not for severe curvature needing plate division; proximal TIP is largely abandoned.

## Extended indications

| Indication | Rationale | Caveat |
|---|---|---|
| **Midshaft hypospadias** | Same principle | Good results |
| **Proximal hypospadias with mild curvature**, good plate | Single stage; no flaps | More fistula, stricture, recurrent curvature; Campbell-Walsh authors now prefer staging |
| **Redo or failed repairs** with preserved plate | Plate often survives | Worse than primary; inlay graft if scarred |
| **MIP (megameatus intact prepuce)** | Wide, deep plate | GAP (glans approximation procedure) alternative |
| **Narrow or shallow plate**: TIP + **dorsal inlay graft** ("Snodgraft") | Preputial or buccal graft fills the incision, so it cannot contract | Less meatal stenosis |
| **Adults, circumcised** boys | No preputial tissue needed | Success similar |

- **Also know:** in redo cases with poor blood supply a dorsal buccal inlay beats onlay or tube; incise-and-graft is also used in adult urethral stricture (Asopa dorsal inlay) and female urethroplasty.

## Contraindications

- Curvature >30° needing plate division; absent or scarred plate; penoscrotal transposition. Use staged repair instead (Bracka, STAG [staged tubularised autograft], Byars).

## Operative steps

1. Glans stay (traction) suture; tourniquet.
2. U-incision around the plate and meatus; circumscribe and deglove; artificial erection.
3. Separate (mobilise) the glans wings deeply from the plate.
4. **Midline incision of the plate** from inside the meatus to the end of the plate, down to the corpora.
5. Tubularise over a **6–8 Fr stent** with subepithelial 6-0/7-0 absorbable sutures, from distal to proximal, placing the first stitch about 3 mm (2–3 mm) short of the plate end to leave an oval meatus.
6. **Dartos flap** (dorsal preputial) over the neourethra as a barrier layer.
7. Two-layer glansplasty; mucosal collar; skin closure with Byars flaps, or preputioplasty.
8. Stent 5–7 days; compressive dressing; oxybutynin for bladder spasm.

## Complications and outcomes

- Pfistermuller meta-analysis (2015, 4,675 boys): complications 10.6%, reoperation 4.5%; **fistula 5.7%**, **meatal stenosis 3.6%**, stricture 1.3%.
- **Meatal stenosis is commoner than after Mathieu**; glans dehiscence also occurs.
- Distal repair success 83–95%.

- **Proximal and redo cases do significantly worse**; a dartos barrier layer improves results and reduces fistula.

### Modifications

- **Dorsal inlay graft (Snodgraft)**: inner preputial skin graft in the incised bed for a shallow plate.
- Lateral plate incisions to 12 mm width.
- **Also know:** MAGPI (meatal advancement and glanuloplasty incorporated), Mathieu, Thiersch–Duplay, onlay flap.

**Diagram to draw:** cross-sections showing plate incision, inlay graft, tubularisation and dartos cover.

> **Key points to remember:**
> - Snodgrass 1994; incise and tubularise; heals by epithelialisation.
> - Slit-like meatus; dartos barrier reduces fistula.
> - Glans <14 mm and poor plate raise complications.
> - Fistula 5.7%, meatal stenosis 3.6% (Pfistermuller 2015).
> - Extended to midshaft, selected proximal, redo, MIP and adults.
> - Key condition: curvature <30° without plate division.
> - Poor plate: dorsal inlay graft ("Snodgraft").
> - Proximal and redo TIP fail more often.

## SAQ 2 Glandular hypospadias

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 11 Paper 1, Q3b

*"Glandular" is read as glanular hypospadias: meatus on the glans.*

### Definition

- The mildest **distal (anterior)** hypospadias: the meatus opens on the ventral glans, proximal to the normal tip.
- Hypospadias occurs in 1 in 150–300 male births; distal forms are 70–85%.
- Usual features: **dorsal hooded foreskin**, ventral glans tilt, little or no curvature. **MIP (megameatus intact prepuce)** variant (up to 5%): wide meatus with normal foreskin.

### Embryology

- Urethral folds close from proximal to distal during weeks 8–16 under **DHT (dihydrotestosterone)**.
- Glanular urethra: ectodermal ingrowth (classic) or endodermal plate ("two-zipper", current view). Arrest at the last step gives a glanular meatus.

### Evaluation

- Examine meatus, plate, glans size, curvature, torsion; retract the foreskin before any **newborn circumcision**.
- **GMS (glans–meatus–shaft) score** (each 1–4, total 3–12) predicts complications.

- No workup needed for isolated distal hypospadias; DSD (difference of sexual development) workup if a testis is undescended.

### Indications and timing

- Deflected or spraying stream, meatal stenosis, curvature, cosmesis. A straight penis that voids standing may need no surgery; counsel parents.
- **Age 6–18 months** (AAP, American Academy of Pediatrics, 6–12 months).

### Operations

| Procedure | Suitable for | Key point |
|---|---|---|
| Urethromeatoplasty | Stenotic meatus, blind pit | Heineke–Mikulicz: vertical incision, horizontal closure |
| **MAGPI** (meatal advancement and glanuloplasty, Duckett 1981) | Glanular meatus, mobile skin | Advance meatus, two-layer glanuloplasty; retracts if used more proximally |
| MIV (M inverted V) glansplasty | Glanular or coronal | Avoids MAGPI retraction |
| **GAP** (glans approximation) | Wide deep groove, fish-mouth meatus (MIP) | Tubularise and close glans wings |
| **TIP** (tubularised incised plate, Snodgrass 1994) | Narrow plate | Most used; slit-like meatus |
| Mathieu flip-flap; Thiersch–Duplay | Selected | Stenosis, fistula |

- Preputioplasty or circumcision; dartos barrier layer.

### Complications

- Meatal stenosis or retraction, fistula, **glans dehiscence**, residual curvature, poor cosmesis. Distal repairs succeed in about 80%.

**Diagram to draw:** MAGPI steps (meatal advancement, glans wings closed beneath).

> **Key points to remember:**
> - Glanular = most distal hypospadias; dorsal hood, minimal curvature.
> - MAGPI only for a truly glanular meatus; GAP for MIP; TIP is the workhorse.
> - Repair at 6–18 months; may be left alone if function is normal.

## SAQ 3 Vanishing testes

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 13 Paper 1, Q3b

### Definition

- A testis that **developed normally and was then lost**, usually from **torsion or a vascular accident in fetal life**, after genital masculinisation but before scrotal fixation.
- Unilateral loss gives **monorchia**; bilateral loss gives **anorchia** (testicular regression syndrome), which is rare.
- Excised "nubbins" contain haemosiderin.

### Incidence

- **15–40% of nonpalpable testes** at surgery.

### Clinical features

- **Contralateral compensatory hypertrophy:** volume >2 mL (sensitivity 72%, specificity 100%) or length about 19–20 mm suggests a solitary testis.
- No palpable intrascrotal "appendage" tissue; sometimes a scrotal nubbin.
- These aid counselling but **do not replace exploration**.
- **Anorchia:** micropenis in nearly half.

### Evaluation

- **Imaging** (ultrasound, MRI [magnetic resonance imaging]) is not recommended: it does not change the operation.
- **Bilateral nonpalpable:** exclude DSD (difference of sex development) and CAH (congenital adrenal hyperplasia) first.
- Anorchia: high FSH (follicle-stimulating hormone) and LH (luteinising hormone); low AMH (anti-Müllerian hormone) and inhibin B; no rise after hCG (human chorionic gonadotrophin). Most reliable during mini-puberty.

### Management: diagnostic laparoscopy (after examination under anaesthesia)

| Laparoscopic finding | Action |
|---|---|
| **Blind-ending vessels and vas** above a closed ring | Diagnosis confirmed; stop |
| Vessels and vas entering the internal ring | Inguinal or scrotal exploration; excise the nubbin |
| Testis seen | Orchidopexy (single stage or Fowler–Stephens) |
| Neither vas nor vessels | Search up to the kidney (agenesis very rare) |

- **Excise the remnant:** viable germ cells or tubules in 5–15%.
- **Scrotal-first exploration** is reasonable with a palpable nubbin and contralateral hypertrophy (length ≥1.8 cm); laparoscopy is then needed in about a quarter.
- **Contralateral fixation:** debated; not required by the AUA (American Urological Association) algorithm.
- **Prosthesis:** offer at least 6 months after scrotal surgery or after puberty, inguinal approach.
- **Anorchia:** testosterone from puberty. **Also know:** agenesis shows ipsilateral Müllerian persistence.

**Diagram to draw:** laparoscopic view of blind-ending vessels at a closed ring.

> **Key points to remember:**
> - Prenatal torsion after masculinisation; 15–40% of nonpalpable testes.
> - Contralateral hypertrophy (>2 mL) suggests it, but explore anyway.
> - Blind-ending vessels confirm; excise remnant (germ cells 5–15%).
> - Anorchia: high FSH/LH, low AMH and inhibin B.

## SAQ 4 Pelvicalyceal anomalies other than PUJ obstruction

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 14 Paper 1, Q3d

## Introduction

- Congenital pelvicalyceal variants occur in about 4% of people.
- **Key question:** is there obstruction? Dilatation without obstruction needs no treatment; stasis causes stones and infection.

## Anomalies

| Anomaly | Features | Management |
|---|---|---|
| **Calyceal diverticulum** | Transitional-lined intrarenal cavity joined to a calyx (Type I, commoner, small) or pelvis (Type II, larger, symptomatic) by a narrow neck; 4.5 per 1000; congenital (persistent 3rd–4th generation ureteric branches) or acquired. **Stones in almost 40%**; milk of calcium | Asymptomatic: observe. Pain, infection, stones: ureteroscopic neck incision, **percutaneous ablation with fulguration**, laparoscopic marsupialisation; SWL (shock wave lithotripsy) ineffective |
| **Hydrocalycosis** | Dilated major calyx connected to the pelvis; infundibular obstruction by vessel, stenosis, scarring or achalasia | Symptomatic only: infundibulotomy or dilatation |
| **Megacalycosis** | **Non-obstructive**; malformed papillae; **12–20 calyces**; pelvis normal; male 6:1; normal diuretic renogram | **None**; may coexist with segmental megaureter |
| **Infundibular / infundibulopelvic stenosis** | Narrow infundibula ± small pelvis; usually bilateral, with VUR (vesicoureteric reflux), dysplasia; over a third reach renal failure (hyperfiltration) | Monitor creatinine and eGFR (estimated glomerular filtration rate); ACE (angiotensin-converting enzyme) inhibitors; surgery only if obstruction harms function: endoscopic, calicocalicostomy, ureterocalicostomy |
| **Extrarenal pelvis** | Large pelvis outside the sinus; mimics dilatation | None |
| **Bifid pelvis** | About 10%; normal variant | None |
| **Unipapillary kidney** | Single papilla and calyx; rare; usually hypoplastic kidney | None |

- **Also know:** acquired infundibular stenosis (tuberculosis, post-PCNL (percutaneous nephrolithotomy)); Fraley syndrome (crossing artery on upper infundibulum).

## Evaluation

- **Ultrasound:** cyst-like space; milk of calcium layers and shifts with position.
- **CT (computed tomography) urography or MRU (magnetic resonance urography): delayed contrast pooling** in a diverticulum.
- **Diuretic renogram (MAG3 (mercaptoacetyltriglycine)):** separates obstruction (hydrocalycosis) from non-obstruction (megacalycosis, extrarenal pelvis).
- **Retrograde pyelography:** maps the neck.

### Differential diagnosis

- Obstructive hydronephrosis, papillary necrosis, pyelonephritic clubbing, tuberculosis, renal cyst.
- **Stones:** retrograde first (find, dilate or laser-incise the ostium); percutaneous for lower-pole diverticula.

**Diagram to draw:** kidney showing diverticulum with narrow neck, hydrocalyx and megacalyces.

> **Key points to remember:**
> - Diverticulum: Type I to calyx, Type II to pelvis; stones in 40%.
> - Megacalycosis: non-obstructive, 12–20 calyces, no treatment.
> - Infundibulopelvic stenosis: bilateral, renal failure risk.

## SAQ 5 Retractile testis?

P2

**Type:** SAQ (10 marks)
**Source:** 2020-Summer Paper 2, Q3b

### Definition

- A **descended testis** that moves up out of the scrotum because of an active **cremasteric reflex**. It can be brought to the bottom of the scrotum by hand and **stays there**, at least for a while, after release.
- It is not undescended, but may be a mild form of the same disease and can later become an acquired undescended testis.

### Epidemiology

- Common: suprascrotal in up to one-third of 4-year-olds; 11–15% of boys up to 11 years. Nearly all are scrotal after 12 years.

### Examination (diagnosis is clinical)

- Warm room and hands, relaxed child, soap or gel; supine frog-leg (thigh abduction inhibits the cremasteric reflex), cross-legged or squatting.
- One hand sweeps from the internal ring down the canal; the other pulls the testis to the scrotal bottom. **Sustained gentle traction tires the cremaster.**
- Note size, mobility, contralateral testis and hernia.

| Testis | Finding | Action |
|---|---|---|
| **Retractile** | Reaches scrotal bottom and stays | Observe yearly |
| Gliding (unstable high scrotal) | Pulled down but springs back | Treat as undescended |
| Undescended (palpable) | Cannot reach mid-scrotum | Orchidopexy by 18 months |
| Ascending (acquired) | Previously scrotal, now not | Orchidopexy |

### Natural history

- Ascent: 2.6% in a prospective birth cohort; 7–32% in specialist series.
- Operated retractile testes show small volume, patent processus vaginalis and epididymal anomalies, suggesting a continuum with cryptorchidism.

### Management

- No ultrasound (adds nothing); no hormones (hCG [human chorionic gonadotrophin] neither separates nor treats).
- Reassure parents; **examine yearly** (AUA [American Urological Association] 2014) until puberty, because some testes ascend.
- **Orchidopexy** if it ascends, cannot be brought down, rests mostly high or becomes smaller. If unsure, examine under anaesthesia.

**Diagram to draw:** inguinoscrotal sketch: testis drawn by traction to the scrotal bottom.

> **Key points to remember:**
> - Retractile = comes down and stays down.
> - Gliding testis springs back: treat as undescended.
> - Yearly examination until puberty; operate if it ascends.

## SAQ 6 Retroileal Ureter

P2

**Type:** SAQ (10 marks)
**Source:** 2025-Winter Paper 2, Q3b

*"Retroileal" is taken as **retroiliac ureter**, the nearest real entity; retrocaval ureter is compared below (full answer: 2017-Winter Paper 3, Q3d).*

### Definition

- The ureter passes **behind the common iliac artery** and is compressed there.
- Better called "**preureteral iliac artery**": the anomaly is presumed vascular; the ureter is trapped.

### Embryology

- Normally the **ventral root of the umbilical artery** is replaced by a more **dorsal** branch, which lies behind the ureter.
- If the **ventral root persists** and the dorsal root fails, the ureter is trapped behind the artery.

### Features

- **Very rare**; either side; **bilateral** cases reported.
- Obstruction at **L5–S1**.
- **Associated anomalies are common**, especially **vasal and mesonephric duct ectopia**: ureter and vas draining together into the posterior urethra; vasa ending in the ureters; ectopic vaginal ureter.

### Presentation and diagnosis

- Flank pain, UTI (urinary tract infection) or hydroureteronephrosis down to the iliac crossing.
- **CT urography (computed tomography urography) with arterial phase** or MRI (magnetic resonance imaging) shows ureter and artery together; **diuretic renogram** for function and drainage.
- Caution: a vessel indenting an already dilated ureter may not be the cause.

### Management

- Non-obstructed: observe.
- Obstructed: **divide the ureter, re-route it in front of the artery** and do a spatulated ureteroureterostomy (or reimplantation) over a double-J stent; open, laparoscopic or robotic.

### Comparison with retrocaval ureter

| | **Retroiliac** | **Retrocaval (circumcaval)** |
|---|---|---|
| Vessel | Common iliac artery | IVC (inferior vena cava) |
| Embryology | Persistent ventral umbilical artery root | Persistent subcardinal vein |
| Side | Either, may be bilateral | Right |
| Level | L5–S1 | L3–L4, "fishhook" |
| Associations | Vasal/mesonephric ectopia | Horseshoe kidney, left renal anomalies |
| - **Also know:** distal ureter obstruction by uterine, umbilical, obturator or hypogastric vessels. | | |

**Diagram to draw:** ureter behind the common iliac artery at L5–S1, dilated above.

> **Key points to remember:**
> - Retroiliac = preureteral iliac artery; persistent ventral umbilical artery root.
> - Obstruction at L5–S1; either side or bilateral.
> - Look for vasal and ureteric ectopia.
> - Retrocaval: right, L3–L4, fishhook, 1 in 1500.

## SAQ 7 Medullary sponge kidney

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 9 Paper 2, Q3e

### Definition

- **MSK (medullary sponge kidney; precalyceal canalicular ectasia):** dilated (ectatic) distal **collecting ducts** with small cysts and diverticula **confined to the medullary pyramids**. Nephrons are initially normal.
- Usually **non-heritable** (sporadic); usually bilateral.

### Pathogenesis of stones and nephrocalcinosis

- **Stasis and infection** in the ectatic ducts favour crystal deposition.
- **Tubular defects:**
- **Hypercalciuria** in one-third to one-half (mostly absorptive; sometimes renal leak).
- **Incomplete distal RTA (renal tubular acidosis)** in 30–40%, shown on acid loading. This causes **hypocitraturia** and alkaline urine.
- Impaired urinary concentration.

- Mutations in $H^+$-ATPase genes (ATP6V1B1, ATP6V0A4) support an MSK–distal RTA link.
- Result: calcium stones within dilated ducts and medullary **nephrocalcinosis**.

### Clinical features

- Often silent for life; presents **after 20 years**.
- **Renal colic 50–60%**, UTI (urinary tract infection) 20–33%, gross haematuria 10–18%.
- Found in **2.6–21% of stone formers**, more often women.

### Diagnosis

- **Urographic triad:** 1. Enlarged kidneys, with or without papillary calcification. 2. Elongated papillary tubules or cavities filling with contrast: **"bristles on a brush"** (paintbrush); when calcified, a **"bouquet of flowers"**. 3. Papillary blush and persistent medullary opacification.
- CT urography now replaces IVU (intravenous urography); plain CT shows medullary nephrocalcinosis.
- **Metabolic work-up:** 24-hour urine (calcium, citrate, pH), serum bicarbonate and potassium.
- **Differentials:** papillary necrosis, other medullary nephrocalcinosis (distal RTA, hyperparathyroidism), calyceal diverticulum.

### Management

| Agent | Target | Note |
|---|---|---|
| **Potassium citrate** | Hypocitraturia, hypercalciuria | **First choice**; reduces calciuria and recurrence |
| Thiazide | Hypercalciuria | Useful even without hypercalciuria |
| Inorganic phosphate | Hypercalciuria | If thiazide unsuitable; avoid with urease infection |
| Fluids, low-sodium diet | Volume, calciuria | All patients |
| - Treat infection; stones by SWL (shock-wave lithotripsy) or ureteroscopy; stones inside ducts may be unreachable. | | |
| - **Prognosis:** usually benign; renal failure is rare. | | |

**Diagram to draw:** pyramid with ectatic ducts filled with contrast ("brush") and small calculi.

> **Key points to remember:**
> - Ectatic collecting ducts; sporadic; "bristles on a brush".
> - Hypercalciuria, incomplete distal RTA, hypocitraturia.
> - Potassium citrate is first-line prevention.

## SAQ 8 Megameatus intact prepuce

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 15 Paper 2, Q3f

### Definition

- **MIP (megameatus intact prepuce)** is a hypospadias variant (up to 5%) with a **wide, "fish-mouth" glanular or coronal meatus**, a **completely formed prepuce** and **no significant curvature**. Described by Duckett and Keating (1989).

### Embryology

- Normally the prepuce closes ventrally only after the urethral folds fuse (about 20 weeks), so 95% of hypospadias have a hooded, incomplete foreskin.
- In MIP the foreskin is complete, showing that **preputial and urethral development can dissociate**.

### Clinical features

- Hidden under a normal foreskin; often found on **retraction before circumcision** or during circumcision.
- Wide, deep glans groove with a large, patulous meatus; well-developed spongiosum.
- Symptoms: spraying or deflected stream; cosmetic concern.
- **Advice:** retract the foreskin before every newborn circumcision. If found, the circumcision may be completed; prior circumcision does not harm later repair.

### Evaluation

- Examine meatal site and size, glans and plate, curvature (artificial erection at surgery). GMS (glans–meatus–shaft) score.
- No routine upper-tract imaging for isolated distal hypospadias.

### Management

- **Indication:** cosmesis and a straight stream; repair is optional. Usual timing 6–18 months.
- **Pyramid procedure (Duckett and Keating):** 1. Glans traction suture; retract prepuce. 2. Incise around the megameatus in a **pyramid shape**, apex on the ventral shaft. 3. Dissect the wide distal urethra free of the glans wings. 4. **Excise a ventral wedge** of redundant urethra to taper it. 5. Tubularise over a catheter; cover with dartos. 6. Approximate glans wings in two layers (glansplasty); preserve or reconstruct the prepuce.
- **Alternatives:** GAP (glans approximation procedure; pyramid = GAP with intact prepuce), TIP (tubularised incised plate) urethroplasty, Thiersch–Duplay tubularisation, Mathieu.

### Complications

- Fistula, meatal stenosis or regression, glans dehiscence; generally low after distal repair.

**Diagram to draw:** fish-mouth glanular meatus with intact prepuce, and the pyramid incision.

> **Key points to remember:**
> - Wide glanular meatus, intact prepuce, no curvature; up to 5% of hypospadias.
> - Often found at circumcision; retract foreskin first.
> - Pyramid repair: dissect, taper and tubularise; glansplasty.

## SAQ 9 Techniques of chordee correction

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 17 Paper 2, Q3b (related: Practice Set 10 Paper 1, Q3d)

## Definition and causes

- Chordee: **ventral penile curvature**, with or without hypospadias. It, not meatal position, decides one-stage versus staged repair.
- Causes: tethered ventral skin, fibrous dysplastic dartos and Buck's fascia, short urethral plate, **corporal disproportion**.
- Devine–Horton (1973) graded the dysplastic layers: I spongiosum, II Buck's and dartos, III dartos only, IV corporal disproportion.

## Assessment

- Parents' photographs; **artificial erection** in theatre: Gittes–McLaughlin (tourniquet at the base plus saline), corporal compression, or prostaglandin. A tourniquet can hide basal curvature.
- Grade: mild <30°, moderate 30–45°, severe ≥45° (goniometer; eye underestimates).

## Stepwise correction

```
Deglove to penoscrotal junction + excise dysplastic dartos → erection test ├─ Straight → urethroplasty ├─ Residual <30° → dorsal plication └─ >30° → divide urethral plate → still >30° → ventral lengthening (stage)
```

| | **Dorsal shortening (plication)** | **Ventral lengthening (corporotomy ± graft)** |
|---|---|---|
| Methods | **Nesbit:** excise dorsal tunical ellipses, close transversely. **Baskin:** single midline plication in the **12 o'clock nerve-free zone**. Heineke–Mikulicz dorsal incision closed transversely; 16-dot plication | 1–3 transverse **full-thickness corporotomies** (STAG, staged tubularised autograft); single **9-to-3 o'clock corporotomy** with graft: **dermis, tunica vaginalis**, SIS (small intestinal submucosa); dermal graft 20% larger than defect |
| Use | ≤15–30° | Moderate–severe after plate division |
| Recurrence | 28% (one series) | 9% |
| Risk | Shortening; dorsal nerve injury off midline | Longer, staged; theoretical erectile dysfunction |

- **Plate division** (transection) releases a short plate; not dividing it is an independent risk factor for recurrence. Repair is then staged.
- **Also know:** "fairy cuts" (superficial tunical incisions); lateral or dorsal curves need multipoint plication; correct **torsion** by dartos flap rotation.
- **Rule:** correct fully the first time and repeat the erection test before urethroplasty. Curvature may recur at puberty.

**Diagram to draw:** penile cross-section with dorsal nerves at 11 and 1 o'clock and midline plication at 12.

> **Key points to remember:**
> - Artificial erection after degloving decides the technique.
> - <30°: dorsal plication; >30°: divide plate ± ventral graft and stage.
> - Plication recurrence 28% vs grafting 9%.

## SAQ 10 Congenital ureteral anomalies

P3

**Type:** SAQ (10 marks)
**Source:** 2017-Summer Paper 3, Q3f

### Embryology basis

- The **ureteric bud** arises from the **mesonephric (wolffian) duct** at about week 5 and induces the metanephric blastema.
- Anomalies arise from **abnormal budding**, **failed distal canalisation**, **abnormal muscle maturation** or **persistent embryonic vessels**.

### Classification

| Category | Anomaly | Key points |
|---|---|---|
| **Number** | **Duplication**: incomplete (bifid, Y-ureter) or complete | About 1 in 125; unilateral 6× bilateral. Lower pole: VUR (vesicoureteric reflux), PUJ (pelviureteric junction) obstruction. Upper pole: ectopia, ureterocele |
| **Termination (orifice)** | **Ectopic ureter** | Opens at or below the bladder neck. **Females**: bladder neck to vagina, uterus; **continuous dribbling with normal voiding**. **Males**: always above external sphincter (posterior urethra, seminal vesicle, vas); **never incontinent**; epididymitis. Zinner syndrome |
| **Structure** | **Ureterocele** | Cystic dilation of distal submucosal ureter (Chwalle membrane). Intravesical vs ectopic (Glassberg); cecoureterocele (Stephens) |
| | **PUJ obstruction** | Intrinsic aperistaltic segment (infants) or crossing lower-pole vessel (older children) |
| | **Megaureter** (≥7 mm) | Obstructed, refluxing, non-refluxing non-obstructed, refluxing-obstructed; primary obstructive = adynamic distal segment |
| | **Primary VUR** | Short submucosal tunnel, lateral orifice |
| **Position** | **Preureteral (retrocaval) vena cava** | Persistent subcardinal vein traps the right ureter; fishhook (S) shape at L3–L4; type I obstructs, type II does not |
| | Retroiliac ureter; herniation | Rare |

### Weigert–Meyer law

- In complete duplication the **upper-pole ureter opens lower and more medial** (ectopic, obstructs, ureterocele); the **lower-pole ureter opens higher and lateral** (refluxes).
- Reason: the upper-pole bud arises more cephalad on the wolffian duct, is incorporated later and ends more caudally.
- **Also know:** blind-ending ureter, inverted-Y ureter, triplication, congenital stricture, ureteral valves, fibroepithelial polyp.

### Evaluation

- Ultrasound, VCUG (voiding cystourethrogram), **DMSA (dimercaptosuccinic acid)** scan, MR (magnetic resonance) urography.

### Management principles

- **Ureterocele:** endoscopic incision, upper-pole heminephrectomy or ureteroureterostomy.
- **Ectopic ureter:** heminephroureterectomy if non-functioning; otherwise ureteroureterostomy or reimplantation.
- **PUJ obstruction:** pyeloplasty. **Megaureter:** observe most; taper and reimplant (5:1) if function falls.
- **Retrocaval ureter:** divide, bring anterior to the cava, excise the aperistaltic segment, reanastomose.

**Diagram to draw:** complete duplication showing Weigert–Meyer orifice positions.

> **Key points to remember:**
> - Upper pole obstructs (ectopia, ureterocele); lower pole refluxes.
> - Female ectopic ureter: continuous dribbling with normal voiding; male never incontinent.
> - Retrocaval ureter is a venous anomaly; always right-sided.

## SAQ 11 Autosomal recessive polycystic kidney disease.

P3

**Type:** SAQ (10 marks)
**Source:** 2018-Summer Paper 3, Q3c

### Definition and genetics

- **ARPKD (autosomal recessive polycystic kidney disease)**: bilateral, symmetrical renal enlargement from **dilated collecting ducts**, **always with congenital hepatic fibrosis**.
- Gene **PKHD1 on chromosome 6p**; protein **fibrocystin (polyductin)** on the **primary cilium** of tubular and bile-duct cells, so it is a **ciliopathy**.
- Incidence about 1 in 10,000–50,000 live births. Siblings have a 25% risk.
- "Infantile" is a misnomer: a **spectrum**; earlier onset means more severe disease.

### Pathology

- Radially arranged dilated collecting ducts (**microcysts**).
- **Liver:** periportal fibrosis, biliary ectasia; **Caroli syndrome** if intrahepatic ducts dilate.

### Clinical features

| Age | Features |
|---|---|
| **Fetus** | Large echogenic kidneys, **oligohydramnios** → Potter facies, limb deformity, **pulmonary hypoplasia** |
| **Neonate** | Huge, smooth, hard flank masses; respiratory distress; rising creatinine; **30–50% die early** |
| **Child** | Hypertension, CKD (chronic kidney disease), growth failure, anaemia, osteodystrophy |
| **Older child** | **Portal hypertension**: varices, splenomegaly, cholangitis; liver function stays normal |

- No association with renal tumours.

### Investigations

- **Ultrasound:** bilateral very large, **homogeneously hyperechoic** kidneys (brighter than liver) from countless microcyst interfaces; discrete cysts appear with age.
- **Parents' ultrasound** normal (excludes ADPKD); **PKHD1 genetic testing**; liver ultrasound.
- **Differential:** neonatal ADPKD (autosomal dominant PKD), bilateral hydronephrosis, MCDK (multicystic dysplastic kidney), glomerulocystic disease, bilateral tumours, renal vein thrombosis.

### Management (no cure)

- **Neonatal:** ventilation, fluid and electrolyte care, antihypertensives (ACE (angiotensin-converting enzyme) inhibitors).
- **CKD care:** nutrition, growth hormone, anaemia and bone treatment; dialysis and **kidney transplantation**.
- **Nephrectomy** (breathing, feeding, blood pressure, room for peritoneal dialysis): cautiously; **very early bilateral nephrectomy** (first 3 months) had the worst neurological outcomes (ARegPKD registry).
- **Liver:** variceal sclerotherapy, portosystemic shunt; **combined liver–kidney transplant** for severe liver disease.
- **Genetic counselling** and prenatal diagnosis.

### ARPKD versus ADPKD

| | ARPKD | ADPKD |
|---|---|---|
| Gene | PKHD1 (chromosome 6) | PKD1 (16), PKD2 (4) |
| Age | Perinatal | 30–50 years |
| Cysts from | Collecting duct | Whole nephron |
| Ultrasound | Symmetric, homogeneous, echogenic | Large discrete cysts, asymmetric |
| Liver | Fibrosis (always) | Cysts |
| Other | None | Berry aneurysm, mitral prolapse |

**Diagram to draw:** cut kidney with radial dilated collecting ducts.

> **Key points to remember:**
> - PKHD1, chromosome 6, fibrocystin; always congenital hepatic fibrosis.
> - Huge homogeneous echogenic kidneys with oligohydramnios.
> - Neonates die of pulmonary hypoplasia; survivors develop portal hypertension.

## SAQ 12 Ureterocele

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 2 Paper 3, Q3c (ectopic ureter also answered in 2018-Summer Paper 3, Q1)

### Definition

- Cystic dilatation of the distal ureter, in the bladder or spanning bladder neck and urethra; **never wholly urethral**, never joined to Wolffian structures.

- Commoner in girls (about 4:1); about 80% drain the **upper pole of a duplex kidney** (Weigert–Meyer rule: upper pole orifice caudal-medial); about 10% bilateral.
- **Cause:** persistent **Chwalla membrane**; upper pole often dysplastic.

## Classification

| System | Basis | Types |
|---|---|---|
| **Glassberg (most useful)** | Position | **Intravesical** vs **ectopic (extravesical)**: part at bladder neck or urethra |
| Stephens | Orifice morphology | Stenotic, sphincteric, sphincterostenotic, **cecoureterocele** (cavity extends under urethra), blind, non-obstructed |
| Churchill | Function | Upper pole only / whole kidney / contralateral at risk |

## Clinical features

- Antenatal hydronephrosis (most); febrile UTI (urinary tract infection) or urosepsis; adult single-system type with stone.
- **Prolapse:** smooth interlabial mass in girls; BOO (bladder outlet obstruction).
- **No incontinence when untreated** (unlike female ectopic ureter).

## Evaluation

- **Ultrasound:** thin-walled cyst in bladder, dilated upper pole and ureter.
- **VCUG (voiding cystourethrogram):** lower pole reflux ("drooping lily"), eversion (weak floor), prolapse.
- **DMSA (dimercaptosuccinic acid):** upper pole function; cystoscopy before incision.

## Management

- **Goals:** preserve function; remove infection, obstruction, reflux; keep continence.
- **CAP (continuous antibiotic prophylaxis)** from birth if hydronephrotic.
- **Observation:** small, non-obstructing, low-grade reflux, non-functioning upper pole.

| Option | Indication / result |
|---|---|
| **TUI (transurethral incision)** | Sepsis, neonates, intravesical type. Low transverse full-thickness incision; relieves obstruction in 78–97%; new reflux 0–50%. Intravesical without reflux: 70–80% cured; ectopic: secondary surgery up to 70% |
| Upper pole heminephrectomy | Non-functioning upper pole, no reflux; secondary surgery 40–50% |
| Ureteroureterostomy / pyeloureterostomy | Functioning upper pole |
| Excision + common sheath reimplant ± bladder neck repair | Persistent reflux, ectopic or cecoureterocele; persisting reflux 5–10% |
| Total reconstruction | Older child, non-functioning upper pole, lower pole reflux |

- **Also know:** "watering-can" laser punctures (less reflux); pseudoureterocele; ureterocele disproportion.

**Diagram to draw:** duplex kidney with upper pole ureterocele and lower pole reflux.

**Key points to remember:**

- Upper pole of duplex; never wholly urethral; Chwalla membrane.
- Intravesical vs ectopic predicts TUI success.
- TUI relieves obstruction in 78–97%; new reflux 0–50%.

## SAQ 13 Primary obst. megaureter.

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 2 Paper 3, Q3f

*Assumed meaning: primary obstructive megaureter (POM).*

### Definition

- **Megaureter:** ureter diameter **≥7 mm** (normal up to 16 years is 5.0–6.5 mm). It is a descriptive term.
- **Classification:** primary (intrinsic) or secondary (neurogenic bladder, outlet obstruction); and **obstructed, refluxing, non-refluxing non-obstructed, or refluxing and obstructed**.
- **POM:** an **adynamic juxtavesical segment** at the UVJ (ureterovesical junction) blocks peristalsis and causes functional obstruction above.

### Pathogenesis

- Ureteric muscle matures cranially to caudally; the juxtavesical segment matures last, even after birth. This explains spontaneous resolution.
- Excess collagen, thick muscle sleeve, abnormal interstitial cells of Cajal.

### Clinical features

- 10–23% of antenatal upper-tract dilatation. Boys, **left side**; **bilateral 25%**; contralateral dysplasia or obstruction 10–15%.
- Mostly asymptomatic; or UTI (urinary tract infection), pain, haematuria, stones.

### Evaluation

- **Ultrasound:** hydroureteronephrosis.
- **VCUG (voiding cystourethrogram):** excludes reflux and PUV (posterior urethral valves).
- **MAG3 (mercaptoacetyltriglycine) renogram:** split function; washout T½ unreliable (false positive up to 44%).
- MRU (magnetic resonance urography) for anatomy.

### Management

#### Conservative (most)

- CAP (continuous antibiotic prophylaxis) in infants; serial ultrasound and renograms; most resolve.
- Diameter <10 mm: virtually never needs surgery; ≥10 mm: a substantial minority do.
- Follow to early adulthood.

#### Indications for surgery

- Recurrent febrile UTI; progressive dilatation; **DRF (differential renal function) <40%**; **fall in DRF ≥5%**.

**Infant (small bladder, huge ureter): temporise**

- Cutaneous ureterostomy; refluxing reimplant; nephrostomy; **endoscopic balloon dilatation with double-J** for 2–6 months (70–80% success; complications up to a third).

**Definitive repair (child)**

- Excise stenotic segment → straighten → **taper** → **non-refluxing reimplant at 5:1** (Cohen or Politano–Leadbetter; extravesical for huge ureters). Stent afterwards.

| Tailoring | Method | Use |
|---|---|---|
| Starr plication | Imbricate lateral wall | Moderate (<1.75 cm) |
| Kalicinski folding | Exclude and fold lumen | Moderate |
| **Hendren excision** | Excise lateral wall | Massive; risk of ischaemia |

- Success over 90%.
- **Complications:** obstruction (stent first), VUR (vesicoureteral reflux; observe or inject), persistent dilatation.

**Diagram to draw:** dilated ureter above narrow adynamic segment; tapered reimplant.

> **Key points to remember:**
> - Megaureter ≥7 mm; POM = adynamic juxtavesical segment.
> - Surgery: UTIs, progression, DRF <40% or fall ≥5%.
> - Excise, taper, reimplant 5:1; success >90%.

# Rapid Revision Sheet · Section 4

### Primary VUR: VUJ embryology/anatomy, pathophysiology, management (incl. bilateral, adult grade 3), reimplantation techniques and STING (LAQ)

- Low (early) bud → high lateral orifice → short tunnel → VUR plus dysplasia; high (late) bud → ectopia.
- One embryological fault: lateral orifice, short tunnel, dysplasia.
- Tunnel length to diameter 5:1 normal versus 1.4:1 refluxing (Paquin); adult intramural ureter 1.2–2.5 cm.
- Dysplasia (boys, congenital) versus scar (girls, infection); sterile reflux rarely scars.
- Grade V loses papillary impressions; grade IV–V and bilateral grade III rarely resolve; bilateral grade IV persists most; bilateral VUR is high risk.
- RIVUR: CAP halves recurrent UTI, not scarring.
- Treat BBD and give CAP first; correct for breakthrough febrile UTI, new scars or persistent grade IV–V.
- Trend: risk-stratified care, fewer VCUGs, BBD first, endoscopic or robotic correction.
- STING: O'Donnell and Puri 1986; needle 2–3 mm below orifice at 6 o'clock.
- Dx/HA 80–250 µm; HIT better than STING; double HIT about 90%; Elder 2006: grade IV 63% after one injection.
- Watch for late obstruction; calcified mound mimics a stone.
- Open reimplantation 95.9%; Cohen cross-trigonal (over 98%) suits bilateral VUR through one cystotomy.
- Bilateral extravesical (Lich–Gregoir) repair risks retention up to 10%.
- Paquin: tunnel 5:1; combined, suprahiatal reimplant. Gil-Vernet: trigonoplasty, medial advancement, no tunnel; for low-grade bilateral VUR.
- 34–38% of scarred children become hypertensive; correction does not lower BP; scars bring 8× pre-eclampsia risk.
- In adults the scar, not the reflux, carries the risk; exclude secondary causes first; correct for recurrent pyelonephritis, new scars or pregnancy plans.

### Testicular descent and undescended testis: evaluation, treatment algorithm and orchidopexy methods (LAQ)

- SRY → SOX9 → Sertoli cells (AMH); Leydig cells → testosterone and INSL3.
- Transabdominal: CSL regression plus INSL3-driven gubernacular swelling.
- Inguinoscrotal: androgen via genitofemoral nerve (GFN) and CGRP; most testes scrotal after 27 weeks.
- Ectopic = off the path; commonest site superficial inguinal pouch.
- Operate from 6 months, preferably by 12 months, and by 18 months at the latest.
- No imaging, no hormones.
- Non-palpable: EUA then diagnostic laparoscopy; laparoscopy decides.
- Bilateral non-palpable: DSD work-up first (karyotype, 17-OHP, AMH) to exclude CAH/DSD.
- Subdartos pouch; Prentiss manoeuvre gives extra length.
- Fowler–Stephens relies on vasal collaterals.

**Retrocaval ureter and IVC anomalous development (LAQ)**

- Infrarenal IVC = right supracardinal; suprarenal = right subcardinal; hepatic = vitelline.
- Circumaortic left renal vein is commonest (2–16%): ligate both limbs.
- Retrocaval ureter: venous anomaly (persistent subcardinal vein); right side only; 1 in 1500; male 3:1; transpose anterior to IVC.
- Type I fishhook obstructs; type II sickle often does not.
- CT urography; repair by anterior transposition and ureteroureterostomy.
- Duplicated IVC: filter both cavae; do not mistake it for nodes.

**Hypospadias: present concepts, single-stage vs staged repairs, and hypospadias fistula (LAQ)**

- Classify after degloving (GMS); curvature, not meatus, decides one versus two stages.
- Curvature >30° after degloving: stage.
- TIP for distal; staged graft repair (STAG/Bracka) for severe proximal; STAG has the lowest proximal complications.
- Stage 2 after ≥6 months.
- Hypospadias plus undescended testis: DSD work-up.
- Follow through puberty; use patient-reported outcomes.
- Fistula: commonest complication, just under 10%; TIP 5.7%.
- Exclude distal obstruction; wait 6–12 months.
- Layered closure with barrier flap; recurrence about 20%.

**Wilms tumour, including bilateral Wilms tumour (LAQ)**

- WT1 at 11p13 (WAGR, Denys–Drash); BWS at 11p15.
- Anaplasia (≥3× nuclei, hyperchromasia, abnormal mitoses) is unfavourable; TP53.
- Any spill, biopsy or positive node = stage III; sample ≥5 nodes.
- COG operates first; SIOP gives chemotherapy (VCR + AMD) first.
- Bilateral: stage V, 5–7%; no upfront nephrectomy or biopsy.
- Bilateral: COG three drugs for 6 weeks (reassess at 6 weeks), surgery by week 12, NSS; SIOP two drugs.
- AREN0534: 4-year OS about 95%.
- Bilateral NSS to avoid ESRD.

**Proximal (perineoscrotal) hypospadias: penile embryology and management including Koyanagi repair (LAQ)**

- DHT at 8–12 weeks; opening then closing zipper; 12 o'clock nerve-free zone.
- Hypospadias (including perineoscrotal) plus undescended testis: DSD work-up.
- <30° plication; >30° divide plate ± ventral graft and stage.
- STAG/Bracka staged repair preferred; fistula commonest complication.
- Koyanagi: one stage; meatus-based parameatal foreskin flap.
- Koyanagi: plate divided; no proximal circular anastomosis.
- Koyanagi: fistula commonest; diverticulum, meatal stenosis.

**Urological evaluation and management of pediatric kidney transplant patient. (LAQ)**

- Bladder target: age-expected capacity at <40 cm $H_2O$; augment if <75% despite CIC.
- PUV bladder heads to myogenic failure; neuropathic bladder needs earlier reconstruction.

- Never test a bladder with a graft; cycle the defunctionalised bladder.
- Stenosis 6%, leak 3%, stones 2%, significant reflux 2%.

**Embryology and management of ectopic ureter. (LAQ)**

- Ectopia = orifice at or below the bladder neck; bud too cranial on the wolffian duct.
- Girls: continuous dribbling with normal voiding; boys: never incontinent, epididymitis.
- DMSA decides keep (UU/reimplant) versus remove (heminephroureterectomy).
- Protect the lower-pole blood supply; stent the recipient ureter for UU.

**TIP (Snodgrass) repair for hypospadias (SAQ)**

- Snodgrass 1994; incise and tubularise; heals by epithelialisation.
- Slit-like meatus; dartos barrier reduces fistula.
- Glans <14 mm and poor plate raise complications.
- Fistula 5.7%, meatal stenosis 3.6% (Pfistermuller 2015).
- Extended to midshaft, selected proximal, redo, MIP and adults.
- Key condition: curvature <30° without plate division.
- Poor plate: dorsal inlay graft ("Snodgraft").
- Proximal and redo TIP fail more often.

**Glandular hypospadias (SAQ)**

- Glanular = most distal hypospadias; dorsal hood, minimal curvature.
- MAGPI only for a truly glanular meatus; GAP for MIP; TIP is the workhorse.
- Repair at 6–18 months; may be left alone if function is normal.

**Vanishing testes (SAQ)**

- Prenatal torsion after masculinisation; 15–40% of nonpalpable testes.
- Contralateral hypertrophy (>2 mL) suggests it, but explore anyway.
- Blind-ending vessels confirm; excise remnant (germ cells 5–15%).
- Anorchia: high FSH/LH, low AMH and inhibin B.

**Pelvicalyceal anomalies other than PUJ obstruction (SAQ)**

- Diverticulum: Type I to calyx, Type II to pelvis; stones in 40%.
- Megacalycosis: non-obstructive, 12–20 calyces, no treatment.
- Infundibulopelvic stenosis: bilateral, renal failure risk.

**Retractile testis? (SAQ)**

- Retractile = comes down and stays down.
- Gliding testis springs back: treat as undescended.
- Yearly examination until puberty; operate if it ascends.

**Retroileal Ureter (SAQ)**

- Retroiliac = preureteral iliac artery; persistent ventral umbilical artery root.
- Obstruction at L5–S1; either side or bilateral.
- Look for vasal and ureteric ectopia.
- Retrocaval: right, L3–L4, fishhook, 1 in 1500.

**Medullary sponge kidney (SAQ)**

- Ectatic collecting ducts; sporadic; "bristles on a brush".

- Hypercalciuria, incomplete distal RTA, hypocitraturia.
- Potassium citrate is first-line prevention.

**Megameatus intact prepuce (SAQ)**

- Wide glanular meatus, intact prepuce, no curvature; up to 5% of hypospadias.
- Often found at circumcision; retract foreskin first.
- Pyramid repair: dissect, taper and tubularise; glansplasty.

**Techniques of chordee correction (SAQ)**

- Artificial erection after degloving decides the technique.
- <30°: dorsal plication; >30°: divide plate ± ventral graft and stage.
- Plication recurrence 28% vs grafting 9%.

**Congenital ureteral anomalies (SAQ)**

- Upper pole obstructs (ectopia, ureterocele); lower pole refluxes.
- Female ectopic ureter: continuous dribbling with normal voiding; male never incontinent.
- Retrocaval ureter is a venous anomaly; always right-sided.

**Autosomal recessive polycystic kidney disease. (SAQ)**

- PKHD1, chromosome 6, fibrocystin; always congenital hepatic fibrosis.
- Huge homogeneous echogenic kidneys with oligohydramnios.
- Neonates die of pulmonary hypoplasia; survivors develop portal hypertension.

**Ureterocele (SAQ)**

- Upper pole of duplex; never wholly urethral; Chwalla membrane.
- Intravesical vs ectopic predicts TUI success.
- TUI relieves obstruction in 78–97%; new reflux 0–50%.

**Primary obst. megaureter. (SAQ)**

- Megaureter ≥7 mm; POM = adynamic juxtavesical segment.
- Surgery: UTIs, progression, DRF <40% or fall ≥5%.
- Excise, taper, reimplant 5:1; success >90%.

# Section 5 · Paediatric Urology 2

## Long answer questions (LAQ)

### LAQ 1 Bladder embryology and exstrophy-epispadias complex

P1 · P2 · P3 **merged from 3**

*Asked as:* P3 LAQ "Bladder and ureter development, exstrophy–epispadias complex and its management" · P1 LAQ "Describe embryology of bladder. What are the proposed theories for extrophy epispadias complex? Describe skeletal abnormalities." · P2 SAQ "Embryogenesis of urinary bladder"

#### Introduction

- Assumed meanings: "ectopia vesicae" means classic bladder exstrophy (CBE); "single stage repair of exstrophy of UB" means single-stage (complete primary) repair of exstrophy of the urinary bladder, mainly CPRE (complete primary repair of exstrophy, Mitchell).
- **EEC (exstrophy–epispadias complex)** is one spectrum: glanular epispadias, then **CBE (classic bladder exstrophy)** (over 50%), then cloacal exstrophy.
- CBE incidence about 1 in 46,000; male:female 2.3:1. Recurrence in offspring of an affected parent: 1 in 70.

#### Development of the ureter

- **Weeks 4–6:** the **ureteric bud** arises from the distal **Wolffian (mesonephric) duct**. GDNF (glial cell line-derived neurotrophic factor) from the metanephric mesenchyme induces a single bud (extra buds cause duplication).
- **Reciprocal induction:** the bud invades the metanephric blastema. The bud stalk forms the ureter; its branches form the pelvis, calyces and collecting ducts; the blastema forms the nephrons.
- **Wall:** urothelium, lamina propria, smooth muscle and serosa, the same plan as the bladder.
- **Chwalla membrane** at the future orifice normally ruptures; failure is the classic explanation of ureterocele.

#### Development of the bladder: division of the cloaca

- The **cloacal membrane** (bilaminar ectoderm–endoderm) occupies the infraumbilical wall. **Mesenchyme grows in** between its layers and forms the lower abdominal muscles and pelvic bones.
- **Cloacal division (weeks 5–6; P2: weeks 4–6):** the **urorectal septum** (Tourneux and Rathke folds) grows caudally and divides the endodermal cloaca into a dorsal **anorectal canal** (rectum, posterior) and a ventral **UGS (urogenital sinus)** (bladder, anterior). It meets the cloacal membrane, which perforates to form the urogenital and anal openings.
- Paired genital tubercles fuse **cephalad** to the membrane.
- **Wolffian (mesonephric) duct entry** is the landmark that divides the UGS:
- **Cranial vesicourethral canal** (continuous with the allantois) forms the **bladder and pelvic urethra**.
- **Caudal (definitive) UGS** forms the phallic urethra in males, and the vestibule and distal vagina in females.

## Development of the bladder: wall layers

- **Urothelium:** UGS endoderm.
- **Lamina propria, detrusor, adventitia / serosa:** surrounding splanchnic mesenchyme, by epithelial–mesenchymal signalling (epithelium and mesenchyme induce each other).

## Trigone and ureterovesical junction (weeks 5–6)

- The **common excretory duct** (Wolffian duct distal to the bud) dilates and is absorbed into the UGS; **apoptosis** separates the ureter from the Wolffian duct.
- The arms rotate (**Mackie–Stephens**): the ureteric orifice moves **cranially and laterally**; the Wolffian duct ends caudally as the ejaculatory duct at the verumontanum.
- **Trigone muscle:** classic view, ureteric extension; current view (mouse lineage tracing), mostly bladder smooth muscle.

| Bud origin | Orifice | Result |
|---|---|---|
| Too low (early contact) | High, lateral, short tunnel | VUR (vesicoureteral reflux) |
| Too high (late contact) | Caudal, medial | Ectopic ureter, often obstructed |
| Either | Abnormal bud–blastema contact | Dysplasia (VUR and dysplasia coexist) |

- **Weigert–Meyer:** in duplex systems the upper-pole orifice is caudal and medial.

## Allantois, urachus and fetal bladder function

- **Months 4–5:** the bladder descends into the pelvis; the allantoic duct elongates and obliterates as the **urachus** (median umbilical ligament).
- Urine production starts at about **week 10**. The fetal bladder is seen in about half at 10 weeks and almost all by 13 weeks; it empties every 15–20 minutes. After **16 weeks** amniotic fluid is mainly fetal urine.

## Bladder anomalies explained by embryology

| Anomaly | Basis |
|---|---|
| **Exstrophy** | Cloacal membrane not reinforced by infraumbilical mesoderm |
| Urachal (patent urachus, sinus, cyst, diverticulum) | Failed obliteration at different levels |
| Agenesis, duplication | UGS or trigone failure; septation |
| Congenital (Hutch) diverticulum | Detrusor deficiency at the ureteric hiatus |
| VUR, ectopic ureter | Abnormal bud position |

## Developmental basis of EEC: core defect

- The **cloacal membrane is not reinforced by mesodermal ingrowth**. This mesoderm normally forms the lower abdominal muscles and pelvic bones, so the thin membrane ruptures (**Marshall–Muecke** overdevelopment theory).
- **Timing and extent of rupture decide the variant:** before septum descent gives cloacal exstrophy; after gives CBE; limited rupture gives epispadias.

## Developmental basis of EEC: theories

| Theory | Proposed defect | Result |
|---|---|---|
| **Cloacal membrane overdevelopment (Marshall–Muecke)** | Abnormally large membrane in week 4 blocks medial mesenchymal migration | Rupture before septum descent: cloacal exstrophy; after: bladder exstrophy; limited: epispadias |
| Lateral body-wall fold failure | Folds fail to meet in midline | Abdomen and pelvis: cloacal; pelvis alone: classic |
| Caudal genital hillocks | Genital tubercles fuse **below** rather than above the membrane | Dorsal urethral defect (epispadias) |
| Abnormal body-stalk insertion | Caudal insertion prevents mesenchyme interposition | Superficial unstable membrane disintegrates |
| **Bony pelvis first** | Pelvic ring primordium fails to rotate | Pelvic structures cannot meet, so the bladder herniates |

- All theories converge on **absent infraumbilical midline mesoderm**.

## Aetiology of EEC

- Multifactorial (inconsistent twin concordance). **22q11.2 duplication** is the commonest variant; **ISL1** susceptibility gene; p63 under-expressed in exstrophic urothelium.
- **IVF (in vitro fertilisation)** raises risk about 7.5-fold and early high-dose progesterone about tenfold; maternal smoking is linked to cloacal exstrophy; folate to the mildest form.

## Resulting anatomy in classic exstrophy

- Exposed (open) bladder plate; epispadias; low umbilicus; anterior anus; wide levator hiatus; inguinal hernias.
- **Male genitalia:** anterior corpora about 50% shorter; short, upturned penis with dorsal chordee.
- **Female:** bifid clitoris, short vagina, anterior stenotic introitus; later prolapse.
- **Bladder:** immature, not neuropathic (raised type III collagen falls after closure). Ureters enter with no obliquity, so **VUR (vesicoureteric reflux) occurs in 100%** after closure.

## Skeletal abnormalities: classic bladder exstrophy

### 3D (three-dimensional) CT (computed tomography) studies

- **Rotational:** posterior pelvis externally rotated (mean 12° each side); anterior segment externally rotated (mean 18° each side); sacroiliac joint rotated; acetabular and femoral retroversion.
- **Dimensional: pubic diastasis** (mean 4.8 cm); **anterior pubic segment shortened by 30%**; intertriradiate distance increased by about 31%; posterior segment length normal; larger sacrum.
- Bone histology is normal, so growth potential is normal.
- **Clinical effects:** short pendular penis (anterior segment short and splayed); wide hips, waddling gait and out-toeing in early childhood. These cause little disability and improve with growth.
- **Pelvic floor:** levator ani displaced posteriorly and flattened; anterior anus; later pelvic organ prolapse.

- **Spine:** minor variants (spina bifida occulta, lumbarisation, sacralisation) fairly common; scoliosis and dysraphism uncommon.

### Skeletal abnormalities: cloacal exstrophy and isolated epispadias

- **Cloacal exstrophy:** widest, **asymmetric** diastasis (about twice CBE), sacroiliac malformation, hip dislocation.
- **Spinal anomalies in 85–100%** (lumbar myelodysplasia commonest; lipomeningocele); MRI (magnetic resonance imaging) for all.
- Limb defects: **clubfoot** (commonest), limb shortening, tibial and fibular deformity.
- **Isolated epispadias:** male symphysis widened (less than CBE); female symphysis usually closed or narrow.

### Management: prenatal and at birth

- **Prenatal:** prenatal diagnosis in only about 25%. Ultrasound shows no bladder filling, low umbilicus, wide pubic rami, small genitalia and an enlarging lower abdominal mass. Counsel; deliver at an exstrophy centre.
- **At birth:** tie the cord with silk (no clamp); cover the plate with plastic wrap; saline irrigation at nappy change. Renal ultrasound; multidisciplinary team.
- **Assess:** template size and polyps (**≥3 × 3 cm** suits newborn closure), diastasis and pelvic malleability under anaesthesia, urethral plate and penile length, associated anomalies.
- Small template, penoscrotal duplication, hypoplastic bladder: **delayed closure at 6–9 months** (delay closure rather than do a bad early closure).

### Pelvic osteotomy and immobilisation

- **Purpose:** closing the pelvic ring places the bladder neck deep in the pelvis, reduces tension and helps continence.
- **Indications:** closure **after 72 hours**, diastasis **>4 cm**, stiff (poorly malleable) pelvis, failed closure, or cloacal exstrophy (always).
- **Combined bilateral anterior innominate + vertical iliac osteotomy** (most used) with external fixator and Buck traction for 4–6 weeks.
- Under 72 hours with malleable pubes: closure without osteotomy, with modified Bryant traction for 4 weeks.
- Spica casting fails more often (failure with external fixation plus traction 7.5% versus spica 47%).

### Modern staged repair of exstrophy (MSRE)

- Aim of closure: convert exstrophy into incontinent epispadias, with enough outlet resistance for bladder growth but not enough to harm the kidneys.

| Stage | Age | Procedure | Purpose |
|---|---|---|---|
| 1 | Newborn | Bladder, posterior urethra and abdominal wall closure; ± osteotomy | Converts to complete epispadias; outlet resistance stimulates bladder growth |
| 2 | 6–12 months | Epispadias repair (**modified Cantwell-Ransley**), after testosterone | Raises capacity; straightens penis |

| Stage | Age | Procedure | Purpose |
|---|---|---|---|
| 3 | 5–7 years (another source: 4–7 years) | BNR (bladder neck reconstruction): modified **Young-Dees-Leadbetter (YDL)** + bilateral **Cohen** reimplantation | Continence; needs capacity **≥100 mL** and a motivated child |

- **Closure keys: radical division of the urogenital diaphragm fibres** so the vesicourethral unit sits deep in the pelvis (missing this is the commonest error); ureteric stents; suprapubic tube 4 weeks; pubic nylon mattress suture.
- **Alternatives:** CPRE (Mitchell) with penile disassembly; Kelly radical soft-tissue mobilisation; Erlangen total repair (single stage); Mainz ureterosigmoidostomy. Most still need an outlet procedure.

## Single-stage repair (CPRE and others)

- Single-stage repairs aim for fewer operations and continence without bladder neck repair.

| Repair | Features |
|---|---|
| **CPRE (Mitchell)** | Newborn closure + penile disassembly + bladder neck tailoring |
| Erlangen ("total") | All in one at about 8 weeks, no osteotomy |
| Combined closure + epispadias repair | Delayed or failed closures, boys >6 months |

- **Selection:** template ≥3 × 3 cm, elastic, few polyps; experienced team. Not for penoscrotal duplication, hypoplastic bladder or bilateral hydronephrosis.
- **CPRE technique:** 1. Combined general and epidural anaesthesia; **osteotomy if >72 hours** (3 days) old, diastasis >4 cm or stiff pelvis. 2. Incise around the plate; free the bladder; **divide the urogenital diaphragm fibres** to the levator hiatus so the unit sits deep. 3. **Penile disassembly:** right and left hemicorporoglanular bodies plus a urethral wedge on spongiosum (now often left attached to the glans). 4. Urethra placed ventrally between the corpora; corpora joined; meatus usually **hypospadiac**. 5. Bladder and neck closed and tapered; ± bilateral ureteric reimplantation (BUR-CPRE). 6. Pubic No. 2 nylon mattress suture; suprapubic catheter 4 weeks, ureteric stents. 7. Immobilise: external fixation plus traction (failure 7.5% versus spica 47%).
- **Complications:** same frequency as MSRE but **more serious**: dehiscence, prolapse, vesicocutaneous fistula, **glans or corporal loss** (ischaemia), early reimplantation for VUR or hydronephrosis; retention in girls.

## Follow-up, failures and salvage

- **Follow-up:** prophylaxis; ultrasound; yearly cystogram and cystoscopy under anaesthesia.
- Dehiscence or prolapse: wait 4–6 months, then reclose with osteotomy.
- Failed BNR (no 3-hour dry interval by 2 years): bulking, redo YDL, or **augmentation + bladder neck closure + Mitrofanoff** (continent catheterisable stoma).

## Cloacal exstrophy (OEIS) management

- OEIS: omphalocele, exstrophy, imperforate anus, spinal defects.
- Stabilise; neurosurgery (spinal defects 85–100%); separate bowel from hemibladders; colostomy; osteotomy always; gender congruent with karyotype; continence mostly by augmentation with CIC (clean intermittent catheterisation).

### Outcomes and long-term issues

- Daytime dryness after MSRE about **70–74%** (another source: 70–75%), versus about 56% after CPRE with YDL; most still need bladder neck surgery after CPRE.
- **Bladder capacity at neck repair is the main predictor**; each failed closure lowers it. **Best continence predictors: template size at birth and a successful first closure.**
- **Also know:** variants (superior vesical fissure, duplicate exstrophy, pseudoexstrophy); adenocarcinoma risk (cytology, cystoscopy; colonoscopy after ureterosigmoidostomy); reduced male fertility and uterine prolapse in adults; female epispadias; psychosocial support.

**Diagrams to draw:** cloaca divided by the urorectal septum (vesicourethral canal, UGS, allantois) with failed mesodermal ingrowth of the cloacal membrane; Wolffian duct with ureteric bud and its absorption into the UGS; exstrophy pelvis with diastasis; MSRE timeline; penile disassembly into two hemicorpora and urethral wedge.

> **Key points to remember:**
> - Bud from Wolffian duct; position decides VUR or ectopia.
> - Urorectal septum weeks 5–6 (P2: 4–6); mesonephric duct entry is the landmark; vesicourethral canal forms the bladder; urachus from allantois.
> - Trigone: common excretory duct absorption; mostly bladder muscle (current view).
> - Urachus forms at months 4–5; urine from week 10; amniotic fluid mainly urine after 16 weeks.
> - Marshall–Muecke: overdeveloped cloacal membrane, failed mesodermal reinforcement; rupture timing decides variant.
> - CBE: external rotation 12°/18°, anterior pubis 30% short, diastasis about 4.8 cm; VUR in 100%.
> - Cloacal exstrophy: spinal anomalies 85–100%; osteotomy always.
> - MSRE: closure at birth, epispadias repair 6–12 months, YDL / BNR at 5–7 years at ≥100 mL.
> - Osteotomy if closure after 72 h or diastasis >4 cm.
> - CPRE: closure + penile disassembly in newborn. Risks: glans loss, hypospadias, VUR.
> - Capacity ≥100 mL, template size and a successful first closure predict continence.

## LAQ 2 Prune belly syndrome

P1 · P3 **merged from 2**

*Asked as:* P1 LAQ "Prune Belly Syndrome. Antenatal diagnosis, pathophysiology and postnatal management." · P3 SAQ "Prune belly syndrome"

### Definition

- **PBS (prune belly syndrome)**, also called Eagle–Barrett or triad syndrome: **deficient abdominal wall muscles**, **bilateral intra-abdominal testes** and a **dilated, poorly contracting urinary tract**.
- 1 in 29,000–44,000 live births; **95% male**.
- **Survival depends on renal dysplasia and lung hypoplasia, not on the abdominal wall.**

### Antenatal diagnosis

- **Ultrasound** may suggest PBS from 11–14 weeks, but the classic signs are not consistent even at 30 weeks.
- **Findings:** huge bladder filling the abdomen; bilateral hydroureteronephrosis; urachal diverticulum; dilated posterior urethra; echogenic kidneys with cortical cysts (dysplasia); irregular abdominal circumference; fetal ascites; oligohydramnios when severe.
- **Differential diagnosis:** PUV (posterior urethral valves), urethral atresia, megacystis–megaureter.
- Fetal dilatation does **not** predict postnatal renal function.
- **Prenatal management:** intervention has no proven renal benefit. Vesicoamniotic shunting is justified only for **urethral atresia with progressive oligohydramnios**, or to decompress the abdomen and avoid dystocia. Termination is hard to justify except with early, severe oligohydramnios. Plan delivery in a centre with neonatal intensive care.

**Diagram to draw:** fetal ultrasound sketch: huge bladder, dilated posterior urethra, bilateral hydroureteronephrosis.

### Pathogenesis (theories, none proven)

| Theory | Mechanism |
|---|---|
| Early (fetal) urethral obstruction | Back-pressure dilates the tract and stretches the wall; but most have no obstruction after birth |
| Lateral plate mesoderm defect | This mesoderm forms ureters, bladder, prostate, urethra and gubernaculum, so one defect explains the triad |
| Intrinsic urinary tract (ureteric) defect | Primary ureteric dilatation and fetal ascites |
| Yolk sac defect | Primary defect of the yolk sac |

- **Genetics:** mostly sporadic, normal karyotype; smooth-muscle genes **MYOCD (myocardin)** and **FLNA (filamin A)** in multiplex families.

### Pathophysiology and features by organ

- **Kidneys:** dysplasia in about 50% (Potter type II from mesenchymal defect; type IV cystic from obstruction).

- **Ureters:** long and tortuous, worse distally. Collagen replaces smooth muscle, so walls do not coapt and peristalsis fails: **a failed pump, not a blocked pipe**. The result is stasis and infection. **VUR (vesicoureteric reflux) in about 75%.** The proximal ureter has better muscle.
- **Bladder:** large, smooth, collagen-rich; urachal pseudodiverticulum (urachus patent in 25–30%); splayed trigone. Normal compliance but poor contractility, so voiding is "unbalanced", with residual urine.
- **Prostate and urethra:** prostatic hypoplasia dilates the posterior urethra; distal obstruction in about 20%; urethral atresia; **megalourethra** (scaphoid: spongiosum deficient; fusiform: all erectile bodies, more severe).
- **Testes:** intra-abdominal; azoospermia usual; paternity only by sperm retrieval and ICSI (intracytoplasmic sperm injection).
- **Extra-genitourinary (about 75%):** pulmonary hypoplasia; cardiac about 10%; gastrointestinal at least 30% (malrotation); orthopaedic up to 65% (talipes, hip dysplasia).

### Classification (Woodard)

| Category | Features | Outlook | Management |
|---|---|---|---|
| I | Oligohydramnios, **pulmonary hypoplasia**, dysplasia, urethral atresia | Perinatal death | Supportive; bladder drainage only |
| II | Full triad, moderate or unilateral dysplasia / moderate uropathy, **no pulmonary hypoplasia** | Risk of CKD (chronic kidney disease) | Individualised (controversial, debated group) |
| III | Mild or incomplete; **stable function** | Good; majority | Surveillance |

- **Also know:** RUBACE score (Renal, Ureter, Bladder, Abdominal wall, Cryptorchidism, Extra-genitourinary) grades morbidity / severity in survivors; "pseudo-prune" presents late with renal failure.

### Postnatal management: neonatal

Team: neonatology, nephrology, urology, cardiology.

1. Heart and lungs first: **chest radiograph** for pneumothorax and hypoplasia.
2. Creatinine, BUN (blood urea nitrogen), electrolytes; follow the **trend**, because the first value reflects the mother.
3. Renal and bladder ultrasound once stable.
4. Antibiotic prophylaxis; **circumcision**; strict asepsis.
5. **VCUG (voiding cystourethrogram)** only with renal insufficiency or suspected outlet obstruction.
6. **DMSA (dimercaptosuccinic acid) at 4–6 weeks** for split function; MAG3 (mercaptoacetyltriglycine) for drainage.
7. Outlet obstruction: catheter or suprapubic tube.

**Diagram to draw:** VCUG showing large bladder and dilated posterior urethra.

### Postnatal management: urinary tract

- **Conservative** for category III: surveillance, treat infection.
- **Surgery for:** recurrent upper tract infection, sepsis, progressive dilatation or rising creatinine / renal decline, high-grade VUR, true obstruction.

- **Temporary diversion:** wide **cutaneous vesicostomy (Blocksom)**. Avoid high ureterostomy, which wastes the good proximal ureter.
- **Reconstruction after 3–6 months** (lung maturity): excise the distal ureter, taper and reimplant the healthier proximal ureter; reduction cystoplasty gives little lasting benefit (capacity recurs). **CIC (clean intermittent catheterisation)** or Mitrofanoff for poor emptying.

### Orchidopexy

- **Bilateral transabdominal at about 6 months**, often with other surgery; later, **staged Fowler–Stephens** (better than single-stage) or laparoscopic.

### Abdominoplasty

- Aims: cosmesis; better cough, voiding / emptying and defecation.

| Technique | Key feature |
|---|---|
| Randolph | Transverse; excises the worst lower wall |
| Ehrlich | Vertical; umbilicus kept; vest-over-pants overlap |
| Monfort | Vertical; umbilicus kept; central fascial plate with lateral flaps overlapped; transperitoneal access, good exposure for combined surgery |

- **Also know:** Furness (extraperitoneal plication); Dénes; corset.

### Prognosis and follow-up

- About 20% die perinatally; **up to 30% develop chronic renal failure**.
- **Nadir creatinine <0.7 mg/dL at 1 year** predicts stable function.
- Lifelong timed voiding, cultures and imaging; transplant when needed.

> **Key points to remember:**
> - Triad; 95% male; dysplasia and lungs decide survival.
> - Ureter is a failed pump (collagen); VUR about 75%; keep the proximal ureter.
> - Woodard I–III; vesicostomy for drainage; orchidopexy at about 6 months.
> - Nadir creatinine <0.7 mg/dL is favourable; up to 30% reach renal failure.

## LAQ 3 Voiding dysfunction in children: neurophysiology and differential diagnosis of frequency/urgency

P1 · P3 **merged from 2**

*Asked as:* P1 LAQ "Describe neurophysiology of lower urinary tract and various types of voiding dysfunction in children" · P3 SAQ "DD of frequency and urgency in 3 yr male child and treatment"

### Introduction

- The LUT (lower urinary tract) has two opposite functions: **storage** at low pressure with a closed outlet, and **emptying** by a coordinated detrusor contraction with a relaxed outlet. Bladder and sphincter act reciprocally.

### Neurophysiology: peripheral innervation

| | Parasympathetic | Sympathetic | Somatic |
|---|---|---|---|
| Origin | **S2–S4** (sacral parasympathetic nucleus) | Thoracolumbar (T10–L2) | **Onuf nucleus** (S2–S4 ventral horn) |
| Nerve | Pelvic | Hypogastric | Pudendal |
| Bladder | Contracts (ACh [acetylcholine] on **M3**; ATP) | Relaxes (noradrenaline on **β3**) | – |
| Outlet | Relaxes urethra (**NO**, nitric oxide) | Contracts bladder neck (**α1**) | Contracts EUS (external urethral sphincter; nicotinic) |
| Role | Voiding | Storage | Voluntary closure |

- All three nerves carry afferents. **Aδ fibres** (tension receptors in muscle) signal normal fullness. **Unmyelinated C fibres** are "silent" normally; after inflammation, SCI (spinal cord injury) or obstruction they are recruited and drive overactivity (the basis of capsaicin and resiniferatoxin).
- Detrusor has more M2 receptors, but **M3 mediates contraction**.

### Neurophysiology: reflex and central control

- **Storage (spinal reflexes):** low afferent activity triggers sympathetic outflow (detrusor inhibition, bladder-neck closure) and the **guarding reflex** (pudendal; sphincter activity rises with filling).
- **Voiding (spinobulbospinal reflex):** high afferent activity reaches the **PAG (periaqueductal grey)** and then the **PMC (pontine micturition centre, Barrington nucleus)**; the sphincter relaxes first, then the detrusor contracts. The PMC is an on–off switch.
- **Forebrain decides:** the insula senses fullness; the **medial prefrontal cortex** tonically holds the switch "off" until voiding is socially appropriate.
- Lesion patterns: above the pons, overactivity with a coordinated sphincter; between pons and sacral cord, overactivity with **DSD (detrusor–sphincter dyssynergia)**; sacral or peripheral, underactive bladder.
- **Central transmitters:** glutamate excitatory; GABA (γ-aminobutyric acid) and glycine inhibitory; serotonin inhibits voiding; dopamine tonically inhibits the PMC (lost in Parkinson disease).

`Filling → Aδ afferents → sacral cord → PAG low traffic: spinal storage reflexes (sympathetic + guarding) high traffic + cortex "permits": PMC → EUS relaxes → detrusor contracts`

**Diagram to draw:** bladder with pelvic, hypogastric and pudendal nerves, plus brainstem loop (PAG–PMC).

### Maturation in children

- Bladder visible on fetal ultrasound by 14 weeks. Infants void reflexly but **wake before voiding** (cortical arousal gates voiding).
- Infant voiding is often interrupted (transient dyscoordination), with detrusor pressure often **>100 cmH2O**; frequent small voids largely disappear by **2 years**.
- Capacity rises: **EBC (expected bladder capacity) = (age + 2) × 30 mL** (infants: weight in kg × 7 mL).
- Toilet training: bowel control, then daytime bladder, then night-time bladder. Persisting dyscoordination after toddlerhood is dysfunction, not immaturity.

### Definitions: frequency and urgency

- **Frequency:** ≥8 voids a day (ICCS, International Children's Continence Society). **Urgency:** sudden, compelling need to void.
- At 3 years toilet training may still be ongoing, so decide first whether the pattern is abnormal.

### Types of voiding dysfunction: non-neurogenic (functional) LUT dysfunction (ICCS)

| Type | Key feature |
|---|---|
| **OAB (overactive bladder)** | Urgency ± frequency (≥8 voids a day), urgency incontinence; tower flow |
| **Voiding postponement** | Habitual holding; holding manoeuvres; infrequent voids (≤3 a day) |
| **Dysfunctional voiding** | Habitual sphincter/pelvic-floor contraction during voiding in a neurologically normal child; **staccato** flow |
| **Underactive bladder** | Needs abdominal straining to void; interrupted flow |
| Giggle incontinence | Complete emptying with laughter |
| Vaginal reflux | Small leak just after voiding in girls |
| **Enuresis** | Wetting in sleep, ≥5 years; monosymptomatic or non-monosymptomatic |

- **BBD (bladder and bowel dysfunction):** LUT dysfunction plus constipation; rectal loading worsens symptoms and UTI (urinary tract infection).
- **Hinman (non-neurogenic neurogenic bladder):** the severe end of dysfunctional voiding, with trabeculated bladder, VUR (vesicoureteric reflux) and renal damage.

### Types of voiding dysfunction: neurogenic LUT dysfunction

- Causes: **myelomeningocele** (commonest), occult dysraphism / **tethered cord**, sacral agenesis, anorectal malformation, cerebral palsy, SCI, transverse myelitis, pelvic tumour surgery (sacrococcygeal teratoma).

- Pattern = detrusor (overactive or underactive) × outlet (competent, incompetent or dyssynergic). Bony level does not predict bladder function.
- **Hostile bladder: DLPP (detrusor leak point pressure) ≥40 cmH2O** (McGuire 1981), poor compliance, or NDO (neurogenic detrusor overactivity) with DSD.

### Types of voiding dysfunction: anatomical

- **PUV (posterior urethral valves)** "valve bladder": overactivity, then poor compliance, then myogenic failure; ectopic ureter.

### Differential diagnosis of frequency and urgency (3-year-old boy)

| Group | Causes and clues |
|---|---|
| **Infection** | **UTI**: fever, dysuria; balanoposthitis; viral haemorrhagic cystitis |
| **Functional** | **BBD**/OAB; voiding postponement; **extraordinary daytime urinary frequency** (sudden very frequent small voids, no nocturia or wetting, often after stress; self-limiting) |
| **Bowel** | **Constipation**: rectal mass presses on bladder; commonest associated cause |
| **Obstruction** | **PUV**, mild late form; **meatal stenosis** (circumcised boys; thin deflected stream); severe phimosis |
| **Neurogenic** | Occult spinal dysraphism, tethered cord (sacral dimple, hair tuft, gait or foot abnormality) |
| **Bladder irritants** | Stone, foreign body, hypercalciuria, pinworms, rarely **rhabdomyosarcoma** |
| **Polyuria** | Diabetes mellitus, diabetes insipidus, chronic kidney disease |
| **Behavioural** | Anxiety, family stress, developmental disorders |

### Evaluation

- **History:** onset, nocturia, wetting, stream, fever, **bowel habit (Bristol stool chart)**, thirst, stressors.
- **Examination:** abdomen, meatus and foreskin, **lumbosacral spine**, lower-limb neurology.
- **Urinalysis and culture**; glucose; specific gravity >1.005 makes diabetes insipidus unlikely.
- **Bladder and bowel diary.**
- **Ultrasound** of kidneys and bladder: wall thickness, residual, dilation, stones.
- **Uroflowmetry with PVR (post-void residual) and patch EMG (electromyography)** if the child cooperates: bell (normal), tower (OAB), staccato (dysfunctional voiding), interrupted (underactive/DSD), plateau (obstruction).
- **Selective:** VCUG (voiding cystourethrogram) for suspected PUV or febrile UTI with abnormal ultrasound; spinal MRI (magnetic resonance imaging) for neurological signs; urine calcium/creatinine ratio.
- Invasive (video)urodynamics only for neurogenic, anatomical or refractory cases.

**Diagram to draw:** flowchart: urinalysis, then ultrasound with residual, then targeted tests.

### Management

- **Treat constipation first:** disimpaction, then maintenance **polyethylene glycol**, diet and toilet routine.

- **Standard urotherapy** for 4 weeks: timed voiding every 2–3 hours, relaxed posture, adequate fluids, avoid caffeine.
- **UTI:** antibiotics per culture.
- **OAB persisting after urotherapy: oxybutynin 0.3–0.6 mg/kg/day** in 2–3 doses (maximum 15 mg/day; watch for constipation); mirabegron; botulinum toxin; parasacral TENS (transcutaneous electrical nerve stimulation).
- **Dysfunctional voiding: biofeedback**; α-blocker; sphincter botulinum toxin if refractory.
- **Extraordinary daytime frequency:** reassurance; resolves in weeks to months.
- **Enuresis:** alarm or desmopressin.
- **Meatal stenosis:** meatotomy. **PUV:** endoscopic valve ablation. **Tethered cord:** neurosurgical release.
- **Neurogenic: CIC (clean intermittent catheterisation; Lapides 1971) + antimuscarinic**, then intradetrusor botulinum toxin, then augmentation.

**Key points to remember:**

- S2–S4 pelvic (voiding), T10–L2 hypogastric (storage), Onuf/pudendal (EUS).
- PMC = switch; medial prefrontal cortex holds it off.
- EBC = (age + 2) × 30 mL; staccato = dysfunctional voiding.
- DLPP ≥40 cmH2O = hostile.
- Always ask about bowels; treat constipation first.
- Frequency/urgency: rule out UTI, PUV, meatal stenosis and occult dysraphism.
- Extraordinary daytime frequency: no nocturia, normal tests, self-limiting.

## LAQ 4 Antenatal hydronephrosis, fetal uropathy and prenatal scan anomalies

P1 · P3 **merged from 2**

*Asked as:* P3 LAQ "Antenatal hydronephrosis (UTD) and fetal uropathy (LUTO)" · P1 SAQ "Anomalies detected on Prenatal scan between 16th and 20th weeks of gestation"

### Introduction and terminology

- Assumed meaning: HN (hydronephrosis) and ANH (antenatal hydronephrosis) are now called **UTD (urinary tract dilation)**.
- Major congenital anomalies occur in about **2–3% of births**. Urinary tract anomalies are among the commonest detected prenatally (commonly quoted as 20–30% of all prenatal anomalies).
- **UTD is the second commonest prenatal anomaly after cardiac defects, in 1–3% of pregnancies.**
- Mostly UTD on the 20-week scan, usually mild and transient: mild 57–88%, moderate 10–30%, severe 1.5–13% of UTD. The serious group is **LUTO (lower urinary tract obstruction)**: PUV (posterior urethral valves), urethral atresia, PBS (prune-belly syndrome).

### The prenatal anomaly scan (16–20 weeks)

- The **anomaly scan is done at 18–20 weeks** (ACOG (American College of Obstetricians and Gynecologists)); by 20 weeks most urological anomalies are visible.
- Screening US (ultrasound) at 18–20 weeks detects dilation well (sensitivity about 90%) but predicts the diagnosis poorly.
- **Why this window:** fetal urine production begins at about 14 weeks, and after about 16 weeks amniotic fluid is mainly fetal urine, so **oligohydramnios reflects renal failure or outlet obstruction** and causes **pulmonary hypoplasia**. The bladder should be seen filling and emptying.
- Survey kidneys, collecting system, ureters, bladder, genitalia, amniotic fluid, abdominal wall and spine. AFI (amniotic fluid index): normal 8–18 cm; oligohydramnios <5–6 cm.

### Anomalies detected and their signs

| Anomaly | Signs |
|---|---|
| **PUV (posterior urethral valves)** | Male; thick bladder; **keyhole sign**; bilateral hydroureteronephrosis; oligohydramnios |
| **Bilateral renal agenesis** | **Anhydramnios after 16 weeks**, no bladder, no renal arteries: lethal Potter sequence |
| **MCDK (multicystic dysplastic kidney)** | Non-communicating cysts, no central pelvis |
| **ARPKD (autosomal recessive polycystic kidney disease)** | Large, homogeneously **bright** kidneys; oligohydramnios |
| **Bladder exstrophy** | **Bladder never seen**, low cord insertion, lower abdominal mass, wide pubis |
| **Prune belly** | Thin-walled large bladder, empty scrotum |
| **Myelomeningocele** | Spinal defect, lemon and banana signs |

- **Also know:** duplex kidney with ureterocele (intravesical cyst); cloacal exstrophy (adds omphalocele, spinal defects); hydrocolpos; ambiguous genitalia; renal ectopia.

**Diagram to draw:** fetal keyhole sign.

### Causes of antenatal UTD (SFU [Society for Fetal Urology] consensus)

| Cause | Incidence |
|---|---|
| Transient/physiological dilation | 50–70% |
| UPJ (ureteropelvic junction; PUJ, pelviureteric junction) obstruction pattern | 10–30% |
| VUR (vesicoureteral reflux) | 10–40% |
| UVJ (ureterovesical junction) obstruction/megaureter | 5–15% |
| MCDK | 2–5% |
| Duplex system, ureterocele, ectopic ureter | 5–7% |
| LUTO (PUV, prune-belly) | 0.1–0.5% |

- The chance of postnatal pathology rises with grade: about 12% (mild) to 88% (severe). UPJ obstruction and PUV track grade; **VUR is not predicted by grade**.

### Grading systems

**APD (anteroposterior renal pelvic diameter, APRPD):** transverse image at the hilum. APD normal <4 mm before 28 weeks, <7 mm after, <10 mm postnatally.

| Grade | Second trimester | Third trimester |
|---|---|---|
| Mild | 4 to <7 mm | 4 to <9 mm |
| Moderate | 7–10 mm | 9–15 mm |
| Severe | >10 mm | >15 mm |

**SFU (postnatal only):** 1 pelvis barely split; 2 pelvis and major calyces; 3 minor calyces uniformly dilated; 4 plus parenchymal thinning.

**UTD consensus 2014 (risk stratification):** six parameters: APD, central/peripheral calyces, parenchymal thickness, parenchymal appearance, ureter, bladder (plus amniotic fluid prenatally).

| Prenatal parameter | A1 low risk | A2–3 increased risk (any one) |
|---|---|---|
| APD 16–27 weeks | 4 to <7 mm | ≥7 mm |
| APD ≥28 weeks | 7 to <10 mm | ≥10 mm |
| Calyces | Central or none | Peripheral |
| Parenchyma, ureter, bladder | All normal | Abnormal |
| Amniotic fluid | Normal | Unexplained oligohydramnios |

- **A1:** one repeat US after 32 weeks. **A2–3:** US every 4 weeks to delivery. Both: postnatal US after 48 hours.

### Prenatal algorithm (unilateral UTD, male fetus)

```
Anomaly scan: unilateral UTD → Detailed scan: contralateral kidney, ureter, bladder wall and emptying, posterior urethra (keyhole), amniotic fluid, other anomalies ├─ Bilateral / thick bladder / keyhole / oligohydramnios ─► LUTO
```

pathway │ (fetal medicine; karyotype; Ruano staging; possible shunt) ├─ A1 ─► repeat US after 32 weeks └─ A2–3 ─► US every 4 weeks to delivery → Normal delivery at local centre; no fetal intervention for unilateral UTD → Postnatal US after 48 hours (earlier if solitary kidney or suspected LUTO)

- Repeat scans because the lesion may progress and a second anomaly is more likely. **Fetal MRI** (magnetic resonance imaging) only for complex anatomy.
- Counsel the parents: most cases resolve; prognosis with a normal opposite kidney is excellent.

### Antenatal management of fetal uropathy (suspected LUTO): evaluation

- Male fetus, thick-walled "keyhole" bladder, bilateral hydroureteronephrosis, oligohydramnios: LUTO until proved otherwise.
- Anomaly scan, **fetal echocardiogram, karyotype** (one-fifth of megacystis is aneuploid); fetal MRI if unclear.
- **Serial fetal urine** (3 samples, 48–72 hours apart). Favourable: Na <100 mEq/L, Cl <90 mEq/L, osmolality <200 mOsm/L, β2-microglobulin <6 mg/L; not validated.
- **Ruano staging:** I mild, normal fluid: observe. **II severe, function preserved** (oligohydramnios, favourable urine): intervene. III cysts, dysplasia, unfavourable urine: intervention for the lungs only.

### Antenatal management of fetal uropathy: options

| Option | Notes |
|---|---|
| Observation, serial scans | Most cases |
| **VAS (vesicoamniotic shunt)** | Dislodgement, blockage, preterm labour |
| **Fetal cystoscopy + laser valve ablation** | Diagnoses and treats; fistula up to 10% |
| Serial amnioinfusion | Lungs in anhydramnios; under trial |
| Early delivery / termination | Mature lungs, falling fluid / poor prognosis |

- Shunt or fetal cystoscopy is done at 18–24 weeks in severe LUTO.
- **PLUTO (2013):** shunt versus conservative (31 fetuses): better perinatal survival, **no renal benefit**.

### Antenatal management of fetal uropathy: counselling and delivery

- Multidisciplinary team: fetal medicine, paediatric urology, nephrology, neonatology.
- Intervention improves survival, not renal function; complications **over 40%**.
- Deliver where the bladder can be decompressed at once; prophylaxis, then VCUG (voiding cystourethrogram).

### Postnatal algorithm

- **First US after 48 hours**, because neonatal dehydration under-reads dilation. Postnatal normal: APD <10 mm, no calyceal or ureteral dilation.

| | P1 (SFU 1–2) | P2 (SFU 3) | P3 (SFU 4) |
|---|---|---|---|
| Findings | APD 10–15 mm, central calyces | APD ≥15 mm, peripheral calyces or dilated ureter | Abnormal parenchyma or bladder |
| Repeat US | 3–6 months | Individualised | 1 month |

| | P1 (SFU 1–2) | P2 (SFU 3) | P3 (SFU 4) |
|---|---|---|---|
| Antibiotic prophylaxis | No | Discretion; **yes if ureter >7 mm** | Yes |
| VCUG | No | Discretion | Yes |
| MAG3 (mercaptoacetyltriglycine) renogram | No | Discretion | If persistent |

- **Male-specific points:** VCUG confirms a normal posterior urethra. **Circumcision** protects (NNT [number needed to treat] 10). UTI (urinary tract infection) risk: female sex, uncircumcised boys, high grade, dilated ureter.
- Prophylaxis: amoxicillin or a first-generation cephalosporin from birth; nitrofurantoin and trimethoprim only after 8 weeks.
- MAG3: DRF (differential renal function) <40%, T½ >20 minutes, or delayed retention suggest obstruction. MRU (magnetic resonance urography) for anatomy; DMSA (dimercaptosuccinic acid) for MCDK or duplex function.
- **Surgery indications (UPJ):** rising APD, DRF <40% with poor drainage, falling DRF, symptoms (UTI, stone, pain). Megaureter surgery: DRF <40%, febrile UTI; defer reimplant beyond 1 year.
- Outcome: SFU 1 resolves in nearly all by 2.5 years; SFU 2 about 95% by 4 years.

### Bilateral dilation and PUV: how they differ

- Bilateral hydroureteronephrosis, thick bladder, keyhole sign and oligohydramnios mean LUTO until proved otherwise.
- At birth: 5 Fr feeding tube, prophylaxis, creatinine, early VCUG; prenatal intervention improves survival but not renal function (PLUTO).

**Diagram to draw:** UTD A1/A2–3 prenatal and P1–P3 postnatal flowchart; keyhole bladder with bilateral hydroureteronephrosis and shunt.

> **Key points to remember:**
> - UTD: 1–3% of pregnancies; second commonest anomaly after cardiac.
> - Anomaly scan 18–20 weeks; after 16 weeks amniotic fluid = fetal urine.
> - ANH = UTD; APD normal <4 mm before 28 weeks, <7 mm after, <10 mm postnatally.
> - UTD A2–3 = APRPD ≥7 mm (16–27 weeks) or any abnormal feature.
> - Transient dilation 50–70% (commonest cause); UPJ obstruction commonest surgical cause; VUR is not predicted by grade.
> - First postnatal US after 48 hours; A1: one scan after 32 weeks; A2–3 scan every 4 weeks.
> - Risk-based care: P1 no prophylaxis or VCUG; prophylaxis if ureter >7 mm or P3; P3 prophylaxis, VCUG, MAG3.
> - Keyhole + oligohydramnios = PUV; absent bladder = exstrophy or agenesis.
> - ARPKD large bright kidneys; MCDK non-communicating cysts.
> - Favourable urine: Na <100, Cl <90, osmolality <200, β2-microglobulin <6.
> - PLUTO: survival up, renal function unchanged.

# LAQ 5 Embryological development of urinary bladder and management of Hutch Diverticulum?

P3

**Type:** LAQ (25 marks)
**Source:** 2021-Summer Paper 3, Q1 (bladder diverticulum also answered in 2016-Summer Paper 2, Q3c)

## A. Embryology of the bladder

### 1. Division of the cloaca (weeks 4–6)

- The **urorectal septum** (Tourneux and Rathke folds) divides the endodermal cloaca into a dorsal **anorectal canal** and a ventral **UGS (urogenital sinus)**. The cloacal membrane perforates to form the urogenital and anal openings.
- The **Wolffian (mesonephric) duct entry** is the landmark:
- **Cranial vesicourethral canal** (continuous with the allantois) → bladder and pelvic urethra.
- **Caudal (definitive) UGS** → phallic urethra in males; distal vagina and vestibule in females. The whole female urethra comes from the pelvic part.

### 2. Wall layers

- **Urothelium:** endoderm of the UGS.
- **Lamina propria, detrusor and adventitia:** surrounding splanchnic mesenchyme. Epithelium and mesenchyme induce each other (epithelial–mesenchymal signalling).

### 3. Trigone (weeks 5–6)

- The **common excretory ducts** (mesonephric duct distal to the ureteric bud) dilate and are absorbed into the UGS. **Apoptosis** separates the ureters from the mesonephric ducts.
- Ureteric orifices migrate **cranially and laterally**; the Wolffian ducts end caudally as ejaculatory ducts at the verumontanum.
- Classic view: trigone muscle is ureteric (Wolffian) tissue. Current view (mouse lineage tracing): **mostly bladder smooth muscle**, minor ureteric contribution.
- **Bud position (Weigert–Meyer):** bud too low → lateral orifice, short tunnel → VUR (vesicoureteral reflux); too high → caudal, medial, ectopic orifice.

### 4. Allantois and urachus

- **Months 4–5:** the bladder descends into the pelvis. The allantoic duct elongates and obliterates into the **urachus** (median umbilical ligament).

### 5. Function

- Urine production starts at **week 10**. Fetal bladder is seen from 10 weeks, almost always by 13 weeks, and empties every 15–20 minutes. After **16 weeks** amniotic fluid is mainly fetal urine.

## B. Anomalies explained by embryology

| Anomaly | Embryological basis |
|---|---|
| Agenesis | Cranial UGS atrophy, or trigone not formed; survival only with ectopic ureters |
| Exstrophy | Cloacal membrane not reinforced by mesoderm |

| Anomaly | Embryological basis |
|---|---|
| Duplication | Complete or incomplete; sagittal or coronal |
| Urachal (patent urachus, sinus, cyst, vesicourachal diverticulum) | Urachus not obliterated at different levels |
| Congenital diverticulum | Focal detrusor deficiency, often at the ureteric hiatus |

## C. Hutch diverticulum: definition

- **Paraureteral diverticulum**: mucosa herniates through the **ureteric hiatus**, between the intravesical ureter and the roof of the hiatus, superolateral to the orifice, sparing the trigone. **Hutch** divided them into primary and secondary.

| | Primary (congenital) | Secondary (acquired) |
|---|---|---|
| Cause | Congenitally deficient detrusor at the hiatus | High pressure: PUV (posterior urethral valves), BOO (bladder outlet obstruction), neurogenic bladder (described in paraplegics), dysfunctional voiding, prior surgery |
| Bladder | Smooth; single, often intermittent | Trabeculated; one of many |

## D. Pathophysiology

- No muscle backing → **VUR**; no muscle in the sac → cannot empty → stasis, UTI (urinary tract infection), stones; tumour in adults.
- An enlarging sac damages **Waldeyer's sheath** and pulls the ureter out → UVJ (ureterovesical junction) dysfunction, reflux, later **obstruction** and renal damage.
- Large sacs compress the bladder neck → retention (vicious circle).

## E. Evaluation

- **US (ultrasound):** sac, residual urine, hydronephrosis.
- **VCUG (voiding cystourethrogram), gold standard:** oblique and post-void films show size, VUR and whether the ureter enters the sac. Repeat if suspicion persists (sac is dynamic).
- **DMSA (dimercaptosuccinic acid):** scars; **MAG3 (mercaptoacetyltriglycine) renogram** if obstruction.
- **Urodynamics** if secondary; cystoscopy (orifice, neck); in adults enter the sac, cytology.

## F. Management

**1. Treat the cause first (secondary):** valve ablation, BOO surgery, CIC (clean intermittent catheterisation) and antimuscarinics. Remodelling may make excision unnecessary.

**2. Primary diverticulum:**

| Situation | Management |
|---|---|
| Small, asymptomatic, no VUR | Observe |
| VUR, ureter beside the sac | As primary VUR (resolves like VUR of the same grade); many excise if it persists. Dx/HA (dextranomer/hyaluronic acid) injection possible |
| Ureter enters the sac | VUR will not resolve → surgery |

| Situation | Management |
|---|---|
| Breakthrough febrile UTI, new scars, large sac with residual, stones, obstruction, retention | **Diverticulectomy + ipsilateral reimplantation** |

**3. Operative steps (open, laparoscopic or robotic):** `Cystoscopy ± ureteric catheter → cystotomy → evert sac or dissect it from its pseudocapsule (0.5 cm mucosal cuff round the neck) → close detrusor defect in two layers (firm backing) → reimplant ureter: Cohen cross-trigonal or Politano–Leadbetter (tunnel 5:1) → catheter ± suprapubic tube, drain`

- Extravesical or combined approaches suit selected cases. In adults, endoscopic outlet surgery is done first.

## G. Complications and follow-up

- Ureteric injury, leak, persistent VUR, obstruction, recurrence (syndromic patients).
- US at 6–12 weeks; VCUG for febrile UTI; urodynamics if voiding is abnormal.

**Also know:** posterolateral multiple diverticula (Ehlers–Danlos, Williams, Menkes); hypoplasia; megacystis.

**Diagram to draw:** cloacal division by the urorectal septum; Hutch sac beside the ureteric orifice.

> **Key points to remember:**
> - Urorectal septum weeks 4–6; vesicourethral canal → bladder; urachus at months 4–5.
> - Hutch: herniation through the ureteric hiatus → VUR.
> - Ureter entering the sac: VUR will not resolve.
> - Secondary: treat the outlet first; else diverticulectomy + reimplantation.

# LAQ 6 Development of the urethra, congenital urethral anomalies and management of epispadias (including isolated penopubic epispadias)

P3

**Asked as:**

- Discuss development of urethra. Classify congenital anomalies and discuss management of epispadias — *Practice Set 6 Paper III Q2 (LAQ)*
- Isolated penopubic Epispadias. — *Summer 2022 Paper III Q3f (SAQ)*

**Type:** LAQ (25 marks); also asked as SAQ (10 marks)
(isolated penopubic epispadias also answered in 2022-Summer Paper 3, Q3f)

## Development of the urethra

### 1. Division of the cloaca (weeks 5–6)

- The urorectal septum divides the cloaca into the dorsal anorectal canal and the ventral **UGS (urogenital sinus)**.
- Wolffian duct entry is the landmark. Above it, the **vesicourethral canal** forms the bladder and pelvic urethra. Below it, the **definitive UGS** forms the phallic urethra in males, and the urethra and vestibule in females.
- UGS also forms the prostate (buds at 10 weeks) and Cowper and Littre glands; the **utricle** is Müllerian.
- The epithelium of the **entire male urethra is endodermal** (UGS origin).

### 2. Penile urethra: the "double zipper" (androgen-dependent)

- Indifferent genital tubercle until 8–9 weeks. Testosterone (Leydig cells, 8 weeks) is converted to DHT (dihydrotestosterone) by **type 2 5α-reductase**.
- The **urethral plate** (endoderm) forms in the tubercle.
- **Opening zipper:** the plate canalises proximal to distal into a urethral groove bounded by the urethral folds.
- **Closing zipper:** the folds fuse in the ventral midline, forming the tubular urethra and median raphe.
- **Glanular urethra:** formed by direct canalisation of the plate inside the glans (current endodermal view; Glenister's ectodermal ingrowth theory is outdated).
- **Timetable:** coronal sulcus 11–12 weeks; fusion complete 16 weeks; urethra complete 18 weeks; fetal curvature gone 20 weeks.
- The prepuce forms with the glanular urethra, so hypospadias has a hooded foreskin.
- **Female:** the vestibular plate canalises but the folds do not fuse, giving an open vestibule and labia minora.

## Classification of congenital urethral anomalies

| Mechanism | Anomaly |
|---|---|
| Failed ventral fold fusion | **Hypospadias** (1 in 250 males; commonest at shaft–glans junction); congenital urethral fistula |
| Dorsal defect | **Epispadias** (exstrophy–epispadias complex) |

| Mechanism | Anomaly |
|---|---|
| Obstruction | **PUV (posterior urethral valves)**, Young type 1 (95%) and type 3; **anterior urethral valves** with diverticulum; urethral atresia or hypoplasia; meatal stenosis |
| Duplication | **Urethral duplication**: sagittal; ventral channel is functional; Effmann type I blind, II complete, III with caudal duplication |
| Dilatation | **Megalourethra**: scaphoid (spongiosum deficient) or fusiform (all erectile tissue deficient, more severe); prune-belly association |
| Cystic or remnant | Cowper's syringocele; enlarged utricle (proximal hypospadias); parameatal cyst |

## Epispadias (including isolated penopubic epispadias): definition, embryology, classification and clinical features

- Part of the **exstrophy–epispadias complex**: a **dorsal urethral defect** (defect of the dorsal urethral wall): the roof is replaced by a broad (dorsal) mucosal strip (urethral plate). Cause (embryology): absent mesodermal reinforcement of the cloacal membrane; the genital tubercles fuse **caudal** to it, giving a dorsal urethral defect.
- **Male (1 in 117,000; incidence of isolated male epispadias about 1 in 117,000), classification by meatus:** glanular, penile, **penopubic** (subsymphyseal) (about 70% of male cases are complete (penopubic), with incontinence). **Female (1 in 484,000), Davis:** vesicular (patulous), subsymphyseal, retrosymphyseal (incontinent).
- **Features (clinical):** penopubic meatus; outlet may admit a finger → **total incontinence**. **Short, broad, upturned penis with dorsal chordee**, like exstrophy; pubic diastasis (less than exstrophy); small bladder from no outlet resistance; **VUR (vesicoureteral/vesicoureteric reflux) 30–40% (male), 30–75% (female)**. Female (Davis retrosymphyseal type): bifid clitoris, depressed mons, incontinence.

## Evaluation

- (Renal) ultrasound; VCUG (voiding cystourethrogram); cystoscopy and capacity under anaesthesia.
- **Key decision:** epispadias or exstrophy pathway. Treat as exstrophy (osteotomy and closure) if pubic diastasis is over 3 cm, the intrasymphyseal bar is poor, sphincter is absent or capacity is tiny (tiny bladder).

## Management of epispadias

- **Aims:** continence, safe upper tracts, a straight functional penis of good length.
- **Preoperative testosterone** 2 mg/kg intramuscularly (intramuscular), 5 and 2 weeks before urethroplasty.

### Stage 1: penile reconstruction and urethroplasty (6–12 months)

- Release chordee: divide suspensory ligaments and scar; free corpora from the inferior pubic rami.
- **Modified Cantwell–Ransley:** outline an 18-mm plate to the tip; reverse MAGPI (meatal advancement and glanuloplasty) brings the meatus to the tip; free the plate from the corpora (distal 1 cm left on the glans); tubularise over 8 Fr; transpose it (the tubularised plate is moved) **ventral to the corpora**, which are rotated medially ± cavernocavernostomy; close glans wings; Z-plasty at the base; stent 10–12 days.
- **Mitchell–Bägli complete penile disassembly:** reliable chordee correction, but often leaves hypospadias.

- Outlet resistance rises, so capacity grows.

**Stage 2: bladder neck reconstruction (4–5 years, capacity about 100 mL)**

- **YDL (Young–Dees–Leadbetter)** bladder neck repair + Marshall–Marchetti–Krantz suspension + **ureteric reimplantation**. Osteotomy usually unnecessary.
- **Good capacity:** single-stage repair possible.

**Female epispadias:** at 6–12 months excise the urethral roof, close over 10 Fr, unite clitoral halves and labia minora, monsplasty; YDL at 4–5 years.

### Outcomes and complications

- Continence after bladder neck reconstruction: **males 47–82%, females 70–87%**; capacity is the main predictor.
- Early fistula up to (about) one-fifth (at the penile base); long-term fistula and stricture about 5% (stricture rare); erections and paternity (usually) preserved.
- Better than exstrophy (bladder never exposed or scarred). Failures (failure): bulking injection (injectables), augmentation, continent stoma.

**Diagram to draw:** double zipper (canalisation then fusion); penopubic epispadias (dorsal penopubic meatus and plate).

> **Key points to remember:**
> - Whole male urethra is endodermal UGS; fold fusion complete at 16, urethra at 18 weeks.
> - About 70% of male epispadias are penopubic and incontinent; VUR 30–40%.
> - Diastasis over 3 cm (>3 cm) → exstrophy pathway.
> - Cantwell–Ransley at 6–12 months; YDL (Young–Dees–Leadbetter) at about 100 mL capacity.
> - Continence after YDL: males 47–82%, females 70–87%.

## Short answer questions (SAQ)

## SAQ 1 Posterior urethral valves: embryology, management and valve bladder

P1 · P2 · P3 **merged from 3**

*Asked as:* P3 SAQ "Embryology and Management of P.U. valve patient." · P1 SAQ "Embryology of Posterior Urethral Valves." · P2 SAQ "Valve bladder: management"

### Introduction

- PUV (posterior urethral valves): the commonest cause of bladder outlet obstruction in boys (1.4–2.2 per 10,000 births).
- Morgagni described them (1769); Hugh Hampton Young named and classified them (1919).

### Normal development

- **Weeks 5–6:** the **urorectal septum** divides the cloaca into the anorectal canal (rectum) and the **urogenital sinus**.
- The **Wolffian (mesonephric) ducts** join the cloaca just before division. Their orifices normally migrate to the **verumontanum**, leaving small folds (plicae colliculi).

- The **urogenital membrane** normally disappears completely.
- Fetal urine starts at 10–14 weeks; after 16 weeks amniotic fluid is mainly fetal urine. So obstruction in this period damages bladder, kidneys and lungs.

### Embryology of the valves: Young types and their origin

| Type | Anatomy | Embryology |
|---|---|---|
| **I (about 95%)** | Leaflets from the verumontanum fuse anteriorly just proximal to the external sphincter | Abnormal anterior / anterolateral insertion of the Wolffian duct ends into the cloacal wall: hypertrophied inferior urethral crest |
| **II** | Folds from verumontanum to bladder neck | Not real (Stephens 1955): secondary bladder-neck hypertrophy |
| **III (5–10%)** | Annular (ring) diaphragm with a central hole | **Persistent urogenital membrane** |

### Other theories and genetics

- Hypertrophied mucosal folds (oldest); cloacal remnants; Wolffian–Müllerian maldevelopment.
- **COPUM (congenital obstructing posterior urethral membrane, Dewan):** one oblique membrane, torn by the first catheter into apparent type I leaflets. Plausible but unproven.
- **Cobb's collar (Moormann's ring):** a congenital ring-like narrowing of the proximal bulbar urethra, just distal to the external sphincter; some authors group it with type III valves.
- **Genetics:** mostly sporadic; chromosome 11 duplications; copy number variants (worse renal outcome); TBX5, PTK7.

**Diagram to draw:** posterior urethra with type I leaflets versus a type III membrane, with keyhole VCUG.

### Consequences of obstruction

- Thick bladder, dilated posterior urethra (**keyhole sign**), hydroureteronephrosis, **renal dysplasia**, oligohydramnios, pulmonary hypoplasia.
- Pop-offs (VUR (vesicoureteral reflux), urinoma, ascites) relieve pressure.
- Dysplasia is fixed in utero; ablation cannot reverse it.

### Antenatal management

- Bilateral hydroureteronephrosis, thick bladder, **keyhole sign**, oligohydramnios.
- **Vesicoamniotic shunt** helps survival (lungs), not kidneys (PLUTO trial, 2013).

### Neonatal management

1. **Drain the bladder:** 5–7 Fr feeding tube (coudé if it curls); minimal balloon inflation.
2. **Stabilise:** ventilation; correct **post-obstructive diuresis**, hyponatraemia, acidosis, hyperkalaemia; antibiotics.
3. **Confirm:** VCUG (voiding cystourethrogram), renal ultrasound, MAG3 (mercaptoacetyltriglycine) at 2–3 months.
4. **Primary endoscopic valve ablation** (preferred): laser, Bugbee or hook at 5 and 7 o'clock ± 12; VCUG within 1 month.
5. **Vesicostomy** if the urethra is too small or drainage fails. **Upper-tract diversion** rarely: no proven renal benefit.

6. **Circumcision:** cuts febrile UTI (urinary tract infection) sharply. - **VUR** (50–80%) is secondary; treat the bladder, not the ureter.

### Prognosis

| Nadir creatinine (first year) | ESRD (end-stage renal disease) risk |
|---|---|
| < 0.4 mg/dL | Nil |
| 0.4–0.69 | 2% |
| 0.7–0.99 | 27% |
| > 1.0 | Up to 100% |

- Lifetime ESRD risk 20–50%.

### Valve bladder: definition

- **Valve bladder syndrome** (Mitchell, 1982): upper tracts and renal function worsen after successful PUV ablation, without residual obstruction, because of bladder dysfunction.

### Valve bladder: pathophysiology (vicious cycle)

``` Detrusor hypertrophy, collagen, low compliance → poor sensation, rising residual

- polyuria (nephrogenic diabetes insipidus) → bladder never empty → pressure transmitted to dilated upper tracts → they refill the bladder → renal damage ```
- Three processes: **polyuria, poor sensation, residual urine**; together they keep the upper tracts dilated.
- Age pattern: hyperreflexia in infancy, better compliance in childhood, then a large atonic bladder (myogenic failure) after puberty.

**Diagram to draw:** vicious-cycle diagram of the valve bladder.

### Valve bladder: clinical features and evaluation

- Late continence (about 5.5 years), incontinence, UTI, worsening hydronephrosis, rising creatinine. **New incontinence after being dry is a red flag.**
- Uroflowmetry and PVR (post-void residual) at each visit; ultrasound; creatinine; voiding diary.
- **Videourodynamics** when the picture changes and before transplantation.

### Valve bladder: management (escalating ladder)

1. **Behavioural:** adequate fluids, timed and double voiding, biofeedback.
2. **Drugs:** anticholinergic (oxybutynin 0.1 mg/kg twice daily) for a small high-pressure bladder; α-blocker for high residual. Stop oxybutynin if residual rises.
3. **CIC (clean intermittent catheterisation)** if residual rises despite drugs.
4. **Overnight bladder drainage** (catheter for 7–10 hours) for worsening hydronephrosis, UTI or renal function; it gives bladder and kidneys a daily rest.
5. **Mitrofanoff appendicovesicostomy** if urethral CIC is difficult (sensate urethra, high bladder neck).
6. **Augmentation** (rare) for a small, high-pressure bladder despite the above: **ureterocystoplasty** with the dilated ureter is preferred, avoiding bowel (mucus, acidosis, stones). - Reflux is treated through the bladder; reimplantation is rarely needed.
```

### Long-term care and transplantation

- **CKD (chronic kidney disease)** care.
- ESRD in 20–50%. Before **transplantation**, optimise the bladder (videourodynamics, CIC or overnight drainage); graft survival is then similar.

> **Key points to remember:**
> - Type I (95%): abnormal Wolffian duct insertion; type III: persistent urogenital membrane; type II not real.
> - Urorectal septum divides the cloaca at weeks 5–6.
> - COPUM: one membrane torn into leaflets by catheterisation.
> - Dysplasia is fixed in utero; ablation cannot reverse it.
> - Catheter, stabilise, then primary valve ablation; vesicostomy if not possible.
> - Nadir creatinine below 0.7 mg/dL predicts good renal outcome.
> - Valve bladder: polyuria, poor sensation, residual urine; overnight drainage and CIC break the cycle.
> - Ureteric augmentation rather than bowel; optimise before transplant.

## SAQ 2 Ultrasonography in neonates

P1 · P3 **merged from 2**

*Asked as:* P1 SAQ "USG in neonates" · P3 SAQ "HSG in neonates"

### Introduction

- Assumed meaning: ultrasonography (USG) of the neonatal urinary tract, including antenatal hydronephrosis follow-up. "HSG" (hysterosalpingography) in the P3 wording is not done in neonates, so it is taken as USG.
- First-line imaging, used for antenatal **UTD (urinary tract dilation)** (1–3% of pregnancies), febrile UTI (urinary tract infection), mass and poor stream.

### Why ultrasound first

- No radiation or sedation; repeatable. Shows **anatomy, not function**.

### Timing and technique

- **First scan after 48 hours**: earlier, physiological (relative) dehydration under-reads dilation.
- **Scan early if outlet obstruction (PUV, posterior urethral valves) is suspected or the kidney is solitary**; catheterise in suspected obstruction.
- Higher-frequency probes than in adults; kidneys supine and prone; length against age nomograms; bladder full and after voiding; Doppler.
- Measure **APD (anteroposterior pelvic diameter)** on a transverse mid-kidney view; assess calyces, parenchyma, ureters and bladder wall (thick if over 3 mm when full).

### Normal neonatal kidney (pitfalls)

- Cortex **iso- or hyperechoic** to liver.
- **Prominent hypoechoic pyramids** are normal and mimic dilated calyces or cysts (not hydronephrosis).
- Normal postnatal **APD <10 mm**, with no calyceal or ureteric dilation.

## Grading

- **SFU (Society for Fetal Urology) grades:** 0 none; 1 pelvis only; 2 major calyces; 3 major and minor calyces; 4 grade 3 plus parenchymal thinning.
- **UTD 2014, prenatal:** A1 (APD 4 to <7 mm before 28 weeks, 7 to <10 mm after) versus A2–3 (larger, or any other abnormality).

## Antenatal hydronephrosis follow-up (UTD 2014, postnatal)

| UTD risk | Findings | Action |
|---|---|---|
| **P1** (SFU 1–2) | APD 10 to <15 mm, central calyces | Repeat US at 3–6 months; no VCUG (voiding cystourethrogram) or prophylaxis |
| **P2** (SFU 3) | APD ≥15 mm, peripheral calyces or dilated ureter | Repeat US; VCUG and prophylaxis at discretion / optional (give prophylaxis if ureter >7 mm) |
| **P3** (SFU 4) | Abnormal parenchyma or bladder | Repeat US at 1 month; VCUG, prophylaxis, MAG3 (mercaptoacetyltriglycine) scan |

**Diagram to draw:** SFU grades 0–4 as kidney outlines.

## Specific neonatal findings

| Condition | Ultrasound finding |
|---|---|
| PUJO (pelviureteric junction obstruction) | Dilated pelvis, normal ureter; pyramids under 3 mm predict pyeloplasty |
| VUR (vesicoureteral reflux) | Normal first scan in up to 25% |
| **PUV** | Thick bladder, dilated posterior urethra (**keyhole sign**), bilateral hydroureteronephrosis; VCUG before discharge |
| Duplex anomalies | Ureterocele (thin-walled intravesical cyst), ectopic ureter, megaureter |
| MCDK (multicystic dysplastic kidney) | Non-communicating cysts of varying size, no central pelvis, scant parenchyma |
| ARPKD (autosomal recessive polycystic kidney disease) | Bilateral large, bright (echogenic) kidneys |
| Renal vein thrombosis | Large echogenic kidney, no venous flow; haematuria, mass, thrombocytopenia, hypertension |
| Adrenal haemorrhage | Avascular, evolving suprarenal mass (neuroblastoma is vascular) |
| Other masses | Mesoblastic nephroma |
| Spine | Occult dysraphism before arch ossification (up to 6 months) |
| DSD (differences of sex development) | Müllerian structures, gonads |

- **Also know:** hydrometrocolpos, scrotal masses, twinkling artefact (stones).

## Limitations

- Normal US does not exclude VUR. Function needs MAG3 renography (after renal maturation) or DMSA (dimercaptosuccinic acid).

### Recent advances

- ceVUS (contrast-enhanced voiding urosonography).

> **Key points to remember:**
> - First postnatal scan after 48 h, except suspected PUV; normal APD <10 mm.
> - Echogenic cortex and prominent pyramids are normal in neonates.
> - UTD: P1 APD 10 to <15 mm; P2 15 mm or more; P3 abnormal parenchyma or bladder.
> - Keyhole sign + thick bladder = PUV; spinal US only up to 6 months.

## SAQ 3 Nocturnal enuresis

P1 · P3 **merged from 2**

*Asked as:* P1 SAQ "Nocturnal enuresis" · P3 SAQ "Enuresis"

### Definition (ICCS [International Children's Continence Society])

- Intermittent urinary incontinence **during sleep** in a child **aged 5 years or more**.
- **Primary:** never dry. **Secondary:** wetting after at least **6 months** dry (look for stressors, diabetes, UTI [urinary tract infection], neurological disease).
- **MNE (monosymptomatic):** no daytime LUTS (lower urinary tract symptoms). **NMNE (non-monosymptomatic):** with daytime LUTS.

### Epidemiology

- About **5–10% at 7 years**; falls with age, with spontaneous resolution about 15% a year; about 2% of adults; familial.

### Pathophysiology: three-way mismatch

1. **Nocturnal polyuria** (poor or reduced night-time ADH [antidiuretic hormone] rise).
2. **Small (reduced) nocturnal bladder capacity** or DO (detrusor overactivity).
3. **Failure to arouse** from sleep. - **Why sleep apnoea matters:** obstructed breathing makes arousal difficult and releases atrial natriuretic peptide, which raises urine output. - Comorbidities: constipation, sleep-disordered breathing, ADHD (attention deficit hyperactivity disorder).

**Diagram to draw:** triad: nocturnal polyuria, small bladder capacity, arousal failure.

### Evaluation

- **History:** day symptoms, bowels, snoring, fluids, family history.
- **Examination:** spine, genitalia, abdomen for faecal loading; urinalysis.
- **Bladder diary:** daytime voided volumes against EBC (expected bladder capacity) = (age + 2) × 30 mL in the chapter (ICCS uses [age + 1] × 30 mL); night urine output by weighing nappies.
- Uroflowmetry, PVR (post-void residual) and ultrasound only for NMNE or before adding anticholinergics. No invasive tests in MNE.

### Management: general

- **Treat daytime LUTS and constipation first** (in NMNE).
- Education removes blame; standard urotherapy. Waking, fluid restriction alone and punishment do not work.

### Management: first line

| | Alarm | Desmopressin |
|---|---|---|
| Target | Arousal (conditioning) | Night urine output |
| Use | Nightly for months; parent wakes child; continue to 14 dry nights | 0.2–0.4 mg tablet or 120–240 µg melt, 1 hour before bed |
| Response | 50–70%; durable | Rule of thirds; immediate but relapses off the drug |
| Caution | Needs motivated family | Restrict / limit evening fluids (hyponatraemia) |

- Overlearning (extra evening fluid once dry) reduces alarm relapse.

### Management: failure and resistant cases

- **Failure:** switch or combine.
- **Second line:** add an anticholinergic (oxybutynin) after checking PVR and bowels.
- **Resistant:** imipramine (cardiotoxic; ECG [electrocardiogram]), off-label β3-agonists; treat sleep apnoea (adenotonsillectomy).

### Adult enuresis

- New enuresis in an adult may signal high-pressure chronic retention: catheterise and treat the obstruction.

> **Key points to remember:**
> - ≥5 years; secondary = after 6 months dry.
> - Alarm (durable) and desmopressin (rule of thirds) are first line.
> - Treat daytime symptoms and constipation first.

## SAQ 4 Paediatric neurogenic bladder (myelodysplasia): assessment, management and bladder myoplasty

P1 · P3 **merged from 2**

*Asked as:* P3 SAQ "Neurogenic bladder: prenatal/neonatal assessment and bladder myoplasty" · P1 SAQ "Neurogenic bladder in myelodysplasia"

### Introduction

- Myelodysplasia (NTD [neural tube defect], mainly **myelomeningocele**, an open NTD) is the commonest cause of neurogenic LUT (lower urinary tract) dysfunction in children; other causes are occult dysraphism, sacral agenesis and anorectal malformation.
- Lesions: lumbosacral 47%, lumbar 26%, sacral 20%.
- **Bony or motor level does not predict bladder function**, and the lesion can progress (re-tethering), so every newborn needs urodynamic testing.
- About 10% of deaths in SB (spina bifida) are urological. Folic acid 400 µg daily before conception prevents 50–70% of NTDs.

### Patterns of dysfunction

- Detrusor: overactive (NDO, neurogenic detrusor overactivity) or underactive. Outlet: competent, incompetent or dyssynergic.

- Newborns: most have detrusor contractions; over a third are acontractile. Needle EMG (electromyography): intact sacral arc about 40%, partial denervation 25%, complete 35%.

### Prenatal assessment

- **Maternal serum AFP (alpha-fetoprotein)** is raised in open defects; amniotic AFP and acetylcholinesterase confirm.
- **Ultrasound:** spinal defect; "lemon" and "banana" signs of Chiari II; ventriculomegaly; hydronephrosis, thick bladder. Fetal MRI (magnetic resonance imaging) for level.
- Multidisciplinary counselling at a fetal centre.
- **MOMS (Management of Myelomeningocele Study) 2011:** closure **before 26 weeks** reduced shunt need (about 40% versus 82%) and improved motor and cognitive scores, but caused prematurity (34 versus 37 weeks) and uterine thinning. **No proven bladder benefit**: CIC (clean intermittent catheterisation) and reconstruction rates unchanged.

### Neonatal assessment (baseline; EAU [European Association of Urology] 2022)

| Test | Timing / purpose |
|---|---|
| Examination | Spine, legs, anal tone, perineal sensation, bladder |
| **Renal and bladder ultrasound** | Before discharge, then 6-monthly for 2 years (serially); hydronephrosis, wall thickness |
| Residual urine | Start CIC if not emptying |
| Creatinine | Within 3 months (cystatin C later; better with low muscle mass) |
| **VCUG (voiding cystourethrogram) or video-UDS** | Reflux; dilating VUR (vesicoureteric reflux) in about 7% |
| **UDS (urodynamics)** with EMG | Within 3 months, once stable after closure |
| DMSA (dimercaptosuccinic acid) | Baseline scarring |

### Urodynamic risk (McGuire 1981; UMPIRE [Urologic and Renal Protocol for the Newborn and Young Child with Spina Bifida])

| Risk | DLPP (detrusor leak point pressure) or end-filling pressure |
|---|---|
| High (hostile) | 40 cm $H_2O$ or more |
| Intermediate | 25–39 cm $H_2O$, NDO or poor compliance |
| Low (safe) | Under 25 cm $H_2O$ |

- Poor compliance and DSD (detrusor–sphincter dyssynergia) also predict upper-tract damage.
- Repeat studies: the lesion can change (re-tethering).

**Diagram to draw:** neonatal UDS trace with DLPP over 40 cm $H_2O$.

### Management: kidneys first, then continence

- **Early CIC (Lapides 1971) plus oxybutynin** for high-risk (hostile) bladders: hydronephrosis 0–15% versus 18–80% expectant; ESRD (end-stage renal disease) falls from about 30% to under 2% (P1: 0–1.6%).
- Other drugs: solifenacin, mirabegron; **botulinum toxin A** about 10 U/kg (maximum 360 U), continence 65–87%.

- Surgery: vesicostomy (bridge); **augmentation cystoplasty**; Mitrofanoff channel; bladder-neck sling, reconstruction or artificial sphincter.
- VUR: CIC and antimuscarinic first; avoid Credé; reimplant during augmentation if grade ≥III.
- **Also know:** latex allergy; bowel programme; sexual dysfunction (about 75% of men have ED [erectile dysfunction]; sildenafil helps); precocious puberty.

### Bladder myoplasty: definition

- The P3 wording "bladder myeloplsty" is assumed to mean "bladder myoplasty" (a muscle procedure on the bladder wall).
- **Detrusor myoplasty:** wrapping a **vascularised, innervated skeletal muscle flap** round the bladder so that voluntary contraction of that muscle empties an **acontractile bladder**.

### Bladder myoplasty: indications

- **Acontractile or underactive detrusor** with catheter dependence: lower motor neuron lesions (spina bifida, cauda equina injury), myogenic failure.
- Needs good bladder capacity and compliance, and a competent but not obstructing outlet.

### Latissimus dorsi detrusor myoplasty (LDDM; Stenzl)

- **Free latissimus dorsi flap:** artery and vein joined to the **inferior epigastric vessels**; motor nerve joined to an **intercostal nerve**.
- Muscle wrapped **spirally over more than three-quarters of the bladder** and anchored to pelvic floor fascia and ligaments.
- The patient voids by contracting the transferred muscle (abdominal effort).
- **Results:** most voided with low residual; bladder contractility index rose to the strong range.
- **Complications** in about one-third: thromboembolism, abscess, wound infection; none long term.
- **Rectus abdominis detrusor myoplasty:** simpler, less morbid; small spinal cord injury series with spontaneous voiding.
- **Status: investigational.** Standard care stays **clean intermittent catheterisation**.

**Diagram to draw:** latissimus flap spiralled round the bladder with vascular and nerve anastomoses.

### Other "myoplasty": detrusor myotomy/myectomy (autoaugmentation)

- Purpose is the opposite: **increase capacity and compliance** in a small, poorly compliant or overactive bladder.
- **Technique (Cartwright and Snow, 1989):** strip and excise the detrusor over the dome, leave the mucosa intact, and anchor the detrusor edges (psoas hitch) so the mucosa bulges as a **wide-mouthed diverticulum**.
- **Pros:** urothelial lining, no mucus, no metabolic change, no bowel.
- **Cons:** seldom adds significant capacity; compliance gain may not last; **not recommended for adult detrusor overactivity**.
- **Also know:** seromuscular enterocystoplasty covers the bulge with demucosalised sigmoid.

### Alternatives

- **Also know:** sacral neuromodulation, sacral anterior root stimulation, reduction cystoplasty (abandoned), muscle-derived stem cells (research).

**Key points to remember:**

- Level does not predict bladder: UDS within 3 months.
- DLPP 40 cm $H_2O$ or more = hostile (McGuire 1981); early CIC + antimuscarinic protects kidneys.
- MOMS: fewer shunts, no bladder benefit.
- LDDM: free latissimus to inferior epigastric vessels and intercostal nerve; for acontractile bladder.
- Investigational; clean intermittent catheterisation remains standard.
- Autoaugmentation: detrusor excised, mucosa bulges; little lasting gain.

## SAQ 5 Anterior urethral valve

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 17 Paper 2, Q3c

### Definition and epidemiology

- AUV (anterior urethral valve): a congenital obstructing fold or cusp in the anterior urethra (bulbar, penoscrotal or penile). It is the **commonest congenital obstructive lesion of the anterior urethra** but **25–30 times less common than PUV (posterior urethral valves)**.
- Often linked to an **anterior urethral diverticulum**: the distal lip of the diverticulum fills and acts as a valve, or a semilunar fold hangs from the roof.

### Embryology (unclear)

- Hypoplastic corpus spongiosum over the segment suggests a place on the **hypospadias spectrum**, or faulty union of urethral mucosa and fossa navicularis epithelium; a ruptured dilated Cowper (bulbourethral) gland duct is also proposed.

### Clinical features (vary with severity and age)

- Antenatal hydronephrosis, LUTO (lower urinary tract obstruction) in severe cases.
- **Ventral penile bulge** during voiding, with urine expressed by compressing the shaft; post-void dribbling, weak stream, incontinence.
- Palpable bladder, UTI (urinary tract infection), renal insufficiency.

### Evaluation

- **VCUG (voiding cystourethrogram):** dilated anterior urethra proximal to the valve, diverticulum, trabeculated bladder, bladder diverticula, often VUR (vesicoureteric reflux).
- Ultrasound, creatinine; urethroscopy confirms.
- **Differential:** Cowper syringocele, megalourethra, acquired diverticulum, stricture.

### Management

- **Premature or small infant:** vesicostomy until the urethra takes a cystoscope.
- **Most cases: endoscopic valve ablation** (Bugbee electrode or laser), usually definitive.
- **Large diverticulum:** open excision and urethroplasty over a catheter.
- Then manage VUR, infection and bladder dysfunction.

### AUV versus PUV

| Feature | AUV | PUV |
|---|---|---|
| Frequency | 25–30 times rarer | Commonest infant outlet obstruction |
| Site | Bulbar, penoscrotal, penile | From the verumontanum |
| Sign | Penile bulge | Weak stream, palpable bladder |
| Bladder dysfunction | Up to 80% | Near universal |
| Renal outcome | 78% normal function | ESRD (end-stage renal disease) 20–50% |

### Prognosis

- Depends on pre-treatment creatinine and GFR (glomerular filtration rate). Azotaemia, VUR and UTI together raise the risk of poor renal outcome 25-fold.

**Diagram to draw:** VCUG sketch: dilated bulbar urethra with diverticulum ending at a valve lip.

> **Key points to remember:**
> - 25–30 times rarer than PUV; diverticulum common.
> - Penile bulge on voiding; diagnosis by VCUG.
> - Endoscopic ablation usually definitive; 78% keep normal renal function.

## SAQ 6 CSIC.

P3

**Type:** SAQ (10 marks)
**Source:** 2018-Summer Paper 3, Q3e

*Taken to mean **CISC – clean intermittent self-catheterisation**.*

### Definition and principle

- Regular, **clean (not sterile)** passage of a catheter by the patient (or carer) to empty the bladder, then removal.
- **Lapides (1971–72)**: infection comes from **stasis and overdistension** (poor bladder-wall blood flow), not from bacteria alone. Regular complete emptying protects the bladder wall and upper tracts, even though bacteriuria develops.
- Gold standard for chronic emptying failure.

### Indications

- **Neurogenic bladder:** spina bifida, spinal cord injury, multiple sclerosis, cauda equina, diabetic cystopathy.
- **Underactive or acontractile detrusor**; chronic retention; post-operative retention.
- **Detrusor–sphincter dyssynergia**: CISC bypasses the obstructed sphincter.
- After **augmentation or continent diversion**; **valve bladder**; high residual after botulinum toxin.
- **Contraindications:** poor hand function or cognition without a carer, urethral false passage or stricture, unwillingness.

### Technique

- Wash hands → clean meatus with water → lubricated catheter (or pre-lubricated **hydrophilic** catheter) → pass until urine flows → drain fully → withdraw slowly.
- Sizes: adults about 12–14 Fr; children by age (8–10 Fr).
- **Frequency:** 4–6 times a day, keeping each volume **below about 400–500 mL** (child: below expected bladder capacity).
- Combine with **antimuscarinics** for overactivity; bladder diary.

### Catheters

| Type | Note |
|---|---|
| Plain PVC (polyvinyl chloride), reusable after washing | Cheap; common in India |
| **Hydrophilic-coated** single-use | Less urethral trauma; longer time to first UTI (urinary tract infection) |
| Compact or closed-system | Convenience, outdoor use |

### Children

- Start **early (infancy)** in spina bifida with hostile bladder; families adapt best; child involved from 3–5 years.
- **Mitrofanoff appendicovesicostomy** (1980) if urethral catheterisation is difficult (wheelchair, sensate urethra, girls).

### Complications

- **Asymptomatic bacteriuria** (common): **do not treat or give prophylaxis**; treat only symptomatic UTI.
- Urethral trauma, bleeding, **false passage**, stricture, epididymitis, bladder stones (hair, mucus), rarely lost catheter.

### Outcome

- Less pyelonephritis, epididymo-orchitis and urosepsis than indwelling catheters; preserves the upper tracts; with drugs, continence in many.

**Diagram to draw:** neurogenic bladder management ladder: CISC + antimuscarinic → botulinum toxin → augmentation + channel.

> **Key points to remember:**
> - Lapides: overdistension, not bacteria, causes infection; clean is enough.
> - 4–6 times daily, under about 500 mL each.
> - Do not treat asymptomatic bacteriuria.
> - Mitrofanoff when urethral access is hard.

## SAQ 7 Discuss diagnostics and operative pediatric laparoscopy in urology

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 3 Paper 3, Q3a

## Evolution

- Diagnostic (non-palpable testis, 1976) → nephrectomy (1994) → pyeloplasty (1995) → robotic (2002).

## Diagnostic laparoscopy

- **Non-palpable testis** (procedure of choice, after examination under anaesthesia): note vessels, vas, testis and internal ring.

| Finding | Meaning |
|---|---|
| Blind-ending vessels and vas, closed ring | Vanishing testis: stop |
| Vessels and vas entering ring | Explore groin; excise remnant |
| Intra-abdominal testis | Orchidopexy (primary or Fowler-Stephens) |
| - **DSD (differences of sex development):** gonadal inspection and biopsy, Müllerian structures. | |

## Operative laparoscopy

- **Testis:** primary orchidopexy (>90% success); Fowler-Stephens one- or two-stage; gonadectomy in DSD.
- **Kidney:** nephrectomy, heminephrectomy, **pyeloplasty** (commonest MIS (minimally invasive surgery) procedure).
- **Ureter and bladder:** extravesical or vesicoscopic (cross-trigonal) reimplantation, Mitrofanoff, augmentation, bladder neck reconstruction, utricle and urogenital sinus.
- **Also know:** RPLND (retroperitoneal lymph node dissection) for paratesticular rhabdomyosarcoma.

## Technical principles

- **Access:** open Hasson or Veress at umbilicus (Palmer's point if scarred); equal safety.
- 3–5 mm ports; robots use 8 mm. Transperitoneal (most space), retroperitoneal (spares peritoneum, useful in peritoneal dialysis), vesicoscopic.
- **Pressures:** $CO_2$ absorption, reduced venous return, raised airway pressure; reversible oliguria (no fluid bolus).

| Age | 0–1 month | 2–12 months | 1–2 years | 2–8 years |
|---|---|---|---|---|
| Max IAP (intra-abdominal pressure) | ≤6 mmHg | ≤8 | ≤10 | ≤12 |
| - **Positioning:** modified flank for kidney; supine steep Trendelenburg for pelvis. Infants: minimise inhalational anaesthetic, rectal tube to reduce bowel distension. | | | | |
| - **Only absolute contraindication:** inability to tolerate pneumoperitoneum. | | | | |

## Complications

- Right common iliac vessel injury (behind umbilicus; commonest major), inferior epigastric (commonest minor), bowel injury, subcutaneous emphysema, **port-site hernia 0.5–3%** (close fascia).

### Pros and cons

- Shorter stay, better cosmesis, deep pelvic access; pain benefit unproven. Robot: cost, learning curve; infant working space only about 1 L.

**Diagram to draw:** laparoscopic view of internal ring with blind-ending vessels.

> **Key points to remember:**
> - Diagnostic laparoscopy is the procedure of choice for non-palpable testis.
> - Pressure ceiling 6–12 mmHg by age.
> - Right common iliac at Veress entry; close port fascia.

## SAQ 8 Vesicostomy.

P4

**Type:** SAQ (10 marks)
**Source:** 2016-Summer Paper 4, Q3f

### Definition

- **Cutaneous vesicostomy**: the bladder dome is brought to the lower abdominal skin as an incontinent stoma draining into the nappy. It "vents" the bladder, protecting the upper tracts; usually a **temporary** bridge to reconstruction.

### Indications

- **PUV (posterior urethral valves)**: infant too small for a resectoscope; upper tract deterioration despite drainage or ablation.
- **Prune-belly syndrome**: acute renal failure, urosepsis, urethral atresia. A larger stoma is needed because stenosis is common.
- **Neurogenic bladder**: high pressure despite CIC (clean intermittent catheterisation) and drugs, or CIC impossible; worsening hydronephrosis; febrile UTI (urinary tract infection) with high-grade reflux.
- **Also know:** urethral atresia; MMIHS (megacystis-microcolon-intestinal hypoperistalsis syndrome); urogenital sinus anomalies.

### Blocksom technique (Duckett modification)

1. Transverse 2 cm incision midway between umbilicus and pubis. (In prune-belly, excise any urachal diverticulum.)
2. Incise the rectus sheath; separate the recti.
3. Fill the bladder; mobilise the **dome** up to the urachus, sweeping peritoneum off.
4. Bring the **dome** (not the anterior wall) to the wound. This reduces prolapse of the posterior wall.
5. Anchor detrusor to rectus fascia; stoma should take about 24 Fr.
6. Open the dome; suture mucosa to skin. No catheter needed.

### Advantages

- Simple; no catheter or appliance.
- The bladder still fills and empties through the stoma at low leak pressure, so compliance is **not** lost; it remains usable for reconstruction or transplantation.

### Complications

- **Stomal stenosis**, mucosal **prolapse**, peristomal dermatitis, stones, UTI; leakage harder to manage as the child grows.

### Closure

- When the cause is treated (valve ablation, urethral growth) or at definitive reconstruction; check bladder function first (cystogram or urodynamics).
- Excise the stoma, free the bladder from fascia, close in two layers, catheter drainage for a few days.

### Alternatives

| Option | Use |
|---|---|
| Cutaneous ureterostomy or pyelostomy | Rare; persistent upper tract obstruction |
| Ileovesicostomy | Unable to catheterise; 10–15 cm ileum |
| Appendicovesicostomy (Mitrofanoff, 1980) | Continent catheterisable channel |

**Diagram to draw:** sagittal view of bladder dome sutured to fascia and skin.

> **Key points to remember:**
> - Blocksom: dome-based stoma midway between umbilicus and pubis.
> - Bladder keeps cycling; compliance preserved.
> - Stenosis and prolapse are the main complications.

## SAQ 9 Management of Defecation Disorders resulting into LUTS

P4

**Type:** SAQ (10 marks)
**Source:** 2025-Winter Paper 4, Q3f

*Assumed meaning: bladder and bowel dysfunction, where constipation or defecation disorders cause LUTS (lower urinary tract symptoms).*

### Why the bowel causes LUTS

- **Shared space:** a loaded rectum compresses the bladder, reducing capacity and obstructing outflow.
- **Shared nerves:** bladder and bowel afferents converge on the same spinal segments; cross-organ sensitisation causes urgency and detrusor overactivity.
- **Shared muscle:** a pelvic floor that contracts during defecation (**dyssynergic defecation**) also contracts during voiding (**dysfunctional voiding**) → high PVR (post-void residual), UTI (urinary tract infection), reflux.
- Result: urgency, daytime incontinence, enuresis, retention, recurrent UTI.

### Evaluation

- Stool frequency, pain, **Bristol chart**; 3-day bladder and bowel diary.
- **Rome IV (child):** at least 2, weekly for 1 month: ≤2 defecations/week; faecal incontinence weekly; retentive posturing; painful or hard stools; large rectal mass; large-diameter stools.
- **Rome IV (adult), for 3 months:** ≥2 of straining, hard stools, incomplete evacuation, blockage, manual manoeuvres (each in >25% of defecations), <3 stools/week.

- Examine abdomen, spine and perineum; exclude organic causes (Hirschsprung disease, hypothyroidism, neurogenic).
- **Also know:** rectal diameter ≥30 mm on ultrasound; no routine X-ray; manometry if refractory.

### Management

**Rule: treat the bowel first (EAU, European Association of Urology/ESPU, European Society for Paediatric Urology).**

1. **Disimpaction:** oral PEG (polyethylene glycol) 3350, 1–1.5 g/kg/day for 3–6 days; enemas if needed.
2. **Maintenance:** PEG 0.4 g/kg/day (range 0.2–0.8), at least 2 months; lactulose alternative; stimulants (senna) for expulsion.
3. **Urotherapy:** toileting after meals (gastrocolic reflex); footstool posture; normal fluid and fibre.
4. **Biofeedback, pelvic floor physiotherapy** for dyssynergia.
5. **Refractory: transanal irrigation** (water via rectal cone; a quarter to a third stop); **ACE (antegrade continence enema, Malone)**; botulinum toxin to the sphincter; sacral or tibial neuromodulation. - **Adults:** laxatives, prucalopride or linaclotide, biofeedback; rectocele repair; irrigation in neurogenic bowel. - **Caution:** anticholinergics worsen constipation; LUTS return.

**Diagram to draw:** pelvic cross-section showing loaded rectum compressing bladder.

> **Key points to remember:**
> - Treat the bowel first.
> - Rome IV: ≥2 criteria for 1 month (child).
> - PEG disimpaction 1–1.5 g/kg/day for 3–6 days.

# Rapid Revision Sheet · Section 5

### Bladder embryology and exstrophy-epispadias complex (LAQ)

- Bud from Wolffian duct; position decides VUR or ectopia.
- Urorectal septum weeks 5–6 (P2: weeks 4–6); mesonephric duct entry is the landmark.
- Vesicourethral canal forms the bladder; urachus from allantois.
- Trigone: common excretory duct absorption; mostly bladder muscle (current view).
- Urachus forms at months 4–5; urine from week 10; amniotic fluid mainly urine after 16 weeks.
- Marshall–Muecke: overdeveloped cloacal membrane, failed mesodermal reinforcement; rupture timing decides variant.
- CBE: external rotation 12°/18°, anterior pubis 30% short, diastasis about 4.8 cm; VUR in 100%.
- Cloacal exstrophy: spinal anomalies 85–100%; osteotomy always.
- MSRE: closure at birth, epispadias repair 6–12 months, YDL / BNR at 5–7 years at ≥100 mL.
- Osteotomy if closure after 72 h or diastasis >4 cm.
- CPRE: closure + penile disassembly in newborn. Risks: glans loss, hypospadias, VUR.
- Capacity ≥100 mL predicts continence; template size and first closure success predict continence.

### Prune belly syndrome (LAQ)

- Triad; 95% male; dysplasia and lungs decide survival.
- Ureter is a failed pump (collagen); VUR about 75%; keep the proximal ureter.
- Woodard I–III; vesicostomy for drainage; orchidopexy at about 6 months.
- Nadir creatinine <0.7 mg/dL is favourable; up to 30% reach renal failure.

### Voiding dysfunction in children: neurophysiology and differential diagnosis of frequency/urgency (LAQ)

- S2–S4 pelvic (voiding), T10–L2 hypogastric (storage), Onuf/pudendal (EUS).
- PMC = switch; medial prefrontal cortex holds it off.
- EBC = (age + 2) × 30 mL; staccato = dysfunctional voiding.
- DLPP ≥40 cmH2O = hostile; treat bowel first.
- Always ask about bowels; treat constipation first.
- Rule out UTI, PUV, meatal stenosis and occult dysraphism.
- Extraordinary daytime frequency: no nocturia, normal tests, self-limiting.

### Antenatal hydronephrosis, fetal uropathy and prenatal scan anomalies (LAQ)

- UTD: 1–3% of pregnancies; second commonest anomaly after cardiac.
- Anomaly scan 18–20 weeks; after 16 weeks amniotic fluid = fetal urine.
- ANH (antenatal hydronephrosis) = UTD; APD normal <4 mm before 28 weeks, <7 mm after, <10 mm postnatally.
- UTD A2–3 = APRPD ≥7 mm or any abnormal feature.
- Transient dilation 50–70% (commonest cause); UPJ obstruction commonest surgical cause; VUR is not predicted by grade.
- First postnatal US after 48 hours; A1: one scan after 32 weeks; A2–3 scan every 4 weeks.

- Risk-based care: P1 no prophylaxis or VCUG; prophylaxis if ureter >7 mm or P3; P3 prophylaxis, VCUG, MAG3.
- Keyhole + oligohydramnios = PUV; absent bladder = exstrophy or agenesis.
- ARPKD large bright kidneys; MCDK non-communicating cysts.
- Favourable urine: Na <100, Cl <90, osmolality <200, β2-microglobulin <6.
- PLUTO: survival up, renal function unchanged.

**Embryological development of urinary bladder and management of Hutch Diverticulum? (LAQ)**

- Urorectal septum weeks 4–6; vesicourethral canal → bladder; urachus at months 4–5.
- Hutch: herniation through the ureteric hiatus → VUR.
- Ureter entering the sac: VUR will not resolve.
- Secondary: treat the outlet first; else diverticulectomy + reimplantation.

**Development of the urethra, congenital urethral anomalies and management of epispadias (including isolated penopubic epispadias) (LAQ)**

- Whole male urethra is endodermal UGS; fold fusion complete at 16, urethra at 18 weeks.
- About 70% of male epispadias are penopubic and incontinent; VUR 30–40%.
- Diastasis over 3 cm (>3 cm) → exstrophy pathway.
- Cantwell–Ransley at 6–12 months; YDL (Young–Dees–Leadbetter) at about 100 mL capacity.
- Continence after YDL: males 47–82%, females 70–87%.

**Posterior urethral valves: embryology, management and valve bladder (SAQ)**

- Type I (95%): abnormal Wolffian duct insertion; type III: persistent urogenital membrane; type II not real.
- Urorectal septum divides the cloaca at weeks 5–6.
- COPUM: one membrane torn into leaflets by catheterisation.
- Dysplasia is fixed in utero; ablation cannot reverse it.
- Catheter, stabilise, then primary valve ablation; vesicostomy if not possible.
- Nadir creatinine below 0.7 mg/dL predicts good renal outcome.
- Valve bladder: polyuria, poor sensation, residual urine.
- Valve bladder: overnight drainage and CIC; overnight drainage breaks the cycle.
- Ureteric augmentation rather than bowel; optimise before transplant.

**Ultrasonography in neonates (SAQ)**

- First postnatal scan after 48 h, except suspected PUV; normal APD <10 mm.
- Echogenic cortex and prominent pyramids are normal in neonates.
- UTD: P1 APD 10 to <15 mm; P2 15 mm or more; P3 abnormal parenchyma or bladder.
- Keyhole sign + thick bladder = PUV; spinal US only up to 6 months.

**Nocturnal enuresis (SAQ)**

- ≥5 years; secondary = after 6 months dry.
- Alarm (durable) and desmopressin (rule of thirds) are first line.
- Treat daytime symptoms and constipation first.

**Paediatric neurogenic bladder (myelodysplasia): assessment, management and bladder myoplasty (SAQ)**

- Level does not predict bladder: UDS within 3 months.
- DLPP 40 cm $H_2O$ or more = hostile (McGuire 1981); early CIC + antimuscarinic.
- Early CIC plus antimuscarinic protects kidneys.
- MOMS: fewer shunts, no bladder benefit.
- LDDM: free latissimus to inferior epigastric vessels and intercostal nerve; for acontractile bladder.
- Investigational; clean intermittent catheterisation remains standard.
- Autoaugmentation: detrusor excised, mucosa bulges; little lasting gain.

**Anterior urethral valve (SAQ)**

- 25–30 times rarer than PUV; diverticulum common.
- Penile bulge on voiding; diagnosis by VCUG.
- Endoscopic ablation usually definitive; 78% keep normal renal function. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**CSIC. (SAQ)**

- Lapides: overdistension, not bacteria, causes infection; clean is enough.
- 4–6 times daily, under about 500 mL each.
- Do not treat asymptomatic bacteriuria.
- Mitrofanoff when urethral access is hard.

**Discuss diagnostics and operative pediatric laparoscopy in urology (SAQ)**

- Diagnostic laparoscopy is the procedure of choice for non-palpable testis.
- Pressure ceiling 6–12 mmHg by age.
- Right common iliac at Veress entry; close port fascia. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Vesicostomy. (SAQ)**

- Blocksom: dome-based stoma midway between umbilicus and pubis.
- Bladder keeps cycling; compliance preserved.
- Stenosis and prolapse are the main complications.

**Management of Defecation Disorders resulting into LUTS (SAQ)**

- Treat the bowel first.
- Rome IV: ≥2 criteria for 1 month (child).
- PEG disimpaction 1–1.5 g/kg/day for 3–6 days. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

# Section 6 · Female Urology

## Long answer questions (LAQ)

### LAQ 1 Female pelvic floor anatomy and pelvic organ prolapse: evaluation and surgical repair

P1 · P2 **merged from 3**

*Asked as:* P1 LAQ "Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia" · P1 SAQ "Describe clinical presentation of pelic floor prolapse. diagnostic evaluation and brief discussion of various surgical options." · P2 SAQ "Vaginal prolapse repair."

#### Introduction and definitions

- Assumed meanings: "procedentia" is read as procidentia (complete uterine prolapse); "fertility sparing" / "fertility preserving" means uterus-preserving treatment for a woman who wants future pregnancy; "pelic floor prolapse" and "vaginal prolapse repair" mean pelvic organ prolapse (POP) in women and the surgical repair of vaginal wall and vault prolapse.
- **SUI (stress urinary incontinence):** involuntary leakage on effort, coughing or sneezing.
- **POP (pelvic organ prolapse):** descent of the anterior wall, posterior wall, uterus/cervix or vault. Anterior is commoner than posterior, which is commoner than apical. It is a disease only if symptomatic: a quality-of-life condition.
- Pelvic organs are supported by the **levator ani**, the **endopelvic fascia and its ligaments** and the **perineal membrane and perineal body**, anchored to the **bony pelvis**. Failure causes POP and SUI. Every SUI operation uses one of these anchors.

#### Bony pelvis and surgical anchors

- **Bony landmarks** and dangers: **sacrospinous ligament** (pudendal bundle), **Cooper's ligament** (bleeding), **sacral promontory** (left common iliac vein, right ureter).
- **Sacrospinous ligament (SSL)**, 7–8 cm, ischial spine to sacrum: the pudendal nerve and vessels pass behind the spine, the sacral plexus just posterior.
- **Sacral promontory:** left common iliac vein and right ureter each about 2.7 cm away.

#### Pelvic floor muscles: pelvic diaphragm (levator ani + coccygeus)

| Muscle | Attachment | Function |
|---|---|---|
| Puborectalis | Back of pubis, loops (as a sling) behind the anorectal junction | Anorectal angle; anorectal sling |
| Pubococcygeus (pubovaginal part) | Posterior pubis and arcus tendineus to viscera (vagina, perineal body, anus) and anococcygeal raphe | Slings urethra and vagina; closes hiatus, lifts urethra |
| Iliococcygeus | ATLA (arcus tendineus levator ani), ischial spine to raphe, coccyx | Horizontal **levator plate** |
| Coccygeus (not levator; separate) | Ischial spine to sacrum, coccyx | Lies on sacrospinous ligament |

- Levator tone (tonic contraction) closes the **urogenital hiatus**, through which all POP passes; the uterus and upper vagina rest on the levator plate.
- Levator is palpable through the lateral vaginal walls for PFMT (pelvic floor muscle training) assessment.
- **Muscle fails first, fascia second:** once injury (birth injury, avulsion) opens the hiatus, ligaments carry the load and fail; the fascia then carries the whole load and stretches.
- Levator defects in up to **20% of primiparous women**; avulsion predicts advanced prolapse.

### Endopelvic fascia and ligaments

- **Cardinal ligament** (side wall to cervix; carries the uterine artery) and **uterosacral ligament** (sacrum to cervix): the **cardinal–uterosacral complex** suspends the apex.
- The ureter is 4 cm from the uterosacral ligament near the sacrum but 0.9 cm near the cervix.
- **Pubocervical (pubovesical) fascia** attaches laterally to the **ATFP (arcus tendineus fasciae pelvis)**, the "white line" from the lower pubis to just above the ischial spine.
- **Rectovaginal fascia** supports the posterior wall.
- **Pubourethral ligaments** suspend the urethra from the pubis and anchor the mid-urethra; **urethropelvic ligaments** anchor it to the ATFP.

### Perineal membrane, perineal body and urethra

- The perineal membrane spans the pubic arch and is pierced by the urethra and vagina; above it lies the striated urogenital sphincter. The perineal body anchors the distal vagina.
- **Striated urogenital sphincter:** sphincter urethrae (mid-urethra), **compressor urethrae** and **urethrovaginal sphincter**.
- Urethra 4–5 cm; the continence zone is the **proximal and mid-urethra**. The **oestrogen-sensitive vascular submucosa** gives up to about 30% of closure force.

### DeLancey's levels of support (1992)

| Level | Region | Structures | Loss produces |
|---|---|---|---|
| **I – suspension** | Cervix/upper third | Cardinal–uterosacral complex | **Uterine or vault prolapse**, enterocele |
| **II – attachment** | Middle third | Paravaginal attachment to ATFP, levator fascia | Cystocele (central or paravaginal), rectocele; anterior or posterior wall prolapse |
| **III – fusion** | Distal third | Urethra, perineal membrane, perineal body | **Urethral hypermobility**, perineal descent, distal rectocele |

- **Central cystocele:** midline weakness, rugae lost. **Paravaginal cystocele:** ATFP detachment (Richardson 1981), rugae preserved.
- Level I is the keystone: always correct the apex.

**Diagram to draw:** sagittal female pelvis with levator plate and DeLancey levels I–III, with sacrospinous and sacrocolpopexy fixation points; hammock cross-section.

### Innervation

- **Pudendal nerve (S2–S4):** sphincters and perineal muscles; runs through Alcock's canal near the ischial spine; stretched in delivery.

- **Nerve to levator ani (S3–S5):** runs on the muscle's pelvic surface, separate from the pudendal; neuropathy in about a quarter of primiparous women.
- Autonomic pelvic splanchnic and hypogastric nerves: injured in radical pelvic surgery.

## Theories of continence and the operations they produced

| Theory | Mechanism | Surgical answer |
|---|---|---|
| **Pressure transmission (Enhorning 1961)** | Hypermobile urethra drops out of the abdominal pressure zone | Restore position: Burch, MMK (Marshall–Marchetti–Krantz) |
| **Hammock (DeLancey 1994)** | Urethra lies on anterior vaginal wall and endopelvic fascia attached to ATFP and levator; abdominal pressure compresses it against this firm backing; repositioning not essential | Restore backing |
| **Integral theory (Petros & Ulmsten 1990)** | Pubourethral ligament laxity at the **mid-urethra** | Tension-free mid-urethral sling |
| **Intrinsic sphincter deficiency (ISD) (McGuire 1978)** | Urethra fails to close | Sling, bulking, AUS (artificial urinary sphincter) |

- ISD: VLPP (Valsalva leak point pressure) <60 $cmH_2O$ or maximal urethral closure pressure <20 $cmH_2O$. Most SUI includes some ISD. So slings work in hypermobility, but Burch does poorly in severe ISD with a fixed urethra.

## Why prolapse occurs: risk factors (Bump & Norton)

- Muscle fails first (birth trauma, levator avulsion, denervation). The hiatus widens and the ligaments carry the whole load and stretch.
- **Predisposing:** genetics, collagen disorders (Ehlers–Danlos, Marfan).
- **Inciting: vaginal delivery** (strongest; forceps worse), parity, hysterectomy.
- **Promoting:** obesity, constipation, cough.
- **Decompensating:** ageing.

## Clinical presentation of prolapse

- **Vaginal bulge** (most specific), pressure, heaviness, worse by evening.
- **Splinting/digitation** to void or defecate (high-grade POP).
- Urinary: voiding difficulty (urethral kinking), OAB (overactive bladder); SUI in about 40%, which may be **occult**.
- Bowel (rectocele): obstructed defecation, stool trapping.
- Sexual dysfunction, discharge, ulceration; procidentia may cause hydronephrosis.

## Examination

- Lithotomy (supine) and **standing** with Valsalva / strain; Sims or split speculum for each compartment; lateral sulci (paravaginal defect); rectovaginal examination.
- **Cough stress test**, repeated with prolapse reduced (**occult SUI**); **Q-tip** angle change >30° = hypermobility; levator strength and avulsion.

## POP-Q (Pelvic Organ Prolapse Quantification, Bump 1996)

- Nine points in cm relative to the **hymen (0)**: inside (above) negative, outside positive; measured on maximal strain, recorded on a 3 × 3 grid.

| Point | Definition |
|---|---|
| Aa / Ap | Anterior / posterior wall 3 cm inside hymen (−3 to +3) |
| Ba / Bp | Most distal point of anterior / posterior wall (−3 if no prolapse) |
| C | Cervix or vault cuff |
| D | Posterior fornix (omitted after hysterectomy) |
| gh, pb, tvl | Genital hiatus, perineal body, total vaginal length (at rest) |

| Stage | Most distal point (leading edge) |
|---|---|
| 0 | No prolapse |
| I | >1 cm above hymen (< −1 cm) |
| II | Within ±1 cm of hymen (−1 to +1 cm) |
| III | >1 cm below hymen (> +1 cm) but < (tvl − 2) cm |
| IV | Complete eversion (≥ tvl − 2) |

- **Also know:** Baden–Walker grades 0–4; stage I–II often normal (POSST study).

## Investigations

- Urinalysis, PVR (post-void residual), bladder diary.
- **Cough test or urodynamics with prolapse reduced** for occult SUI; urodynamics also if ISD or mixed symptoms.
- Renal ultrasound in procidentia (hydronephrosis).
- Dynamic MRI or translabial ultrasound if complex or recurrent.
- A pessary trial confirms that the POP causes the symptoms.

## Principles of prolapse repair

- Operate for bother, not anatomy. The first operation gives the best chance.
- Identify each compartment by POP-Q. **The apex is the cornerstone:** failing to support it greatly raises recurrence; uncorrected level I loss causes recurrent prolapse after SUI repair.
- **Three categories: restorative** (native tissue), **compensatory** (graft or mesh, for weak tissue or failed repair), **obliterative**.
- Vaginal route: fewer complications, quicker recovery. Abdominal route: more durable.
- Screen for **occult SUI** with prolapse reduced and consider a concomitant sling (OPUS, CARE trials). Finish with **cystoscopy** to confirm ureteric efflux.

## Surgical options by compartment

| Compartment | Operations | Key points |
|---|---|---|
| **Anterior** (cystocele) | **Anterior colporrhaphy** (midline plication of pubocervical fascia); **paravaginal repair** (fascia reattached to the ATFP) | Central versus lateral defect; does not support the apex |

| Compartment | Operations | Key points |
|---|---|---|
| **Apical, vaginal** (uterus, vault) | **Uterosacral suspension** (sutures; ureter injury 1–11%; cystoscopy); **sacrospinous fixation** (suture 1.5–3 cm medial to the ischial spine, avoiding the pudendal bundle; gluteal pain 15%); iliococcygeus suspension; McCall culdoplasty; vaginal hysterectomy; uterus-sparing hysteropexy; Manchester | OPTIMAL trial: uterosacral = sacrospinous at 2 years |
| **Apical, abdominal** (compensatory) | **Sacrocolpopexy** (open, laparoscopic, robotic): macroporous polypropylene mesh from vaginal apex to the anterior longitudinal ligament over S1 / sacral promontory | Most durable (>90%); mesh exposure about 3–10% (P2: about 10% long term, CARE) |
| **Posterior** (rectocele) | **Midline fascial plication** without levatorplasty; site-specific repair; perineorrhaphy (levels II–III) | Levator plication causes dyspareunia; constipation may persist |
| **Obliterative** | **Le Fort** (partial, uterus in situ, lateral channels) or total colpocleisis | Elderly, not sexually active; success >90%; coital regret ≤13% |

- Cochrane 2016: vaginal apical repair has more recurrence and dyspareunia than sacrocolpopexy, but suits many older women.
- **April 2019:** the US FDA (Food and Drug Administration) ordered transvaginal synthetic mesh for (anterior) POP withdrawn; slings unaffected.
- Anterior colporrhaphy is not an SUI operation.

## Importance of anatomy in SUI surgery

| Operation | Anatomy used | Anatomical hazards |
|---|---|---|
| **Burch colposuspension** | Space of Retzius (avascular: symphysis in front, bladder behind, endopelvic fascia below). Paravaginal fascia at mid-urethra and bladder neck sutured to **Cooper's (iliopectineal) ligament** | Bleeding from Cooper's ligament and perivesical veins; ureteric kinking (cystoscopy); needs a mobile vagina |
| Paravaginal repair / VOS (vagino-obturator shelf) | Lateral vaginal sulcus to **ATFP** or obturator fascia (level II) | Obturator vessels and nerve |
| **Retropubic MUS (mid-urethral sling), TVT (tension-free vaginal tape; Ulmsten 1995)** | Tape under mid-urethra (integral theory), through **endopelvic fascia** and Retzius; suprapubic exits about 2 cm from midline | Bladder perforation (4.5%), vessels, bowel |
| **TOT (transobturator tape; Delorme 2001)** | Through obturator foramen / membrane, **obturator internus and externus**; exit at adductor longus tendon notch | Groin pain (6.4%), obturator neurovascular bundle; bladder injury rare (0.6%) |
| **SIMS (single-incision mini-sling)** | Anchors in obturator internus or endopelvic fascia | Works by restricting mobility |
| **Autologous PVS (pubovaginal sling)** | Rectus fascia at the **bladder neck** (proximal urethra), arms to rectus | Voiding dysfunction; best for ISD (compression) |
| Bulking agents | **Submucosa** coaptation | Repeated injections |

- **Why mid-urethra:** it is the zone of maximal closure pressure and pubourethral support. A loose tape lets the posterior urethral wall be compressed and kinked during stress.
- **TOMUS (2010):** retropubic 80.8% versus transobturator 77.7% objective success, not different.
- **SISTEr:** autologous sling more successful than Burch at 2 years, with more voiding dysfunction.
- **Also know:** needle suspensions abandoned (sutures pulled through).

**Diagram to draw:** cross-section of urethra on the vaginal hammock (DeLancey) with Burch, bladder-neck sling and mid-urethral sling sites.

### Procidentia: definition and assessment

- **Procidentia:** complete uterine (or vault) prolapse with the uterus wholly outside the hymen, **POP-Q stage IV**.
- Assess all compartments, occult SUI with prolapse reduced, PVR, hydronephrosis (ureteric kinking at the hiatus), cervical length and ulceration.
- Abnormal bleeding or cervical pathology needs gynaecological evaluation first, because it may argue against keeping the uterus.

### Fertility-sparing treatment of procidentia

**Principle:** restore **level I** support while keeping uterus and cervix. Defer definitive surgery until the family is complete where possible, because pregnancy and delivery cause recurrence.

#### 1. Conservative (first line)

- PFMT, weight loss, treat constipation.
- **Pessary:** space-filling type (Gellhorn, cube) for procidentia; removed for conception; review about 6-monthly for discharge and erosion.

#### 2. Uterus-preserving surgery

| Operation | Technique | Fertility points |
|---|---|---|
| **Sacrospinous hysteropexy** (vaginal) | Cervix sutured into the SSL **1.5–3 cm medial to the ischial spine** | Non-inferior to vaginal hysterectomy with uterosacral suspension (SAVE U trial); later data suggest superiority. Buttock pain, pudendal injury |
| **Uterosacral hysteropexy** (vaginal or laparoscopic) | Shorten and reattach uterosacral ligaments to cervix, sutures lateral to medial | No mesh; cystoscopy for ureteric kinking |
| **Sacrohysteropexy** (open, laparoscopic, robotic) | Mesh from posterior cervix to anterior longitudinal ligament over the promontory | Most durable apical support; later delivery usually by caesarean |
| **Manchester (Fothergill)** | Cervical amputation, cardinal ligaments sutured in front of the stump, anterior colporrhaphy | Suits elongated cervix, but risks **cervical stenosis, incompetence, miscarriage, preterm labour**: avoid if pregnancy wanted |

- **Also know:** Indian abdominal (cervicopexy) slings for young nulliparous prolapse (Shirodkar, Purandare, Khanna, Virkud). **Le Fort colpocleisis** keeps the uterus but not fertility (elderly only).

**3. Choosing** `Procidentia, wants pregnancy -> PFMT + pessary -> fails -> sacrospinous hysteropexy or laparoscopic sacrohysteropexy (avoid Manchester) -> recurrence after childbearing -> hysterectomy + vault suspension or sacrocolpopexy`

## Complications and follow-up

- Bladder, ureter or rectal injury; retention; de novo SUI; dyspareunia.
- Recurrence (mostly posterior overall; anterior after sacrospinous fixation).
- Mesh exposure: 3.4–5.4% after sacrocolpopexy in meta-analyses (P1 SAQ: about 3–10%; P2: about 10% long term, CARE); ureteric kinking.
- The retained uterus needs cervical screening.

**Key points to remember:**

- Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–S5.
- DeLancey: I suspension (uterosacral–cardinal; failure causes uterine prolapse), II attachment (ATFP), III fusion (urethral hypermobility); level I is the keystone: always restore apical support.
- Enhorning 1961, DeLancey 1994, Petros & Ulmsten 1990, McGuire ISD 1978.
- Integral theory: mid-urethral sling; hammock: backing, not repositioning.
- Burch to Cooper's ligament; TOT through obturator foramen; PVS at bladder neck.
- Bulge is most specific; splinting means high grade.
- Central cystocele rugae lost, paravaginal preserved; POP-Q stage II = −1 to +1 cm (±1 cm of hymen).
- Sacrocolpopexy most durable; colpocleisis for frail elderly.
- Cystoscopy after repair; screen for occult SUI.
- Pessary first; sacrospinous hysteropexy 1.5–3 cm medial to the ischial spine.
- Manchester amputates the cervix: poor choice for future pregnancy.

## LAQ 2 Vesicovaginal fistula: management, timing of repair, eponymous techniques and recurrent VVF

P1 · P2 · P3 **merged from 3**

*Asked as:* P3 LAQ "Vesicovaginal fistula (VVF): management options and early vs delayed repair" · P2 SAQ "VVF repair: early versus delayed timing, and surgery for recurrent VVF" · P1 SAQ "Eponymous techniques of VVF repair"

### Definition and aetiology

- VVF (vesicovaginal fistula): an epithelialised bladder–vagina tract (communication) causing continuous leakage; the commonest acquired urinary fistula (>75% of genitourinary fistulae).
- **Developing world (low-resource settings):** obstetric (obstructed labour) in >90%; the fetal head impacted for days causes pressure necrosis.
- **Developed world (high-resource settings):** iatrogenic in >75%, mainly after hysterectomy, usually supratrigonal at the vaginal cuff.
- Other causes: radiation, malignancy, mesh, foreign body.
- Common mechanism: **ischaemic necrosis**; continuous urine flow keeps the tract open until it epithelialises with time.
- Time to presentation: cystotomy, days; thermal injury, 7–10 days; radiation, months to years.

### Classification

| Goh (distance from meatus) | Size | Tissue |
|---|---|---|
| 1: >3.5 cm; 2: 2.5–3.5; 3: 1.5–2.5; 4: <1.5 cm | A <1.5 cm; B 1.5–3 cm; C >3 cm | i mild fibrosis; ii moderate/severe; iii radiation, ureteric, circumferential, previous repair |
| - **Also know:** Waaldijk (closing mechanism involvement); WHO (World Health Organization) simple vs complex. | | |

### Evaluation (guides choice)

- Speculum; fluid creatinine.
- **Dye tests:** single (blue backfill, tampon); **double** (oral phenazopyridine plus blue): orange = UVF (ureterovaginal fistula), blue = VVF.
- **Cystoscopy and vaginoscopy:** number, size, relation to ureteric orifices and bladder neck, mesh; **biopsy the tract if any history of cancer** (the fistula may signal recurrence).
- **Upper tract imaging (CTU, computed tomography urography) is mandatory:** to exclude a coexisting ureteric (ureterovaginal) fistula (up to 12–25%).
- Urodynamics: capacity, compliance, SUI (stress urinary incontinence) in 10–15%.
- Prepare: treat UTI (urinary tract infection), optimise nutrition, vaginal oestrogen for atrophy; counsel about residual incontinence, dyspareunia and possible reoperation.

### Principles of management

- Optimise nutrition, treat infection, drain continuously, relieve obstruction, exclude malignancy.
- Surgery: excise scar, wide mobilisation, tension-free, multilayer, **non-overlapping**, watertight closure; remove foreign material; consider interposition; drain afterwards. **The first repair has the best chance.**
- At repair: treat infection; ureteric stents if within 1 cm of a ureter; vaginal or abdominal route (open, laparoscopic, robotic); **interposition flap** (Martius, omentum, peritoneum) if irradiated or recurrent; catheter drainage for about 2 weeks (≥10–14 days).

### Option 1: Conservative and endoscopic (conservative trial while waiting)

- **Catheter drainage** cures about 15%. Best if small (<1 cm, ideally <3 mm), fresh, started (catheterised) within **3 weeks**, and non-irradiated. Once epithelialised (≥6 weeks), closure is unlikely. EAU (European Association of Urology): if tried, at least 12 weeks.
- **Fulguration** plus 2 weeks' drainage for tiny tracts <3.5 mm.
- **Sealants** (fibrin glue, cyanoacrylate, collagen): tiny series; not recommended by EAU.

### Timing: early versus delayed repair (the debate)

| | **Early repair** | **Delayed repair** |
|---|---|---|
| Definition (varies) | Days, <6 weeks or <3 months | Traditionally ≥3 months |
| Rationale | Clean iatrogenic tissue heals well; ends wetness and distress early | Inflammation settles, necrotic tissue sloughs, margins become clear |
| Evidence | **Comparable closure** in most series; Waaldijk: immediate obstetric repair gave very high first-attempt closure | One small series: more recurrence when repaired <6 weeks |
| Drawback | Friable tissue if inflamed | Prolonged distress, no better outcome |

- Some authors avoid the **4–12 week** window (active healing phase).
- **EAU consensus:** no minimum delay if there is **no active infection and no recent irradiation**.
- **Timing by cause:**
- **Clean iatrogenic / post-hysterectomy:** early repair.
- **Obstetric with necrosis (necrotic edges):** catheter trial, then repair at about 3 months.
- **Radiation:** wait **≥12 months** after radiotherapy; biopsy to exclude recurrence; interposition flap; consider diversion if the bladder is poor.
- **Recurrent (redo) fistula:** wait until inflammation settles (commonly about 3 months); radiation ≥12 months after radiotherapy; failed radiation repairs were redone after **≥6 months**.

### Option 2: Surgical route

| **Vaginal** | **Abdominal (open, laparoscopic, robotic)** |
|---|---|
| Low, distal or trigonal, accessible apex | Supratrigonal, narrow vagina, no descent |

| Vaginal | Abdominal (open, laparoscopic, robotic) |
|---|---|
| Ureters distant | Ureter involved, needs reimplantation |
| Less pain, blood loss, shorter stay | Augmentation, **Tanagho bladder flap** or diversion possible |
| Flap-splitting, offset flaps | Large (≥4 cm primary, **≥2 cm recurrent**), radiation |
| Failed **Latzko** can be cured by a second Latzko | **O'Conor** bivalving finds the scarred plane |
| Risk: vaginal shortening | |
| - First-attempt closure about **90% by either route**; choose the route the surgeon knows best. | |

## Eponymous and named techniques: transvaginal

- **Sims:** first successful systematic vaginal repair (silver wire).
- **Flap-splitting (Raz):**
- Ureteric stents if within 1 cm; Foley through the tract for traction.
- Inverted J or U incision with a 1–3 mm cuff; raise anterior and posterior vaginal flaps.
- Layer 1: bladder (3-0 or 4-0), including the tract edge. Layer 2: perpendicular perivesical layer. Fill test 200–300 mL.
- Optional Martius or peritoneal flap; advance the opposite vaginal flap ≥2 cm beyond. Closure ≥90%.
- **Latzko (1942), high partial colpocleisis:**
- Denude vaginal epithelium 1–2 cm around the tract; the bladder is **not opened or dissected**.
- Close the vaginal walls over the tract in layers.
- ≥90% success in non-irradiated post-hysterectomy apical VVF; favoured for radiation VVF and tracts near the ureters. Drawback: **vaginal shortening and dyspareunia**.
- **Also know:** vaginal cuff excision (all apical scar excised as an inverted cone).

## Eponymous and named techniques: abdominal

- **O'Conor and Sokol (1951), suprapubic extraperitoneal:**
- Midline cystotomy at the dome, **bivalved down** to the fistula; ureteric catheters.
- Excise the tract; separate bladder from vagina 2–3 cm beyond.
- Close the vagina; **omental or peritoneal flap** between; close the bladder in layers; suprapubic catheter.
- Modifications: **transvesical** (limited cystotomy) and **extravesical** (no cystotomy; fewer LUTS [lower urinary tract symptoms]).
- **Laparoscopic (1994) and robotic (2005):** equal closure (85–100%) with less blood loss, shorter stay and less morbidity; vaginal stent, backfill, ICG (indocyanine green) perfusion check. Robotic repair of recurrences gives high closure.

### Interposition flaps (radiation, recurrent, ischaemic or large VVF)

| Flap | Blood supply | Route | Use |
|---|---|---|---|
| **Martius (1928)** (labial fibrofatty pad) | Internal pudendal (inferior), external pudendal (superior) | Vaginal | Mid–distal VVF, bladder neck; does not reach the apex |
| Peritoneal (vaginal) | Local; pouch of Douglas fold | Vaginal or abdominal | High apical post-hysterectomy VVF |
| **Omentum (J-flap)** | Gastroepiploic arteries | Abdominal | Radiation, recurrent, large |
| Gracilis, Singapore, rectus | Myocutaneous | Vaginal/perineal | Refractory, large tissue loss |
| - RCT (randomised controlled trial): Martius added nothing in simple VVF, with donor-site pain in nearly a third. | | | |
| - **Interposition is widely used in recurrent fistula** (accepted, though not proven superior). | | | |
| - **Also know:** rectus abdominis flap (last resort). | | | |

### Recurrent VVF

- Persists after a failed repair; redo tissue is scarred and poorly vascularised, so the first repair has the best chance.
- Failure causes: tension, ischaemia, overlapping suture lines, infection, poor drainage, **malignancy, radiation**, missed second tract.
- Before redo: biopsy, cystoscopy and vaginoscopy, CT urography and dye test to exclude a **ureterovaginal fistula**, urodynamics; treat infection, optimise nutrition.
- Route: abdominal for recurrent ≥2 cm; a failed Latzko can be redone vaginally; robotic repair gives high closure.
- Interposition flap; catheter ≥10–14 days, then cystogram.
- More than one repair predicts residual incontinence.

### Postoperative care

- Catheter ≥10–14 days (urethral and/or suprapubic); antimuscarinic or β3-agonist for spasm; cystogram before removal; stents out at 4–6 weeks.

### Option 3: Urinary diversion (when repair is unwise)

- Multiple failures, severe radiation damage, massive tissue loss: conduit or continent reservoir/diversion.
- Unfit: ureteric occlusion with nephrostomies.
- Resource-poor: ureterosigmoidostomy as a last resort.

### Outcomes

- Residual urgency or incontinence despite closure (Goh 3–4 predicts; more than one repair predicts); retention (vertical bladder closure); dyspareunia (narrow vagina, fistula ≥3 cm).

ICS (International Continence Society): pelvic floor physiotherapy.

### Diagrams to draw

- Flap-splitting repair: perpendicular bladder layers, Martius flap, advanced vaginal flap.
- Latzko denuded area versus flap-splitting layers.
- Timing timeline: early window, 4–12 week zone, 3 months (obstetric), 12 months (radiation).
- Martius flap between bladder and vaginal closures.

**Key points to remember:**

- Image upper tracts; exclude ureteric fistula and cancer (biopsy before redo).
- Catheter trial only if small, fresh (<3 weeks), non-irradiated; works only if started within 3 weeks.
- Vaginal and abdominal routes both close about 90%.
- Early repair for clean iatrogenic VVF; closure comparable to delayed.
- Delay ~3 months for obstetric, ≥12 months for radiation fistula.
- First repair has the highest success.
- Interposition flap (Martius, peritoneum, omentum) for complex and recurrent VVF.
- Diversion after multiple failures or severe radiation.
- Latzko: colpocleisis, bladder not opened, vagina shortened.
- O'Conor: bivalve bladder to the fistula, interpose omentum.
- Martius for distal, omentum or peritoneum for apical VVF.

# LAQ 3 Urological complications of obstetric-gynaecological surgery: uretero- and urethrovaginal fistula

P3 **merged from 2**

*Asked as:* P3 LAQ "Urological complications in OBS-GYN and uretero-vaginal fistula" · P3 LAQ "Urological complications of OBGY surgery and urethrovaginal fistula"

*Assumed meaning: OBS-GYN / OBGY = obstetric and gynaecological practice and surgery.*

## Part A: Urological complications in OBS-GYN

- The bladder, distal ureter and urethra lie against the cervix, uterine vessels and anterior vaginal wall. Risk rises with distorted anatomy: fibroids, endometriosis, cancer, previous caesarean, radiation, heavy bleeding, prolapse or mesh surgery.

### By setting

| Setting | Complications |
|---|---|
| **Obstetric** | **Obstructed labour** (fetal head impacted for days): pressure necrosis, so large VVF (vesicovaginal fistula), urethral loss, ureteric involvement; over 90% of VVF in the developing world. **Caesarean**: bladder injury, ureteric injury, **vesicouterine fistula** (Youssef syndrome: cyclical haematuria, amenorrhoea, continence preserved). Post-partum or post-epidural **retention** (overdistension) |
| **Gynaecological surgery** | **Bladder injury** (commonest organ). **Ureteric injury** (over 90% distal third; ligation commonest). **Fistulae**: VVF (over 75% follow hysterectomy in the developed world), UVF (ureterovaginal fistula), urethrovaginal fistula |
| **Radical hysterectomy** | Pelvic plexus injury: retention, poor compliance; manage with CIC (clean intermittent catheterisation) |
| **Prolapse and continence surgery** | Ureteric kinking at apical suspension; mesh or sling exposure, perforation, outlet obstruction, pain |
| **Radiotherapy** | Radiation cystitis, VVF, ureteric stricture |

### By complication: key facts

| Complication | Key facts |
|---|---|
| **Bladder injury** (commonest organ injured) | Caesarean 0.27–0.3%; hysterectomy about 0.3–0.75% (laparoscopic highest); retropubic tape 3–9%. Repair: debride, 2-layer closure, fill test, catheter 7–14 days; check ureters if trigonal |
| **Ureteric injury** | Gynaecology causes over half of iatrogenic injuries; over 90% distal third. Hysterectomy overall 0.2–2.4%: vaginal 0.02–0.5%, abdominal 0.03–2%, minimally invasive 0.13–6%; prolapse surgery 1.7–4.3%. **Ligation** is the commonest gynaecological mechanism |
| **Fistulae** | VVF about 0.8 per 1000 hysterectomies; UVF; vesicouterine fistula after caesarean (Youssef syndrome); urethrovaginal fistula |
| **Voiding dysfunction** | Radical hysterectomy (pelvic plexus injury) 16–80% historically: retention, poor compliance; CIC and wait. Post-partum or post-epidural overdistension |
| **Mesh and sling** | Exposure, erosion into bladder or urethra, outlet obstruction, pain, dyspareunia |
| **Radiation** | Radiation cystitis; VVF in 0.6–2% after cervical cancer radiotherapy; ureteric stricture |

- **Ureteric danger sites (four):** pelvic brim (near the infundibulopelvic ligament); **under the uterine artery** in the cardinal ligament; lateral to (beside) the uterosacral ligament; anterolateral vaginal fornix near the UVJ (ureterovesical junction). Most injuries are in the **distal 6 cm**.
- **Prevention and recognition:** haemostasis and anatomy; know the danger sites; prophylactic stents only selectively, if high risk (they aid recognition but do not prevent injury); cystoscopy with dye (IV indigo carmine or oral phenazopyridine) after hysterectomy where indicated, minimally invasive hysterectomy, and prolapse or continence surgery.
- Only about one-third of ureteric injuries are recognised at open surgery. A missed injury presents with fever, ileus, flank pain, urinoma or leakage; do CTU (computed tomography urography) with delayed films, then retrograde stenting. Distal repair: ureteroneocystostomy ± psoas hitch or Boari flap (see Part B).

## Part B: Ureterovaginal fistula (UVF)

### Definition

- Abnormal communication between the ureter and the vagina, causing continuous leakage.

### Aetiology and pathogenesis

- Usually an **unrecognised injury to the distal third of the ureter** during pelvic surgery. About **three-quarters follow an often uneventful abdominal hysterectomy**.
- **Mechanisms:** ligation, crush, laceration, thermal injury, **devascularisation** or ischaemia. The injured segment necroses and leaks into the vaginal cuff, which is the weakest point.
- Diagnosis is delayed in 18–88%; **delayed recognition raises fistula risk about fourfold**.
- Other causes: obstetric, cancer, radiation, endometriosis, foreign body.

### Clinical features

- **Continuous leakage with normal voiding** (the other kidney fills the bladder), usually appearing **within 4 weeks**, often with fever, flank pain or ileus, as thermal or ischaemic injury necroses.

| | VVF | UVF |
|---|---|---|
| Voiding | Reduced | Normal |
| Upper tract | Normal | Hydronephrosis common |
| First line | Catheter, repair | Stent |

### Investigations (confirm before treatment)

| Test | Method and reading |
|---|---|
| **Double dye test** | Oral phenazopyridine 100–200 mg (orange urine); after 1 hour, blue dye instilled into the bladder; tampon; walk about. **Orange = UVF**; blue = VVF; both = both |
| **Three-swab test** | Blue dye in the bladder; three swabs stacked in the vagina. Blue upper swab = VVF; **wet, unstained swab = UVF** |
| **CT urography, delayed phase** | Hydronephrosis, extravasation; MR (magnetic resonance) urography if contrast is contraindicated |
| **Cystoscopy + RGP (retrograde pyelogram)** | Abrupt cut-off or contrast leak; stent at the same sitting |

| Test | Method and reading |
|---|---|
| - **Exclude a coexisting VVF** (up to 25% of VVF have a UVF) and bilateral injury. | |

**Management**

- **Goals:** dryness and preservation of renal function. Management has moved from nephrectomy, through reimplantation, to endourology.
- **1. Endourological (first line): stent first**
- **Retrograde double-J stent as soon as possible** if a wire passes. Stenting **within 2 weeks** of detection works best, succeeding far more often than at 2–6 weeks, because the ureter heals over the stent before fibrosis.
- Wire will not pass: **PCN (percutaneous nephrostomy)** with antegrade stent, or rendezvous.
- Stent for **4–6 weeks**, then remove and re-image. Most stentable fistulae close; a stricture may follow.
- **2. Surgical repair (stent fails or impossible)**
- **Timing:** no consensus. Early repair equals delayed: early repair took less time and blood loss with equal success; a systematic review found equal success before and after 6 weeks. Delay only if tissue is still inflamed.
- **Ureteroneocystostomy** (distal third, so usually feasible):
  1. Transperitoneal or extraperitoneal approach; identify the ureter above the fistula.
  2. Divide above the diseased segment; dissect as distally as possible for length; **preserve periureteric adventitia**, especially laterally where vessels enter.
  3. Mobilise the bladder; spatulate; tension-free mucosa-to-mucosa anastomosis over a double-J stent; refluxing or tunnelled.
  4. Drain; catheter 7–14 days.
- **Adding length:**

| Technique | Extra reach |
|---|---|
| **Psoas hitch** (sutures parallel to psoas fibres, spare genitofemoral nerve) | Up to about 5 cm |
| **Boari flap** (base ≥4 cm, length:base ≤3:1) | 10–15 cm |
| Downward nephropexy; ileal ureter | Longer defects |
| - Open, **laparoscopic and robotic repair give equal results**; vaginal reimplantation reported. **Success ≥90%.** | |
| - The vaginal opening is left to heal; no separate vaginal repair is needed. | |

**Algorithm** `Leak + normal voiding -> exclude VVF (dye test, cystoscopy) -> CT urography / RGP -> wire/stent passes? Yes -> double-J 4-6 wk -> re-image -> healed / persistent No -> PCN + antegrade stent -> fails -> ureteroneocystostomy + psoas hitch / Boari flap`

**Complications and follow-up**

- Hydronephrosis, infection, ischaemic stricture.

- Ultrasound 1–2 weeks after stent removal; some add a renal scan at 6 weeks. Stricture usually appears within a year but may appear **up to 30 months** later, so follow beyond this.

## Part C: Urethrovaginal fistula

### Aetiology

- Developing world: **obstructed labour**, often with a VVF; also trauma, radiation, catheter injury.
- Developed world (iatrogenic): **urethral diverticulectomy** (fistula in 1–8%; causes 20–25% of urethrovaginal fistulae), slings, bulking agents, anterior repair or mesh, indwelling catheters (pressure necrosis), pessaries, radiation, trauma; also gender-affirming and paediatric genital surgery.

### Clinical features

- Depend on site relative to the sphincter. **Distal** fistula: often asymptomatic, split stream, vaginal voiding, post-void dribble. **Proximal** (bladder neck): continuous leakage or SUI (stress urinary incontinence).
- Dyspareunia, recurrent UTI (urinary tract infection), perineal irritation; partner discomfort with exposed mesh.

### Evaluation

- **Speculum examination** usually suffices; **methylene blue test** if rugae hide it (distal stain on tampon). Assess prolapse, SUI, atrophy.
- **VCUG (voiding cystourethrogram), lateral view;** VUDS (videourodynamics) for associated SUI or detrusor overactivity.
- **Cystourethroscopy** (flexible is better: outflow at the lens distends the urethra): size, site, bladder neck, exposed mesh, other fistulae. **About 20% have a separate VVF.** Image the upper tract.
- Goh location (distance from external meatus): type 1 >3.5 cm, 2: 2.5–3.5, 3: 1.5–2.5, 4: <1.5 cm. Closer means worse closure and continence.
- Optimise: treat UTI, vaginal oestrogen, nutrition; biopsy if previous cancer.

### Management

- Asymptomatic distal fistula: may need no repair. Timing: early if tissue is healthy; otherwise after inflammation settles.
- **Standard: transvaginal rotational-flap repair:**

| Step | Details |
|---|---|
| 1. Exposure | Lithotomy (prone gives better anterior exposure); cystourethroscopy, cannulate tract; suprapubic catheter (Lowsley tractor), urethral catheter |
| 2. Incision | Hydrodissect; circumscribe tract with 1–2 mm cuff, minimal excision; inverted **J- or U-shaped** vaginal flap incision |
| 3. Flaps | Raise vaginal flap; mobilise periurethral tissue widely to bladder neck; open retropubic space if sling planned |
| 4. Closure | **Close urethra** over a urethral sound: **4-0 absorbable** sutures, transverse or longitudinal; **second layer perpendicular**; buccal graft or flap onlay for large defects; leak test |

| Step | Details |
|---|---|
| 5. Interposition | **Martius flap** (labial fat pad, external pudendal pedicle) most often; omentum, gracilis, rectus fascia occasionally |
| 6. Cover | Advance vaginal flap beyond closure (2-0) |

- **Autologous pubovaginal sling** for SUI or bladder-neck involvement; **never synthetic** (erodes the ischaemic urethra). Remove any mesh in the field.
- **Near-total urethral loss:** tubularised anterior bladder flap neourethra. Irreparable cases: diversion.
- **After repair:** urethral catheter out at 2–3 days; suprapubic catheter ≥14 days; lateral VCUG before removal; nothing per vagina for 6 weeks.

**Outcomes and complications**

- Closure **90–95%**, but continence only **50–80%**; postoperative SUI in up to half. Sling then or staged later.
- Urethral stricture, recurrence, dyspareunia.

## Diagrams to draw

- Ureter passing under the uterine artery; double dye test; ureteroneocystostomy with psoas hitch.
- Urethrovaginal repair: inverted-U vaginal flap, two perpendicular urethral layers, Martius flap.

> **Key points to remember:**
> - Bladder is the commonest organ injured; obstructed labour causes most fistulae worldwide.
> - Ureteric injury is mostly distal and missed.
> - UVF: distal ureteric injury at hysterectomy; continuous leak within 4 weeks with normal voiding.
> - Orange tampon = UVF on double dye test; exclude coexisting VVF.
> - Stent early (within 2 weeks); remove at 4–6 weeks.
> - Reimplant with psoas hitch or Boari flap if stenting fails; success ≥90%; follow 30 months.
> - Urethrovaginal fistula: diverticulectomy is the commonest Western cause; obstetric in developing countries.
> - Look for a coexisting VVF (about 20%).
> - Layered closure plus Martius flap; autologous sling only.
> - Closure 90–95% (closure is easy); continence only 50–80% (continence is hard).

# LAQ 4 Female SUI: continence mechanism, classification, evaluation (UDS) and management, including male SUI surgery

P3 **merged from 2**

*Asked as:* P3 LAQ "Female pelvic floor, continence mechanism, classification and management of SUI" · P3 LAQ "SUI: evaluation in females, role of UDS, and surgical therapy in males"

## Definition and mechanism

- **SUI (stress urinary incontinence)**: involuntary leakage on effort, exertion, coughing or sneezing (ICS, International Continence Society).
- Continence needs **urethral pressure greater than bladder pressure** at rest and during stress.
- Support comes from the **levator ani, endopelvic fascia and ligaments, perineal membrane and perineal body**.
- Two mechanisms: **urethral hypermobility** (loss of the suburethral support, the DeLancey "hammock") and **ISD (intrinsic sphincter deficiency)**. Most women have some ISD.

## Anatomy of the female pelvic floor

### 1. Pelvic diaphragm

| Muscle | Attachment | Function |
|---|---|---|
| Puborectalis | Back of pubis, loops behind anorectal junction | Anorectal angle |
| Pubococcygeus (pubovaginal) | Pubis, ATFP (arcus tendineus fasciae pelvis) to viscera, raphe | Slings urethra and vagina |
| Iliococcygeus | Arcus tendineus levator ani, ischial spine to coccyx | Horizontal **levator plate** |
| Coccygeus (not levator) | Ischial spine to sacrum, coccyx | Lies on sacrospinous ligament |
| - Nerve to levator ani **(S3–5)** on its pelvic surface. Resting tone closes the **urogenital hiatus**; widening of the hiatus or tilting of the plate causes prolapse. Muscle fails first, then fascia carries the load. | | |

### 2. Perineal membrane and perineal body

- The perineal membrane spans the pubic arch; urethra and vagina pierce it. Above it lies the **striated urogenital sphincter**: sphincter urethrae (mid-urethra), **compressor urethrae** and **urethrovaginal sphincter**. The perineal body anchors the distal vagina.

### 3. Endopelvic fascia and ligaments

- **Pubocervical fascia** attaches laterally to the **ATFP** (pubis to ischial spine); **rectovaginal fascia** posteriorly; **cardinal–uterosacral complex** at the apex; **pubourethral ligaments** anchor the mid-urethra.

### 4. DeLancey levels (1992)

| Level | Structures | Defect |
|---|---|---|
| I suspension | Cardinal–uterosacral | Apical prolapse, enterocele |
| II attachment | ATFP, levator fascia | Cystocele (central or paravaginal), rectocele |
| III fusion | Perineal membrane, perineal body | **Urethral hypermobility**, perineal descent |

## Continence mechanism in the female

**1. Intrinsic urethral factors** (smooth and striated sphincters, vascular submucosa)

- **Mucosal seal**: folded urothelium coapts. The **oestrogen-dependent submucosal vascular plexus** provides up to **30%** of closure force.
- **Smooth muscle** (internal sphincter, bladder neck and proximal urethra): involuntary tone. The female bladder neck is weak and often open in continent women.
- **Striated urogenital sphincter**: the rhabdosphincter, mostly **slow-twitch** fibres, gives tonic mid-urethral closure. Distal parts are the **compressor urethrae** and **urethrovaginal sphincter**.
- The **continence zone is the proximal and mid-urethra.**

**2. Support (extrinsic)**

- The urethra lies on the **anterior vaginal wall and endopelvic fascia**, attached laterally to the **ATFP** and levator ani.
- **Pubourethral ligaments** anchor the mid-urethra; **levator ani** (pubococcygeus) tone keeps the backing firm.

**3. Neural**

- **Pudendal nerve (S2–4)** drives the sphincter (**guarding reflex**). Urethral pressure rises **just before** a cough reaches it, so closure is active, not just transmitted.

**4. Theories of continence and SUI**

| Theory | Mechanism | Surgical implication |
|---|---|---|
| **Pressure transmission (Enhorning 1961)** | Hypermobile urethra drops out of the abdominal pressure zone | Restore position (Burch) |
| **Hammock (DeLancey 1994)** | Abdominal pressure compresses the urethra against a firm suburethral layer; a lax layer fails | Restore backing |
| **Integral theory (Petros & Ulmsten 1990)** | Pubourethral ligament laxity at the mid-urethra | Tension-free MUS (mid-urethral sling) |
| **ISD (McGuire & Lytton 1978)** | Urethra itself fails to close (VLPP [Valsalva leak point pressure] <60, MUCP [maximum urethral closure pressure] <20 $cmH_2O$) | Sling, bulking, AUS (artificial urinary sphincter) |
| - Most SUI has some ISD. **Hypermobility alone never causes SUI.** | | |

## Classification of SUI

**1. Blaivas & Olsson (1988), on video-urodynamics**

| Type | Bladder neck at rest | On stress |
|---|---|---|
| **0** | Closed, above lower border of symphysis | Descends and opens, but no leak seen (history of SUI) |
| **I** | Closed, above symphysis | Descends **<2 cm**; leaks; little or no cystocele |
| **IIA** | Closed, above symphysis | Descends **>2 cm** with cystourethrocele; leaks |
| **IIB** | Closed but **at or below** lower border of symphysis | Proximal urethra opens; leaks with or without further descent |
| **III** | **Open at rest** without detrusor contraction | Leaks with minimal stress = **ISD** |

### 2. Functional: hypermobility versus ISD

| | Urethral hypermobility | ISD |
|---|---|---|
| Q-tip test | Angle change **>30°** | Often fixed urethra |
| VLPP / ALPP | **≥90 cm$H_2O$** | **<60 cm$H_2O$** (60–90 equivocal) |
| MUCP | Normal | **≤20 cm$H_2O$** |
| Cause | Childbirth, ageing, obesity | Previous surgery, radiation, neuropathy, ageing |

### 3. Other

- **Severity (Stamey):** grade 1 leak on cough or sneeze; grade 2 on minimal effort (walking, rising); grade 3 at rest.
- **Index (uncomplicated)** versus **complicated** SUI (recurrent, prolapse, neurogenic, high PVR [post-void residual], radiation); pure versus mixed incontinence.

## Evaluation of SUI in women

### History

- Leakage with cough or activity means SUI; after urgency, UUI (urgency urinary incontinence); both, MUI (mixed). A positive SUI history predicts urodynamic SUI in about three-quarters.
- Severity (pads/day), bother; voiding symptoms, prolapse, obstetric history, prior surgery or radiation, neurological disease, drugs.
- **Questionnaires:** ICIQ-UI SF (International Consultation on Incontinence Questionnaire–UI Short Form); UDI-6 (Urogenital Distress Inventory: bother); IIQ-7 (Incontinence Impact Questionnaire: impact). **Bladder diary** (3 days).

### Examination (comfortably full bladder)

- Neurological (S2–S4 sensation, anal tone, bulbocavernosus reflex); atrophy; prolapse by **POP-Q (Pelvic Organ Prolapse Quantification)**.
- **Cough stress test:** cough supine, then standing if negative. Leakage synchronous with cough confirms SUI. Reduce prolapse to unmask **occult SUI**.
- **Q-tip test:** swab to bladder neck, Valsalva; rotation **>30°** = hypermobility. Visual mobility examination is now routine.

### Tests

| Test | Method and use |
|---|---|
| Urinalysis | Required; exclude infection, haematuria |
| PVR | Required before surgery; ultrasound; repeat if high (no agreed cut-off) |

| Test | Method and use |
|---|---|
| 1-hour pad test (ICS) | 500 mL fluid, standard exercises; >1.4 g positive; 1–10 g mild, 11–50 g moderate, >50 g severe |
| 24-hour pad test | Normal activities; more reproducible |
| Cystoscopy | Haematuria, irritative symptoms, previous mesh |

## Role of urodynamics (UDS) in SUI

- UDS (urodynamic studies) assess storage and voiding. The key question in SUI is: **will the result change the operation?** Pure, demonstrable SUI gains little. **UDS is not routine.**

| Parameter | Method and meaning |
|---|---|
| USI (urodynamic stress incontinence) | Leak with raised abdominal pressure, no detrusor contraction |
| ALPP/VLPP (abdominal/Valsalva leak point pressure) | Standing, from 150 mL, Valsalva or cough; **<60 cm$H_2O$ = ISD**, 60–90 equivocal, ≥90 = little ISD (McGuire 1993) |
| MUCP | Urethral pressure profile; a low MUCP (≤20 cm$H_2O$) also suggests ISD; does not diagnose or grade SUI |
| Filling cystometry | DO (detrusor overactivity), compliance, capacity; cough-induced DO mimics SUI |
| Pressure–flow study | Voiding dysfunction, poor contractility (retention risk after sling) |
| Prolapse reduction | Occult SUI |
| - If no leak with the catheter in place, repeat with it removed. | |

### Evidence

- **VALUE (ValUE) trial (2012), 630 women** with uncomplicated SUI: office evaluation alone was **non-inferior** to office evaluation plus UDS before surgery (success 77.2% vs 76.9%).
- **Dutch VUE study:** immediate midurethral sling was non-inferior to UDS-tailored treatment.
- **SISTEr:** UDS did not predict outcome.

### Guidelines

- **AUA (American Urological Association)/SUFU (Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) 2023:** UDS may be omitted in the index patient with demonstrable SUI.
- **UDS advised in non-index patients:** unclear diagnosis or SUI not demonstrated; mismatch between symptoms and findings; urgency-predominant MUI; neurogenic bladder; raised PVR or voiding dysfunction; prolapse stage ≥3 without demonstrable SUI; failed previous anti-incontinence or prolapse surgery.
- **EAU (European Association of Urology):** no routine UDS for uncomplicated SUI; do not use leak point or urethral pressure to grade severity.
- **Men:** before AUS or sling for post-prostatectomy incontinence, UDS confirms USI and finds DO, poor compliance or weak contractility. The MASTER trial recruited men with USI.

## Conservative management (first line)

- Weight loss, fluid and caffeine adjustment.
- **Supervised PFMT (pelvic floor muscle training)** for at least 3 months (Cochrane: more cure than no treatment).

- **Pessary**; vaginal oestrogen for atrophy.
- **Duloxetine** (serotonin–noradrenaline reuptake inhibitor): increases Onuf's nucleus drive to the sphincter; approved in Europe, not the USA.

## Surgical options for female SUI: principles

| Operation | Principle (how it works) | Best for | Key facts |
|---|---|---|---|
| **MUS**: retropubic (TVT, tension-free vaginal tape, Ulmsten 1995), transobturator (TOT; Delorme 2001; inside-out de Leval), single-incision | Tension-free type I polypropylene tape under the **mid-urethra** acts as a new pubourethral ligament; the urethra kinks against it during stress | First-line index SUI | **First line** (AUA/SUFU strong recommendation). **TOMUS**: retropubic 80.8% vs transobturator 77.7% cure (equivalent); retropubic often preferred in ISD. Retropubic: more bladder perforation; transobturator: more groin pain |
| **SIMS (single-incision mini-sling)** | Short tape, vaginal only | Option | Non-inferior to TOT for subjective cure in most studies; conditional recommendation (2023) |
| **Autologous pubovaginal sling (PVS)** (rectus fascia or fascia lata) | At the **bladder neck**, minimal tension; restores the hammock, giving bladder-neck support and compression | **All types, including ISD**, recurrent SUI, neurogenic SUI, failed MUS, after mesh removal, mesh-averse | Gold standard. **SISTEr**: sling beat Burch (47% vs 38%) but more voiding dysfunction and UTI (urinary tract infection) |
| **Burch colposuspension** (open, laparoscopic or robotic) | Paravaginal fascia beside the mid-urethra and bladder neck sutured to **Cooper's ligament**; lifts the bladder neck back into the pressure zone and gives a stable backing | Hypermobility; women declining mesh; at abdominal surgery | 85–90% at 1 year, about 70% at 5 years. Needs a mobile urethra; poor in ISD |
| Paravaginal repair, VOS (vagino-obturator shelf) | Vagina to ATFP or obturator fascia | | Flatter, less obstructive |
| **MMK (Marshall–Marchetti–Krantz)** | Bladder neck to symphysis periosteum | Historical | Osteitis pubis 0.7–2.5%; abandoned |
| **Urethral bulking agents** | Submucosal injection at bladder neck or proximal urethra improves coaptation | ISD with fixed urethra; frail patients; minimal surgery | Less durable; repeat injections |
| **AUS** | Bladder-neck cuff; constant circumferential closure | Severe refractory ISD, fixed urethra | Second-line; erosion, infection, revision |

| Operation | Principle (how it works) | Best for | Key facts |
|---|---|---|---|
| - **Also know:** needle suspensions abandoned (sutures pull through); anterior colporrhaphy is not an SUI operation; CARE (prophylactic Burch at sacrocolpopexy); Ward–Hilton (TVT = colposuspension). | | | |

**Burch steps (algorithm):** Pfannenstiel → space of Retzius → stay lateral to urethra → 2–4 sutures each side (mid-urethra, then bladder neck) → Cooper's ligament → tie loosely (two fingers behind pubis) → **cystoscopy** for ureteric efflux.

**Retropubic vs transobturator MUS**

| | Retropubic | Transobturator |
|---|---|---|
| Bladder perforation | 4.5% | 0.6% |
| Voiding dysfunction (TOMUS surgery) | More (2.7% vs 0%) | Less |
| Groin pain | 1.3% | 6.4% |
| ISD, long-term durability | Often preferred | Possibly more repeat surgery |

## Complications of female SUI surgery

- **MUS complications:** perforation, retention, de novo urgency, mesh exposure (about 2%), pain.
- Bladder trocar injury: re-pass and continue (cystoscopy after every pass); **urethral injury: abandon mesh**.
- Voiding dysfunction (sling too tight or proximal): urethrolysis or sling incision; de novo urgency; UTI; haematoma.
- **Mesh complications** (IUGA/ICS [International Urogynecological Association/International Continence Society] terms): exposure, extrusion, perforation into bladder or urethra, pain; manage by excision. FDA (Food and Drug Administration) 2019 stopped transvaginal mesh for prolapse, not slings.

## Male SUI: surgical therapy

- **Aetiology:** commonest after **radical prostatectomy (RP)**, which removes the internal sphincter, leaving continence to the rhabdosphincter. Also TURP (transurethral resection of the prostate), radiation, pelvic fracture.
- **Work-up:** 24-h pad weight (mild <150 g, moderate <400 g, severe above); standing cough test; **cystoscopy** (anastomotic stricture, sphincter coaptation); UDS if mixed or poor contractility.
- **Timing:** PFMT first; no surgery for 6–12 months, as continence keeps improving (other P3 answer: continence improves for up to 2 years, but do not delay surgery beyond 6 months for severe leak).

| Device | Mechanism | Use and results |
|---|---|---|
| **Transobturator sling (AdVance XP)** | Repositions the bulb 3–4 cm proximally; needs residual sphincter | Mild–moderate leakage, no radiotherapy; cure about 60% |
| Quadratic sling (Virtue) | Compression plus elevation | Mild; limited data |
| Adjustable sling (ATOMS, adjustable transobturator male system) | Cushion adjusted via scrotal port | Mild–moderate; about two-thirds dry |
| ProACT balloons | Bladder-neck compression | More erosion |
| Bulking agents | Submucosal bulk | Small leaks, unfit men; rarely durable; not advised |
| **AUS** | Circumferential compression | **Gold standard**; all severities, moderate–severe, after radiation; about 80% continent |

**AMS 800 AUS**

- **Components:** urethral **cuff** (bulbar, usually 4.0–4.5 cm); **PRB (pressure-regulating balloon) 61–70 cm$H_2O$** (51–60 high risk; 71–80 bladder neck) in a preperitoneal pocket; scrotal **pump**. Filled with 23 mL.
- **Technique:** perineal incision; cuff on the proximal bulbar urethra; **deactivate for 4–6 weeks**.
- **Needs** dexterity and cognition. Social continence (0–1 pad) 55–77%; satisfaction about 90%; lifespan 7–10 years.
- **Complications:** infection 1–3% (explant; reimplant after 3–6 months); **erosion** 2–15% (explant, catheter about 3 weeks); urethral **atrophy** (late recurrent leak); mechanical failure; retention.
- **Erosion risks:** radiation, prior erosion, urethroplasty; use a **transcorporal cuff** (tunica adds bulk) or lower PRB. Atrophy: smaller or tandem cuff.

**Algorithm (AUA/SUFU 2019)** `Severe SUI or any radiotherapy -> AUS Moderate, no radiotherapy -> AUS or male sling Mild, no radiotherapy -> sling (or PFMT, AUS) Sling failure -> AUS`

- **MASTER RCT (randomised controlled trial):** sling was **non-inferior** to AUS for continence at 12 months; AUS remains preferred for heavy leak (>250 g/day) or radiation.

## Diagrams to draw

- DeLancey levels and hammock cross-section.
- Sites of Burch sutures, bladder-neck sling and mid-urethral tape; sagittal pelvis showing retropubic and transobturator tape paths.
- Cystometrogram trace with abdominal pressure spike and leak (ALPP).
- AMS 800 (cuff, pump, PRB) and male transobturator sling path.

**Key points to remember:**

- Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–5.
- DeLancey: I suspension, II attachment (ATFP), III fusion.
- Continence zone is the mid-urethra; submucosal plexus gives up to 30% of closure.
- Blaivas types 0, I, IIA, IIB, III; type III = ISD.
- Q-tip >30° = hypermobility; VLPP / ALPP <60 $cmH_2O$ = ISD; MUCP ≤20 $cmH_2O$.
- Burch corrects position; slings and AUS correct function.
- VALUE (ValUE) trial: UDS not needed in uncomplicated (index) SUI.
- Use UDS when it changes treatment: mixed, failed surgery, voiding dysfunction.
- PFMT first; MUS first-line surgery.
- SISTEr (fascial sling > Burch); TOMUS (retropubic = transobturator).
- PVS for ISD; urethral injury → no mesh.
- Men: AUS gold standard; sling for mild, non-irradiated.
- AUS PRB 61–70 $cmH_2O$; deactivate 4–6 weeks.
- Radiation or severe male SUI: AUS first; MASTER: sling non-inferior.

## LAQ 5 Recto-urethral (recto-prostatic) fistula: etiopathogenesis, diagnosis and management

P3

**Asked as:**

- Etio-pathogenesis, diagnosis and management of Recto-urethral fistula? — *Summer 2020 Paper III Q1 (LAQ)*
- Etiopathogenesis, diagnosis and management of Recto-Prostatic fistula — *Winter 2025 Paper III Q1 (LAQ)* (also answered in 2020-Summer Paper 3, Q1, as recto-urethral fistula)

**Type:** LAQ (25 marks)

*Assumed meaning (recto-prostatic): rectourethral fistula at the level of the prostatic urethra or the vesicourethral anastomosis.*

### Definition

- **RUF (rectourethral fistula):** a communication between the posterior urethra / prostatic urethra (or the vesicourethral anastomosis after prostatectomy) and the distal rectum.
- Rare but very morbid. Almost never seen in females, because the vagina lies between urethra and rectum.

### Aetiology

| Group | Causes |
|---|---|
| Congenital | Anorectal malformation in boys (rectobulbar or rectoprostatic fistula) |
| Surgical (commonest acquired) | **RP (radical prostatectomy)**; historically TURP (transurethral resection of the prostate) and open (simple) prostatectomy |
| Radiation and energy | **EBRT (external beam radiotherapy) plus brachytherapy** (commonest), then brachytherapy alone, then pelvic radiotherapy; cryotherapy; HIFU (high-intensity focused ultrasound): low after primary, up to about 9% after salvage HIFU |
| Others | Trauma (penetrating injury, pelvic fracture); Crohn disease; advanced rectal or prostate cancer |

### Pathogenesis

- **Surgical:** the rectum is torn during apical or posterior dissection behind Denonvilliers fascia, especially with small glands and salvage surgery. An unrecognised tear (near the apex) leaks urine (urine leaks into it), an abscess forms and the tract epithelialises.
- Rectal injury at RP: laparoscopic about 1%, open and robotic under 0.5%, salvage RP about 7%. (Another source: rectal injury is commonest with retropubic and laparoscopic RP, lowest with robotic RP, and was over 15% in historical open salvage series.)
- **Radiation or ablation:** obliterative endarteritis makes the tissue hypoxic and fibrotic. Necrosis follows months to years later. These fistulae are large and heal poorly.
- **Congenital:** abnormal cloacal development leaves the rectum ending in the posterior urethra.

## Classification: Martini (after prostate cancer treatment)

| Axis | Category | Definition |
|---|---|---|
| Stage (size on CT (computed tomography) urography or cystography) | I / II / III | ≤1.5 cm / >1.5 cm / any size with sphincter damage |
| Grade (tissue) | 0 | Surgical injury, no prior non-surgical treatment |
| | 1 | Prior radiation, brachytherapy, HIFU or other energy |
| | 2 | Salvage RP or salvage ablation |
| - **Tissue quality (grade), not size, chiefly decides the repair.** | | |

## Clinical features

- **Faecaluria and pneumaturia** (most typical); urine per rectum on voiding.
- Recurrent UTI (urinary tract infection), dysuria, urosepsis; coexisting SUI (stress urinary incontinence).

## Diagnosis

- **DRE (digital rectal examination):** may feel the defect on the anterior rectal wall.
- **Cystoscopy and proctosigmoidoscopy:** show the tract and tissue vitality; **biopsy** to exclude recurrent cancer.
- **VCUG (voiding cystourethrogram) or RUG (retrograde urethrogram):** usually conclusive; gives size and site for planning.
- Upper-tract imaging to exclude ureteric injury; pelvic MRI (magnetic resonance imaging) for complex cases.
- **UDS (urodynamics)** with rectal packing: fistula repair alone will not cure severe SUI; an AUS (artificial urinary sphincter) may be needed later; a later AUS erodes more often.

## Prevention

- At RP: a recognised tear without gross spillage → two-layer (multilayer) closure, air test, **omental or peritoneal interposition**, drain, antibiotics; colostomy if (previously) irradiated.
- Radiotherapy: a **hydrogel rectal spacer** lowers rectal dose and late rectal toxicity (randomised trial: late rectal toxicity 2.0% vs 7.0%).

## Management

- **Principles:** control sepsis; divert urine ± faeces; let inflammation settle (about 12 weeks); exclude cancer; then excise the tract and close urethra and rectum separately in **tension-free, non-overlapping layers with vascularised (interposition) tissue interposed**.
- **Conservative:** no sepsis and no faecaluria → catheter (urethral or suprapubic), low-residue diet ± colostomy. A simple fistula may close in about 12 weeks; irradiated or complex ones rarely do.
- **Surgery:** no single accepted operation; choice depends on site, radiation and prior repair. **A diverting colostomy usually comes first.**

| Approach | Essentials | Strengths | Weaknesses |
|---|---|---|---|
| **Perineal** (Young; Goodwin) | Separate and close urethra and rectum separately; **gracilis flap** between (or dartos; buccal graft) | High closure; most stomas reversed | Urinary incontinence common |
| **Transsphincteric (York-Mason)** | Prone jackknife; sphincter divided in tagged layers; through anterior rectal wall; three-layer closure | Avoids scarred plane; spares sexual function | Poor membranous exposure; no flap; faecal incontinence up to 20% |
| **Abdominal** (transvesical, extravesical, O'Conor-like) / robotic | Omental flap | Large or irradiated fistulae, extensive resection | Morbid; poor deep-pelvic view; small robotic series |
| **Laparoscopic / robotic** | Open pouch of Douglas; close rectum in two layers; omental or peritoneal flap; can combine with perineal gracilis | Deep-pelvic vision | Small series only |
| **Salvage** | Cystoprostatectomy, ileal conduit ± permanent colostomy | Unreconstructable irradiated field | Major surgery |

`RUF → sepsis control, catheter ± colostomy, biopsy small, non-irradiated, no faecaluria → conservative ~12 weeks persistent, grade 0 → perineal (gracilis) or York-Mason irradiated or large → abdominal/robotic (omentum) or perineal (gracilis) unreconstructable → urinary + faecal diversion Congenital → newborn colostomy → posterior sagittal anorectoplasty`

- **PSARP (posterior sagittal anorectoplasty)** separates the rectum from the urethra and places it within (in) the sphincter complex.

### Complications

- **Recurrence** (the greatest challenge), urinary or faecal incontinence, urethral stricture, ED (erectile dysfunction), wound infection, permanent stoma.

**Diagram to draw:** sagittal pelvis showing (with) the fistula and a gracilis flap between urethra and rectum.

> **Key points to remember:**
> - Most acquired RUFs follow prostate cancer treatment.
> - Martini: stage = size; grade = tissue (surgery, radiation, salvage).
> - VCUG or RUG plus endoscopic biopsy to exclude cancer.
> - Divert first; perineal (gracilis) or York-Mason for surgical fistulae; irradiated ones need imported tissue or diversion.

## Short answer questions (SAQ)

## SAQ 1 Leak point pressures and ALPP

P1 · P2 **merged from 2**

*Asked as:* P1 SAQ "Leak point pressures (VLPP, ALPP, DLPP) in SUI: clinical implications" · P2 SAQ "ALPP"

## Definition

- **LPP (leak point pressure):** the pressure at which leakage is first seen, spontaneous or provoked. There are two independent types:

| Feature | ALPP (abdominal LPP), incl. VLPP (Valsalva LPP) and CLPP (cough LPP) | DLPP (detrusor LPP) |
|---|---|---|
| Definition | Total Pves (vesical pressure) at leak during raised abdominal pressure, **no detrusor contraction** | Lowest Pdet (detrusor pressure) at leak, **without contraction or straining** |
| Measures | Sphincter strength against abdominal pressure | Bladder pressure against outlet resistance |
| Bad result | **Low = bad** (weak sphincter) | **High = bad**: >40 $cmH_2O$ means upper-tract risk; hydronephrosis or reflux in 85% (McGuire 1981) |
| Used in | SUI (stress urinary incontinence) | Neurogenic or low-compliance bladder |

- **ALPP (abdominal leak point pressure):** the **intravesical pressure** at which urine leaks because of raised abdominal pressure, **without a detrusor contraction**. It measures the urethra's ability to resist abdominal pressure: **lower ALPP = weaker sphincter**.
- Provoked by Valsalva (**VLPP**, the intravesical pressure at which urine leaks during Valsalva) or cough (**CLPP**).
- Seen only in SUI; there is **no normal value** (continent women never leak).
- Newer term: "urodynamic stress test" (ICS, International Continence Society).

## Technique (VLPP/ALPP)

1. Small dual-lumen urethral catheter plus rectal catheter; zero to atmosphere.
2. Patient **standing or sitting**; fill to **150 mL** (original method; some use 200 mL).
3. Ask for a **graded Valsalva** or cough; record the **lowest total Pves at visible leak** (include baseline, not the change).
4. No leak: repeat every 50 mL to capacity; then, at capacity, **remove the urethral catheter** and read pressure from the rectal line.
5. Report volume, catheter size and provocation (Valsalva or cough).

## Interpretation (McGuire 1993)

| VLPP / ALPP ($cmH_2O$) | Inference |
|---|---|
| **<60** | **ISD (intrinsic sphincter deficiency)** |
| **60–90** | Equivocal; mixed |
| **≥90** (>90) | **Urethral hypermobility**, little ISD |

## Clinical implications and uses of ALPP

- Separates ISD from hypermobility; a low VLPP/ALPP (ISD) traditionally favours a sling or bulking agent over colposuspension.
- Post-prostatectomy incontinence: many men leak only once the urethral catheter is removed.

## Limitations

- Not standardised: falls with **larger catheters** and **higher volumes**.

- **VLPP is lower than CLPP** in the same woman (VLPP < CLPP), so state the method.
- Invalid during DO (detrusor overactivity) or poor compliance; prolapse raises it or masks the leak (reduce it).
- ISD and hypermobility coexist; VLPP/ALPP alone **does not predict surgical outcome** well.
- **VALUE trial (Nager 2012):** for uncomplicated SUI, office evaluation was non-inferior to urodynamics before surgery. AUA (American Urological Association)/SUFU (Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) 2023 SUI guideline: urodynamics optional in the index patient.
- **EAU (European Association of Urology):** do not use leak point pressure to grade severity.

### DLPP: clinical implications

- **McGuire (1981):** DLPP **>40 $cmH_2O$** in myelodysplastic children leads to hydronephrosis or VUR (vesicoureteral reflux) in **85%**.
- A strong outlet keeps the patient dry but transmits pressure to the kidneys; a weak outlet "pops off" safely.
- **Hostile bladder** (≥40 $cmH_2O$): CIC (clean intermittent catheterisation) plus antimuscarinic; botulinum toxin, augmentation or vesicostomy if needed.
- **Caveat:** 40 is not a validated "safe" value; 20 $cmH_2O$ is more sensitive in children.

### Also know

- DO leak point pressure (ICS 2017); MUCP (maximum urethral closure pressure) ≤20 $cmH_2O$ suggests ISD.

### Diagram to draw

- Pves, Pabd and Pdet traces marking ALPP (leak on Valsalva with flat Pdet) and DLPP.

> **Key points to remember:**
> - VLPP/ALPP: Pves at leak with Valsalva or strain, no detrusor contraction; start at 150 mL.
> - ALPP low = weak sphincter; DLPP high = upper-tract danger.
> - VLPP <60 ISD; 60–90 equivocal; ≥90 (>90) hypermobility.
> - VLPP < CLPP; catheter size and volume alter it.
> - DLPP >40: upper-tract danger; 85% upper-tract change (McGuire 1981).
> - Lower DLPP is safer; no validated safe cut-off.

## SAQ 2 Mid-urethral slings: TOT and TVT

P1 · P2 **merged from 2**

*Asked as:* P2 SAQ "Mid-urethral slings for SUI: TOT, TVT and TOT versus TVT" · P1 SAQ "TOT in SUI"

### Definition and history

- Both are polypropylene **MUS (mid-urethral slings)** for female SUI (stress urinary incontinence), based on the **integral theory** (Petros and Ulmsten, 1990): a loose mid-urethral tape replaces the pubourethral ligaments as a backboard on straining.
- **TVT (tension-free vaginal tape, retropubic):** retropubic polypropylene MUS; Ulmsten, 1995–1998.

- **TOT (transobturator tape):** a synthetic MUS of macroporous monofilament polypropylene, passed through the obturator foramina instead of the retropubic space. Delorme (2001) designed it to avoid bladder, bowel and vessel injury.

### Mechanism

- **Integral theory:** pubourethral ligaments, the suburethral vaginal hammock and pubococcygeus maintain continence at the mid-urethra. Hypermobility is a sign of their damage, not the cause.
- The loose tape at the mid-urethra supports the **posterior urethral wall**, which is compressed against it on straining (dynamic kinking); it is not tight at rest.
- TOT anchors in the obturator muscles (**obturator internus and externus**).

### Material

- **Type 1 (Amid) macroporous monofilament polypropylene**, pores >75 µm: allows macrophage entry and fibrous ingrowth, so it resists infection.

### Indications and evaluation

- SUI or stress-predominant mixed incontinence after failed PFMT (pelvic floor muscle training); index SUI with urethral hypermobility.
- TVT preferred over TOT in **ISD (intrinsic sphincter deficiency)** or recurrent SUI.
- History, examination, **objective stress test**, urinalysis, PVR (post-void residual); urodynamics only if complicated.
- Counsel about permanent mesh, mesh risks and alternatives (colposuspension, autologous sling, bulking) (AUA/SUFU [American Urological Association/Society of Urodynamics and Female Urology]).

### Technique: differences

| | **TVT (retropubic)** | **TOT (transobturator)** |
|---|---|---|
| Path | Vagina to retropubic space to suprapubic skin, hugging the pubis; cystoscopy essential | Labiocrural fold to obturator membrane to vagina (outside-in), or reverse (inside-out) |
| Structures at risk | Bladder, bowel, iliac and epigastric vessels | Obturator nerves and vessels, vaginal fornix |

### TVT technique

1. Local, spinal or general anaesthesia; lithotomy.
2. Two suprapubic stab incisions 2 cm lateral to midline; 1.5 cm midline vaginal incision 1.5 cm from the meatus; minimal dissection, endopelvic fascia not opened.
3. **Bottom-up** 5 mm trocar hugging the back of the pubis, through the retropubic space to the skin (catheter guide deflects the bladder).
4. **Cystoscopy with a 70° lens, full bladder**, after each pass; if perforated, re-pass more laterally.
5. Tension **loosely** with a clamp between tape and urethra; remove sheath.

### TOT technique (outside-in)

1. Lithotomy with hips hyperflexed; 1.5 cm midline vaginal incision 1.5 cm below the meatus; dissect laterally to the ischiopubic ramus.

2. Groin stab incision in the labiocrural fold, at the notch between inferior pubic ramus and **adductor longus tendon**.
3. Trocar perforates the obturator membrane (felt as a give), turns medially onto the finger and exits the vaginal incision.
4. Cystourethroscopy; check the vaginal fornices (sulci).
5. Tension-free: a clamp or scissors pass easily between tape and urethra. - **Inside-out** (de Leval; vagina to thigh): fewer vaginal perforations, more voiding dysfunction.

### Outcomes

- TVT: Ulmsten (1998): 91% cured at 1 year. **Ward–Hilton:** equal to colposuspension at 5 years. Long-term cure varies 48–97% by definition.
- **TOMUS (Richter, 2010):** 597 women; 12-month objective cure **80.8% retropubic vs 77.7% transobturator** (not significantly different, equivalent); voiding dysfunction needing surgery 2.7% vs 0%.
- TOMUS at 5 years: 51.3% vs 43.4%; equivalence not shown; satisfaction similar.
- **Cochrane review (Ford, 2017):** cure similar up to and beyond 5 years; repeat surgery possibly commoner after TOT.
- TOT is poorer with **ISD** or a **fixed urethra**: retropubic sling preferred.

### Complications: comparison (Cochrane, 2017)

| Complication | TVT (retropubic) | TOT |
|---|---|---|
| Bladder perforation | **4.5%** | **0.6%** |
| Groin pain | 1.3% | **6.4%** |
| Voiding dysfunction, bleeding, vascular injury | More | Less |
| Mesh exposure | About 2% | About 2% |
| Long-term repeat surgery | Less | Possibly more |

### TVT complications

| Complication | Notes |
|---|---|
| Bladder perforation | 4.5% (TOT 0.6%); re-pass and continue |
| Urethral injury | Abandon mesh; staged repair |
| Bleeding/haematoma | 2–3%; rarely bowel or vascular injury |
| Voiding dysfunction | Surgery needed 2.7%; incise tape in midline within 3 months |
| Mesh exposure | About 2% |
| Others | De novo urgency, suprapubic pain (groin pain less than TOT) |

### TOT complications

- Groin or thigh pain (usually temporary; NSAIDs [non-steroidal anti-inflammatory drugs]), vaginal mesh exposure, voiding dysfunction (incise sling within 4 weeks), de novo urgency, UTI (urinary tract infection), rarely thigh abscess.
- Urethral injury during either sling: **do not place mesh** (AUA guideline).

### Choosing between TVT and TOT

- **TVT preferred: ISD** (ALPP [abdominal leak point pressure] <60 $cmH_2O$ or MUCP [maximum urethral closure pressure] <20 $cmH_2O$), fixed urethra, recurrent SUI. Failure is 12 times likelier after TOT with low leak point pressure (O'Connor, 2006).
- **TOT preferred:** previous retropubic surgery, obesity, avoiding bladder injury.
- AUA: either may be offered to the index patient.
- **Also know:** SIMS (single-incision mini-slings; non-inferior to TOT); mesh warnings (FDA [Food and Drug Administration] 2010; UK pause 2018): counsel about alternatives (Burch colposuspension, autologous fascial sling, bulking agents).

### Diagram to draw

- Coronal pelvis: U-shaped retropubic TVT at mid-urethra versus flat TOT, the tape under the mid-urethra through both obturator foramina.

> **Key points to remember:**
> - Integral theory; loose tape at mid-urethra.
> - Type 1 macroporous monofilament polypropylene.
> - Delorme 2001; TOT anchors in obturator muscles; fewer bladder injuries than retropubic.
> - TOMUS: cure equivalent at 1 year (80.8% retropubic vs 77.7% transobturator).
> - TOT: less bladder injury (0.6% vs 4.5%), more groin pain (6.4% vs 1.3%); worse in ISD.
> - Cystoscopy after each TVT trocar pass; bladder perforation 4.5%.
> - Prefer retropubic TVT in ISD or fixed urethra.

## SAQ 3 Burch colposuspension

P1

**Type:** SAQ (10 marks)
**Source:** 2022-Summer Paper 1, Q3b (related: 2017-Winter Paper 3, Q2)

### Definition

- Retropubic suspension (**Burch, 1961**) of the paraurethral and para-bladder-neck vaginal wall to **Cooper's (iliopectineal) ligament**; Tanagho (1976) modified it.

### Principle

- SUI (stress urinary incontinence) from **urethral hypermobility**: the urethra drops out of the retropubic position. Elevation restores **pressure transmission** and a backstop; scarring holds it.
- It treats hypermobility, not ISD (intrinsic sphincter deficiency).

### Indications

- SUI or stress-predominant MUI (mixed urinary incontinence) with hypermobility, especially with **concomitant abdominal surgery** (sacrocolpopexy), **mesh refusal**, or poor vaginal access.
- **Contraindications:** immobile urethra or vagina, ISD alone, pure urgency incontinence.

### Work-up

- Cough stress test, PVR (post-void residual), urinalysis; urodynamics only if SUI is not demonstrated.

### Steps (open)

1. Low lithotomy, Foley; Pfannenstiel incision; enter the **space of Retzius**.
2. Vaginal finger elevates the vagina; sweep bladder medially. **Avoid midline dissection** over urethra.
3. Two to four 0 or 1-0 delayed-absorbable or non-absorbable sutures each side through vaginal wall (not lumen): distal at mid-urethra, proximal at bladder neck, placed laterally.
4. Pass through Cooper's ligament; tie with vaginal elevation, without tension; leave about **two fingerbreadths** between bladder neck and pubis.
5. **Cystoscopy:** ureteric efflux, no suture in bladder. - Laparoscopic and robotic versions match open short-term results.

### Outcomes

- Cure 85–90% at 1 year, about 70% at 5 years (Cochrane).
- **SISTEr (2007):** autologous fascial sling more effective but more voiding dysfunction and UTI.
- **Ward and Hilton:** TVT (tension-free vaginal tape) equivalent to colposuspension; meta-analyses favour midurethral slings.
- **CARE (2006):** prophylactic Burch at sacrocolpopexy reduced postoperative SUI.

### Complications

- Voiding dysfunction 2–27%, de novo storage symptoms 8–27%, **de novo prolapse** 2.5–27%; bladder or ureteric injury; bleeding.
- **Also know:** MMK (Marshall–Marchetti–Krantz; osteitis pubis 0.7–2.5%), paravaginal repair, VOS (vagino-obturator shelf); failure predictors: BMI (body mass index) >25, age >60, prior surgery.

**Diagram to draw:** sagittal pelvis with sutures from paravaginal fascia to Cooper's ligament.

> **Key points to remember:**
> - Burch 1961: vaginal wall to Cooper's ligament; for hypermobility.
> - Cure 85–90% at 1 year, ~70% at 5 years.
> - Niche: concomitant abdominal surgery, mesh avoidance.

## SAQ 4 Management of perforated mid urethral sling

P1

**Type:** SAQ (10 marks)
**Source:** 2023-Winter Paper 1, Q3a

### Terminology (IUGA/ICS, International Urogynecological Association / International Continence Society)

- **Perforation:** mesh in a hollow organ (bladder, urethra). **Exposure:** mesh seen through unhealed vaginal epithelium. **Extrusion:** passage out of a tissue. "Erosion" is no longer used.

- Two settings: **intraoperative trocar injury** and **late mesh perforation**.

### A. Intraoperative trocar injury

- Bladder injury: 4.5% retropubic versus **0.6% transobturator**; registries 2.7–3.8%.
- **Detection:** cystoscopy with a **70° lens and full bladder distension** after every trocar pass.

| Injury | Management |
|---|---|
| **Bladder** | Withdraw trocar, **repass more laterally**, confirm on cystoscopy, complete the sling; catheter overnight to 72 h. Benign; no later mesh perforation. |
| **Urethra** | **Do not place mesh** (AUA/SUFU, American Urological Association / Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction). Repair, catheter drainage; **staged** anti-incontinence surgery after healing. |

### B. Late perforation of bladder or urethra

- **Presentation:** recurrent UTI (urinary tract infection), haematuria, urgency, pain, stones on mesh, voiding dysfunction, fistula. Often months later.
- **Evaluation:** examination, **cystourethroscopy**, translabial ultrasound (best imaging for mesh); exclude fistula.

#### Management

- **Bladder:** endoscopic excision (scissors, resection or holmium laser) flush with the wall; or transvesical laparoscopic or robotic excision; open cystotomy if large.
- **Urethra:** transvaginal excision of mesh in the field and layered urethral repair; **Martius flap** if complex or fistula.
- **Continence after excision:** SUI recurs in many (44–100% after synthetic sling revision). Options: concomitant **autologous pubovaginal sling** (Shah 2013: good results) or staged. A damaged urethra predicts poor continence.
- Avoid urethral dilation after mesh slings; it can cause perforation.

### Prevention

- Empty bladder; trocar hugging the pubis; bottom-up retropubic route; transobturator route when suitable.

**Also know:** Amid type I macroporous polypropylene integrates best; chronic pain or infection may need excision from both sides of the bone.

**Diagram to draw:** retropubic and transobturator trocar paths with bladder at risk.

> **Key points to remember:**
> - Bladder trocar injury: repass laterally, continue.
> - Urethral injury: abandon mesh; stage the repair.
> - Late perforation: excise mesh, repair, consider autologous sling.

## SAQ 5 Female urethral diverticula

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 13 Paper 1, Q3e

## Definition

- **UD (urethral diverticulum):** a urine-filled periurethral sac within the periurethral fascia, connected to the urethra by an **ostium**.
- Third to seventh decades; prevalence up to 1–6%.

## Aetiology

- Mostly **acquired**. **Routh theory** (most accepted): infection of **periurethral glands** leads to duct obstruction, suburethral abscess, rupture into the lumen and an epithelialised sac.
- Hence the ostium is **posterolateral at mid or distal urethra** in ≥90%.
- **Also know:** childbirth trauma theory (but 20–30% are nulliparous).

## Clinical features

- "Three Ds" (dysuria, dyspareunia, post-void dribbling) in only about 5%.
- Commonest: frequency, urgency, pain, **recurrent UTI (urinary tract infection)**.
- Tender anterior vaginal wall mass; about one-third incontinent.
- **Complications:** stones (4–10%); 10% show metaplasia, dysplasia or carcinoma (**adenocarcinoma** commonest).

## Evaluation

- Urine culture; **cystourethroscopy** (ostium seen in under half).
- **Imaging sensitivity: MRI (magnetic resonance imaging) 97%**, double-balloon urethrography 95%, ultrasound 82%, VCUG (voiding cystourethrogram) 74%.
- MRI shows saddlebag or circumferential extension.
- **Videourodynamics:** true SUI (stress urinary incontinence) versus dribbling from the sac.
- **Differential diagnosis:** Skene cyst (distal, distorts meatus, no communication), Gartner cyst, leiomyoma, caruncle, bulking agent.

## Classification

- **L/N/S/C3 (Leach):** location, number, size, configuration, communication, continence.
- **Also know:** Leng–McGuire (absent fascia = pseudodiverticulum).

## Management

- **Asymptomatic or unfit:** suppressive antibiotics, post-void stripping; counsel about cancer.
- **Symptomatic: transvaginal diverticulectomy.**
- **Principles:** 1. Sterile urine; topical oestrogen if atrophic. 2. **Inverted-U** anterior vaginal wall flap. 3. **Preserve periurethral fascia** as a separate layer. 4. Excise the **entire sac** and close the ostium. 5. **Watertight** urethral closure over a catheter (as small as 12 Fr). 6. **Multilayered, non-overlapping** closure; obliterate dead space. 7. **Martius flap** if poor tissue.
- **Bothersome SUI:** concomitant **autologous pubovaginal sling**; **never synthetic mesh**.
- **Also know:** Spence–Duckett marsupialisation for distal lesions only.
- Pericatheter VCUG at 14–21 days.

## Complications

- Recurrence 10–22%; de novo SUI 7–16%; urethrovaginal fistula 1–8%; stricture; recurrent UTI.

**Diagram to draw:** sagittal section of a mid-urethral sac and its ostium.

**Key points to remember:**

- Periurethral gland infection (Routh); posterolateral ostium.
- MRI most sensitive (97%).
- Preserve fascia; non-overlapping layers; Martius flap.
- No synthetic mesh at diverticulectomy.

## SAQ 6 Endoscopic procedure of SUI

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 13 Paper 1, Q3f

*Read as endoscopic injection of urethral bulking agents.*

### Introduction

- **SUI (stress urinary incontinence)** results from urethral hypermobility and **ISD (intrinsic sphincter deficiency)**.
- **Bulking agents**, injected submucosally under cystoscopic vision, are the least invasive surgical option.

### Mechanism

- Bulk at the proximal urethra improves **mucosal coaptation** and resistance.
- It does not correct hypermobility, but most SUI includes some ISD.

### Indications

- ISD with a **fixed, immobile urethra** (low VLPP [Valsalva leak point pressure] <60 $cmH_2O$, historically).
- Women wanting a **low-risk** procedure; elderly or unfit; future pregnancy.
- Persistent leakage after a sling; neurogenic SUI (safer for bladder compliance than a sling).
- **Avoid in:** active UTI (urinary tract infection), untreated detrusor overactivity.

### Agents

| Agent | Note |
|---|---|
| Polyacrylamide hydrogel (Bulkamid) | Non-particulate |
| Polydimethylsiloxane (Macroplastique) | Silicone particles |
| Calcium hydroxylapatite; carbon-coated zirconium beads | Particulate |
| Bovine collagen (skin test); PTFE (polytetrafluoroethylene); autologous fat | Withdrawn or abandoned: allergy, migration, granuloma, embolism |

### Technique

1. Sterile urine; local anaesthesia; cystoscope (0–30° lens) in the urethra.
2. **Transurethral:** needle into the submucosa about 1–1.5 cm distal to the bladder neck; or **periurethral:** needle alongside the urethra, watched cystoscopically.
3. Inject at 2–4 sites until the mucosal cushions **coapt in the midline**.
4. Avoid an indwelling catheter (it moulds the bulk); use CISC (clean intermittent self-catheterisation) for retention.

### Outcomes

- Early improvement is common, but cure is lower than with slings and **declines with time**; repeat injections are often needed.
- **Cochrane review (Kirchin 2017):** limited evidence of benefit.
- **AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction):** an option, with counselling on durability.
- **Men:** EAU (European Association of Urology): **do not offer** for post-prostatectomy incontinence.

### Complications

- Transient retention, UTI, dysuria, haematuria, urgency.
- Sterile abscess, migration, erosion; may mimic a urethral diverticulum on imaging.

**Diagram to draw:** cystoscopic view of coapted mucosal cushions after injection.

> **Key points to remember:**
> - Bulking improves coaptation; does not correct hypermobility.
> - Best for fixed urethra with ISD, or unfit patients.
> - Less durable than slings; repeat injections common.
> - Not recommended for post-prostatectomy incontinence.

## SAQ 7 Endometriosis in urology

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 22 SAQ Sheet, Q3d

### Definition and epidemiology

- Endometriosis is endometrial glands and stroma outside the uterus. Urinary tract involvement occurs in about 1–5% of women with endometriosis, mostly with deep disease.
- Sites: **bladder** (commonest, about 80%) > ureter (about 15%) > kidney, urethra (rare).

### Pathogenesis

- Retrograde menstruation (Sampson), coelomic metaplasia, lymphovascular spread, and **iatrogenic implantation** at pelvic surgery (e.g. caesarean section).
- Bladder lesions start on the **serosa** (vesicouterine pouch) and invade the detrusor.
- Ureteric disease: **extrinsic** (periureteric fibrosis, commoner) or intrinsic (muscularis or mucosa); usually distal, left more often.

### Clinical features

- **Bladder:** cyclical haematuria or frequency, urgency and dysuria with menses; suprapubic pain. Differential: BPS (bladder pain syndrome), vesicouterine fistula (menouria).
- **Ureter:** often **silent**; hydronephrosis and loss of renal function may go unnoticed.

### Evaluation

- Transvaginal and transabdominal ultrasound gauge size and depth; **renal ultrasound in every woman with deep endometriosis**.

- **MRI (magnetic resonance imaging):** extent; high T1 signal from blood products.
- **Cystoscopy:** limited value (disease is extravesical); may show a bluish submucosal lesion. Biopsy excludes carcinoma.
- CTU (computed tomography urography) and MAG3 (mercaptoacetyltriglycine) renogram for obstruction and split function.
- Staging: rASRM (revised American Society for Reproductive Medicine) I–IV; #Enzian records bladder and ureter.

### Management (with gynaecologist)

- **Medical:** progestins (dienogest), combined pill, GnRH (gonadotrophin-releasing hormone) analogues. Symptomatic relief; not for obstructive ureteric disease.
- **Bladder: partial cystectomy** (full-thickness excision with margin), laparoscopic or robotic; stent ureters if near. TUR (transurethral resection) is incomplete, with recurrence and perforation risk.
- **Ureter:**
- Extrinsic, mild: ureterolysis.
- Intrinsic or stricture: segmental resection with ureteroureterostomy, or **reimplantation with psoas hitch or Boari flap**.
- Non-functioning kidney: nephrectomy.
- Treat all pelvic disease at the same operation to lower recurrence.
- **Also know:** risk factor for VVF (vesicovaginal fistula) and ureteric injury at hysterectomy; rare malignant change (clear-cell, endometrioid).

**Diagram to draw:** endometriotic nodule in the vesicouterine pouch invading the detrusor.

> **Key points to remember:**
> - Bladder commonest; cyclical haematuria.
> - Ureteric disease silent: screen kidneys.
> - Partial cystectomy, not TUR.

## SAQ 8 Urethral caruncle.

P2

**Type:** SAQ (10 marks)
**Source:** 2016-Summer Paper 2, Q3f

### Definition

- A benign, red, fleshy, **inflammatory polypoid lesion of the posterior lip of the distal female urethra**. Not a true tumour, but a benign mimic.
- Common; seen mainly in **postmenopausal women**.

### Aetiology and pathogenesis

- **Oestrogen deficiency (GSM, genitourinary syndrome of menopause)**: atrophy and laxity of distal urethral mucosa.
- Related to **mucosal prolapse**: the everted posterior mucosa is chronically irritated, causing haemorrhage, necrosis and inflammatory growth.

## Pathology

- Classical types: **papillomatous, angiomatous and granulomatous**; inflamed vascular stroma.

## Clinical features

- Often asymptomatic, found on examination.
- **Spotting on underwear**, bleeding, pain, dysuria; may thrombose into a dark, tender mass.
- Examination: small, red, soft mass at the **6 o'clock** meatal margin; the rest of the meatus looks normal.

## Differential diagnosis

| Lesion | Distinguishing feature |
|---|---|
| **Urethral mucosal prolapse** | **Circumferential "doughnut"** around whole meatus; prepubertal girls and postmenopausal women |
| **Urethral carcinoma** | Firm, ulcerated, indurated, fixed; bleeding mass; biopsy |
| **Urethral diverticulum** | Mid or proximal anterior vaginal wall mass; dysuria, dyspareunia, dribbling; MRI (magnetic resonance imaging) |
| Skene gland cyst or abscess | Lateral to meatus; distorts it; does not communicate |
| Others | Melanoma, lymphoma, TB (tuberculosis), condyloma, intestinal metaplasia; postmenopausal bleeding from the uterus |
| - Caruncle is listed as a possible risk factor for female urethral cancer, so atypical lesions must be biopsied. | |

## Evaluation

- Clinical diagnosis; exclude a gynaecological bleeding source.
- **Excision biopsy** if atypical, indurated, ulcerated, enlarging or refractory; cystourethroscopy if in doubt.

## Treatment

- **Conservative (first line)**: **topical vaginal oestrogen** (shrinks most), anti-inflammatory creams, sitz baths. Ask about breast or endometrial cancer before oestrogen.
- **Surgical excision** for large, symptomatic, thrombosed, refractory or suspicious lesions: 1. Grasp the tip and apply traction to expose the base. 2. Excise at the base. 3. Reapproximate mucosa with absorbable sutures if the defect is large. 4. Foley catheter for a few days.
- Complications: bleeding, recurrence, meatal stenosis.

**Diagram to draw:** meatus with a 6 o'clock caruncle compared with circumferential prolapse.

> **Key points to remember:**
>
> - Postmenopausal posterior-lip lesion; oestrogen deficiency.
> - Prolapse is circumferential; caruncle is focal.
> - Topical oestrogen first; excise if atypical or refractory.

## SAQ 9 Female sexual dysfunction

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 1 Paper 3, Q3f

### Definition and prevalence

- FSD (female sexual dysfunction): persistent desire, arousal, orgasm or pain problems causing **personal distress**.
- About **40%** of women in their lifetime; **HSDD (hypoactive sexual desire disorder) commonest**. Almost half of women with UI (urinary incontinence) have FSD.

### Physiology and response models

- Arousal: pudendal afferents, pelvic parasympathetic efferents → **NO (nitric oxide)–cGMP (cyclic guanosine monophosphate)** → engorgement and lubrication. Oestrogen maintains tissues; testosterone drives desire.
- **Masters and Johnson** (linear): excitement, plateau, orgasm, resolution; Kaplan added desire. **Basson** (circular): desire is often responsive.

### Classification

| DSM-5 (Diagnostic and Statistical Manual, 2013) | ICSM (International Consultation on Sexual Medicine)/ISSWSH (International Society for the Study of Women's Sexual Health) |
|---|---|
| Female sexual interest/arousal disorder (≥3 of 6 criteria) | HSDD and arousal disorder kept separate (cognitive vs genital arousal) |
| Female orgasmic disorder | Orgasm disorders |
| Genito-pelvic pain/penetration disorder | Genital-pelvic pain dysfunction |
| – | PGAD (persistent genital arousal disorder) |
| - ≥6 months with distress; lifelong or acquired; generalised or situational. | |

### Causes

- Hormonal (menopause, GSM [genitourinary syndrome of menopause]), drugs (SSRIs [selective serotonin reuptake inhibitors], antihypertensives), diabetes, depression, relationship factors.
- Urological: incontinence, prolapse, sling dyspareunia, cystectomy nerve or clitoral injury.

### Evaluation

- Clinician must ask. **PLISSIT** (permission, limited information, specific suggestions, intensive therapy); BSSC-W (Brief Sexual Symptom Checklist for Women) screen; **FSFI (Female Sexual Function Index, 19 items, six domains)**.
- Examination: atrophy, lichen sclerosus, cotton-swab test, pelvic floor tone. Laboratory tests rarely help.

### Management

- Treat causes; education; sex therapy (sensate focus, mindfulness).

- **HSDD (premenopausal, FDA [US Food and Drug Administration] approved): flibanserin** 100 mg at bedtime (5-HT1A agonist/5-HT2A antagonist; alcohol caution); **bremelanotide** 1.75 mg subcutaneously on demand (melanocortin agonist). Postmenopausal: transdermal testosterone (off-label).
- **Arousal:** testosterone; PDE5 (phosphodiesterase type 5) inhibitors only with neurovascular disease.
- **Orgasm:** education, directed masturbation.
- **Pain/vestibulodynia:** pelvic floor physiotherapy, dilators, biofeedback, topical lidocaine, vaginal oestrogen; vestibulectomy if refractory.
- **Also know:** ospemifene, intravaginal prasterone (menopausal dyspareunia).

**Diagram to draw:** Basson circular sexual response model.

> **Key points to remember:**
> - About 40% prevalence; HSDD commonest; distress required.
> - DSM-5 merges desire and arousal, dyspareunia and vaginismus.
> - Flibanserin and bremelanotide: only FDA-approved HSDD drugs (premenopausal).

## SAQ 10 Duloxetin

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 4 Paper 3, Q3e

*Read as: duloxetine, mainly its role in stress urinary incontinence.*

### Introduction

- **Duloxetine** is an SNRI (serotonin–noradrenaline reuptake inhibitor), also an antidepressant.
- Licensed in Europe (not USA) for **moderate to severe SUI (stress urinary incontinence) in women**; off-label in men.

### Mechanism

- During storage, bladder afferents drive glutamate excitation of pudendal motor neurons in **Onuf's nucleus (S2–S4)**, which keep the striated urethral sphincter closed.
- Serotonin (5-HT2) and noradrenaline (α1) amplify this glutamate drive. Duloxetine blocks their reuptake, so **sphincter tone and the sneeze-induced continence reflex increase**.
- In voiding, glutamate drive switches off, so retention does not occur.
- Central effects also reduce bladder activity.

### Dose

- **40 mg twice daily**; starting at 20 mg twice daily for 2 weeks reduces nausea. Taper on stopping.

### Evidence

- Phase III RCTs (randomised controlled trials) in women: about 50% median fall in incontinence episodes versus about 30–40% with placebo; better quality of life; **cure rare**.
- Benefit is temporary and lasts only while taken.
- Male PPI (post-prostatectomy incontinence): small, short-term benefit.

### Guideline position

- **EAU (European Association of Urology):** may be offered to women for **temporary** improvement, not cure; warn about side effects. Men: offer for SUI (weak), and **inform of off-label use and adverse effects** (strong).
- **NICE (National Institute for Health and Care Excellence):** not first line; not routinely second line; only if surgery is unwanted or unsuitable.
- Adding PFMT (pelvic floor muscle training): evidence conflicting.

### Adverse effects

- **Nausea** (commonest, early), dry mouth, dizziness, insomnia, constipation; **high discontinuation**.
- Raised blood pressure, hyponatraemia (elderly), suicidal ideation, discontinuation syndrome.

### Contraindications and interactions

- MAOIs (monoamine oxidase inhibitors), including **methylene blue** (serotonin toxicity); CYP1A2 inhibitors (fluvoxamine, ciprofloxacin); hepatic impairment; severe renal impairment; uncontrolled hypertension or narrow-angle glaucoma.

**Diagram to draw:** Onuf's nucleus with 5-HT and noradrenaline terminals driving the pudendal nerve.

> **Key points to remember:**
> - SNRI acting on Onuf's nucleus to increase sphincter tone.
> - 40 mg twice daily; titrate from 20 mg to reduce nausea.
> - Temporary benefit only; approved in Europe, not USA.

## SAQ 11 Acute urinary retention in females.

P4

**Type:** SAQ (10 marks)
**Source:** 2017-Summer Paper 4, Q3f

### Introduction

- AUR (acute urinary retention) is sudden inability to void. It is uncommon in women and is often non-obstructive, so search for the cause.

### Causes

| Group | Examples |
|---|---|
| Obstructive | Large cystocele or prolapse; tight sling after incontinence surgery; urethral diverticulum, stricture; pelvic mass, **retroverted gravid uterus**, haematocolpos; cancer |
| Functional | **Fowler syndrome**; dysfunctional voiding; primary bladder neck obstruction (rare) |
| Neurogenic | MS (multiple sclerosis; may present first with retention), cauda equina, transverse myelitis, diabetes |
| Inflammatory | Cystitis, vulvovaginitis, **genital herpes**, Bartholin abscess |
| Drugs | Anticholinergics, opioids, tricyclics, calcium-channel blockers |

| Group | Examples |
|---|---|
| Postoperative/postpartum; psychogenic | Epidural, pain; diagnosis of exclusion |

## Fowler syndrome

- Young women (20–35) with painless retention and bladder capacity over 1 litre.
- **Urethral sphincter EMG (electromyography):** complex repetitive discharges, so the sphincter fails to relax. Urethral closure pressure >100 cm $H_2O$.

## Evaluation

- **History:** drugs, surgery, delivery, neurological and bowel symptoms, genital lesions.
- **Examination:** pelvic (prolapse, mass); neurological (perineal sensation, anal tone).
- **Tests:** drained volume; urine culture; creatinine; pregnancy test; ultrasound.
- MRI (magnetic resonance imaging) spine if neurological signs; cystoscopy.
- **Later:** video-urodynamics (female obstruction, e.g. Qmax ≤12 mL/s with Pdet ≥20 cm $H_2O$); sphincter EMG.

## Management

- **Immediate:** urethral catheter; suprapubic if urethral access fails.
- **Treat the cause:** stop drugs; treat infection or herpes; pessary or prolapse repair; reposition the uterus; sling division; diverticulectomy.
- **TWOC (trial without catheter)** after correction.
- **Persistent retention:**
- **CISC (clean intermittent self-catheterisation)** rather than an indwelling catheter.
- **SNM (sacral neuromodulation):** about 70% success in Fowler syndrome.
- Bethanechol not recommended. Bladder neck incision only with caution (incontinence).

**Diagram to draw:** flowchart: AUR → catheterise → find cause → treat → TWOC → CISC or SNM.

> **Key points to remember:**
> - Look for prolapse, sling obstruction, pelvic mass, neurology and drugs.
> - Fowler: abnormal sphincter EMG; SNM about 70%.
> - CISC for persistent retention.

# Rapid Revision Sheet · Section 6

### Female pelvic floor anatomy and pelvic organ prolapse: evaluation and surgical repair (LAQ)

- Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–S5.
- DeLancey: I suspension (uterosacral–cardinal; failure causes uterine prolapse), II attachment (ATFP), III fusion (urethral hypermobility); level I is the keystone.
- Always restore apical support.
- Enhorning 1961, DeLancey 1994, Petros & Ulmsten 1990, McGuire ISD 1978.
- Integral theory: mid-urethral sling; hammock: backing, not repositioning.
- Burch: Cooper's ligament; TOT: obturator foramen; PVS: bladder neck.
- Bulge is most specific; splinting means high grade.
- Central cystocele rugae lost, paravaginal preserved; POP-Q stage II = −1 to +1 cm (±1 cm of hymen).
- Sacrocolpopexy most durable; colpocleisis for frail elderly.
- Cystoscopy after repair; screen for occult SUI.
- Pessary first; sacrospinous hysteropexy 1.5–3 cm medial to the ischial spine.
- Manchester amputates the cervix: poor choice for future pregnancy.

### Vesicovaginal fistula: management, timing of repair, eponymous techniques and recurrent VVF (LAQ)

- Image upper tracts; exclude ureteric fistula and cancer (biopsy before redo).
- Catheter trial only if small, fresh (<3 weeks), non-irradiated; works only if started within 3 weeks.
- Vaginal and abdominal routes both close about 90%.
- Early repair for clean iatrogenic VVF; closure comparable to delayed.
- Delay ~3 months for obstetric, ≥12 months for radiation fistula.
- First repair has the highest success.
- Interposition flap (Martius, peritoneum, omentum) for complex and recurrent VVF.
- Diversion after multiple failures or severe radiation.
- Latzko: colpocleisis, bladder not opened, vagina shortened.
- O'Conor: bivalve bladder to the fistula, interpose omentum.
- Martius for distal, omentum or peritoneum for apical VVF.

### Urological complications of obstetric-gynaecological surgery: uretero- and urethrovaginal fistula (LAQ)

- Bladder is the commonest organ injured; obstructed labour causes most fistulae worldwide.
- Ureteric injury is mostly distal and missed.
- UVF: distal ureteric injury at hysterectomy; continuous leak within 4 weeks with normal voiding.
- Orange tampon = UVF on double dye test; exclude coexisting VVF.
- Stent early (within 2 weeks); remove at 4–6 weeks.
- Reimplant with psoas hitch or Boari flap if stenting fails; success ≥90%; follow 30 months.
- Urethrovaginal fistula: diverticulectomy is the commonest Western cause; obstetric in developing countries.

- Look for a coexisting VVF (about 20%).
- Layered closure plus Martius flap; autologous sling only.
- Closure 90–95% (closure is easy); continence only 50–80% (continence is hard).

**Female SUI: continence mechanism, classification, evaluation (UDS) and management, including male SUI surgery (LAQ)**

- Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–5.
- DeLancey: I suspension, II attachment (ATFP), III fusion.
- Continence zone is the mid-urethra; submucosal plexus gives up to 30% of closure.
- Blaivas types 0, I, IIA, IIB, III; type III = ISD.
- Q-tip >30° = hypermobility; VLPP / ALPP <60 $cmH_2O$ = ISD; MUCP ≤20 $cmH_2O$.
- Burch corrects position; slings and AUS correct function.
- VALUE (ValUE) trial: UDS not needed in uncomplicated (index) SUI.
- Use UDS when it changes treatment: mixed, failed surgery, voiding dysfunction.
- PFMT first; MUS first-line surgery.
- SISTEr (fascial sling > Burch); TOMUS (retropubic = transobturator).
- PVS for ISD; urethral injury → no mesh.
- Men: AUS gold standard; sling for mild, non-irradiated.
- AUS PRB 61–70 $cmH_2O$; deactivate 4–6 weeks.
- Radiation or severe male SUI: AUS first; MASTER: sling non-inferior.

**Recto-urethral (recto-prostatic) fistula: etiopathogenesis, diagnosis and management (LAQ)**

- Most acquired RUFs follow prostate cancer treatment.
- Martini: stage = size; grade = tissue (surgery, radiation, salvage).
- VCUG or RUG plus endoscopic biopsy to exclude cancer.
- Divert first; perineal (gracilis) or York-Mason for surgical fistulae; irradiated ones need imported tissue or diversion.

**Leak point pressures and ALPP (SAQ)**

- VLPP/ALPP: Pves at leak with Valsalva or strain, no detrusor contraction; start at 150 mL.
- ALPP low = weak sphincter; DLPP high = upper-tract danger.
- VLPP <60 ISD; 60–90 equivocal; ≥90 (>90) hypermobility.
- VLPP < CLPP; catheter size and volume alter it.
- DLPP >40: upper-tract danger; 85% upper-tract change (McGuire 1981).
- Lower DLPP is safer; no validated safe cut-off.

**Mid-urethral slings: TOT and TVT (SAQ)**

- Integral theory; loose tape at mid-urethra.
- Type 1 macroporous monofilament polypropylene.
- Delorme 2001; TOT anchors in obturator muscles; fewer bladder injuries than retropubic.
- TOMUS: cure equivalent at 1 year (80.8% retropubic vs 77.7% transobturator).
- TOT: less bladder injury (0.6% vs 4.5%), more groin pain (6.4% vs 1.3%); worse in ISD.
- Cystoscopy after each TVT trocar pass; bladder perforation 4.5%.
- Prefer retropubic TVT in ISD or fixed urethra.

**Burch colposuspension (SAQ)**

- Burch 1961: vaginal wall to Cooper's ligament; for hypermobility.
- Cure 85–90% at 1 year, ~70% at 5 years.
- Niche: concomitant abdominal surgery, mesh avoidance.

**Management of perforated mid urethral sling (SAQ)**

- Bladder trocar injury: repass laterally, continue.
- Urethral injury: abandon mesh; stage the repair.
- Late perforation: excise mesh, repair, consider autologous sling.

**Female urethral diverticula (SAQ)**

- Periurethral gland infection (Routh); posterolateral ostium.
- MRI most sensitive (97%).
- Preserve fascia; non-overlapping layers; Martius flap.
- No synthetic mesh at diverticulectomy.

**Endoscopic procedure of SUI (SAQ)**

- Bulking improves coaptation; does not correct hypermobility.
- Best for fixed urethra with ISD, or unfit patients.
- Less durable than slings; repeat injections common.
- Not recommended for post-prostatectomy incontinence.

**Endometriosis in urology (SAQ)**

- Bladder commonest; cyclical haematuria.
- Ureteric disease silent: screen kidneys.
- Partial cystectomy, not TUR. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Urethral caruncle. (SAQ)**

- Postmenopausal posterior-lip lesion; oestrogen deficiency.
- Prolapse is circumferential; caruncle is focal.
- Topical oestrogen first; excise if atypical or refractory.

**Female sexual dysfunction (SAQ)**

- About 40% prevalence; HSDD commonest; distress required.
- DSM-5 merges desire and arousal, dyspareunia and vaginismus.
- Flibanserin and bremelanotide: only FDA-approved HSDD drugs (premenopausal).

**Duloxetin (SAQ)**

- SNRI acting on Onuf's nucleus to increase sphincter tone.
- 40 mg twice daily; titrate from 20 mg to reduce nausea.
- Temporary benefit only; approved in Europe, not USA. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Acute urinary retention in females. (SAQ)**

- Look for prolapse, sling obstruction, pelvic mass, neurology and drugs.
- Fowler: abnormal sphincter EMG; SNM about 70%.
- CISC for persistent retention. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

# Section 7 · Voiding Dysfunction

## Long answer questions (LAQ)

### LAQ 1 Overactive bladder: pathophysiology, pharmacotherapy and non-pharmacological management

P1 · P2 · P3 **merged from 4**

*Asked as:* P1 LAQ "Describe pathophysiology of OAB. Briefly outline management &describe nonmedical non-pharmacological management" · P2 SAQ "Overactive bladder and its drug therapy" · P3 SAQ "OAB Pharmacotherapy: Darifenacin vs Solifenacin vs Tolterodine, and Mirabegron" · P1 SAQ "Compare and contrast various detrusor relaxants in management of OAB"

#### Definition

- **OAB (overactive bladder), ICS (International Continence Society): urgency**, usually with frequency and nocturia, with or without UUI (urgency urinary incontinence), in the absence of UTI (urinary tract infection) or other obvious pathology.
- **OAB dry** versus **OAB wet** (with UUI).
- OAB is a **symptom** diagnosis. DO (detrusor overactivity) is a **urodynamic** finding; only 44–69% of patients with urgency show DO.

#### Pathophysiology

All mechanisms converge on **more afferent signal reaching awareness**: the bladder turns the signal up, or the CNS (central nervous system) fails to turn it down.

| Mechanism | Explanation |
|---|---|
| **Afferent (sensory)** | Aδ fibres normally sense filling. Unmyelinated **C fibres** are "silent", but become spontaneously active and hypersensitive in pathological states |
| **Urothelial** | Urothelium acts as a transducer: mechanical, inflammatory or chemical stimuli release ATP (adenosine triphosphate), acetylcholine and other mediators onto suburothelial afferents |
| **Myogenic** | Patchy denervation causes receptor upregulation and spontaneous contraction; exaggerated **micromotions** cause urgency, and coordinated spread across the wall gives DO |
| **Neurogenic / central** | Forebrain and PAG–PMC (periaqueductal grey–pontine micturition centre) normally inhibit voiding; loss of this inhibition lets "afferent noise" be felt as urgency |
| **Other** | Bladder–bowel crosstalk (constipation), ischaemia–reperfusion, chronic inflammation, altered urinary microbiome |

- **Risk factors and associations:** ageing, female sex, BOO (bladder outlet obstruction: an animal model, but only an indirect cause in humans; DO persists in up to 60% after TURP [transurethral resection of the prostate]), anxiety and depression, metabolic syndrome, poor toileting habits.

#### Evaluation

- **Minimum:** history, examination, **urinalysis**.

- Bladder diary (3–7 days; P2: 3 days), questionnaires, PVR (post-void residual) when indicated.
- Urodynamics and cystoscopy only for complicated or refractory cases, or if urodynamics will change an invasive decision.

### Outline of management (AUA/SUFU [American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction]; not a compulsory sequence)

`Education + behavioural therapy (first line) ↓ (4–12 weeks) Antimuscarinic or β3-agonist (mirabegron, vibegron); combinations ↓ PTNS / onabotulinumtoxinA / SNM (third line) ↓ Augmentation cystoplasty or urinary diversion`

| Line | Treatment |
|---|---|
| Behavioural | Education; reduce fluids by about 25%; cut caffeine; weight loss; **bladder training** (timed voids, interval increased by 15–30 minutes); **PFMT (pelvic floor muscle training)** with urge suppression: 50–80% fewer leaks |
| Drugs | **Antimuscarinics** (oxybutynin, solifenacin, extended-release trospium, darifenacin, fesoterodine, tolterodine); **β3-agonists** (mirabegron, vibegron); combinations |
| Third line | **PTNS (percutaneous tibial nerve stimulation)** weekly × 12; **onabotulinumtoxinA 100 U** intradetrusor; **SNM (sacral neuromodulation)** at S3 (success about 70–80%; revision >30% at 5 years) |
| Last resort | Augmentation cystoplasty or urinary diversion |

### Non-pharmacological management

#### 1. Education and lifestyle

- Explain normal bladder function; set realistic goals.
- Reduce fluid intake by about 25% if excessive; cut caffeine; treat constipation.
- **Weight loss:** about 8 kg loss in obese women reduced UUI episodes.

#### 2. Bladder training (BT) (breaks the cycle of urgency, frequent voiding and smaller capacity)

1. Keep a bladder diary.
2. Start a fixed interval (often 1 hour while awake).
3. Void by the clock, not by urge.
4. Increase by 15–30 minutes a week as control improves.
5. Target interval 3–4 hours. - Works by **uncoupling the void from the urge**.

#### 3. PFMT and urge suppression

- Kegel exercises: quick 1–2 s flicks and 5–10 s holds, twice daily.
- A pelvic floor contraction **reflexly inhibits the detrusor**. Urge suppression: stop, stay still, squeeze quickly several times, relax, let the urge pass, then walk calmly to the toilet.
- PFMT and BT reduce incontinence episodes by **50–80%**.
- **Also know:** biofeedback helps patients who cannot find the muscle; electrical stimulation at 5–20 Hz inhibits the detrusor; timed and prompted voiding for cognitively impaired patients.

#### 4. PTNS

- 34-gauge needle 3–4 cm above the medial malleolus; 20 Hz, 200 µs; confirm great-toe flexion or sole tingling. **30 min weekly for 12 weeks**, then maintenance.

- The posterior tibial nerve (L4–S3) shares sacral roots with the bladder. **SUmiT trial:** 55% vs 21% (sham) responders.

**5. SNM**

- Mechanism: stimulates **afferents** that modulate brainstem and forebrain control.
- **Test phase:** PNE (percutaneous nerve evaluation, 5–7 days) or **staged tined lead** at **S3** (anal "bellows" and great-toe flexion). ≥50% improvement on diary leads to implanting the pulse generator in the buttock.
- Success about 70–80%. **InSite trial:** 61% vs 42% (standard medical therapy). Revision >30% at 5 years; MRI (magnetic resonance imaging)-conditional rechargeable devices now available.

**6. Intradetrusor onabotulinumtoxinA**

- Cleaves SNAP-25 (synaptosomal-associated protein 25), blocking acetylcholine release; also reduces afferent signalling.
- **100 U in 10 mL at 20 sites** cystoscopically; effect lasts about 6–9 months.
- About 25% become completely dry. Risks: UTI about 20%, retention needing CISC (clean intermittent self-catheterisation). **ROSETTA:** greater UUI reduction than SNM at 6 months, but more UTI.

**7. Last resort:** augmentation cystoplasty or urinary diversion.

## Drug therapy: place and mechanisms

- Drugs are second line, after behavioural therapy, as first-line pharmacotherapy (antimuscarinics and β3 agonists alongside each other; AUA/SUFU). Two main classes: **antimuscarinics** and **β3-adrenoceptor agonists**.
- **Antimuscarinics:** acetylcholine acts on **M3** receptors (raising intracellular calcium) to contract the detrusor; the commoner M2 enhances it. Blockade reduces involuntary contractions and afferent urgency signalling.
- Muscarinic side effects by subtype: **M1** (brain) cognition; **M2** heart rate; **M3** (glands, gut) dry mouth, constipation.
- Selectivity is by **receptor subtype** (M3) or **tissue** (bladder over salivary gland); tissue selectivity matters more.
- **β3 agonists:** β3 is the commonest (most highly expressed) adrenoceptor in human detrusor. Activation raises **cAMP (cyclic adenosine monophosphate)** and relaxes the detrusor during storage, sparing the voiding contraction.

## Agents

| Drug | Distinguishing feature | Usual dose |
|---|---|---|
| Oxybutynin | Non-selective; direct relaxant; metabolite causes most dry mouth; patch, gel | 5 mg two to three times daily; ER (extended-release) |
| Tolterodine | Bladder (over salivary) selective; QT prolongation at high dose | 2 mg twice daily (IR, immediate-release); ER 4 mg daily |
| Fesoterodine | Prodrug of tolterodine's active metabolite | 4–8 mg daily |
| Solifenacin | 10-fold M3 > M2; long half-life | 5–10 mg daily |
| Darifenacin | About 30-fold M3-selective | 7.5–15 mg daily |

| Drug | Distinguishing feature | Usual dose |
|---|---|---|
| Trospium | Quaternary; does not cross the blood–brain barrier (little brain entry) | 20 mg twice daily |
| Mirabegron (β3) | Inhibits CYP2D6 (a liver enzyme): metoprolol interaction; small blood-pressure rise | 25–50 mg daily |
| Vibegron (β3) | No CYP2D6 interaction | 75 mg daily |

## Darifenacin vs solifenacin vs tolterodine

| Feature | Darifenacin | Solifenacin | Tolterodine |
|---|---|---|---|
| Selectivity | **Most M3-selective (about 30-fold)**; not tissue-selective | M3 about 10-fold over M2; bladder-selective | Non-subtype-selective; **bladder-selective** |
| Dose | 7.5–15 mg daily (ER) | 5–10 mg daily | 2 mg twice daily (IR); 4 mg daily (ER) |
| Half-life | Moderate (ER once daily) | **Long** (about 2–3 days): once daily | Short; active metabolite 5-HMT (5-hydroxymethyl tolterodine) |
| Metabolism | CYP3A4 (a liver enzyme) and CYP2D6 | CYP3A4 | CYP2D6 (and 3A4) |
| Dry mouth | Dose-related | Dose-related | Lower with ER |
| Constipation | **Highest** (gut M3) | Moderate–high | Lower |
| CNS | Spares M1: **preferred in elderly** | Better persistence | High dose: QT prolongation |
| Dose limit | Lower dose with CYP3A4 inhibitors or liver disease | 5 mg in severe renal or moderate liver impairment | Halve with renal or liver impairment |

- **Efficacy is similar**; response is individual, so switch agent or dose.
- **STAR trial:** flexible solifenacin beat tolterodine ER for urgency and incontinence.

## Mirabegron (β3 agonist)

- First **β3-adrenoceptor agonist** for OAB; FDA (Food and Drug Administration) approval 2012.
- **Mechanism:** activation of adenylyl cyclase raises **cAMP**, relaxing the detrusor during **storage**. Reduces non-voiding contractions and afferent firing. The voiding contraction (muscarinic M3) is spared, so PVR barely rises.
- **Dose:** oral, extended release, once daily: start **25 mg**, increase to **50 mg** after 4–8 weeks. **25 mg maximum** in severe renal impairment (eGFR [estimated glomerular filtration rate] 15–29) or moderate hepatic impairment; avoid in end-stage renal disease and severe hepatic impairment.
- **Moderate CYP2D6 inhibitor:** raises levels of metoprolol, desipramine, flecainide; monitor digoxin.
- **Efficacy:** phase III trials (SCORPIO, ARIES, CAPRICORN): fewer incontinence episodes and micturitions than placebo. **TAURUS** (12 months): efficacy similar to tolterodine, with less dry mouth.
- **Adverse effects:** hypertension, tachycardia, headache, nasopharyngitis, UTI. **Dry mouth and constipation near placebo rates**, far below antimuscarinics.

- **Contraindicated:** severe uncontrolled hypertension (≥180/110 mmHg). Caution with arrhythmia, and with BOO when combined with antimuscarinics (retention).
- Other uses: add-on to α-blocker in men with BPH (benign prostatic hyperplasia); neurogenic detrusor overactivity.
- **Also know:** vibegron 75 mg (no CYP2D6 interaction).

## Antimuscarinics versus β3 agonists

| Feature | Antimuscarinics | β3 agonists |
|---|---|---|
| Efficacy | Similar | Similar (TAURUS: mirabegron ≈ tolterodine) |
| Dry mouth, constipation, blurred vision | Common; main reason for stopping | Near placebo |
| Cognition | Central M1 blockade; **dementia risk 10–65% higher with use >3 months** | No anticholinergic burden |
| Retention | Up to 3% in men | Low |
| Contraindications | Urinary or gastric retention, uncontrolled narrow-angle glaucoma | Uncontrolled (severe) hypertension (≥180/110 mmHg) |

## Choice of drug and special groups

- **Elderly and cognitive impairment:** antimuscarinic use >3 months is linked to 10–65% more new-onset dementia (SUFU white paper). Prefer a **β3 agonist** (no anticholinergic burden), or ER trospium, darifenacin or fesoterodine. Beers criteria: avoid antimuscarinics in dementia.
- ER forms cause less dry mouth. Poor persistence is the main problem; switch dose or drug before third-line therapy.
- **Combination:** solifenacin plus mirabegron beats each alone (**SYNERGY, SYNERGY II**), without extra cardiovascular effects; in non-responders it equals solifenacin 10 mg with less dry mouth (**BESIDE**).
- **Men:** α-blocker first; add an antimuscarinic (if PVR is low) or β3 agonist; retention up to 3%, so check PVR.
- **Also know:** vaginal oestrogen in atrophy; desmopressin for nocturnal polyuria; imipramine (children, mixed incontinence); intravesical oxybutynin; refractory OAB: onabotulinumtoxinA 100 U and neuromodulation.

## Diagrams to draw

- Bladder showing urothelial afferents, C fibres and micromotions, with brainstem and forebrain (cortical) inhibitory pathways.
- Detrusor cell: acetylcholine → M3 → contraction; noradrenaline → β3 → cAMP → relaxation.
- Muscarinic subtypes M1 (brain), M2 (heart), M3 (bladder, gland, gut) with each drug's selectivity.

**Key points to remember:**

- OAB is defined by urgency; DO is a urodynamic finding.
- Final pathway: increased afferent signal (urothelial, myogenic) or failed central inhibition.
- Behavioural therapy first: BT and PFMT with urge suppression give 50–80% fewer leaks; antimuscarinic or β3-agonist next.
- M3 mediates contraction; β3 raises cAMP for storage-phase relaxation; voiding spared.
- Equal efficacy; β3 avoids dry mouth and dementia risk (10–65% with antimuscarinics >3 months).
- Darifenacin most M3-selective; tolterodine bladder-selective; solifenacin longest half-life.
- STAR: solifenacin beat tolterodine ER.
- Mirabegron 25 mg → 50 mg; avoid if blood pressure ≥180/110; CYP2D6 interaction (metoprolol).
- Elderly: mirabegron (β3), trospium or darifenacin preferred over other antimuscarinics.
- Solifenacin + mirabegron combination works (SYNERGY).
- Refractory OAB: PTNS (SUmiT 55% vs 21%), SNM (InSite 61% vs 42%), Botox 100 U.

## LAQ 2 Neuromodulation for voiding dysfunction, including sacral neuromodulation

P2 · P3 · P4 **merged from 3**

*Asked as:* P4 LAQ "Neuromodulation for voiding dysfunction, including sacral neuromodulation" · P2 SAQ "Neuromodulation" · P3 SAQ "Sacral neuromodulation"

*Taken as neuromodulation for lower urinary tract dysfunction.*

### Definition and history

- **Neuromodulation:** electrical stimulation of a peripheral nerve that alters the activity of central (reflex) pathways controlling the bladder. It acts on the nervous system (**afferents**), not directly on muscle.
- Direct muscle stimulation failed (painful, local). Stimulating **afferents** needs less current and affects the whole organ.
- **SNM (sacral neuromodulation)** (Tanagho and Schmidt, about 1985): continuous low-amplitude **S3** stimulation through a tined lead and IPG (implantable pulse generator). PTNS (percutaneous tibial nerve stimulation): FDA (Food and Drug Administration) approval 2000.
- **Rule:** exclude anatomical causes first (obstruction, stones, cancer, fistula). Stimulation cannot correct anatomy or restore muscle.

### Mechanism

- Sacral roots are mixed nerves. Current excites **afferents to the spinal cord and forebrain** (areas of awareness and attention), altering PMC (pontine micturition centre) output. It acts on **afferent** nerves, not motor nerves.
- **OAB (overactive bladder):** heightened central control (afferent input) damps PMC firing, so there are fewer involuntary contractions.
- **Non-obstructive retention:** restores awareness of filling (sensation) and pelvic floor tone, so the PMC switches on and the **EUS (external urethral sphincter) relaxes**. Hence one therapy treats **both** opposite disorders.
- **Tibial:** the nerve carries **S2–S3** fibres; inhibition of OAB-type (C-fibre) reflexes is **opioid-mediated** (naloxone blocks it); long-term cortical plasticity.

### Modalities

| Modality | Target and method | Main use |
|---|---|---|
| **SNM** | Tined lead in **S3** foramen + implanted generator | Refractory OAB, NOUR (non-obstructive urinary retention), faecal incontinence |
| **PTNS** | 34-gauge needle above medial malleolus | Refractory OAB |
| TTNS (transcutaneous tibial) | Surface patch, home use | OAB (less effective) |
| Implantable tibial | Small subcutaneous device over tibial nerve | OAB, avoids clinic visits |
| Pudendal | Tined lead in Alcock's canal | IDO (idiopathic detrusor overactivity), NDO (neurogenic DO), SNM failure (off-label) |

| Modality | Target and method | Main use |
|---|---|---|
| DGNS (dorsal genital nerve stimulation) | Dorsal penile/clitoral nerve | Experimental; NDO |
| IVES (intravesical electrical stimulation) | Intravesical cathode | Underactive bladder in children |
| **SARS (sacral anterior root stimulator)** | Anterior S2–S4 roots + dorsal rhizotomy | Complete SCI (spinal cord injury) |

## A. Sacral neuromodulation: indications

- **Third line** for OAB after behavioural and drug therapy (AUA [American Urological Association] and EAU [European Association of Urology] agree).

| FDA approved | Not approved |
|---|---|
| Refractory UUI (urgency urinary incontinence) 1997 | Bladder pain syndrome |
| Urgency–frequency 1999 | Neurogenic bladder (small studies positive; low-level evidence) |
| **Non-obstructive urinary retention** (e.g. Fowler's syndrome) 1999 | Pelvic pain |
| Faecal incontinence 2011 | |

- **Fowler's syndrome**: young women in retention with abnormal sphincter EMG (electromyography) activity; responds well to SNM.
- Retention needs some detrusor function; success falls with age.

### SNM: technique

**Test phase** (with a bladder diary)

| Test phase | PNE (percutaneous nerve evaluation) | Staged tined lead |
|---|---|---|
| Lead | Temporary, office, local anaesthesia (dislodges) | Permanent tined lead, theatre, fluoroscopy |
| Duration | 5–7 days | Up to 4 weeks |
| Use | OAB | More reliable; higher conversion; preferred for retention (NOUR) |

- **Landmark:** prone; needle into S3 about **9 cm above the coccyx tip, 1–2 cm lateral**.

| Root | Pelvic response | Foot response |
|---|---|---|
| S2 | — | Calf and foot (external) rotation |
| **S3 (target)** | **Bellows**, perineal sensation | **Great-toe flexion** |
| S4 | Bellows/perineal only | None |

- **Implant if ≥50% improvement** in a diary parameter (retention: ≥50% fewer catheterisations); IPG in the buttock.
- **Also know:** 10–16 Hz at sensory threshold; bilateral stimulation no better.

### SNM: outcomes

- **InSite:** success 61% vs 42% with standard medical therapy at 6 months; about 70–80% success at 5 years (another source: about half keep benefit at ≥5 years).
- **ROSETTA:** onabotulinumtoxinA 200 U reduced UUI slightly more (−3.9 vs −3.3 episodes/day) with higher satisfaction, but caused more UTI (urinary tract infection) (24% vs 10%) and CIC (clean intermittent catheterisation, self-catheterisation); SNM removal 9% at 2 years.
- NOUR: most stop catheterising.

### SNM: complications and contraindications

- Pain at the IPG site, lead migration, loss of effect, infection; **revision over 30% at 5 years**; costly (costliest option); battery replacement.
- Contraindications: failed test, inability to use the programmer, sacral anomaly.

### B. Tibial nerve stimulation

- **PTNS technique:** 34-gauge needle **3–4 cm (three fingerbreadths) above the medial malleolus**, behind the tibia; ground pad on the same leg; 20 Hz, 0–10 mA. Confirm by **toe flexion/fanning** or sole tingling. **30 min weekly × 12 weeks**, then maintenance.
- **SUmiT** (n = 220, validated sham): response **55% vs 21%**. Extension: 77% relapse-free at 3 years with tapering maintenance.
- **OrBIT:** PTNS comparable to extended-release tolterodine, with fewer side effects.
- **TTNS:** home self-treatment; skin impedance limits amplitude, so less effective.
- **Implantable devices** remove clinic visits: **BlueWind** (miniature, external control; benefit sustained at 3 years); **eCoin** (coin-sized, battery-powered, office insertion under local anaesthesia).

### C. Other targets

- **Pudendal nerve stimulation:** tined lead via a posterior approach to Alcock's canal. Better than SNM in a blinded crossover; **salvage after failed SNM**. Off-label.
- **DGNS:** purely sensory; acute DO suppression in SCI; **conditional (on-demand)** stimulation possible. Electrode fixation is difficult.
- **IVES (Katona):** intravesical cathode in saline activates Aδ afferents, producing detrusor contraction. Needs some afferent and muscle function; daily sessions; used in children; the sham-controlled trial was negative.
- **Brindley SARS:** stimulates anterior S2–S4 roots for voiding, defecation and erection. **Dorsal rhizotomy** abolishes reflex incontinence and DSD (detrusor–sphincter dyssynergia). The sphincter relaxes faster than the detrusor between bursts, giving **post-stimulus voiding**. Needs intact efferents.
- **Also know:** intravaginal stimulation (SUI, stress urinary incontinence); sacral dermatome stimulation; closed-loop SNM.

### Comparison of third-line OAB options

| | PTNS | OnabotulinumtoxinA 100 U | SNM |
|---|---|---|---|
| Invasiveness | Lowest | Moderate | Implant |
| Drawback | 12 weekly visits | UTI, retention/CIC | Revision, cost |
| Retention | No role | Worsens | **Treats** |

## Recent advances

- Rechargeable stimulators (FDA 2019; ≥15 years); non-rechargeable devices lasting ≥10 years (2021).
- **Full-body (whole-body) MRI (magnetic resonance imaging)-conditional** systems at 1.5 T and 3 T.
- **Implantable tibial nerve stimulators** (BlueWind, eCoin) and transcutaneous tibial stimulation as less invasive alternatives; PTNS and pudendal stimulation as alternatives to SNM.
- Standardised tined-lead placement improves outcomes and lowers revision.
- Bilateral stimulation no better.

## Diagram to draw

- Lead in the S3 foramen with IPG in the buttock, and PTNS needle above the medial malleolus.

> **Key points to remember:**
>
> - Afferent modulation explains SNM benefit in both OAB and retention.
> - S3: bellows plus great-toe flexion; implant if the test improves ≥50%.
> - SNM revision >30% at 5 years.
> - SUmiT: PTNS 55% vs sham 21%; 12 weekly sessions.
> - ROSETTA: Botox 200 U slightly better for UUI, more UTI and CIC.
> - Rechargeable and long-life non-rechargeable stimulators, lasting at least 10 years.
> - Full-body MRI-conditional devices at 1.5 T and 3 T.
> - Implantable tibial nerve stimulators (BlueWind, eCoin) and transcutaneous tibial stimulation as less invasive alternatives.
> - Standardised tined-lead placement improves outcomes and lowers revision.

# LAQ 3 Pathophysiology, classification and management of LUTS.

P2

**Type:** LAQ (25 marks)
**Source:** 2024-Winter Paper 2, Q1 (medical and surgical management of BPH also answered in 2015-Summer Paper 2, Q1)

## Definition

- **LUTS (lower urinary tract symptoms):** symptoms of the storage, voiding and post-micturition phases; not disease-specific; up to half of older men.
- **BPH (benign prostatic hyperplasia)** is histology; BPE is enlargement; BPO is prostatic obstruction; **BOO (bladder outlet obstruction)** has any cause. Not every man with LUTS has BPO.

## Classification

### 1. By phase (ICS, International Continence Society)

| Storage | Voiding | Post-micturition |
|---|---|---|
| Frequency, urgency, **nocturia** (waking ≥1 time to void), urgency incontinence | Weak or split stream, hesitancy, intermittency, straining, terminal dribble | Feeling of incomplete emptying, post-micturition dribble |

- Storage symptoms form **OAB (overactive bladder)**: urgency ± urgency incontinence, usually with frequency and nocturia; most bothersome.

### 2. Functional (Wein)

| | Because of the bladder | Because of the outlet |
|---|---|---|
| Failure to store | DO (detrusor overactivity), low compliance, hypersensitivity | Sphincter weakness (SUI, stress urinary incontinence) |
| Failure to empty | DUA (detrusor underactivity) | BPO, stricture, bladder neck contracture, sphincter dyssynergia |

**3. Severity:** IPSS (International Prostate Symptom Score), 7 questions scored 0–5 plus a quality-of-life question: mild 0–7, moderate 8–19, severe 20–35.

**4. By cause:** prostate, bladder, urethra, neurological, polyuria, infection, stone, tumour.

## Pathophysiology

- **Prostate:** BPH is an increase in cell number of stroma and glands in the **transition zone** and periurethral region. **Type 2 5α-reductase** in stromal fibroblasts makes **DHT (dihydrotestosterone)**; androgens are permissive, not mitogenic. Ageing, inflammation and metabolic syndrome contribute.
- **Static component:** gland bulk; the **capsule** transmits expansion to the urethra (TUIP, transurethral incision of the prostate, relieves obstruction without removing tissue).
- **Dynamic component: α1A-adrenergic** smooth-muscle tone; the basis of α-blockers.
- Size does not correlate with symptoms or obstruction.
- **Bladder response:** hypertrophy → compensation → decompensation. Collagen causes trabeculation and low compliance; DO causes storage symptoms; late detrusor failure causes retention. One-third keep storage symptoms after surgery.

- **Also know:** nocturnal polyuria (night output >33% of 24 h over age 65); BPO complications: AUR (acute urinary retention), UTI (urinary tract infection), stones, haematuria, renal failure.

### Evaluation (EAU, European Association of Urology, men ≥40)

- **Basic:** history (comorbidity, drugs, sexual function), IPSS, DRE (digital rectal examination) and neurological check, **urinalysis (mandatory)**, PVR (post-void residual), PSA (prostate-specific antigen) if life expectancy >10 years.
- **Frequency–volume chart, 3 days:** for storage symptoms or nocturia.
- **Before treatment:** uroflowmetry (Qmax, maximum flow rate, <10 mL/s suggests BOO); ultrasound for volume and median lobe.
- **Selected:** creatinine and renal ultrasound if large PVR; cystoscopy for haematuria, stricture or before PUL (prostatic urethral lift).
- **Pressure–flow study** (separates BOO from DUA) before surgery if voided volume <150 mL, PVR >300 mL, age <50 or >80, or failed surgery.

### Management: EAU algorithm

```
Bothersome LUTS? No -> watchful waiting, lifestyle advice Absolute surgical indication? Yes -> surgery Otherwise: Nocturnal polyuria -> lifestyle +/- desmopressin Storage predominant -> antimuscarinic or beta3-agonist Prostate >40 mL -> 5ARI (5-alpha-reductase inhibitor) +/- alpha1-blocker Others -> alpha1-blocker or PDE5 inhibitor Residual storage -> add antimuscarinic or beta3-agonist Failure or complication -> surgery
```

- **Watchful waiting:** education, fluid 1.5–2 L/day (none 2 h before bed), less caffeine and alcohol, double voiding, bladder training; yearly review.

| Drug | Example | Mechanism and notes |
|---|---|---|
| α1-blocker | Tamsulosin 0.4 mg, silodosin 8 mg, alfuzosin 10 mg | Relax prostatic smooth muscle; acts in days; dizziness, ejaculatory dysfunction, IFIS (intraoperative floppy iris syndrome) |
| 5ARI | Finasteride 5 mg, dutasteride 0.5 mg | Lower DHT; volume falls about 20% over 6–12 months; **PSA halves**; reduces AUR and surgery |
| Antimuscarinic | Solifenacin, tolterodine | Block detrusor M3 receptors; for storage symptoms if PVR <200 mL |
| β3-agonist | Mirabegron 50 mg, vibegron | Relax detrusor during storage |
| PDE5I (phosphodiesterase-5 inhibitor) | Tadalafil 5 mg daily | Smooth-muscle relaxation; useful with ED (erectile dysfunction) |

- **MTOPS:** progression cut by doxazosin 39%, finasteride 34%, combination 66%; **CombAT** confirmed. Combine when volume >30–40 mL and PSA >1.5 ng/mL.

### Surgery

- **Indications:** refractory retention, recurrent UTI, bladder stones, recurrent haematuria, BPO-related renal insufficiency; failed medical therapy.

| Prostate volume | Options |
|---|---|
| <30 mL | TUIP, TURP (transurethral resection of the prostate) |
| 30–80 mL | **TURP (reference standard)**, bipolar TURP, PVP (photoselective vaporisation of the prostate), enucleation, PUL, Rezum (water vapour), Aquablation |
| >80 mL | **HoLEP (holmium laser enucleation)** or thulium enucleation, open or robotic simple prostatectomy |

- Ejaculation preserved: PUL, Rezum. Anticoagulated: PVP, HoLEP. PAE (prostatic artery embolisation) in selected men.
- **TURP complications:** bleeding, **TUR syndrome** (hyponatraemia with glycine; avoided by bipolar saline), retrograde ejaculation, stricture, incontinence.
- **Also know:** iTind; female LUTS follow OAB and SUI guidelines.

**Diagram to draw:** BPH → BPE → BPO → BOO → bladder changes → LUTS.

> **Key points to remember:**
> - ICS: storage, voiding, post-micturition; IPSS 0–7, 8–19, 20–35.
> - Static (bulk, capsule) plus dynamic (α1A) obstruction; size ≠ symptoms.
> - 5ARI for >40 mL; MTOPS combination cut progression 66%.
> - TURP 30–80 mL; HoLEP or simple prostatectomy for >80 mL.

## LAQ 4 Describe neuromuscular dysfunction and management of bladder dysfunction following cervical spine surgery in young adult male

P2

**Type:** LAQ (25 marks)
**Source:** Practice Set 1 Paper 2, Q2

*Assumed meaning: cervical spinal cord injury (traumatic, operated, or from the surgery itself), giving a suprasacral lesion above T6.*

### Introduction

- A cervical cord lesion lies **between the PMC (pontine micturition centre) and the sacral cord**. It causes tetraplegia and the classic suprasacral pattern.
- Complete transection is rare; incomplete lesions are commonest. Manage by **urodynamics, not neurology**.

### Pathophysiology

- The PMC normally coordinates detrusor contraction with sphincter relaxation. Cutting the descending pathway leaves a **spinal reflex**.
- **Why NDO (neurogenic detrusor overactivity) appears:** unmyelinated C-fibre afferents become sensitised (NGF (nerve growth factor)), sprout and drive a new sacral reflex. Bulbospinal inhibition is lost.

| Phase | Detrusor | Outlet | Clinical |
|---|---|---|---|
| **Spinal shock** (usually 6–12 weeks; up to 1–2 years) | Areflexic, smooth bladder | Bladder neck closed; reduced closure pressure | **Retention**; overflow only |
| **Recovery (suprasacral pattern)** | **NDO**, no sensation below lesion | **Striated DSD (detrusor–sphincter dyssynergia)**; lesion above T7–T8 adds **smooth sphincter dyssynergia** | Reflex leakage, incomplete emptying, high pressure |

- **Blaivas DSD types:** type 1, sphincter relaxes at peak pressure (incomplete lesions); type 2, sporadic contractions; type 3, crescendo–decrescendo obstruction (types 2–3 with complete lesions).
- **Consequences:** high storage and voiding pressure, VUR (vesicoureteric reflux) in 17–25%, hydronephrosis, stones (about 7% in 10 years), UTI (urinary tract infection). Untreated, about half of men with DSD deteriorate (worse if complete, continuous DSD).
- Upper tract risk also rises with low compliance (<12.5 mL/cm $H_2O$) and NDO for over a third of filling.
- **VUR:** treat the bladder first (lower the pressure); reflux surgery in a thick bladder is difficult.
- **Also know:** cervical myelopathy may instead give an underactive detrusor.

### Autonomic dysreflexia

- Occurs with lesions **above T6** (cervical 60%), after spinal shock. Bladder or bowel distension triggers unopposed spinal sympathetic outflow, as medullary inhibition is lost.
- **Features:** pounding headache, severe hypertension (systolic rise ≥20 mmHg defines it at urodynamics), bradycardia, sweating and flushing above the lesion. It can cause stroke or

seizures.
- **Treatment:** sit up, **drain the bladder** (check catheter), then **nitroglycerin 2% paste** above the lesion; intravenous nitroprusside or labetalol if persistent. Prophylaxis with terazosin or prazosin. Monitor blood pressure during cystoscopy and urodynamics; use spinal or general anaesthesia.

## Evaluation

- History, examination, **hand function** (decides self-catheterisation), transfers, carers.
- Urinalysis, creatinine, **renal ultrasound**. Men with good hand function can self-catheterise; others need carers or a suprapubic catheter.
- **VUDS (videourodynamics) after spinal shock**, with blood pressure monitoring: compliance, NDO, DSD, VUR, **DLPP (detrusor leak point pressure) ≥40 cm $H_2O$** = upper tract risk.
- AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) 2021: suprasacral lesions are "unknown risk" until tested; DSD or poor compliance = high risk.

## Management

**Goals:** protect kidneys, low-pressure storage, low-pressure emptying, no infection, continence, social acceptability. Start with the simplest reversible option.

- **Spinal shock: CIC (clean intermittent catheterisation)** preferred; indwelling catheter early is acceptable. Defer irreversible procedures.

| Aim | Options |
|---|---|
| **Emptying** | **CIC: gold standard**, volumes <500 mL; carer CIC in tetraplegics. **Suprapubic catheter** better than long-term urethral catheter. Credé/Valsalva not recommended |
| **Storage (lower pressure)** | **Antimuscarinics** ± mirabegron (β3 agonist); **intradetrusor onabotulinumtoxinA 200 U** if refractory (repeat when the effect wears off) |
| **Refractory small bladder** | **Augmentation ileocystoplasty** ± continent catheterisable channel (Mitrofanoff) |
| **Men unable to do CIC** | **External sphincterotomy** or intrasphincteric botulinum toxin → reflex voiding into condom catheter. High failure; DLPP ≥40 cm $H_2O$ afterwards = failure. Urethral stents abandoned |
| **Neuromodulation** | **SARS (sacral anterior root stimulator, Brindley) + posterior sacral rhizotomy**: abolishes NDO and dysreflexia, allows voiding on demand |
| **Last resort** | Ileovesicostomy; bladder neck closure + suprapubic catheter; ileal conduit |

- **Also know:** sacral neuromodulation is used for neurogenic bladder without SCI (spinal cord injury); hydrophilic catheters delay first UTI.

## Complications and follow-up

- **UTI:** treat only symptomatic bacteriuria (5–14 days); no routine prophylaxis.
- Stones, VUR (treat the bladder first), bladder cancer (no screening cystoscopy, AUA/SUFU 2021).
- Lifelong: yearly review and renal function, upper tract imaging every 1–2 years, repeat urodynamics if anything changes.

`Cervical SCI → spinal shock (retention): CIC → VUDS at 3 months: NDO + DSD → antimuscarinic + CIC → Botox → augmentation → cannot CIC: sphincterotomy + condom / SARS / ileovesicostomy`

**Diagram to draw:** neural axis: PMC, lesion at cervical cord, sympathetic T10–L2, sacral S2–S4, with the resulting NDO + DSD.

> **Key points to remember:**
> - Suprasacral lesion = NDO + striated DSD; above T7–T8 also smooth sphincter dyssynergia.
> - Autonomic dysreflexia above T6: drain bladder first, nitrates next.
> - CIC + bladder relaxation (antimuscarinic, Botox 200 U) is the mainstay.
> - DLPP ≥40 cm $H_2O$ threatens the kidneys.

## Short answer questions (SAQ)

### SAQ 1 Botulinum toxin in urology

P1 · P2 · P3 · P4 **merged from 4**

*Asked as:* P1 SAQ "Botulinum toxin in urology: mechanism, indications and doses" · P2 SAQ "Botulinum neurotoxin and its role in urology" · P3 SAQ "Botulinum toxin in urology" · P4 SAQ "Botulinum toxin in urology"

*"Botulimum" / "Botulinim" in the questions is read as botulinum toxin.*

#### Introduction

- Neurotoxin of *Clostridium botulinum*, serotypes A–G. Urology uses **onabotulinumtoxinA (BoNT-A)**, the only licensed formulation; type B is short-acting.
- 150 kDa: a heavy chain (100 kDa; binding and entry) and a zinc-dependent light chain (50 kDa).

#### Mechanism of action

1. The heavy chain binds the cholinergic nerve terminal and the toxin is internalised.
2. The light chain (a zinc protease) enters the cytosol.
3. It **cleaves SNAP-25** (synaptosomal-associated protein 25), a SNARE (vesicle-docking) protein. Vesicles cannot fuse, so **acetylcholine release stops**: chemical denervation. - **Afferent action:** less ATP (adenosine triphosphate), substance P and CGRP (calcitonin gene-related peptide) release; fewer TRPV1 (capsaicin) and P2X3 (purinergic) sensory receptors on C fibres, so urgency and pain fall. - The effect lasts about 6–9 months and wanes as nerve terminals sprout, so injections are repeated; repeat injections stay effective and response is maintained.

#### Technique

- Flexible or rigid cystoscopy under local anaesthesia; diluted toxin injected into the detrusor at **about 20 sites (100 U, in 10 mL) or 30 sites (200 U, in 30 mL)**, usually sparing the trigone (reflux concern).
- Sterile urine, antibiotic cover; check PVR (post-void residual) at 2 weeks.

## Urological uses: indications and doses

| Indication | Dose and evidence |
|---|---|
| **Idiopathic OAB (overactive bladder)**, third line | **100 U** (EMBARK): incontinence down about 59%, about 25% dry. **ABC** trial: equal to antimuscarinic, more complete dryness. **ROSETTA**: comparable to SNM (sacral neuromodulation), more UTI (urinary tract infection) (P2, P3: 200 U used, equal to or ≥ SNM; another source: slightly better than SNM) |
| **NDO (neurogenic detrusor overactivity):** SCI (spinal cord injury), MS (multiple sclerosis) | **200 U** (DIGNITY trials): less incontinence, larger capacity, lower detrusor pressure; works with or without antimuscarinics |
| **Paediatric NDO** (spina bifida) | About **10 U/kg (maximum 360 U)**, about 30 sites, in published (Campbell) series; FDA (Food and Drug Administration) label (2021, age ≥5) 6 U/kg, maximum 200 U; continence 65–87%, most avoid augmentation |
| **DSD (detrusor–sphincter dyssynergia)**, underactive detrusor | Intrasphincteric injection 50–100 U, mainly in men unable to do CISC (clean intermittent self-catheterisation) |
| **IC/BPS (interstitial cystitis/bladder pain syndrome)** | 100 U intradetrusor, often trigonal; AUA (American Urological Association) option, off-label |
| **BPH (benign prostatic hyperplasia)** | 100–300 U intraprostatic: an early small trial was positive, but large placebo-controlled RCTs (randomised controlled trials) showed **no benefit over placebo (saline/sham)**; not recommended |

- **Also know:** pelvic floor spasm; paediatric dysfunctional voiding (sphincter); chronic pelvic pain; chronic orchialgia (RCT negative); ED (erectile dysfunction) trials. Units of different toxin products are not interchangeable.

## Complications

- **Retention needing CISC** (the main risk): 0–12% in OAB trials, but about 20% of women and up to 35% of men in practice; commoner in NDO. Each 10 mL rise in baseline PVR raises the risk about 27%. **The patient must accept CISC beforehand.**
- **UTI** about 20%; haematuria; rarely generalised weakness.

## Contraindications

- Active UTI, myasthenia gravis, aminoglycosides, pregnancy, unwillingness to self-catheterise.

## Recent advances

- Paediatric NDO: FDA approved onabotulinumtoxinA for children aged ≥5 years (2021); used from about 2 years of age.
- Other formulations: abobotulinumtoxinA and incobotulinumtoxinA under study for detrusor overactivity; doses are not interchangeable between products.
- Injection-free delivery under research: intravesical liposome-encapsulated toxin and other instillation vehicles.

## Diagram to draw

- Cholinergic synapse (nerve terminal) showing SNAP-25 cleavage by the light chain.

**Key points to remember:**

- Cleaves SNAP-25; acts on efferent and afferent nerves.
- 100 U idiopathic OAB; 200 U NDO; spare trigone.
- Children: 10 U/kg (max 360 U) in series; FDA label 6 U/kg (max 200 U), approved for age ≥5 years (2021); used from about 2 years of age.
- Main risks: retention needing CISC, UTI.
- Other formulations (abobotulinumtoxinA, incobotulinumtoxinA) under study; doses are not interchangeable between products.
- Injection-free delivery under research: intravesical liposome-encapsulated toxin and other instillation vehicles.

## SAQ 2 Urodynamic studies, including video urodynamics

P2 · P4 **merged from 3**

*Asked as:* P2 SAQ "Uro Dynamic Studies." · P2 SAQ "Vesicourodynamics" · P4 SAQ "Video Urodynamics"

*"Vesicourodynamics" is taken as videourodynamics (VUDS): urodynamics with simultaneous fluoroscopic imaging of the bladder and urethra.*

### Definition and principles

- UDS (urodynamic studies) test bladder and urethral function in storage and voiding.
- ICS (International Continence Society): **decide the question first** and **reproduce the symptom**; a normal study does not exclude disease.
- Before invasive UDS: symptom score, 3-day **frequency–volume chart**, free uroflow and PVR (post-void residual); treat UTI (urinary tract infection) first.

### Components of conventional UDS

| Test | Method | Main values |
|---|---|---|
| **Uroflowmetry** | Voided volume ≥150 mL | Bell curve normal; Qmax <10 mL/s (men) suggests obstruction or weak detrusor; plateau = stricture |
| **PVR** | Bladder scan or catheter | Emptying |
| **Filling cystometry** | Bladder and rectal lines, fill with saline 30–100 mL/min (adults); **Pdet = Pves − Pabd**; zero at atmosphere, transducer at symphysis | Sensation, capacity, **DO (detrusor overactivity)**, **compliance** = ΔV/ΔPdet (<20 mL/cm$H_2O$ impaired) |
| **Leak point pressures** | ALPP (abdominal LPP): cough or Valsalva at 150 mL | Low ALPP = weak sphincter (ISD, intrinsic sphincter deficiency) |
| | DLPP (detrusor LPP) | **>40 cm$H_2O$** risks upper tracts (McGuire) |
| **Pressure–flow study** | Void with lines in | High pressure–low flow = obstruction; low–low = underactivity |
| **EMG (electromyography)** | Perineal surface electrodes | DESD (detrusor–external sphincter dyssynergia) |

| Test | Method | Main values |
|---|---|---|
| **UPP (urethral pressure profile)** | Catheter withdrawn | Limited value |
| **Videourodynamics** | Above with contrast and fluoroscopy | VUR (vesicoureteric reflux), bladder neck, site of obstruction |

### Indices (men)

- **BOOI (bladder outlet obstruction index)** = PdetQmax − 2Qmax: **>40 obstructed**, 20–40 equivocal, <20 unobstructed.
- **BCI (bladder contractility index)** = PdetQmax + 5Qmax: **>150 strong**, 100–150 normal, **<100 weak**.
- Women: no validated nomogram (Nitti video criteria).

### Indications for UDS

- Male LUTS (lower urinary tract symptoms) before surgery when unclear (young men, equivocal flow, failed TURP [transurethral resection of the prostate]); complex, mixed or recurrent incontinence.
- **Neurogenic bladder** (spinal injury, MS [multiple sclerosis], spina bifida), radiation cystitis, before augmentation.
- Children: neurogenic bladder, PUV (posterior urethral valves); fill at **5–10% of expected capacity per minute**; expected capacity = (age + 2) × 30 mL.

### Video urodynamics (VUDS): definition and set-up

- **VUDS:** multichannel urodynamics (pressure–flow, with or without EMG) recorded at the same time as fluoroscopy, using **contrast as the filling fluid**.
- Vesical and rectal catheters (Pdet [detrusor pressure] = Pves − Pabd), flowmeter, EMG; C-arm (most flexible) or a tilting table.
- Position for the question: **SUI (stress urinary incontinence) views standing**; voiding in the usual position (women sitting, men standing).
- **Screen only at key moments** (provocation, pressure rise, voiding) to limit radiation.

### What fluoroscopy adds

| Situation | VUDS finding |
|---|---|
| NLUTD (neurogenic lower urinary tract dysfunction) | **Gold standard** (EAU); DESD and **DISD** (detrusor–internal sphincter dyssynergia, seen only on VUDS); trabeculation, diverticula |
| **VUR** | Pressure, volume and grade at onset; reflux "pops off" pressure, so compliance looks falsely safe |
| **Bladder neck** | Only way to see and assess it; **PBNO** (primary bladder neck obstruction) in young men and women |
| Female obstruction | **Nitti criteria:** radiographic narrowing between bladder neck and meatus during voiding, with a sustained detrusor contraction |
| Dysfunctional voiding | "Spinning-top" urethra at the external sphincter |
| SUI | Leakage with ALPP; open bladder neck at rest suggests ISD |
| Male obstruction | Localises prostatic obstruction, bladder neck contracture or stricture |

- **PBNO types (young men):** I high pressure, low flow; II normal pressure, low flow with bladder-neck narrowing; III delayed bladder-neck opening.

### Indications for VUDS

- Known or suspected NLUTD; unexplained retention in women; young men with LUTS; complex incontinence; prior radical pelvic surgery or radiation; diversion; renal transplant.
- **Children:** PUV, spinal dysraphism; contrast-enhanced ultrasound VUDS avoids radiation.

### Urodynamic risk factors (upper-tract danger)

- Impaired compliance; DESD; DISD; high-pressure DO throughout filling; DLPP >40 $cmH_2O$; poor emptying with high storage pressure.

### Limitations of VUDS

- Radiation, cost, space, availability, contrast allergy.
- EAU (European Association of Urology): optimum test in NLUTD. AUA (American Urological Association) guideline: lowest radiation dose that answers the question.

### Also know

- Ambulatory UDS; bethanechol test; antibiotic prophylaxis for risk groups.

### Diagrams to draw

- Normal trace: Pves, Pabd, Pdet, flow, EMG.
- VUDS trace beside a voiding cystourethrogram image.

> **Key points to remember:**
> - Pdet = Pves − Pabd.
> - Ask a question first; reproduce the symptom.
> - BOOI >40 obstructed; BCI <100 weak.
> - DLPP >40 $cmH_2O$: upper-tract risk.
> - VUDS = urodynamics + fluoroscopy with contrast.
> - VUDS is the only test for bladder neck dysfunction, PBNO and DISD.
> - Reflux pop-off makes compliance look falsely good.

## SAQ 3 Uroflowmetry

P2 · P3 **merged from 2**

*Asked as:* P2 SAQ "Uroflowmetry." · P3 SAQ "Uroflowmetry"

### Definition

- A non-invasive urodynamic test that records the **volume of urine voided per unit time** (mL/s). It reflects detrusor contraction and outlet resistance together.

### Method

- Patient comes with a **comfortably full bladder** (normal desire) and voids in private, in the usual (preferred) position.
- **Flowmeters:** weight transducer (load cell), rotating disc, capacitance dipstick.
- **Voided volume ≥150 mL** (Qmax is unreliable below 125–150 mL). Repeat 2–3 times; ask whether the void was typical and compare with the frequency–volume chart.

- Measure **PVR** (post-void residual) by ultrasound afterwards.

## Parameters

| Parameter | Meaning |
|---|---|
| **Qmax** (maximum flow rate) | Most useful (key) value; normal men >15 mL/s, women >20 mL/s; read from the smoothed curve, ignoring spikes |
| Qave (average flow rate) | Volume ÷ flow time; less useful and less accurate for obstruction |
| Voided volume (VV) | Must be ≥150 mL |
| Flow time vs voiding time | Voiding time includes interruptions |
| Time to Qmax | Normally early, in the first third of the void |
| Voiding efficiency | VV ÷ (VV + PVR) × 100 |

## Curve patterns (ICCS, International Children's Continence Society)

| Pattern | Suggests |
|---|---|
| **Bell-shaped** | Normal |
| Tower (high, short) | Overactive bladder |
| **Plateau (flattened, box, prolonged)** | Fixed obstruction: **urethral stricture**, PUV (posterior urethral valves) |
| Prolonged, low, skewed | BPO (benign prostatic obstruction) or detrusor underactivity |
| Staccato (never reaches zero) | Sphincter overactivity, dysfunctional voiding |
| **Intermittent / interrupted (reaches zero), straining** | Detrusor underactivity with abdominal straining, obstruction, or DSD (detrusor–sphincter dyssynergia) |

## Interpretation

- **Qmax <10 mL/s:** obstruction likely (high specificity). **Qmax <15 mL/s:** more sensitive, less specific.
- **Nomograms / volume correction: Siroky** (men), **Liverpool** (men and women, centiles). Corrected Qmax = Qmax ÷ √VV.
- **Artefacts:** "wag" (moving stream), knocking the flowmeter, straining spikes. Read Qmax from the smoothed curve, not single spikes.
- Children: valid if voided volume ≥50% of expected capacity; accept Qmax only if the peak lasts ≥2 s.

## Limitations

- **Cannot separate BOO (bladder outlet obstruction) from DUA (detrusor underactivity)**; only a pressure–flow study can.
- Varies with volume, position, anxiety and between days.

## Uses

- BPH (benign prostatic hyperplasia): evaluation before active treatment and follow-up after medical or surgical treatment.
- Follow-up after **urethroplasty** or optical urethrotomy (stricture recurrence).

- Children: dysfunctional voiding, BBD (bladder and bowel dysfunction), after posterior urethral valve ablation; with patch EMG (electromyography).
- Women: voiding dysfunction, including before stress incontinence surgery.
- Pressure–flow study before BPH surgery if Qmax >10 mL/s (P2: EAU [European Association of Urology]; P3: AUA [American Urological Association], "may be done").

### Diagram to draw

- Flow curves: normal bell, tower, plateau (stricture), prolonged low (BPO), interrupted/intermittent.

> **Key points to remember:**
> - Voided volume ≥150 mL; Qmax is the key value.
> - Plateau = fixed obstruction (stricture); staccato = sphincter; interrupted = weak detrusor or straining.
> - Qmax <10 mL/s suggests BOO but cannot exclude underactivity; Siroky and Liverpool nomograms.
> - Cannot distinguish BOO from DUA.

## SAQ 4 Detrusor underactivity and its management

P2 · P4 **merged from 2**

*Asked as:* P2 SAQ "Management of detrusor underactivity" · P4 SAQ "Detrussor underactivity."

### Definition

- **DUA (detrusor underactivity), ICS (International Continence Society) 2002:** "a contraction of reduced strength and/or duration, resulting in prolonged bladder emptying and/or failure to achieve complete bladder emptying within a normal time span."
- **Acontractile detrusor:** no contraction.
- **UAB (underactive bladder):** the symptom complex of slow stream, hesitancy and straining, with or without incomplete emptying.
- DUA is common in elderly men and women having urodynamics: found in 40–48% of men over 70.

### Pathophysiology and causes

- Failure may be **myogenic** (myocyte loss, fibrosis), **efferent** (poor nerve activation) or **afferent** (reduced sensation cannot sustain the voiding reflex).

| Type | Examples |
|---|---|
| Neurogenic | Cauda equina, disc prolapse, spinal injury, diabetic neuropathy, multiple system atrophy, early stroke, radical pelvic surgery |
| Myogenic | Long-standing BOO (bladder outlet obstruction), diabetes, ischaemia, ageing |
| Drugs | Anticholinergics, opioids |
| Functional | Fowler syndrome, dysfunctional voiding |

### Evaluation

- Bladder diary, flow rate, **PVR (post-void residual)**, sacral neurological examination, spinal MRI if indicated (deficit), drug review.

- **Pressure-flow study is the only accepted (diagnostic) test:** low detrusor pressure with low Qmax.
- **BCI (bladder contractility index) = Pdet@Qmax (detrusor pressure at maximum flow) + 5 × Qmax:** <100 weak, 100–150 normal, >150 strong. Also PIP (projected isovolumetric pressure), Watts factor, stop test.
- Ambulatory urodynamics if no contraction is seen on the standard study (the "bashful bladder").

## Management

- **Goals:** protect kidneys, prevent UTI (urinary tract infection) and stones, improve symptoms. No drug reliably improves contraction.

| Aim / option | Details |
|---|---|
| Conservative drainage | Timed and double voiding; treat constipation; stop causative drugs |
| **CISC (clean intermittent self-catheterisation)** | **Mainstay** for high PVR; fewer infections than an indwelling catheter. Suprapubic catheter if CISC impossible |
| Lower outlet resistance | α-blockers (modest benefit); pelvic-floor biofeedback |
| Outlet surgery (TURP, transurethral resection of the prostate) | In selected obstructed men; little help in low-PVR men; may free catheter-dependent fit men |
| **SNM (sacral neuromodulation)** | For non-obstructive retention, especially Fowler syndrome; needs some detrusor function; intact sensation predicts response |
| Sacral anterior root stimulator (Brindley) | Complete spinal cord injury, with posterior rhizotomy |
| Parasympathomimetics (bethanechol, distigmine) | **Not recommended / not routine**: weak evidence; systemic side effects (bronchospasm, rarely cardiac arrest) |
| Intrasphincteric botulinum toxin | Experimental |
| Diversion | Mitrofanoff when urethral CISC fails; rarely ileal conduit |

- Avoid Credé or Valsalva voiding (reflux risk).
- SNM and parasympathomimetics need some intact detrusor and sensory function; a fibrosed detrusor responds to nothing.

## Investigational

- Intravesical electrotherapy, intravesical prostaglandins, TAC-302, latissimus dorsi detrusor myoplasty, muscle-derived stem cells.

## Diagram to draw

- Pressure-flow plot on the Schäfer nomogram with BCI lines at 100 and 150.

> **Key points to remember:**
> - BCI = Pdet@Qmax + 5Qmax; <100 weak.
> - CISC is the mainstay; no effective drug; parasympathomimetics not recommended.
> - SNM helps non-obstructive retention (Fowler syndrome); intact sensation predicts response.

## SAQ 5 Detrusor–external sphincter dyssynergia (DESD / DSD)

P1

**Asked as:**

- External Sphincter Dyssynergia — *Summer 2016 Paper I Q3a (SAQ)*
- Detrusor sphincter dyssynergia — *Practice Set 10 Paper I Q3f (SAQ)* (also answered in 2016-Summer Paper 1, Q3a)

**Type:** SAQ (10 marks)

### Definition

- **DESD / DSD (detrusor–external sphincter dyssynergia):** the striated sphincter contracts involuntarily (involuntary contraction) during detrusor contraction, causing functional obstruction, high pressure and poor emptying.
- **It requires a lesion between the pons and the sacral cord:** suprasacral spinal cord injury (SCI), MS (multiple sclerosis), transverse myelitis, myelomeningocele. Brain lesions (stroke, Parkinson's) do not cause it (**never with brain lesions**).
- Lesions (at or) above T6 may add bladder-neck (**smooth sphincter**) dyssynergia and autonomic dysreflexia.

### Blaivas classification

| Type | EMG (electromyography) pattern | Lesion |
|---|---|---|
| 1 | Activity rises with detrusor pressure, then relaxes suddenly at peak | Incomplete |
| 2 | Sporadic contractions throughout voiding | Complete |
| 3 | Crescendo–decrescendo contraction; obstruction throughout | Complete |
| - Alternative: intermittent versus continuous (continuous means a complete lesion). Type has no relation to lesion level (type is unrelated to level). | | |

### Diagnosis

- **Pressure–flow study with EMG:** EMG rises during the detrusor contraction, with high pressure and interrupted flow. Needle EMG is more reliable than surface.
- **VUDS (videourodynamics):** narrowed (narrow) membranous urethra with dilated posterior urethra; shows VUR (vesicoureteral reflux).
- **Exclude:** guarding reflex (normal); **pseudodyssynergia** (straining or suppressing a contraction); dysfunctional voiding (learned, no neurological lesion; Hinman syndrome in children).

### Significance

- Worse when complete, continuous and **in men**: about half of untreated men get VUR, hydronephrosis, stones or urosepsis. Other risk factors: high filling pressure, indwelling catheter.
- **Management logic:** DSD is both a storage and an emptying failure. Lower bladder pressure, then empty it.

- DLPP (detrusor leak point pressure) ≥40 $cmH_2O$ endangers the kidneys.

### Management

- **First line:** CIC (clean intermittent catheterisation) plus antimuscarinic or β3-agonist; then intradetrusor botulinum toxin.
- **CIC impossible (men):** sphincterotomy or sphincteric botulinum toxin, with condom drainage. The goal is DLPP <40; DLPP ≥40 afterwards means failure; repeat procedures are common.
- Others: suprapubic catheter, Brindley stimulator with deafferentation, diversion. Urethral stents are abandoned; oral drugs disappoint. Credé and Valsalva voiding contraindicated.
- **Also know:** α-blockers for bladder-neck dyssynergia; lifelong upper-tract surveillance.

**Diagram to draw:** urodynamic trace with rising detrusor pressure (Pdet), EMG bursts and interrupted flow.

> **Key points to remember:**
> - Lesion between pons and sacral cord; never brain lesions.
> - Blaivas 1 incomplete; 2 and 3 complete.
> - Worst in men; DLPP ≥40 $cmH_2O$ hostile.
> - CIC plus detrusor relaxation first; sphincterotomy if CIC impossible.

## SAQ 6 EMG

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 9 Paper 1, Q3b

*EMG is taken as urodynamic electromyography of the striated urethral/anal sphincter and pelvic floor.*

### Definition and principle

- EMG (electromyography) records the electrical potentials of depolarising striated muscle membranes, here the **external urethral sphincter and pelvic floor**, during filling and voiding.
- It shows **coordination** between detrusor and sphincter, not sphincter strength. The bladder neck (smooth muscle) is invisible to EMG.
- External anal sphincter activity mirrors the urethral sphincter, except in some neurological diseases.

### Electrodes

| Surface (patch) | Concentric needle |
|---|---|
| Perianal patches at 3 and 9 o'clock (ICS [International Continence Society] standard: two skin electrodes plus reference) | Needle into the sphincter (girls beside the meatus; boys transperineal) |
| Comfortable; poorer signal; movement artefact | Best signal; painful; dislodges |
| Coordination only | Also motor unit analysis: acute versus chronic denervation |

### Normal pattern

- Filling: activity **rises gradually** (recruitment, guarding reflex).
- Voiding: EMG **falls silent before** detrusor pressure rises and stays silent until flow ends.
- **Lag time** (uroflow with patch EMG in children): relaxation to flow 2–6 s is normal; <2 s suggests DO (detrusor overactivity); >6 s suggests primary bladder neck dysfunction.

### Abnormal patterns

| Finding | Meaning |
|---|---|
| **DSD (detrusor–sphincter dyssynergia)** | Sphincter contracts during detrusor contraction; needs a **suprasacral** cord lesion (SCI [spinal cord injury], MS [multiple sclerosis]). Blaivas types 1–3 |
| Dysfunctional voiding | Same tracing without neurological lesion (learned) |
| Pseudodyssynergia | EMG rise from straining or trying to suppress a contraction |
| Denervation potentials | Sacral cord or cauda equina lesion; MSA (multiple system atrophy) |
| Complex repetitive discharges, decelerating bursts | **Fowler syndrome** (retention in young women); predicts SNM (sacral neuromodulation) response |

### Clinical uses

- Neurogenic bladder (spina bifida, SCI); voiding dysfunction in children; female retention; before sphincter surgery.
- **Limitation:** surface EMG misdiagnoses some dysfunctional voiders; **VUDS (videourodynamics)** is more accurate.

**Diagram to draw:** tracing of Pdet, flow and EMG: EMG silent during normal voiding versus active in DSD.

> **Key points to remember:**
> - EMG judges coordination, not strength.
> - True DSD needs a suprasacral lesion.
> - Lag time 2–6 s normal.
> - Fowler syndrome: complex repetitive discharges on needle EMG.

## SAQ 7 Bladder dysfunction non genitourinary systemic disease

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 2 Paper 2, Q3a

*Taken as lower urinary tract dysfunction caused by systemic (non-genitourinary) diseases.*

### Principle

- The **site of the lesion**, not the disease name, predicts the bladder:
- Brain above the pons: **DO (detrusor overactivity), no DSD (detrusor–sphincter dyssynergia)**.
- Spinal cord: DO **with DSD**.
- Peripheral nerves or roots: **areflexia, reduced sensation, high PVR (post-void residual)**.

- **Urodynamics** before treatment, because symptoms do not predict findings.

### Diabetes mellitus (most important)

- **Diabetic cystopathy:** sensory then motor autonomic neuropathy. Impaired sensation leads to large capacity, poor contractility, low flow and later high PVR and UTI (urinary tract infection).
- **DO is in fact the commonest urodynamic finding** (early disease or coexisting cerebrovascular disease); hypocontractility comes late.
- Mechanisms: glycation end products, sorbitol pathway, oxidative stress, polyuria.
- Differentiate from BOO (bladder outlet obstruction) by pressure–flow study. Treatment: **timed voiding**, CIC (clean intermittent catheterisation), cautious antimuscarinics.

### Other systemic diseases

| Group | Disease | Usual finding |
|---|---|---|
| Neurological | Stroke | Acute retention, then DO; no true DSD |
| | Parkinson disease | DO; pseudodyssynergia; TURP (transurethral resection of prostate) possible after excluding MSA (multiple system atrophy) |
| | Multiple sclerosis | DO about 38%, DO with DSD 29%, areflexia 26%; changes over time |
| Infective | HIV/AIDS | DO, DSD; areflexia with CMV (cytomegalovirus) |
| | Herpes zoster/simplex, Guillain–Barré | Retention, areflexia; reversible |
| | Spinal tuberculosis, Lyme, HTLV-1 | DO with DSD or areflexia |
| Autoimmune and connective tissue | SLE (systemic lupus erythematosus) | DO; lupus cystitis |
| | Systemic sclerosis | Areflexia, low compliance |
| Infiltrative | Amyloidosis (transthyretin) | Low sensation, underactivity, SUI (stress urinary incontinence) |
| Nutritional | B12 deficiency, alcohol (thiamine) | Mixed; reversible with replacement |
| Endocrine | Hyperthyroidism | Poor emptying; resolves with treatment |

**Also know:** myasthenia gravis; schizophrenia; gastroparesis.

### Management principles

- Treat the disease; reversible therapy.
- Storage: antimuscarinics (trospium, or mirabegron in elderly brains), botulinum toxin, neuromodulation.
- Emptying: CIC; avoid Credé with DSD.

**Diagram to draw:** neuraxis with lesion levels and the bladder pattern at each.

**Key points to remember:**

- Lesion level predicts pattern: brain DO; cord DO + DSD; peripheral areflexia.
- Diabetic cystopathy: low sensation, large capacity, high PVR; DO common early.
- Urodynamics before treating; reversible treatment first.

## SAQ 8 Urinary bladder compliance

P3

**Type:** SAQ (10 marks)
**Source:** 2017-Summer Paper 3, Q3e

### Definition

- **Compliance** is the change in volume for a change in detrusor pressure: **$C = \Delta V / \Delta P_{det}$**, in **$mL/cmH_2O$**.
- It reflects **low-pressure storage** during filling.

### Physiological basis

- **Viscoelastic wall**: smooth muscle, collagen (about 50% of the wall) and elastin (about 2%). Coiled collagen straightens as the bladder fills.
- More collagen or less elastin makes the wall stiff.

### Measurement (filling cystometry)

- ICS (International Continence Society) points: Pdet at **start of filling** and at **cystometric capacity** (or before a leak-causing contraction), **excluding detrusor contractions**.
- **No agreed normal value**; compliance depends on capacity. Below **20 $mL/cmH_2O$** (some use 10–20) suggests poor accommodation. In children, normal is often taken as **>15 $mL/cmH_2O$**.

### Pitfalls

| Pitfall | Effect | Fix |
|---|---|---|
| Fast filling | False low compliance | Slow or stop filling; pressure falls |
| Sustained low-amplitude detrusor contraction | Mimics low compliance | Stop filling: pressure returns to baseline |
| **Pop-off**: VUR (vesicoureteric reflux), diverticulum, leaking outlet | Falsely good compliance | Video-urodynamics; occlude outlet |
| - A small bladder can give a low number without danger (**compliance paradox**). **Absolute storage pressure matters more than the ratio.** | | |

### Causes of low compliance

- **Neurogenic**: spina bifida, spinal cord injury, lower motor neuron and decentralisation injury (radical pelvic surgery).
- **Outlet obstruction**: PUV (posterior urethral valves), long-standing BPO (benign prostatic obstruction).
- **Structural**: radiation cystitis, tuberculosis, schistosomiasis, multiple bladder operations.

### Consequences and DLPP

- **DLPP (detrusor leak point pressure)**: lowest Pdet at which leakage occurs without contraction or strain.
- **McGuire (1981)**: DLPP **>40 cm$H_2O$** in myelodysplastic children led to hydronephrosis or VUR in 85%. Not a validated safe limit (20 cm$H_2O$ is more sensitive); aim as low as possible.
- A strong outlet keeps the patient dry but transmits high pressure to the kidneys.

### Management

- **Clean intermittent catheterisation + antimuscarinics** (first line).
- **Intradetrusor botulinum toxin**.
- **Augmentation cystoplasty** if refractory; vesicostomy in infants.
- Repeat urodynamics to confirm safety.

**Diagram to draw:** cystometrogram: normal flat versus steep rising filling curve.

> **Key points to remember:**
> - C = ΔV/ΔPdet; <20 mL/cm$H_2O$ suggests poor compliance.
> - McGuire: DLPP >40 cm$H_2O$, 85% upper-tract damage.
> - Pop-offs (VUR, diverticulum) hide true pressure; use video-urodynamics.

# Rapid Revision Sheet · Section 7

### Overactive bladder: pathophysiology, pharmacotherapy and non-pharmacological management (LAQ)

- OAB is defined by urgency; DO is a urodynamic finding.
- Final pathway: increased afferent signal (urothelial, myogenic) or failed central inhibition.
- Behavioural therapy first: BT and PFMT with urge suppression give 50–80% fewer leaks; antimuscarinic or β3-agonist next.
- M3 mediates contraction; β3 raises cAMP for storage-phase relaxation; voiding spared.
- Equal efficacy; β3 avoids dry mouth and dementia risk (10–65% with antimuscarinics >3 months).
- Darifenacin most M3-selective; tolterodine bladder-selective; solifenacin longest half-life.
- STAR: solifenacin beat tolterodine ER.
- Mirabegron 25 mg → 50 mg; avoid if blood pressure ≥180/110; CYP2D6 interaction (metoprolol).
- Elderly: mirabegron (β3), trospium or darifenacin preferred over other antimuscarinics.
- Solifenacin + mirabegron combination works (SYNERGY).
- Refractory OAB: PTNS (SUmiT 55% vs 21%), SNM (InSite 61% vs 42%), Botox 100 U.

### Neuromodulation for voiding dysfunction, including sacral neuromodulation (LAQ)

- Afferent modulation explains SNM benefit in both OAB and retention.
- S3: bellows plus great-toe flexion; implant if the test improves ≥50%.
- SNM revision >30% at 5 years.
- SUmiT: PTNS 55% vs sham 21%; 12 weekly sessions.
- ROSETTA: Botox 200 U slightly better for UUI, more UTI and CIC.
- Rechargeable and long-life non-rechargeable stimulators, lasting at least 10 years.
- Full-body MRI-conditional devices at 1.5 T and 3 T.
- Implantable tibial nerve stimulators (BlueWind, eCoin) and transcutaneous tibial stimulation as less invasive alternatives.
- Standardised tined-lead placement improves outcomes and lowers revision.

### Pathophysiology, classification and management of LUTS. (LAQ)

- ICS: storage, voiding, post-micturition; IPSS 0–7, 8–19, 20–35.
- Static (bulk, capsule) plus dynamic (α1A) obstruction; size ≠ symptoms.
- 5ARI for >40 mL; MTOPS combination cut progression 66%.
- TURP 30–80 mL; HoLEP or simple prostatectomy for >80 mL.

### Describe neuromuscular dysfunction and management of bladder dysfunction following cervical spine surgery in young adult male (LAQ)

- Suprasacral lesion = NDO + striated DSD; above T7–T8 also smooth sphincter dyssynergia.
- Autonomic dysreflexia above T6: drain bladder first, nitrates next.
- CIC + bladder relaxation (antimuscarinic, Botox 200 U) is the mainstay.
- DLPP ≥40 cm $H_2O$ threatens the kidneys.

### Botulinum toxin in urology (SAQ)

- Cleaves SNAP-25; acts on efferent and afferent nerves.

- 100 U idiopathic OAB; 200 U NDO; spare trigone.
- Children: 10 U/kg (max 360 U) in series; FDA label 6 U/kg (max 200 U), approved for age ≥5 years (2021); used from about 2 years of age.
- Main risks: retention needing CISC, UTI.
- Other formulations (abobotulinumtoxinA, incobotulinumtoxinA) under study; doses are not interchangeable between products.
- Injection-free delivery under research: intravesical liposome-encapsulated toxin and other instillation vehicles.

**Urodynamic studies, including video urodynamics (SAQ)**

- Pdet = Pves − Pabd.
- Ask a question first; reproduce the symptom.
- BOOI >40 obstructed; BCI <100 weak.
- DLPP >40 $cmH_2O$: upper-tract risk.
- VUDS = urodynamics + fluoroscopy with contrast.
- VUDS is the only test for bladder neck dysfunction, PBNO and DISD.
- Reflux pop-off makes compliance look falsely good.

**Uroflowmetry (SAQ)**

- Voided volume ≥150 mL; Qmax is the key value.
- Plateau = fixed obstruction (stricture); staccato = sphincter; interrupted = weak detrusor or straining.
- Qmax <10 mL/s suggests BOO but cannot exclude underactivity; Siroky and Liverpool nomograms.
- Cannot distinguish BOO from DUA.

**Detrusor underactivity and its management (SAQ)**

- BCI = Pdet@Qmax + 5Qmax; <100 weak.
- CISC is the mainstay; no effective drug; parasympathomimetics not recommended.
- SNM helps non-obstructive retention (Fowler syndrome); intact sensation predicts response.

**Detrusor–external sphincter dyssynergia (DESD / DSD) (SAQ)**

- Lesion between pons and sacral cord; never brain lesions.
- Blaivas 1 incomplete; 2 and 3 complete.
- Worst in men; DLPP ≥40 $cmH_2O$ hostile.
- CIC plus detrusor relaxation first; sphincterotomy if CIC impossible.

**EMG (SAQ)**

- EMG judges coordination, not strength.
- True DSD needs a suprasacral lesion.
- Lag time 2–6 s normal.
- Fowler syndrome: complex repetitive discharges on needle EMG.

**Bladder dysfunction non genitourinary systemic disease (SAQ)**

- Lesion level predicts pattern: brain DO; cord DO + DSD; peripheral areflexia.
- Diabetic cystopathy: low sensation, large capacity, high PVR; DO common early.
- Urodynamics before treating; reversible treatment first.

**Urinary bladder compliance (SAQ)**

- C = ΔV/ΔPdet; <20 mL/cm$H_2O$ suggests poor compliance.
- McGuire: DLPP >40 cm$H_2O$, 85% upper-tract damage.
- Pop-offs (VUR, diverticulum) hide true pressure; use video-urodynamics.

# Section 8 · Nocturia & Male Incontinence

## Short answer questions (SAQ)

### SAQ 1 Post-prostatectomy incontinence

P2

**Asked as:**

- Post prostatectomy incontinence. — *Summer 2021 Paper II Q3a (SAQ)*
- Prostatectomy incontinence — *Practice Set 11 Paper II Q3f (SAQ)*

**Type:** SAQ (10 marks)

(Practice Set 11 Paper II Q3f: also answered in 2021-Summer Paper 2, Q3a)

#### Definition and incidence

- PPI (post-prostatectomy incontinence) is mainly **SUI (stress urinary incontinence)** after RP (radical prostatectomy), the commonest cause of male SUI. Rates vary with definition ("no pad" vs "safety pad") and reporting; expert no-pad continence is in the mid-90s % at 1 year; continence improves for up to 2 years.

#### Pathophysiology

- RP removes the internal sphincter (bladder neck, prostatic urethra), so continence depends (entirely) on the external **rhabdosphincter**.
- Causes: apical injury to rhabdosphincter or pudendal branches (sphincter or nerve injury at the apex), short membranous urethra, afferent denervation; also detrusor overactivity, low compliance, **VUAS (vesicourethral anastomotic stenosis)**.
- Nerve sparing and posterior reconstruction (Rocco stitch) give **earlier, not better final** continence.

#### Evaluation

- History (type, severity, radiotherapy, cancer status), bladder diary, **24-hour pad weight**: mild <200 g, moderate 200–400 g, severe >400 g (another source: mild <150 g, moderate <400 g, severe more; thresholds vary).
- **Standing cough test**, grade (grades) 0 to 4 (0–4: leak reported but not seen; delayed drops; early drops; delayed stream; early persistent stream).
- Urinalysis, creatinine, PVR (post-void residual), PSA (prostate-specific antigen). **Cystoscopy** excludes VUAS and judges (shows) residual sphincter coaptation; urodynamics when needed (if mixed symptoms).

#### Management

1. **Wait 6–12 months** (improves up to 2 years). Supervised **PFMT (pelvic floor muscle training)** with biofeedback, ideally preoperative (started before surgery); pads, condom catheter or (penile) clamp.
2. **Duloxetine** (off-label). **Bulking agents are not advised.**
3. **Surgery** (AUA/SUFU [American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction] 2019, EAU [European Association of Urology]):

| Situation | Choice |
|---|---|
| Mild–moderate, no radiotherapy | **Male sling**: AdVance / AdVance XP (repositions the bulbar urethra; needs a coapting sphincter) or adjustable systems (ATOMS [adjustable transobturator male system]) |
| Severe, or any radiotherapy | **AUS (artificial urinary sphincter), AMS 800**: gold standard; cuff pressure 61–70 cm $H_2O$ |
| ProACT balloons | Expert centres; avoid after radiotherapy |

- **MASTER trial:** male sling non-inferior to AUS at 12 months.
- AUS: social continence (0–1 pad) 55–77%; revision 10–29% at 3 years; lasts (lifespan) 8–10 years. Erosion risks: radiotherapy (risk higher after radiotherapy), prior erosion, urethroplasty (use a **transcorporal cuff** for fragile urethra).
- **Also know:** climacturia; treat VUAS first.

**Diagram to draw:** AMS 800 with cuff, scrotal pump and pressure-regulating balloon.

> **Key points to remember:**
> - Wait 6–12 months with PFMT before surgery.
> - Sling for mild–moderate; AUS for severe or irradiated.
> - MASTER: sling non-inferior to AUS at 12 months.

## SAQ 2 Desmopressin (DDAVP)

P3

**Asked as:**

- Desmopressin. — *Winter 2023 Paper III Q3b (SAQ)*
- DDAVP — *Practice Set 3 Paper III Q3c (SAQ)*

**Type:** SAQ (10 marks)
(also answered in 2023-Winter Paper 3, Q3b)

### Pharmacology

- DDAVP (**1-deamino-8-D-arginine vasopressin**, desmopressin), a synthetic analogue of AVP (arginine vasopressin/antidiuretic hormone).
- Deamination and D-arginine substitution prolong action (give longer action) and make it a **selective V2 agonist (selective V2 agonism) without pressor (V1) effect**.
- **Mechanism:** V2 receptor on collecting-duct cells raises cAMP (cyclic adenosine monophosphate), inserting **aquaporin-2**; water is reabsorbed, so urine is less and concentrated (V2 receptor → cAMP → aquaporin-2 insertion → water reabsorption → less, concentrated urine).
- Endothelial V2 action releases von Willebrand factor and factor VIII (haemostatic use).

### Formulations

| Form | Dose |
|---|---|
| Tablet | 0.2–0.4 mg at bedtime, about 1 h before bed (enuresis) |
| Oral lyophilisate (melt) | 120–240 µg, about 1 h before bed |
| Low-dose nasal spray (nocturia) | 0.83 or 1.66 µg, 30 min before bed |

| Form | Dose |
|---|---|
| Sublingual ODT (orally disintegrating tablet) | 27.7 µg women, 55.3 µg men |
| IV / subcutaneous | 0.3 µg/kg (bleeding) |

## Uses (in urology)

1. **Monosymptomatic nocturnal enuresis:** first line with the alarm. **Rule of thirds**: one-third full, one-third partial, one-third no response. Acts (works) at once but relapses on stopping. Restrict fluids in the evening (avoid evening fluid); add oxybutynin if partial.
2. **Nocturia from nocturnal polyuria** (NPi [nocturnal polyuria index] >20–33%, age-dependent). EAU (European Association of Urology): offer below (under) 65; low dose above (over) 65 with ≥2 voids. Add to α-blocker for refractory nocturia in BPH (benign prostatic hyperplasia).
3. **Neurogenic nocturia** (multiple sclerosis, MSA [multiple system atrophy]).
4. **Bleeding:** uraemic platelet dysfunction, von Willebrand disease (including acquired, in Wilms tumour). Tachyphylaxis after about 3 (repeated) doses.
5. **Diagnosis:** central versus nephrogenic DI (diabetes insipidus).

## Adverse effects

- **Hyponatraemia** (most important): headache, nausea, confusion, seizures.
- Risk factors: **age ≥65**, baseline sodium <135 mmol/L, CrCl (creatinine clearance) <60 mL/min, higher dose; women are more sensitive.
- Most events occur in the **first 2–3 weeks**.

## Monitoring and contraindications

- Sodium at baseline, then within 1 week and at 1 month (EAU adds day 3), and after dose change.
- **Avoid:** hyponatraemia, polydipsia, heart failure, SIADH (syndrome of inappropriate antidiuretic hormone), significant renal impairment, **loop diuretics**.

**Diagram to draw:** V2 receptor–cAMP–aquaporin-2 pathway in the collecting duct.

> **Key points to remember:**
> - Selective V2 agonist; no pressor effect.
> - Enuresis: 0.2–0.4 mg tablet or 120–240 µg melt; rule of thirds.
> - Hyponatraemia: age ≥65, low sodium; check sodium early.

# Rapid Revision Sheet · Section 8

### Post-prostatectomy incontinence (SAQ)

- Wait 6–12 months with PFMT before surgery.
- Sling for mild–moderate; AUS for severe or irradiated.
- MASTER: sling non-inferior to AUS at 12 months. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

### Desmopressin (DDAVP) (SAQ)

- Selective V2 agonist; no pressor effect.
- Enuresis: 0.2–0.4 mg tablet or 120–240 µg melt; rule of thirds.
- Hyponatraemia: age ≥65, low sodium; check sodium early. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

# Section 9 · Andrology & Sexual Medicine

## Long answer questions (LAQ)

### LAQ 1 Spermatogenesis, azoospermia and non-obstructive azoospermia

P1 · P2 · P3 **merged from 3**

*Asked as:* P1 LAQ "Spermatogenesis, azoospermia, non-obstructive azoospermia and maturation arrest" · P2 SAQ "Azoospermia and non-obstructive azoospermia" · P3 SAQ "Spermatogenesis"

*Maturation arrest is taken as spermatogenic maturation arrest on testicular histology.*

#### Spermatogenesis: definition

- Production of haploid spermatozoa from diploid spermatogonia in the seminiferous tubules: about **64 days** (42–76); about 123 million sperm a day.

#### Hormonal control: HPG (hypothalamic–pituitary–gonadal) axis

- The hypothalamus releases **GnRH (gonadotropin-releasing hormone)** in pulses; pulsatility is essential (constant agonist exposure shuts the axis down). Kisspeptin triggers puberty.
- **LH (luteinising hormone)** acts on Leydig cells to make testosterone (about 5 mg/day). **FSH (follicle-stimulating hormone)** acts on Sertoli cells.
- Feedback: testosterone (and oestradiol) on LH; **inhibin B** from Sertoli cells on FSH. So FSH and inhibin B report the tubules, and LH reports the Leydig cells.
- **Intratesticular testosterone** is about 50-fold higher than serum (held by **ABP, androgen-binding protein**) and is essential for postmeiotic maturation. Exogenous testosterone suppresses LH and so stops spermatogenesis.
- The testis is kept 2–4 °C below core temperature by the pampiniform countercurrent exchange.

#### Sertoli cells and the blood–testis barrier

- Nurse cells: nourish germ cells; secrete ABP, inhibin and fluid; phagocytose residual bodies.
- **Sertoli–Sertoli tight junctions** divide the tubule into a **basal** compartment (spermatogonia) and an **adluminal** compartment (spermatocytes, spermatids). Post-meiotic cells are "foreign" to the immune system; the barrier and local tolerance protect them.

#### Phases of spermatogenesis

| Phase | Events |
|---|---|
| **1. Proliferation (spermatocytogenesis)** | **Dark type A (Ad)** reserve stem cells self-renew (GDNF [glial cell line–derived neurotrophic factor] niche); **pale type A (Ap)** divide every 16 days into type B spermatogonia, then primary spermatocytes. Daughter cells stay joined by cytoplasmic bridges |
| **2. Meiosis** | Primary spermatocytes cross into the adluminal compartment. Meiosis I (reduction division; synapsis and crossing-over) gives secondary spermatocytes; meiosis II gives 4 haploid round spermatids |

| Phase | Events |
|---|---|
| **3. Spermiogenesis** | No division: acrosome from Golgi, flagellum from centriole, nucleus condenses to about 10%, mitochondria form the midpiece, cytoplasm shed |
| **4. Spermiation** | Sertoli cell releases sperm into the lumen |

- **Cycle, stage, wave:** six stages in humans; cycles are arranged as a helical wave, so production is continuous.
- Defective meiotic recombination causes maturation arrest and sperm aneuploidy.

### Epididymal maturation and transport

- Testicular sperm are immotile. During 2–12 days of epididymal transit they gain **progressive motility** and fertilising ability; they are stored in the cauda.
- **Capacitation** and the acrosome reaction occur in the female tract.

### Clinical relevance of spermatogenesis

- A semen sample reflects events about 3 months earlier; reassess 3 months after treatment.
- Disruptors: heat (varicocele, cryptorchidism), exogenous testosterone, chemotherapy, radiation.

### Azoospermia: definition

- No sperm in the **centrifuged** ejaculate on **at least two** semen analyses; 10–15% of infertile men.
- **NOA (non-obstructive azoospermia):** azoospermia due to **failure of spermatogenesis** (testicular failure); about 60% of azoospermic men.

### Classification

| Type | Causes | Clues |
|---|---|---|
| **Pre-testicular** (hypogonadotropic) | Kallmann syndrome, pituitary tumour, hyperprolactinaemia, **exogenous testosterone or anabolic steroids** | Low FSH, LH, testosterone |
| **Testicular (NOA, spermatogenic failure)** | Klinefelter, Y microdeletion, cryptorchidism, mumps, chemotherapy or radiation, idiopathic | Small soft testes, **FSH >7.6 IU/L**, long axis ≤4.6 cm |
| **Post-testicular (OA, obstructive azoospermia)** | CBAVD (congenital bilateral absence of the vas deferens), epididymal obstruction, vasectomy, hernia repair, EDO (ejaculatory duct obstruction) | Normal testes, **FSH ≤7.6**, dilated epididymis |
| Ejaculatory | Retrograde ejaculation, anejaculation | Low volume; sperm in post-ejaculate urine |

### Causes of NOA

- **Genetic: Klinefelter (47,XXY)**, the commonest; **Y microdeletion** (AZF [azoospermia factor] a, b, c); other karyotype anomalies.
- **Acquired:** cryptorchidism, mumps orchitis, torsion, **chemotherapy or radiation**, trauma, varicocele.
- **Idiopathic:** commonest overall.

- *Note:* hypogonadotropic hypogonadism is pre-testicular and treatable medically (hCG [human chorionic gonadotropin] ± FSH).

### Evaluation

- History (mumps, surgery, chemotherapy, drugs, testosterone use); examination (testis size, **vasa**, epididymis, varicocele).
- **Hormones:** FSH (reflects germ-cell mass), LH, testosterone, oestradiol, prolactin.
- NOA: small soft testes, palpable vasa, normal ejaculate volume. **FSH >7.6 IU/L and testis long axis ≤4.6 cm** point to NOA; FSH ≤7.6 with normal testes (>4.6 cm) suggests obstruction.

```
Azoospermia (2 centrifuged samples) ├─ Vasa absent → CBAVD → CFTR testing, both partners; renal scan/US ├─ Volume <1 mL → post-ejaculate urine: sperm = retrograde; none → TRUS for EDO └─ Normal volume: FSH ≤7.6 IU/L + testis long axis >4.6 cm → obstructive FSH >7.6 + testis ≤4.6 cm (small) → spermatogenic failure
```

- **Genetics (AUA/ASRM, American Urological Association/American Society for Reproductive Medicine): karyotype** and **Y microdeletion** in all azoospermic men with NOA, with counselling. **AZFa or AZFb complete deletion: no sperm, do not operate**; AZFc: sperm often found. **CFTR (cystic fibrosis transmembrane conductance regulator)** testing in CBAVD.
- **TRUS (transrectal ultrasound):** seminal vesicle >1.5 cm wide or a midline cyst suggests EDO.
- **Testis biopsy:** combined with sperm extraction and freezing.

#### Histology patterns in NOA

| Pattern | Histology / meaning | Micro-TESE retrieval |
|---|---|---|
| Hypospermatogenesis | All stages present, fewer cells | Best |
| **Maturation arrest** | Halts at one germ-cell stage | About 40% |
| SCO (Sertoli cell–only) | No germ cells | About 30% |

## Maturation arrest

### Definition

- A histological pattern of **spermatogenic failure** in which germ cells develop normally up to one stage and **the sequence halts** at a premeiotic, meiotic or postmeiotic step. No mature sperm, or very few, are made.

### Classification

- **Early:** arrest at spermatogonia or **primary spermatocyte** (meiotic failure).
- **Late:** arrest at the **spermatid** stage.
- **Complete** (all tubules) versus **incomplete** (foci progress to sperm): incomplete gives better retrieval.

### Causes

- **Genetic:** complete **AZFb deletion** causes arrest; meiotic recombination defects in about half of men with maturation arrest; karyotype anomalies.
- **Acquired:** heat (varicocele, cryptorchidism), chemotherapy, radiation, drugs, orchitis; often idiopathic.

### Clinical features and diagnosis

- Azoospermia or severe oligospermia, normal semen volume, palpable vasa.
- **Testis size and FSH are often normal**, so it can mimic obstruction (FSH ≤7.6 IU/L, long axis >4.6 cm).
- **Diagnosis needs testicular biopsy** (Bouin's fixative, never formalin), usually done as sperm extraction.
- Karyotype and Y microdeletion testing in all azoospermic men (AUA/ASRM).

### Management

- Optimise: treat varicocele, stop exogenous testosterone; off-label hormonal stimulation (clomiphene, hCG) unproven.
- **Micro-TESE** with ICSI: retrieval **about 40%** (versus 30% in SCO).
- **Complete AZFb deletion: retrieval highly unlikely; do not operate.**
- No sperm: donor sperm or adoption. Round spermatid injection is experimental.
- Genetic counselling: sons inherit Y deletions.

## Management of azoospermia

### 1. Pre-testicular (hypogonadotropic)

- Stop exogenous testosterone. **hCG 1500–5000 IU 2–3 times weekly ± FSH**; pulsatile GnRH in hypothalamic failure; dopamine agonist for prolactinoma.

### 2. Obstructive

- Microsurgical **vasovasostomy** or **vasoepididymostomy** (clear watery fluid or sperm parts in vasal fluid: vasovasostomy; thick toothpaste-like fluid without sperm: vasoepididymostomy).
- **TURED (transurethral resection of the ejaculatory ducts)** for EDO.
- CBAVD or unreconstructable: sperm retrieval by **MESA (microsurgical epididymal sperm aspiration)**, PESA (percutaneous epididymal sperm aspiration) or TESA + **ICSI (intracytoplasmic sperm injection)**; freeze sperm at reconstruction.

### 3. Non-obstructive

- **Optimisation (before retrieval):** stop exogenous testosterone; repair a clinical varicocele (evidence weak; no definitive evidence in NOA); off-label hormonal therapy (clomiphene, anastrozole, hCG) for low testosterone.
- **Micro-TESE (microdissection testicular sperm extraction) + ICSI:**
- **Principle:** spermatogenesis in NOA is **focal**. Under ×20–25 magnification, **larger, dilated, opaque tubules** are chosen; the least tissue is removed and the blood supply is preserved.
- Sperm found in **about 50%**; better than conventional TESE (Klinefelter 68%, Sertoli cell–only 30%).

| Predictor | Retrieval (%) |
|---|---|
| Post-orchidopexy | 74 |
| AZFc deletion | 70 |
| Klinefelter | 68 |
| Post-chemotherapy | 53 |
| **AZFa or AZFb complete deletion** | **0: do not operate** |

- FSH and testis size **do not** reliably predict retrieval.
- **ICSI:** fresh or frozen testicular sperm. Live birth per cycle is modest; miscarriage is higher (sperm DNA damage). Male offspring of AZF deletion carriers inherit the deletion.

**4. Ejaculatory**

- Retrograde: sympathomimetics, or sperm from alkalinised urine. Anejaculation: vibratory or electroejaculation.

**5. Failure (no sperm):** donor sperm or adoption; genetic counselling.

### Also know

- Spermatogonial stem cell renewal by GDNF; in-vitro spermatogenesis and stem-cell transplantation are experimental.

### Diagrams to draw

- Cross-section of a seminiferous tubule showing basal and adluminal compartments with germ cells maturing towards the lumen.
- Hypothalamic–pituitary–testis axis with the three levels of failure.
- Opened testis under the microscope: plump tubules versus thin Sertoli cell–only tubules.
- Maturation arrest: seminiferous tubule showing germ cells up to spermatocytes with no spermatids or sperm.

> **Key points to remember:**
>
> - Spermatogenesis about 64 days; Ap divide every 16 days; six stages.
> - Intratesticular testosterone 50 × serum; exogenous testosterone causes azoospermia.
> - FSH → Sertoli, LH → Leydig; tight junctions between Sertoli cells form the blood–testis barrier.
> - Confirm azoospermia on two centrifuged samples.
> - FSH ≤7.6 IU/L and testis >4.6 cm → obstructive; FSH >7.6 and testis ≤4.6 cm suggest NOA.
> - Karyotype and Y microdeletion in all.
> - AZFa/b complete deletion: do not operate (no retrieval).
> - NOA: micro-TESE + ICSI; sperm in about half; SCO 30%.
> - Maturation arrest: arrest at one stage; early (spermatocyte) or late (spermatid).
> - FSH and testis size may be normal in maturation arrest: biopsy diagnoses.
> - Micro-TESE retrieval in maturation arrest about 40%; none with complete AZFb deletion.

# LAQ 2 Azoospermia: testicular biopsy and surgical sperm retrieval techniques

P1 · P2 · P3 · P4 **merged from 4**

*Asked as:* P4 LAQ "Define Azoospermia and describe types of testicular biopsies and surgical sperm retrival." · P2 SAQ "Sperm retrieval techniques (sperm extraction in obstructive azoospermia; testicular aspiration and assisted reproduction)" (asked as "Sperm extraction in Obstructive azoospermia", "Testicular aspiration of sperms and assisted reproduction" and "Sperm Retrival techniques") · P1 SAQ "Testicular biopsy and TESA (testicular sperm aspiration)" (asked as "TESA" and "Testicular biopsy") · P3 SAQ "Sperm retrieval techniques"

## Definitions

- **Azoospermia:** no sperm in the **centrifuged pellet** of the ejaculate on **at least two** semen analyses. Found in about 1% of men and 10–15% of infertile men.
- Not **cryptozoospermia** (rare sperm in pellet) or **aspermia** (no ejaculate).
- **OA (obstructive azoospermia):** no sperm in the centrifuged ejaculate despite **normal spermatogenesis**, from blockage of the ducts.
- **TESA (testicular sperm aspiration):** percutaneous needle aspiration of seminiferous tubules under local anaesthesia to retrieve sperm for **ICSI (intracytoplasmic sperm injection)**, which needs only one sperm per oocyte.

## Classification

| Type | Causes | Clues |
|---|---|---|
| Pre-testicular | Kallmann syndrome, pituitary disease, **exogenous testosterone** | Low FSH (follicle-stimulating hormone), LH (luteinising hormone) |
| Testicular: **NOA (non-obstructive azoospermia)** | Klinefelter, Y microdeletion, cryptorchidism, chemotherapy, mumps | Small testes, high FSH |
| Post-testicular: **OA** | CBAVD (congenital bilateral absence of the vas deferens), vasectomy, epididymal or ejaculatory duct obstruction, hernia-repair injury | Normal testes and FSH |
| Ejaculatory | Retrograde ejaculation, anejaculation | Low volume; sperm in post-ejaculate urine |

## Evaluation (brief)

- Absent vasa (CBAVD): CFTR (cystic fibrosis transmembrane conductance regulator) testing of both partners. Volume <1 mL: post-ejaculate urine, then TRUS (transrectal ultrasound).
- **FSH ≤7.6 IU/L and testis long axis >4.6 cm: obstruction likely;** FSH >7.6 and axis ≤4.6 cm: spermatogenic failure.
- **Karyotype and Y microdeletion** in NOA. **AZF (azoospermia factor) a or b complete deletion: no sperm, do not operate.**

## Testicular biopsy

### Indications

- Azoospermia with **normal testicular size, palpable vasa and normal FSH**: separates OA from NOA (AUA/ASRM, American Urological Association/American Society for Reproductive Medicine).
- Since ICSI needs one sperm, biopsy is now **diagnostic and therapeutic**: plan it as TESE (testicular sperm extraction) and **cryopreserve** sperm.
- **Not needed:** CBAVD with normal FSH (spermatogenesis always present); high antisperm antibodies after iatrogenic vasal injury. Do not operate with complete AZFa or AZFb deletion.

### Types

| Type | Method | Use | Limits |
|---|---|---|---|
| **Open biopsy** | Small incision in the albuginea, pea-sized tissue | **Gold standard** for histology and retrieval | Open procedure |
| **Percutaneous core biopsy** | 14 G biopsy gun under cord block | Fresh retrieval in OA; core of tissue | Blind; may injure epididymis or testicular artery; avoid in scarred scrotum |
| **FNA (fine-needle aspiration)**, i.e. TESA | 23 G needle, suction, several passes | OA retrieval; backup to epididymal methods | Few tubules, poor architecture; low yield in NOA |
| **Microdissection biopsy** (micro-TESE) | Operating microscope, plump tubules | NOA | Long operation |

- **Also know:** multi-site FNA "mapping" in NOA.

### Open biopsy technique

1. Local anaesthesia with **spermatic cord block** (or general/spinal); choose the larger testis.
2. 2 cm transverse scrotal incision; open tunica vaginalis.
3. Stay sutures; 3–4 mm albugineal incision avoiding surface vessels (loupes).
4. Excise extruding tubules with iris scissors, **no forceps**.
5. Fix in **Bouin's, Zenker's or glutaraldehyde, never formalin** (distorts histology).
6. **Touch prep** at once, then wet squash prep for sperm; if negative, contralateral biopsy.
7. Close albuginea and tunica vaginalis separately; no antibiotics.

### Histology patterns

| Pattern | Meaning |
|---|---|
| Normal spermatogenesis | Suggests obstruction |
| Hypospermatogenesis | All stages, fewer cells |
| Maturation arrest | Stops at a germ-cell stage |
| SCO (Sertoli cell–only) | No germ cells |
| Tubular sclerosis/hyalinisation | End-stage (e.g. Klinefelter) |

## Surgical sperm retrieval: indications

- **OA with normal spermatogenesis:** CBAVD, unreconstructable obstruction, failed or declined vasectomy reversal, back-up at reconstruction (vasoepididymostomy; frozen for IVF [in vitro fertilisation]), or failed epididymal aspiration.
- **NOA (testicular failure):** only testicular retrieval works; TESA gives poor yield; **micro-TESE (microdissection testicular sperm extraction)** preferred.

- **Anejaculation** (spinal cord injury): vibratory stimulation or electroejaculation first; retrieval after these fail.
- All retrieved sperm need **ICSI**, because sperm from obstructed systems have poor motility; cryopreserve spare sperm.

## Techniques of sperm retrieval

| Technique | Method / site | Best for | Merits, limits and notes |
|---|---|---|---|
| **PESA** (percutaneous epididymal sperm aspiration) | Fine needle into caput epididymis, local anaesthesia | OA | Simple; motile but few sperm; may obstruct epididymis |
| **MESA** (microsurgical epididymal sperm aspiration) | Single tubule opened under microscope at 16–25×; micropipette by **capillary action**, never suction | **OA (best)** | Many (most) motile sperm, multiple frozen vials; needs microsurgical skill |
| **TESA** | Testis, 23 G needle with suction, many passes | OA | Least invasive; immature, immotile sperm, small yield; poor in NOA |
| Percutaneous core biopsy | 14 G biopsy gun | OA | Fresh retrieval; core of tissue; blind; vessel injury |
| **Conventional TESE** | Open biopsy, one or more sites | OA, NOA | When epididymis is absent or empty; gold standard for tissue diagnosis; blind sampling |
| **Micro-TESE** (Schlegel 1999) | Testis bivalved (opened in an avascular plane); **larger, opaque tubules** picked under the microscope | **NOA (best)** | Most sperm, least tissue; sperm in about half; long operation |
| Vasal aspiration | Vasotomy, testicular end flushed | At reconstruction | |

### MESA technique

1. Testis delivered; epididymis inspected at 16–25×.
2. Dilated tubule opened with a microknife; fluid checked at once.
3. No motile sperm: close and **move proximally** (start distally), up to the efferent ductules (proximal sperm are more motile; the most motile are often in the efferent ductules).
4. Collect by **capillary action** into micropipettes, never suction; 4–12 pipettes.
5. **Cryopreserve** in several aliquots; close tubule with 10-0 suture.

### TESA technique

1. Spermatic cord block with local anaesthetic.
2. Testis fixed; **23 G needle** (or angiocatheter) on a high-suction syringe.
3. Multiple passes in different directions draw tubules into the needle (often 10–20 passes needed; P1 table: 3–4 passes).
4. Immediate wet examination for sperm; repeat until sperm are found. **Cryopreserve** spare sperm.

### TESA: advantages and limitations

- **Advantages:** office procedure, no microsurgery, cheap, repeatable.

- **Limitations:** few tubules with poor architecture (not diagnostic histology); small numbers of immature, immotile sperm; often too few to freeze; blind.

**Micro-TESE logic**

- Spermatogenesis in NOA is **focal**. Plump, larger tubules hold sperm; thin white ones are Sertoli cell–only.
- Embryologist examines tissue in theatre; map positive sites; if negative, explore the other testis.
- It removes the least tissue for the most sperm, and is better than conventional TESE.

**Choice:** epididymal sperm first (MESA) in OA; testicular (TESA/TESE) if epididymal sperm are absent or retrieval fails; micro-TESE in NOA.

## Results

- OA: retrieval almost always succeeds; good pregnancy rates.
- NOA: micro-TESE finds sperm in **about half**; fine-needle aspiration yields far fewer sperm than micro-TESE. Retrieval depends on the cause:

| Condition | Micro-TESE retrieval |
|---|---|
| Post-orchidopexy | 74% |
| AZFc deletion | 70% |
| Klinefelter | 68% |
| Post-chemotherapy | 53% |
| Maturation arrest | 40% |
| SCO | 30% |
| **AZFa/AZFb deletion** | **0%: do not operate** |

- **Before retrieval in NOA:** stop testosterone; repair clinical varicocele; off-label clomiphene, anastrozole or hCG (human chorionic gonadotropin).

## Assisted reproduction and use of retrieved sperm

- **IUI (intrauterine insemination):** washed sperm into the uterus; mild male factor.
- **IVF:** oocytes and sperm combined in the laboratory.
- **ICSI:** one sperm injected into each oocyte; essential for surgically retrieved (testicular) sperm (poor motility).
- Frozen retrieved sperm equal fresh, except possibly in spinal cord injury.
- NOA pregnancies: more miscarriage (sperm aneuploidy, DNA damage); genetic counselling. No sperm: donor sperm or adoption.
- ICSI bypasses natural selection: risk of transmitting genetic defects. Offer karyotype, Y-microdeletion and CFTR testing as indicated before ICSI.
- Female burden: ovarian stimulation, multiple pregnancy.

## Complications

- **Haematoma** (commonest), pain, infection (rare), testicular vascular injury (TESA, core biopsy).
- **Inadvertent epididymal biopsy** (most serious: causes obstruction); epididymal obstruction (PESA).

- Testicular atrophy or transient fall in testosterone (TESE).

### Also know

- Testicular sperm may beat ejaculated sperm when DNA fragmentation is high.
- Post-mortem retrieval needs prior consent.
- Microfluidic sperm sorting.

### Diagrams to draw

- Testis and epididymis showing PESA, MESA and TESA sites; testis with 23 G needle on a suction syringe and multiple passes; flow to ICSI.
- Open biopsy: small albugineal window with extruding tubules snipped off.
- Micro-TESE: bivalved testis with plump (dilated) spermatogenic tubules beside thin Sertoli cell–only tubules.

> **Key points to remember:**
>
> - Azoospermia: no sperm in centrifuged pellet on two samples.
> - Biopsy indication: azoospermia, normal testes, vasa and FSH.
> - Open biopsy is the gold standard; fix in Bouin's, never formalin; no forceps; touch prep.
> - OA: normal FSH and testes; MESA is best (capillary action, never suction; move proximally): many motile sperm, multiple vials.
> - TESA: 23 G needle, suction, local anaesthesia; for OA; TESA/TESE when epididymal sperm absent.
> - NOA: micro-TESE best (sperm in about half); TESA yield poor.
> - AZFa/AZFb deletion: 0% retrieval, do not operate.
> - All surgically retrieved sperm need ICSI; cryopreserve extras.
> - Haematoma commonest; epididymal biopsy most serious.

## LAQ 3 Surgical management of ED and penile prostheses (types, biofilms)

P2 · P3 · P4 **merged from 3**

*Asked as:* P2 LAQ "Surgical management of erectile dysfunction and penile prosthesis" (asked as "Discuss surgical management of erectile dysfunction", "Penile Prosthesis" and "Describe various penile prosthesis in ED") · P3 SAQ "Types of penile prosthesis. What are biofilms in penile prosthesis?" · P4 SAQ "Penile implants."

### Introduction

- Surgery for ED (erectile dysfunction) means **penile prosthesis** (main operation) and, rarely, **penile vascular surgery**.
- Prosthesis: most effective, but last-line, treatment of ED; **satisfaction 80–90%**. First devices: Scott inflatable (1972), Small–Carrion rod (1978).
- Lue's goal-directed approach: surgery is for men who fail, cannot use or decline oral PDE5Is (phosphodiesterase type 5 inhibitors), intracavernosal injection, intraurethral alprostadil or a VED (vacuum erection device).
- Detailed vascular testing is needed only before vascular surgery.

### Indications and contraindications for a prosthesis

- **Indications:** ED not responding to, contraindicated for, or unacceptable with (refusal of) oral PDE5Is, injections or vacuum devices; common after prostatectomy, pelvic radiotherapy, diabetes.
- Corporal (cavernosal) fibrosis: after ischaemic priapism (**>36 h**), infection or implant removal; early implant preserves length.
- **Peyronie disease with ED**; penile or pelvic trauma.
- **Contraindications:** active skin, urinary (UTI [urinary tract infection]) or systemic infection; uncontrolled diabetes; unrealistic expectations; reversible ED.
- In insensate men (spinal cord injury) use an inflatable device, because rods erode.

### A. Penile vascular surgery

| | **Arterial revascularisation** | **Venous surgery** |
|---|---|---|
| Defect | Arteriogenic (poor inflow) | Veno-occlusive (venous leak) |
| Work-up | Duplex; selective pudendal angiography (also checks the inferior epigastric donor) | DICC (dynamic infusion cavernosometry and cavernosography) locates the leak |
| Technique | Inferior epigastric artery to dorsal artery (revascularisation) or to deep dorsal vein (arterialisation) | Ligation or embolisation of deep dorsal, superficial dorsal or crural veins; crural plication |
| Selection | Age <55, non-smoker, non-diabetic, no venous leak, focal internal pudendal stenosis; best in young men after pelvic or perineal trauma | Not defined |
| Status | Strictly selected men | Investigational |
| Complications | Glans hyperaemia (commonest), shunt thrombosis, inguinal hernia | Glans numbness, shortening, curvature, skin necrosis |

## B. Penile prosthesis: types

| Type | Devices / examples | Features | Best for |
|---|---|---|---|
| **Malleable (semirigid) rods** | Tactra (nitinol core), Genesis (braided silver), Rigi 10 (stainless steel) | Two bendable rods; simple, cheap; almost never fail (mechanical failure exceedingly rare; P3: no mechanical failure); always firm; erode more in insensate men | Poor manual dexterity; cost |
| Mechanical rods | Spectra (discontinued) | Articulating segments | – |
| 2-piece inflatable | Ambicor (discontinued) | Cylinders + scrotal pump; no reservoir; poor flaccidity | – |
| **3-piece inflatable** | AMS 700 CX/LGX/CXR; Coloplast Titan, Titan Touch, Titan Narrow; Rigicon Infla 10 | Two cylinders, scrotal pump, abdominal reservoir; **best rigidity and true flaccidity**; most popular | Most men |

### Three-piece inflatable details

- **Expansion:** girth only (CX, Titan) or girth and length (LGX, up to 25%; Infla 10 AX). **Narrow** cylinders (CXR, Titan Narrow) for fibrotic corpora.
- **Material:** AMS and Rigicon use silicone–fabric–silicone layers; Titan uses polyurethane (Bioflex).
- **Coatings:** InhibiZone (rifampicin + minocycline) on AMS; hydrophilic coating on Titan, dipped in antibiotic.
- **Reservoir:** space of Retzius, or **ectopic submuscular** after pelvic surgery or cystectomy; lockout valve prevents autoinflation.

## Consent

- The erection may be shorter (pseudocapsule scar); the aim is a firm penis for intercourse; sensation and ejaculation may reduce; infection or mechanical failure may need reoperation.

## Preoperative preparation

- Treat phimosis and skin, fungal and urine infection; sterile urine; stop anticoagulants and aspirin.
- Blood glucose **<200 mg/dL on the day of surgery** (more important than HbA1c [glycated haemoglobin]).
- Shave in theatre; chlorhexidine–alcohol skin preparation (about 40% fewer wound infections than povidone-iodine); MRSA (methicillin-resistant *Staphylococcus aureus*) decolonisation if a carrier.
- IV vancomycin or a cephalosporin plus gentamicin, 1 hour before incision, for up to 24 hours.

## Operative steps (3-piece, penoscrotal approach)

`Catheter → upper scrotal incision → expose corpora lateral to urethra → stay sutures → corporotomy → dilate proximally to ischial tuberosity, distally under glans (instruments dorsolateral; 7→13 mm) → goalpost test for crossover → measure → cylinders equal to or 1 cm shorter → test inflation → close`

`corporotomies → pump in dependent subdartos (dartos) pouch → reservoir in space of Retzius → connect tubing → test inflation → irrigate → close in layers`

- Dilate with Hegar or Brooks dilators, instruments pointing dorsolaterally; check for crossover or perforation; cylinders sized to corporal length (equal to or 1 cm shorter than measured length).
- **Approaches:** penoscrotal (commonest; secures the pump; urethra at risk); infrapubic (reservoir placed under direct vision; dorsal nerve at risk); subcoronal or ventral penile for rods.
- Malleable rods are placed slightly shorter: 0.5–1 cm shorter than the measured length; girth chosen with the two-dilator test.
- **Surgical principles:** antibiotics, glucose control, no-touch technique.
- **Intraoperative problems:** crossover (redilate laterally); crural perforation (suture sling); distal urethral perforation or urethral injury (abandon that side for 5–6 months); fibrotic corpora (cavernotomes, narrow cylinders); curvature (modelling for Peyronie's disease); floppy glans (glans fixation).
- **Postoperative:** mummy wrap with glans visible; cylinders partly inflated; drain out by 24 hours; activation at 4–6 weeks.

### Biofilm

- **Definition:** a structured microbial community in a self-made **polysaccharide matrix**, attached to the implant.
- **Stages:** 1. Conditioning film of host proteins coats the device within minutes. 2. Reversible, then irreversible bacterial adhesion. 3. Microcolonies. 4. Maturation into a 3D structure with water channels and quorum sensing. 5. Dispersal of free bacteria, causing clinical infection.
- **Organisms:** coagulase-negative staphylococci commonest; also *S. aureus*, Gram-negatives, *Candida*.
- **Why antibiotics fail:** the matrix blocks penetration, bacteria are slow-growing (persisters), and they survive up to **1000–1500 times** normal killing levels.
- **Relevance:** organisms from the first operation lie dormant; revising an uninfected device gives about 10% infection, cut to about 3% by **antiseptic washout**.

### Complications

- **Mechanical failure** (commonest; inflatables): repair in about **15% at 5 years, 30% at 10 years**.
- **Infection (1–3%)**: biofilm on the device.
- **Prevention:** glucose <200 mg/dL; treat skin and urine; decolonise MRSA carriers; intravenous vancomycin or cephalosporin plus gentamicin (up to 24 h); chlorhexidine–alcohol preparation; **antibiotic-coated devices halve infection** (InhibiZone rifampin–minocycline; hydrophilic Titan dipped in antibiotic); **no-touch technique** (under 0.5%).
- **Infected implant:** pus, fixed pump or exposed parts. Remove **all** components.
- **Mulcahy salvage:** remove all parts; wash with dilute povidone-iodine (3 minutes) then saline or antibiotic; change gloves, drapes and instruments; insert a new device at once. Success **84%**; fails with early onset, aggressive organisms, cellulitis. Delayed reimplantation loses length.
- **Intraoperative:** crural perforation, crossover, urethral injury.

- **Others:** erosion or extrusion (catheterisation, rods in insensate men); reservoir erosion into bladder or bowel; reservoir herniation; pump migration; autoinflation; penile shortening; **glans necrosis** (diabetes, smoking, radiation, previous implant); glans hypermobility; cylinder aneurysm.

### Special situations (also know)

- Ischaemic priapism >36 hours: early implantation preserves length.
- Transplant recipients: no extra infection.
- After phalloplasty: single cylinder, high erosion.
- Peyronie's disease: modelling.

### Outcomes

- Patient and partner satisfaction **80–90%**, the highest of any ED treatment; the main complaint is shorter length.

### Recent advances

- Minimal-dilation technique; flat ectopic reservoirs; penile transplantation. Regenerative therapies (Li-ESWT [low-intensity shock wave therapy], stem cells) remain experimental.

### Diagrams to draw

- 3-piece inflatable prosthesis: two cylinders, scrotal pump, reservoir in the space of Retzius.
- Biofilm life cycle.

> **Key points to remember:**
>
> - Prosthesis is the last line but gives 80–90% satisfaction.
> - 3-piece inflatable gives the best rigidity and flaccidity.
> - Malleable rods: nitinol (Tactra), silver (Genesis); almost never fail.
> - Glucose <200 mg/dL, chlorhexidine–alcohol, coated device, no-touch technique.
> - Biofilm: microbes in a matrix; antibiotics fail.
> - Infection 1–3%; infected implant: remove all parts; Mulcahy salvage 84%.
> - Repairs 15% at 5 years.
> - Arterial revascularisation only for young men with post-traumatic focal stenosis.

## LAQ 4 Priapism: diagnosis and management

P2 · P3 **merged from 2**

*Asked as:* P3 LAQ "Priapism: Diagnosis and Management" (also asked as P3 SAQ "Priapism") · P2 SAQ "Management of priapism"

### Definition

- Full or partial erection lasting **≥4 hours** beyond sexual stimulation, or unrelated to it (AUA/SMSNA [American Urological Association/Sexual Medicine Society of North America] 2021–2022).

### Classification (types)

| | **Ischaemic (low flow)** | **Stuttering (recurrent ischaemic)** | **Non-ischaemic (high flow)** |
|---|---|---|---|
| Share | **95%** | Subset of ischaemic | <5% |
| Lesion | Veno-occlusion | Repeated (recurrent) ischaemic episodes | Cavernosal artery–sinusoid (arterio-sinusoidal) fistula after trauma |
| Pain, rigidity | Painful, fully rigid | Painful, recurrent, often nocturnal | Painless, partial |
| Typical cause | ICI (intracavernosal injection), drugs, SCD (sickle cell disease) | SCD | Perineal or needle trauma |
| Urgency | **Emergency** | Emergency per episode + prophylaxis | Not an emergency |

### Aetiology

- **Ischaemic:** ICI (papaverine, phentolamine, PGE1 [prostaglandin E1]), including recreational use; α-blockers, trazodone, antipsychotics, cocaine; heparin or warfarin withdrawal; **SCD** (29–42% of affected men in their lifetime; 63% of paediatric ischaemic priapism); leukaemia (1–5%, high white cell count); penile metastases; scorpion and spider venom.
- **Non-ischaemic:** straddle or coital injury, pelvic fracture, needle injury. Onset is often delayed, when the clot over the arterial tear lyses.

### Pathophysiology

- **Ischaemic:** failed outflow acts like a compartment syndrome. Hypoxia, hypercarbia and acidosis cause sinusoidal thrombosis, smooth-muscle necrosis and TGF-β (transforming growth factor beta)-driven fibrosis, so ED (erectile dysfunction) follows. Irreversible damage starts within hours.
- **Stuttering (molecular):** low basal NO (nitric oxide) from eNOS (endothelial NO synthase) dysfunction downregulates PDE5 (phosphodiesterase 5). When an erection starts, cGMP (cyclic guanosine monophosphate) is not broken down, so relaxation is unopposed. Low Rho-kinase and high adenosine add to this. This explains why daily PDE5 inhibitors prevent episodes (they reset PDE5 regulation).
- **Non-ischaemic:** unregulated, well-oxygenated arterial inflow, so fibrosis is unusual.

## Diagnosis

- **History:** duration, pain, drugs, trauma, SCD or blood disease, prior episodes, baseline erectile function.
- **Examination:** rigid tender corpora with a soft glans and spongiosum (ischaemic); partial, non-tender (high flow); perineal bruising; vital signs before phenylephrine.
- **Corporal blood gas** (cornerstone; decides the type); high flow resembles arterial blood:

| | Colour | pO2 (mmHg) | pCO2 (mmHg) | pH |
|---|---|---|---|---|
| Ischaemic | Dark red | **<30** | **>60** | **<7.25** |
| Flaccid (mixed venous) | Medium red | 40 | 50 | 7.35 |
| Non-ischaemic (arterial) | Bright red | >90 | <40 | 7.40 |

- **PDUS (penile duplex ultrasound, colour Doppler)** when the gas is equivocal: ischaemic = absent cavernosal flow or PSV (peak systolic velocity) <50 cm/s with no diastolic flow; high flow = PSV >50 cm/s, may show the fistula. Angiography only for embolisation.
- Blood count, reticulocytes, haemoglobin electrophoresis, coagulation, toxicology: in parallel, never delaying treatment.

## Management of ischaemic priapism (AUA/SMSNA 2021)

``` Counsel on ED risk → dorsal penile block <36 h: aspiration (19 G, penoscrotal junction, 3 or 9 o'clock) ± saline irrigation + phenylephrine 100–500 µg/mL, 1 mL every 3–5 min, up to 1 h (monitor blood pressure, pulse: reflex bradycardia) → fails → distal shunt (Winter, T-shunt, Al-Ghorab) ± tunnelling → proximal shunt (Quackels) or penoscrotal decompression

> 36 h or refractory → early (immediate) penile prosthesis (or analgesia and counselling) ```

- Aspiration alone succeeds in under 1 in 4. Cold (10 °C) saline irrigation was better than body temperature in an RCT (randomised controlled trial); avoid chilled saline in SCD.
- Ice, oral agents, hydration and IV (intravenous) fluids are adjuncts only; in SCD, intracavernosal treatment still comes first, with concurrent haematology care.

| Shunt | Method |
|---|---|
| **Winter** (distal) | Tru-Cut (biopsy) needle through glans; least effective |
| **Ebbehoj / T-shunt** (distal) | Scalpel through glans; T-shunt rotates blade 90° |
| **Al-Ghorab** (distal, open) | Dorsal glans incision; excise corporal (tunica) tips |
| **Snake tunnelling** | 7/8 Hegar dilator passed proximally through distal shunt |
| **Quackels / Sacher** (proximal) | Cavernosum to spongiosum at bulb; stagger ≥1 cm |
| **Grayhack / Barry** (proximal) | Cavernosum to saphenous / dorsal vein |
| **Penoscrotal decompression** | Proximal corporotomy, clot suction (evacuation); allows prosthesis |

- **ED by duration** (Bennett and Mulhall, SCD series): ≤12 h 0%; 12–24 h 22%; 24–36 h 56%; ≥36 h 100%.
- **Early prosthesis** (Zacharakis 2014): less shortening and fewer revisions than delayed insertion: shortening 3% vs 40%, revision 9% vs 27%, satisfaction 96% vs 60%.

## Stuttering priapism

- An episode lasting ≥4 hours is treated as ischaemic; home self-injection of phenylephrine.
```

- **Prophylaxis:** daily PDE5 inhibitor (Burnett 2014 RCT: primary endpoint missed, but promising), pseudoephedrine, etilefrine; hormonal agents (GnRH [gonadotropin-releasing hormone] agonists, antiandrogens, finasteride) at the cost of libido. **Also know:** digoxin; baclofen.

### Non-ischaemic priapism (AUA/SMSNA 2022)

- Observation, perineal compression and ice; about 60% resolve. Reassess at 4 weeks.
- Persistent: PDUS, then **selective cavernosal artery embolisation** (resolution ≥75%, recurrence 30–40%). For recurrence, repeat embolisation with non-resorbable PVA (polyvinyl alcohol), not surgery. Surgical ligation is a last resort (ED about 50%).

### Complications

- ED, corporal fibrosis, penile shortening; shunt-related oedema, haematoma, urethral injury or fistula, skin necrosis, infection.

**Diagram to draw:** distal (glans) and proximal shunt sites on a penile outline.

> **Key points to remember:**
> - Blood gas decides the type: ischaemic pO2 <30, pCO2 >60, pH <7.25.
> - Aspiration plus phenylephrine first; distal shunt ± tunnelling before proximal.
> - ED risk 0% ≤12 h, 100% ≥36 h; early prosthesis beyond 36 h or if refractory.
> - High flow: observe (about 60% resolve), then embolise; repeat embolisation rather than surgery.

## LAQ 5 Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED

P1

**Asked as:**

- Write in detail about physiology of erection. How will you evaluate a diabetic with ED? — *Practice Set 5 Paper I Q2 (LAQ)*
- Neurotransmitters in erectile function — *Practice Set 12 Paper I Q3e (SAQ)*

**Type:** LAQ (25 marks); also asked as SAQ (10 marks)

(erection physiology also in Practice Set 5 Paper 1, Q2)

### Introduction

- Erection is **cavernous smooth-muscle relaxation**: low cytosolic $Ca^{2+}$ dilates helicine arteries and sinusoids; expanding sinusoids compress emissary veins (veno-occlusion).
- Pathways: parasympathetic **S2–S4** cavernous nerves (erection), sympathetic **T11–L2** (detumescence), pudendal (sensation, ischiocavernosus rigidity).

### Functional anatomy

- **Corpora cavernosa:** sinusoids lined by endothelium, separated by **trabecular smooth muscle**. Flaccid tone depends on smooth-muscle contraction.
- **Tunica albuginea:** inner circular and outer longitudinal layers. **Emissary veins** pass obliquely between them and are compressed when the corpora expand: the **veno-occlusive mechanism**.
- **Arteries:** internal pudendal → common penile → **cavernous (helicine branches)**, dorsal and bulbourethral arteries. Accessory pudendal arteries in about one-third.
- Spongiosum and glans have a thin or absent tunica, so no veno-occlusion; they stay low-pressure.

### Haemodynamic phases

| Phase | Events | Intracavernous pressure |
|---|---|---|
| Flaccid | Smooth muscle contracted; low inflow | Low |
| Latent (filling) | Arterial dilation; **inflow peaks** | Barely rises |
| Tumescence | Sinusoids fill; venous outflow compressed | Rising |
| Full erection | Maximal veno-occlusion | About 100 mmHg |
| Rigid | **Ischiocavernosus** contraction (pudendal nerve) | Several hundred mmHg |
| Detumescence | Initial, slow, fast phases | Falls to baseline |

### Molecular mechanism and peripheral neurotransmitters

- **Relaxation (erection):** NO (nitric oxide) from **nNOS (neuronal nitric oxide synthase)** in cavernous nerves initiates erection; endothelial NOS (eNOS) sustains it.
- NO → soluble guanylyl cyclase → **cGMP (cyclic guanosine monophosphate)** → protein kinase G (PKG) → $K^+$ channels open, $Ca^{2+}$ channels close, $Ca^{2+}$ sequestered → **low cytosolic $Ca^{2+}$** → relaxation.

- Second route: PGE1 (prostaglandin E1), VIP (vasoactive intestinal peptide) → **cAMP (cyclic adenosine monophosphate)** → protein kinase A.
- **PDE5 (phosphodiesterase type 5)** degrades (breaks down) cGMP and ends relaxation (ends / terminates erection); PDE5 inhibitors block it.

### Peripheral pro-erectile transmitters

| Transmitter | Source / mechanism |
|---|---|
| **NO (nitric oxide)** | Principal. **nNOS (neuronal NOS)** initiates; **eNOS (endothelial)** sustains. NO → soluble guanylyl cyclase → **cGMP** → PKG → $K^+$ channels open, $Ca^{2+}$ channels close → relaxation |
| ACh (acetylcholine) | Indirect: endothelial NO release, inhibits adrenergic nerves; atropine does not block erection |
| PGE1 (prostaglandin E1), VIP (vasoactive intestinal peptide), CGRP (calcitonin gene-related peptide) | **cAMP** → PKA (protein kinase A); VIP role minimal in man |
| CO, H2S, EDHF (endothelium-derived hyperpolarising factor) | Gaseous/endothelial relaxants via cGMP or $K^+$ channels |

- O2 is a NOS substrate: low flaccid Po2 (~35 mmHg) limits NO; erect Po2 ~90 mmHg.
- **Also know:** carbon monoxide and hydrogen sulfide also raise cGMP; acetylcholine acts indirectly through endothelial NO (atropine does not block erection).

### Anti-erectile transmitters (flaccidity and detumescence)

- **Contraction (flaccidity): noradrenaline (α1)** (principal), endothelin-1, angiotensin II → raised $Ca^{2+}$ → calmodulin → myosin light-chain kinase → contraction.
- **Noradrenaline (α1)**: principal; α2 prejunctional reduces noradrenaline and NO release.
- **Endothelin-1** (ET-A), angiotensin II (AT1), PGF2α, thromboxane A2: Gq → IP3 → $Ca^{2+}$ rise → $Ca^{2+}$–calmodulin → MLC (myosin light-chain) kinase → contraction.
- **Rho-kinase** inhibits myosin (MLC) phosphatase (calcium sensitisation) and maintains (sustaining) tone.
- **Detumescence:** NO release stops, PDE5 degrades cGMP and sympathetic discharge contracts the smooth muscle, so emissary veins reopen.
- Gap junctions (connexin-43) synchronise the smooth muscle.

## Neural control

| Pathway | Origin | Role |
|---|---|---|
| Parasympathetic | **S2–S4** → pelvic plexus → cavernous nerves | Erection |
| Sympathetic | **T11–L2** → hypogastric nerves | Detumescence, emission |
| Somatic | Pudendal nerve (Onuf nucleus) | Penile sensation; ischiocavernosus rigidity |

- **Three types: psychogenic** (brain, chiefly by reducing sympathetic tone), **reflexogenic** (genital touch, sacral arc; preserved in upper cord injury), **nocturnal** (REM [rapid eye movement] sleep).
- **Central:** MPOA (medial preoptic area) and **PVN (paraventricular nucleus)**. Dopamine and oxytocin promote; serotonin, prolactin, GABA (γ-aminobutyric acid) and opioids inhibit.

**Central neurotransmitters (PVN [paraventricular nucleus], MPOA [medial preoptic area])**

| Promote | Inhibit |
|---|---|
| **Dopamine** (D2 in PVN; apomorphine) | **Serotonin** (5-HT1A, 5-HT2; but 5-HT2C pro-erectile) |
| **Oxytocin** | GABA (γ-aminobutyric acid) |
| Central NO | Opioids (high dose), cannabinoids |
| Central noradrenaline; melanocortins (MC4R) | **Prolactin** |

- Dopamine and oxytocin converge on NO in the PVN. Noradrenaline is pro-sexual centrally but anti-erectile peripherally.

**Clinical relevance**

- PDE5 inhibitors, intracavernosal alprostadil (cAMP), papaverine, phenylephrine for priapism (α1), SSRIs (selective serotonin reuptake inhibitors) cause dysfunction, hyperprolactinaemia lowers libido.

**Diagram to draw:** NO–cGMP pathway in the cavernous smooth-muscle cell.

## Evaluation of a diabetic with ED (erectile dysfunction)

**Why diabetics:** ED is commoner, earlier and more severe in DM (diabetes mellitus). Mechanisms: endothelial NO loss, autonomic neuropathy, arteriopathy, smooth-muscle loss and fibrosis (veno-occlusive failure), AGEs (advanced glycation end products) and low testosterone. ED is a **sentinel of cardiovascular disease**.

**1. History**

- Onset, duration (≥3 months), severity, morning erections, libido, ejaculation; partner.
- **IIEF-5 (International Index of Erectile Function, SHIM):** 5–7 severe, 8–11 moderate, 12–16 mild-to-moderate, 17–21 mild, 22–25 no ED.
- Diabetes: type, duration, **glycaemic control**, complications (retinopathy, nephropathy, neuropathy, foot disease), autonomic symptoms (postural giddiness, gastroparesis, bladder dysfunction).
- Cardiovascular: angina, exercise tolerance, smoking, hypertension, dyslipidaemia, nitrates.
- Drugs: antihypertensives, antidepressants. Depression screen.

**2. Examination**

- BMI, waist, blood pressure, peripheral pulses.
- Genitalia: Peyronie plaques, testicular size, gynaecomastia.
- Neurological: genital sensation, **bulbocavernosus reflex**, foot sensation. Penile neuropathy can exist without foot neuropathy.

**3. Laboratory**

- **HbA1c (glycated haemoglobin)** and fasting glucose, **lipid profile**, creatinine/eGFR (estimated glomerular filtration rate), urine albumin.
- **Morning total testosterone**; if low, repeat with free testosterone, LH (luteinising hormone), prolactin. TSH (thyroid-stimulating hormone) if indicated.

**4. Cardiovascular risk stratification (Princeton Consensus)**

- **High risk** (unstable angina, recent infarction, uncontrolled hypertension, some arrhythmias): defer sexual activity and treatment; cardiology referral. **Indeterminate:**

cardiac assessment (e.g. stress test), then restratify. **Low:** start treatment.

**5. Specialised tests (selected cases)**

- **CIS (combined intracavernous injection and stimulation):** alprostadil 10–20 µg; a rigid erection excludes major vascular disease.
- **Penile duplex Doppler:** PSV (peak systolic velocity) **<25 cm/s = arterial insufficiency** (>35 normal); EDV (end-diastolic velocity) **>5 cm/s** and RI (resistive index) **<0.75 = veno-occlusive dysfunction**.
- **NPTR (nocturnal penile tumescence and rigidity), RigiScan:** separates psychogenic from organic ED.
- **Neurophysiological:** bulbocavernosus reflex latency, dorsal nerve conduction velocity, biothesiometry.
- DICC (dynamic infusion cavernosometry and cavernosography) and angiography: rarely, as diffuse diabetic vascular disease is not suited to revascularisation.

`Diabetic with ED → history + IIEF-5 + examination → HbA1c, lipids, testosterone, renal function → Princeton cardiac risk → treat risk factors → PDE5 inhibitor trial → failure → duplex/CIS → ICI or prosthesis`

> **Key points to remember:**
> - nNOS initiates, eNOS maintains; PDE5 ends (terminates) erection.
> - Noradrenaline α1 is the main anti-erectile transmitter.
> - Dopamine/oxytocin promote; serotonin/prolactin inhibit.
> - Rigid phase: ischiocavernosus, suprasystolic pressure.
> - Duplex: PSV <25 cm/s arterial; EDV >5 cm/s, RI <0.75 venous leak.
> - Diabetic ED: check HbA1c, testosterone and cardiac risk.

## LAQ 6 Discuss the role of urologist in reproductive endocrinology and male infertility practice.

P2

**Type:** LAQ (25 marks)
**Source:** 2022-Summer Paper 2, Q1

### Introduction

- **Infertility:** no conception after 12 months of regular unprotected intercourse; about **15% of couples** (WHO, World Health Organization). The male contributes about half, yet many men are never examined.
- The urologist (andrologist) has five roles: **diagnose** the male factor, **treat endocrine causes**, **correct surgical causes**, **retrieve sperm for ART (assisted reproductive technology)**, and protect **general health and future fertility**.
- Commonest diagnoses: idiopathic, **varicocele** (commonest correctable), obstruction.

### 1. Diagnostic role

- **History:** duration, coital timing (semen is best after 1–2 days' abstinence), partner's age (fecundity falls after 35), cryptorchidism, mesh herniorrhaphy, RPLND (retroperitoneal lymph node dissection), diabetes, spinal injury, smoking, cannabis, opioids, **exogenous testosterone or anabolic steroids**.
- **Examination:** androgenisation; **testis size** (reflects spermatogenic mass); epididymis; **both vasa present?**; varicocele (standing, Valsalva).
- **Semen analysis**, at least two. WHO 6th edition (2021) 5th centiles: volume 1.4 mL, concentration 16 million/mL, total count 39 million, progressive motility 30%, normal forms 4%. These describe fertile men, not fertility cut-offs.
- **Genetics:** karyotype and **Y-chromosome microdeletion** (AZF, azoospermia factor) in non-obstructive azoospermia or severe oligospermia; **CFTR (cystic fibrosis transmembrane conductance regulator)** testing of both partners in CBAVD (congenital bilateral absence of the vas deferens).
- **Imaging:** scrotal US (ultrasound); **TRUS (transrectal US)** if volume <1 mL (ejaculatory duct obstruction, EDO), after post-ejaculate urine excludes retrograde ejaculation; renal US in vasal agenesis.

### 2. Endocrine evaluation (the urologist's reproductive endocrinology)

- Spermatogenesis needs a **high intratesticular testosterone (T)**. Morning total T + SHBG (sex hormone-binding globulin) + albumin gives bioavailable T; low is below about **280–300 ng/dL**; repeat borderline values (T is pulsatile and circadian).

| Test | What it tells |
|---|---|
| LH (luteinising hormone) | Localises low T: high → testicular (Leydig) failure; low/normal → pituitary or hypothalamus |
| FSH (follicle-stimulating hormone) / inhibin B | Germ-cell mass. Azoospermia with **FSH >7.6 IU/L and testis long axis ≤4.6 cm → spermatogenic failure**; FSH ≤7.6 and >4.6 cm → obstruction |
| Oestradiol (E2) | **T (ng/dL) : E2 (pg/mL) <10** → aromatase excess (obesity) |

| Test | What it tells |
|---|---|
| Prolactin | Only with low T, low libido/ED (erectile dysfunction), headache or field defect |
| TSH (thyroid-stimulating hormone), cortisol, 17-hydroxyprogesterone | Only if signs; CAH (congenital adrenal hyperplasia) |

## 3. Medical (endocrine) treatment

| Condition | Mechanism | Treatment |
|---|---|---|
| **Hypogonadotropic hypogonadism** (Kallmann, pituitary) | Testis normal, signal missing: most treatable cause | **hCG (human chorionic gonadotropin) 1500–5000 IU 2–3 times weekly**, add hMG (human menopausal gonadotropin) 75 IU or recombinant FSH; pulsatile GnRH (gonadotropin-releasing hormone) pump for hypothalamic failure. Sperm may take 1–2 years |
| **Anabolic steroids / exogenous T** | Negative feedback suppresses LH → intratesticular T falls → azoospermia | Stop; hCG ± clomiphene to speed recovery. **Never give T to a man wanting fertility** (AUA/ASRM, American Urological Association/American Society for Reproductive Medicine) |
| **Hyperprolactinaemia** | Prolactin inhibits GnRH | Repeat level; pituitary MRI (magnetic resonance imaging) if high; **cabergoline** (fewer side effects than bromocriptine) |
| Partial hypogonadism, obesity | Aromatisation of T to E2 suppresses LH | Weight loss; **clomiphene 25 mg daily** (off-label), **anastrozole 1 mg daily** or letrozole if T:E2 <10 |
| **CAH** | Adrenal androgens suppress gonadotropins; testicular adrenal rest tumours | Glucocorticoids |
| Primary testicular failure (Klinefelter 47,XXY) | Germ cells absent or few | No drug restores sperm; exogenous T worsens it; **micro-TESE** (microdissection testicular sperm extraction) |

## 4. Surgical role

| Problem | Operation |
|---|---|
| Clinical varicocele + abnormal semen | **Microsurgical subinguinal varicocelectomy** or embolisation: improves semen, DNA fragmentation and pregnancy. Not for US-only varicocele |
| Vasal or epididymal obstruction | **Vasovasostomy / vasoepididymostomy** (microsurgical intussusception) |
| EDO | **TURED** (transurethral resection of ejaculatory ducts) |
| Non-obstructive azoospermia | **Micro-TESE**, best technique; no operation for **AZFa/AZFb** deletions (no sperm found) |
| Unreconstructable obstruction | PESA/MESA (percutaneous/microsurgical epididymal sperm aspiration), TESA (testicular sperm aspiration) |
| Retrograde ejaculation; anejaculation | Sympathomimetics, urine sperm retrieval; **penile vibratory stimulation**, electroejaculation |

### 5. Partner of the ART team, counselling and men's health

- Choose the least invasive route: natural conception, IUI (intrauterine insemination), IVF/ICSI (in vitro fertilisation/intracytoplasmic sperm injection). **Cryopreserve sperm** at retrieval.
- **Fertility preservation:** sperm banking before chemotherapy, radiotherapy or RPLND; **onco-TESE** if azoospermic.
- **Genetic counselling:** AZFc deletion passes to sons; CFTR risk to offspring; paternal age ≥40 years raises offspring risk.
- **Infertility is a marker of general health:** higher risk of **testis cancer** (scan every azoospermic man), later comorbidity and mortality.

**Recent advances:** sperm DNA fragmentation testing; microfluidic sperm selection; genomic panels; prepubertal testicular tissue cryopreservation (experimental).

**Also know:** antisperm antibodies; globozoospermia and ciliary dyskinesia (ICSI); androgen receptor defects (high T and LH).

**Diagram to draw:** hypothalamic–pituitary–testis axis with feedback loops (T and E2 on LH; inhibin B on FSH).

> **Key points to remember:**
> - FSH >7.6 IU/L + testis ≤4.6 cm = spermatogenic failure.
> - Exogenous testosterone is a contraceptive: never give it for infertility.
> - Hypogonadotropic hypogonadism: hCG ± FSH; most treatable cause.
> - Varicocelectomy only for palpable varicocele with abnormal semen; micro-TESE for non-obstructive azoospermia.

## Short answer questions (SAQ)

## SAQ 1 Male reproductive immunology and immunobead test

P1 · P3 **merged from 2**

*Asked as:* P1 SAQ "Male reproductive immunology" · P3 SAQ "Immunobead test for semen"

### Introduction

- Haploid germ cells appear at puberty, after self-tolerance is established, so their antigens are "foreign". The testis is therefore **immune-privileged**.

### Immune privilege

#### Anatomical barrier: blood–testis barrier (BTB)

- **Tight junctions between Sertoli cells**, formed at puberty, divide a **basal compartment** (spermatogonia, plasma-like fluid) from an **adluminal compartment** (later spermatocytes, spermatids). If the barrier is breached, sperm antigens meet the immune system.

#### Functional barrier

- A small antigen leak at the **rete testis, efferent ductules and epididymis** induces suppressor T cells and tolerance; a large load provokes immunity.
- Sertoli and local mediators: galectin-1, TGF-β (transforming growth factor-β), interferon-γ, soluble Fc receptor, **androgens**.

- Insult before puberty does not induce antibodies; the barrier also limits chemotherapy (isolated testicular relapse).

### Antisperm antibodies (ASA)

- **Causes:** obstruction (**vasectomy** commonest); torsion, trauma, biopsy; infection (orchitis, prostatitis, STI [sexually transmitted infection]); heat (varicocele, cryptorchidism); genetic (human leukocyte antigen HLA-B28).
- **Effects:** agglutination, poor motility, failure to cross (poor penetration of) cervical mucus, failure to bind the zona and impaired fertilisation; head-bound antibody matters most.
- **Indications to test:** sperm agglutination or isolated asthenozoospermia; unexplained infertility; after vasectomy reversal.

### Tests for ASA: immunobead test (IBT) and MAR

- The IBT detects ASA bound to sperm (direct) or free in body fluids (indirect).
- **Principle:** polyacrylamide beads coated with rabbit anti-human **IgG, IgA or IgM**.

| Test | Method | Use |
|---|---|---|
| **Direct IBT** | Patient's motile sperm washed free of seminal plasma; mixed with beads coated with anti-human IgG/IgA; motile sperm with attached beads counted | Sperm-bound antibody: % bound, class and site (head/tail) |
| **Indirect IBT** | Antibody-free donor sperm incubated with heat-inactivated patient serum, seminal plasma or female serum/cervical mucus; then direct IBT done | Free antibody; patient sperm too few or immotile; female partner |
| **MAR (mixed antiglobulin reaction)** | Fresh unwashed semen + immunoglobulin (Ig) G (or IgA)-coated latex beads + antiglobulin (anti-IgG); beads stick to antibody-coated motile sperm; mixed sperm–bead clumps counted | Quick, cheap screen; IgG/IgA only |

- Report: **% of motile sperm with beads**, immunoglobulin class, and **site** (head, midpiece, tail). Tail-tip binding is ignored.
- **WHO (World Health Organization) 2010 (5th edition) threshold: ≥50% of motile sperm bound is clinically significant.**
- **Head binding** and **IgA** matter most (IgA is produced locally and blocks mucus penetration; IgM is too large to enter semen).
- Both direct tests need motile sperm.
- **Also know:** ELISA (enzyme-linked immunosorbent assay) and flow cytometry; serum tests correlate poorly with sperm-bound antibody.

### Management of ASA

| Treatment | Comment |
|---|---|
| Condoms | Lack good evidence |
| Corticosteroids | Two controlled trials conflict; risk of avascular necrosis; not recommended |
| Sperm washing + intrauterine insemination (IUI) | Bypasses cervical mucus; for mild binding; modest benefit, lacks good evidence (washing removes unbound, not bound, antibody) |

| Treatment | Comment |
|---|---|
| **IVF/ICSI (in vitro fertilisation/intracytoplasmic sperm injection)** | Preferred, usual treatment; fertilisation largely unaffected, bypassing antibody effects |

### Infection and inflammation

- **Leukocytospermia:** >1 × $10^6$ peroxidase-positive leukocytes/mL. Reactive oxygen species and cytokines damage sperm.
- **Orchitis:** mumps (post-pubertal, atrophy); bacterial (*Chlamydia*/gonococcus if young, *E. coli* if older); tuberculous; **autoimmune**, primary (antibodies with azoospermia) or secondary to vasculitis.
- **Also know:** sperm DNA fragmentation testing; prostatitis harms sperm via oxidative stress and ASA; seminal plasma suppresses female tract immunity.

### Diagrams to draw

- Seminiferous tubule with Sertoli tight junctions and compartments.
- Motile sperm with beads attached at head and tail.

> **Key points to remember:**
> - BTB = Sertoli tight junctions; plus active tolerance (galectin-1, TGF-β, androgens).
> - ASA: vasectomy commonest; ≥50% bound significant; head-bound IgA matters most.
> - Direct IBT = sperm-bound antibody; indirect = antibody in serum, seminal plasma or mucus.
> - Treatment of choice: IVF/ICSI.
> - Leukocytospermia >1 million/mL.

## SAQ 2 Premature ejaculation

P1 · P3 **merged from 2**

*Asked as:* P1 SAQ "Premature Ejaculation" · P3 SAQ "Premature ejaculation."

### Definition

- **ISSM (International Society for Sexual Medicine) unified definition (2014):** all three needed: 1. Ejaculation always or nearly always within **about 1 minute** of vaginal penetration (lifelong), or a clinically significant, bothersome fall in latency, often to **≤3 minutes** (acquired). 2. Inability to delay ejaculation on all or nearly all penetrations. 3. Negative personal consequences: distress, frustration, avoidance of intimacy.
- **AUA/SMSNA (American Urological Association/Sexual Medicine Society of North America) 2020:** lifelong PE (premature ejaculation) within about **2 minutes**.
- **Types:** lifelong, acquired, variable and subjective (normal latency).

### Physiology and aetiology

- Ejaculation = **emission** (sympathetic T10–L2) then **expulsion** (somatic S2–S4, pudendal). **Serotonin brakes and dopamine accelerates**.
- **Lifelong PE:** neurobiological and genetic; 5-HT2C hyposensitivity and/or 5-HT1A hypersensitivity.

- **Acquired:** performance anxiety, relationship problems, **ED (erectile dysfunction)**, chronic prostatitis, hyperthyroidism, drug withdrawal.

### Evaluation

- Diagnosis is by **history**: latency, control, bother; lifelong or acquired.
- **IELT (intravaginal ejaculatory latency time):** self-estimate in clinic; stopwatch in trials. Population median 5.4 minutes.
- **PEDT (Premature Ejaculation Diagnostic Tool):** 5 items; score ≥11 suggests PE, 9–10 probable PE. Also PEP (Premature Ejaculation Profile) and IPE (Index of Premature Ejaculation).
- IIEF-5 (International Index of Erectile Function) for ED; examination; TSH (thyroid-stimulating hormone) only if suggested.

**Office algorithm (diagram to draw):**

1. Confirm PE.
2. Secondary to ED? → treat the cause.
3. Otherwise → classify lifelong or acquired.

### Management

- **Psychosexual education** for all; behavioural therapy (stop–start, squeeze).
- Acquired PE: treat the cause first (ED, prostatitis, thyroid).

| Drug | Dose and timing |
|---|---|
| **Dapoxetine** (short-acting SSRI [selective serotonin reuptake inhibitor]; only licensed oral drug) | 30–60 mg, 1–3 h before intercourse |
| Paroxetine (strongest delay) 10–40 mg, sertraline, fluoxetine, citalopram | Daily, off-label; full effect 2–3 weeks; do not stop abruptly |
| Clomipramine | 12.5–50 mg daily or 3–4 h before |
| **Topical lidocaine–prilocaine** spray (licensed in Europe 2014) | 5–30 min before (P1: at least 5 min before); condom avoids partner numbness |
| PDE5Is (phosphodiesterase type 5 inhibitors) | With ED, alone or with SSRI |
| Tramadol | Low evidence; addiction risk |

- SSRIs raise synaptic serotonin, which delays ejaculation. Avoid in bipolar disorder.
- **Also know:** high dropout from drug therapy.

> **Key points to remember:**
> - ISSM: about 1 min (lifelong), ≤3 min (acquired) plus poor control plus distress.
> - Lifelong is neurobiological; acquired is usually secondary.
> - Dapoxetine 30–60 mg on demand; paroxetine gives the greatest delay.

## SAQ 3 Oligospermia and oligoasthenozoospermia

P2 · P3 **merged from 2**

*Asked as:* P3 SAQ "Oligospermia and oligoasthenozoospermia" (asked as "Oligoanaesthospermia" and "Oligospermia") · P2 SAQ "Oligospermia."

*Assumed meaning (oligoanaesthospermia): oligoasthenozoospermia (low sperm count with poor motility).*

## Definition (WHO [World Health Organization] 2021, 6th edition, 5th centiles)

- **Oligozoospermia:** sperm concentration below the lower reference limit: **WHO 2021 (6th edition) 16 million/mL** (5th centile; total count 39 million per ejaculate). Earlier thresholds were 20 million/mL (1999, the legacy cut-off) and 15 million/mL (2010).
- **Severity:** mild to moderate 5–16 million/mL; **severe oligozoospermia <5 million/mL**; cryptozoospermia: sperm only in a centrifuged pellet.

| Parameter (WHO 2021 5th centiles) | Lower reference value |
|---|---|
| Concentration | 16 million/mL (total 39 million) |
| Volume | 1.4 mL |
| Progressive motility | 30% |
| Total motility | 42% |
| Normal forms (strict) | 4% |
| Vitality | 54% |

- These centiles come from fertile men; they are a reference, not pass/fail fertility cut-offs. Define oligospermia in context: read a single value with testis size, FSH (follicle-stimulating hormone) and repeated values.

## Causes

- **Varicocele** (commonest correctable; heat and oxidative stress) and **idiopathic** (largest group).
- **Pre-testicular (hormonal):** hypogonadotropic hypogonadism, **exogenous testosterone or anabolic steroids**, obesity (raised oestradiol), hyperprolactinaemia, opioids.
- **Testicular: varicocele**, cryptorchidism, mumps orchitis, torsion, chemotherapy or radiotherapy, gonadotoxins; genetic (Klinefelter, Klinefelter mosaic, **AZFc [azoospermia factor c] deletion**).
- **Post-testicular (motility):** infection or **leucocytospermia** (over 1 million/mL), antisperm antibodies, partial ejaculatory duct obstruction (EDO; low volume), PCD (primary ciliary dyskinesia).
- **Lifestyle:** smoking, alcohol, cannabis, heat, drugs, gonadotoxins.

## Evaluation

- **At least two semen analyses, 2–3 weeks apart**, same abstinence (WHO 2–7 days of abstinence; parameters peak at 1–2 days).
- History and examination: testis size/volume, varicocele, vasa, secondary sex characters.
- Morning **testosterone, LH (luteinising hormone), FSH**; oestradiol and prolactin if testosterone is low.
- **Under 5 million/mL** (especially with raised FSH or small testes): **karyotype and Y-microdeletion** testing (AUA/ASRM [American Urological Association/American Society for Reproductive Medicine] 2020 guideline).
- Selected: vitality test (eosin or hypo-osmotic swelling), antisperm antibodies, semen culture, scrotal Doppler, TRUS (transrectal ultrasonography) if volume is low (<1.0 mL), DNA fragmentation.

### Management

- **General:** stop smoking, weight loss, avoid heat; **stop exogenous testosterone** and gonadotoxins; treat infection.
- **Varicocelectomy** (microsurgical subinguinal) for a **palpable** varicocele with abnormal semen; it improves semen parameters and pregnancy rates. Never treat a subclinical one.
- **Medical:**
- Hypogonadotropic: hCG (human chorionic gonadotropin) ± FSH.
- Idiopathic with low testosterone: off-label clomiphene 25 mg daily, or anastrozole 1 mg daily if the testosterone:oestradiol ratio is low (<10).
- Antioxidants: weak evidence.
- **ART (assisted reproductive technology):** mild: IUI (intrauterine insemination); severe or failed IUI: **IVF/ICSI (in vitro fertilisation/intracytoplasmic sperm injection)**. Bank sperm in AZFc carriers because counts may fall.
- Spermatogenesis takes about 64 days, so judge response (reassess) after 3–6 months.

**Diagram to draw:** hypothalamic–pituitary–testis axis with sites of treatment.

> **Key points to remember:**
> - WHO 2021: concentration 16 million/mL, progressive motility 30%; severe <5 million/mL.
> - Karyotype and Y-microdeletion if under 5 million/mL.
> - Repair palpable varicoceles only; never give testosterone.

## SAQ 4 Management of Peyronie's disease

P1

**Type:** SAQ (10 marks)
**Source:** 2017-Winter Paper 1, Q3a

### Definition

- **PD (Peyronie's disease)** (AUA (American Urological Association) 2015): acquired fibrosis of the tunica albuginea, with pain, deformity, ED (erectile dysfunction) and/or distress. Prevalence 3–20%; up to half worsen untreated.

### Phases

- **Acute (active, first 12–18 months):** pain, worsening deformity; aim to stop progression. **Stable:** pain gone, deformity unchanged for ≥3–6 months; aim to straighten.

### Evaluation

- History: pain, deformity, Dupuytren disease, ability to penetrate. **Key question:** "If your penis were straight, would your current rigidity allow intercourse?" (predicts postoperative function).
- Plaque, SPL (stretched penile length); **office intracavernosal injection ± duplex Doppler ultrasound** before any invasive treatment (curvature by goniometer, calcification, vascular status).
- **Kelami grading:** ≤30° mild, 31–60° moderate, >60° severe; note hourglass or hinge. PDQ (Peyronie's Disease Questionnaire).

### Non-surgical treatment

| Option | Evidence/position |
|---|---|
| NSAIDs (non-steroidal anti-inflammatory drugs) | For pain (only oral agent AUA supports) |
| Oral vitamin E, tamoxifen, potaba, pentoxifylline | No reliable curvature benefit; not recommended |
| **CCH (collagenase *Clostridium histolyticum*)** | Stable disease, curve 30–90°, intact erections. **IMPRESS I/II:** 34% improvement vs 18% placebo. Up to 4 cycles (2 injections of 0.58 mg) with modelling. Risk: corporal fracture |
| Verapamil, interferon α-2b (intralesional) | May offer; stabilise and modestly improve |
| **Traction therapy** | Gains length, reduces curvature; useful in acute phase |
| ESWT (extracorporeal shock wave therapy) | Pain only; no effect on curvature |

### Surgery

- **Indications:** stable disease (≥12 months, stable ≥6 months), deformity preventing intercourse, failed conservative therapy.

```
Erection adequate (± PDE5I (phosphodiesterase-5 inhibitor))? ├─ No → penile prosthesis (inflatable) + manual modelling; │ residual >30° → tunical incision; graft if defect >2 cm └─ Yes → curve <60–70°, no hourglass/hinge → plication (Nesbit, 16-dot, tunica albuginea plication) curve >60–70°, hourglass, hinge, short penis → plaque incision/partial excision + graft (pericardium, small intestinal submucosa)
```

- **Plication** shortens the convex side: simple, low ED risk, but loses length.
- **Grafting** restores length and girth but carries higher ED risk (5–54%).

**Diagram to draw:** Nesbit wedge excised opposite the plaque.

> **Key points to remember:**
> - Ask the "straight penis" question; duplex before surgery.
> - CCH: IMPRESS, stable 30–90° curve, intact erections.
> - Surgery: stable ≥12 months; plication <60–70°, graft complex curves, prosthesis if ED.

## SAQ 5 Haematospermia

P1

**Asked as:**

- Hematospermia — *Winter 2025 Paper I Q3c (SAQ)*
- Hematospermia — *Practice Set 9 Paper I Q3d (SAQ)* (also answered in 2025-Winter Paper 1, Q3c)

**Type:** SAQ (10 marks)

### Definition

- Blood in the ejaculate. It alarms patients but is usually **benign and self-limiting**, most often from non-specific prostatic or seminal vesicle inflammation.

- **Primary (idiopathic):** no cause found, often young men. **Secondary:** a cause is identified.
- **Pseudo-haematospermia:** blood from the partner (partner's blood; condom test) or haematuria.

## Causes

| Group | Examples |
|---|---|
| **Iatrogenic (commonest identifiable)** | Prostate biopsy (about 10–50%, lasts about a month), TURP (transurethral resection of the prostate), radiotherapy, BPH (benign prostatic hyperplasia) procedures |
| Inflammation and infection | Prostatitis, vesiculitis, urethritis, STI (sexually transmitted infection), **genitourinary tuberculosis**, schistosomiasis |
| Obstruction and cysts | Ejaculatory duct obstruction or calculi; seminal vesicle cyst (Zinner syndrome) or utricle cyst |
| Tumours | **Prostate cancer**; rarely urethral, bladder or seminal vesicle |
| Vascular | Prostatic varices, BPH vessels, AV (arteriovenous) malformation |
| Systemic | Hypertension, bleeding disorders, anticoagulants, liver disease; also abstinence, trauma |

## Evaluation

- **History:** episodes, recent biopsy, LUTS (lower urinary tract symptoms), haematuria, TB (tuberculosis) contact, anticoagulants.
- **Examination:** blood pressure, genitalia, **DRE (digital rectal examination)**.
- **Basic tests:** urinalysis and culture, semen culture, STI screen, **PSA (prostate-specific antigen)** in men over 40, coagulation if indicated.

**Who needs full work-up (full work-up if):** age **≥40**, persistent beyond several weeks or recurrent, haematuria, abnormal DRE or PSA, risk factors for TB or cancer, or a distressed patient.

- **TRUS (transrectal ultrasound):** cysts, calculi, dilated seminal vesicles (>1.5 cm) or ejaculatory ducts (>2.3 mm) suggest obstruction.
- **MRI (magnetic resonance imaging) pelvis:** best for seminal vesicle haemorrhage, cysts and prostate cancer.
- **Cystoscopy and urine cytology**; early-morning urine for acid-fast bacilli.

## Management

- **Young, single episode, normal tests: reassure**.
- **Treat the cause:** antibiotics or antitubercular drugs; control BP (blood pressure); correct coagulopathy.
- **Persistent prostatic bleeding:** 5-ARI (5-alpha-reductase inhibitor), e.g. finasteride.
- **Refractory or obstructive:** TURED (transurethral resection of ejaculatory ducts), cyst unroofing or aspiration, **seminal vesiculoscopy** (4–6 Fr small scope through the ejaculatory duct or utricle, irrigation, laser for stones).

**Diagram to draw:** algorithm: <40 and single episode → reassure; ≥40 or persistent → PSA, TRUS/MRI, cystoscopy.

**Key points to remember:**

- Usually benign; prostate biopsy is the commonest identifiable cause.
- Full work-up if ≥40, persistent or recurrent, or associated haematuria (or with haematuria).
- Exclude prostate cancer and genitourinary tuberculosis.
- TRUS: seminal vesicle >1.5 cm or ejaculatory duct >2.3 mm suggests obstruction.

## SAQ 6 Avanafil

P1

**Type:** SAQ (10 marks)
**Source:** Practice Set 21 SAQ Sheet, Q3e

### Introduction

- Second-generation oral **PDE5I (phosphodiesterase type 5 inhibitor)**; **FDA (US Food and Drug Administration) approved 2012**; Europe 2013. Structure differs most from the other PDE5Is.

### Mechanism

- Sexual stimulation releases NO (nitric oxide), which raises cGMP (cyclic guanosine monophosphate) and relaxes cavernosal smooth muscle. PDE5 breaks down cGMP; avanafil blocks this.
- **Enhances, does not initiate, erection**: stimulation is needed.
- **More selective for PDE5** over PDE6 (retina) and PDE1 than sildenafil, so **no visual disturbance** in the comparison table.

### Pharmacokinetics (Campbell Table 72.7)

| Feature | Avanafil | Sildenafil | Tadalafil |
|---|---|---|---|
| Time to peak | **0.3–0.5 h (fastest)** | 0.8 h | 2 h |
| Half-life | 3–5 h | 3–5 h | 17.5 h |
| Fatty food | Reduces absorption | Reduces | No effect |
| Doses | **50, 100, 200 mg** | 25–100 mg | 5–20 mg |
| Blue vision | No | Yes | Rare |

- Taken about 15–30 minutes before sex; start 100 mg; maximum once daily. Metabolised by CYP3A4 (cytochrome P450 3A4).

### Efficacy

- Effective in general ED (erectile dysfunction), **diabetes** and **after nerve-sparing radical prostatectomy** (REVIVE trial programme): improves IIEF (International Index of Erectile Function) erectile scores and successful-intercourse rates, some within 15 minutes. Efficacy equivalent across PDE5Is; no head-to-head superiority.

### Adverse effects

- Headache, flushing, nasal congestion, dyspepsia; back pain rare.

### Contraindications and interactions

- **Nitrates in any form**: absolute; no nitrate within 12 hours of a dose.
- Avoid strong CYP3A4 inhibitors (ketoconazole, ritonavir); maximum 50 mg with moderate inhibitors.
- **α-blockers:** start 50 mg once stable (additive hypotension).
- Caution: aortic stenosis, recent myocardial infarction or stroke, hypotension, retinitis pigmentosa. Not recommended in severe renal impairment, dialysis or severe hepatic impairment.
- **Also know:** proposed prophylaxis for recurrent ischaemic priapism; can cause priapism rarely.

### Advantages

- **Fastest onset**, fewer visual and muscle effects; suits on-demand spontaneous use.

**Diagram to draw:** NO–cGMP pathway in cavernosal smooth muscle with the PDE5 block.

> **Key points to remember:**
> - Peak 0.3–0.5 h; doses 50, 100, 200 mg.
> - Highly PDE5-selective: no blue vision.
> - Nitrates absolutely contraindicated.

## SAQ 7 CRE & BEP

P2

**Type:** SAQ (10 marks)
**Source:** Practice Set 14 Paper 2, Q3f

*"CRE" is read as carbapenem-resistant Enterobacterales, and BEP as bleomycin, etoposide and cisplatin chemotherapy for germ cell tumours; the two are answered separately.*

### Part A: CRE (carbapenem-resistant Enterobacterales)

- **Definition:** Enterobacterales (Klebsiella pneumoniae, Escherichia coli, Enterobacter) resistant to at least one carbapenem, or producing a carbapenemase.

| Mechanism | Example |
|---|---|
| Class A carbapenemase | KPC (Klebsiella pneumoniae carbapenemase) |
| Class B metallo-β-lactamase | **NDM (New Delhi metallo-β-lactamase)**: common in India |
| Class D | OXA-48 |
| Non-enzymatic | ESBL (extended-spectrum β-lactamase) or AmpC plus porin loss |

- **Urological risk:** catheters, stents, prolonged stay, prior carbapenem or fluoroquinolone use, transrectal biopsy, ICU (intensive care unit) care.
- **Treatment** (by culture and carbapenemase type; ID (infectious disease) input):
- KPC, OXA-48: ceftazidime–avibactam; meropenem–vaborbactam or imipenem–relebactam for KPC.
- **NDM:** ceftazidime–avibactam plus aztreonam, or cefiderocol.
- Cystitis: single-dose aminoglycoside, fosfomycin; polymyxins as last resort. Avoid tigecycline for UTI (urinary tract infection) (low urine levels).

- **Prevention:** antibiotic stewardship, contact precautions, rectal screening of high-risk patients, early catheter removal, transperineal rather than transrectal biopsy.

### Part B: BEP (bleomycin, etoposide, cisplatin)

- Standard first-line chemotherapy for metastatic **GCT (germ cell tumour)**; replaced PVB (less neuromuscular toxicity).
- **Usual 21-day cycle:** cisplatin 20 mg/m² days 1–5, etoposide 100 mg/m² days 1–5, bleomycin 30 units days 1, 8, 15.

| IGCCCG (International Germ Cell Cancer Collaborative Group) risk | Regimen |
|---|---|
| Good | BEP ×3 or EP ×4 |
| Intermediate/poor | BEP ×4 (VIP ×4 if lungs poor) |
| Stage I NSGCT (non-seminomatous GCT), high risk | BEP ×1 adjuvant |

- **Toxicity:** bleomycin lung fibrosis (keep $FiO_2$ low at later RPLND (retroperitoneal lymph node dissection)) and Raynaud phenomenon; cisplatin nephrotoxicity, ototoxicity, neuropathy; myelosuppression; infertility (bank sperm); late cardiovascular disease and second cancers.

**Diagram to draw:** BEP 21-day cycle timeline.

> **Key points to remember:**
> - CRE: NDM common in India; ceftazidime–avibactam + aztreonam for metallo-β-lactamase.
> - BEP ×3 good risk, ×4 intermediate/poor; EP ×4 if bleomycin avoided.
> - Bleomycin lung: low $FiO_2$ and restricted fluids at surgery.

## SAQ 8 Genetic causes of male infertility. Enumerate and classify.

P3

**Type:** SAQ (10 marks)
**Source:** 2017-Summer Paper 3, Q3d

### Introduction

- Genetic defects underlie many cases of **azoospermia** and **severe oligozoospermia**. They decide **sperm retrieval odds** and **transmission to offspring** through ICSI (intracytoplasmic sperm injection).

### Classification

| Group | Condition | Key features |
|---|---|---|
| **1. Chromosomal, numerical** | **Klinefelter syndrome (47,XXY)** | **Commonest genetic cause**; 1 in 500–1000; small testes, high FSH (follicle-stimulating hormone), low testosterone, usually azoospermia. Non-mosaic in 80–90%. Testicular sperm in about 50% (micro-TESE, microdissection testicular sperm extraction) |
| | 47,XYY; **46,XX male** (SRY gene translocated to X) | XX male: azoospermia, no sperm retrievable |

| Group | Condition | Key features |
|---|---|---|
| **2. Chromosomal, structural** | **Robertsonian and reciprocal translocations**, inversions, ring Y | Oligozoospermia; recurrent miscarriage |
| **3. Y chromosome microdeletions (Yq, AZF (azoospermia factor))** | **AZFa** (USP9Y, DDX3Y), **AZFb**, **AZFc** (DAZ genes) | Complete AZFa or AZFb deletion: **no sperm at retrieval**. AZFc: sperm often in ejaculate or testis; sons inherit the deletion |
| **4. Single-gene: excurrent ducts** | **CFTR (cystic fibrosis transmembrane conductance regulator) mutations**: CBAVD (congenital bilateral absence of vas deferens); idiopathic ejaculatory duct obstruction | Low-volume azoospermia, normal spermatogenesis; **test both partners** (ΔF508) |
| **5. Single-gene: hypothalamic–pituitary** | **Kallmann syndrome** (KAL1, anosmia); GnRH (gonadotropin-releasing hormone) receptor mutations | Low LH (luteinising hormone) and FSH; treat with hCG ± FSH |
| **6. Androgen action** | **Androgen receptor** mutation (long CAG repeat); 5α-reductase type 2 deficiency | High testosterone and LH, poor sperm |
| **7. Sperm structure and motility** | **Globozoospermia** (SPATA16, DPY19L2: no acrosome); macrocephalic sperm (AURKC); **PCD (primary ciliary dyskinesia)**/Kartagener (DNAH5, DNAI1: immotile sperm) | ICSI; macrocephalic sperm: ICSI not advised (aneuploidy) |

### Genetic testing indications

- **Karyotype and Y microdeletion:** non-obstructive azoospermia and severe oligozoospermia (**<5 million/mL**, ASRM/AUA (American Society for Reproductive Medicine/American Urological Association) 2021; EAU (European Association of Urology) uses <10 million/mL for karyotype).
- **CFTR:** CBAVD, idiopathic obstructive azoospermia; screen the partner; counsel before ICSI.
- **Also know:** Noonan syndrome, myotonic dystrophy.

**Diagram to draw:** Y chromosome long arm showing AZFa, b, c.

> **Key points to remember:**
> - Klinefelter is the commonest genetic cause; micro-TESE finds sperm in about half.
> - AZFa or AZFb complete deletion: do not attempt retrieval.
> - CBAVD = genital form of cystic fibrosis; test both partners.

## SAQ 9 Ejacultory duct obstruction

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 2 Paper 3, Q3b

*Assumed meaning: ejaculatory duct obstruction (EDO).*

## Anatomy

- Intraprostatic continuations of the seminal vesicles, joined by the vasal ampullae; they lack muscle and open at the **verumontanum**. Seminal vesicles give about 60% of the ejaculate and its fructose.

## Aetiology

- **Congenital (commonest):** midline (Müllerian/utricular) cyst, Wolffian duct cyst, atresia. Idiopathic EDO carries CFTR (cystic fibrosis transmembrane conductance regulator) mutations in about half.
- **Acquired:** infection, instrumentation, duct stones, compression.
- **Zinner syndrome:** seminal vesicle cyst, ipsilateral renal agenesis, EDO.
- Complete bilateral EDO causes under 5% of azoospermia. Partial EDO gives oligoasthenospermia.

## Clinical features

- Infertility, painful ejaculation, haematospermia, perineal pain.

## Diagnosis

### Semen (complete bilateral EDO)

- Volume **<1.0 mL**, **acidic pH**, low or absent **fructose**, azoospermia.
- Palpable vas, normal FSH (follicle-stimulating hormone) and testes.
- Exclude retrograde ejaculation and CBAVD (congenital bilateral absence of the vas deferens).

### Investigations

| Test | Method / finding |
|---|---|
| **TRUS (transrectal ultrasound)** | Seminal vesicle AP diameter **>1.5 cm**, duct >2.3 mm, midline cyst. Low specificity |
| **Seminal vesicle aspiration** | **≥3 sperm per high-power field** confirms obstruction |
| Vesiculography | Contrast into vesicles; invasive |
| Chromotubation | Indigo carmine into vesicles; dye at the veru on cystoscopy |

## Management

### TURED (transurethral resection of the ejaculatory ducts)

- For azoospermic or severely oligoasthenospermic men, after the couple declines IVF (in vitro fertilisation).
- **Steps:** TRUS-guided aspiration (cryopreserve sperm) → instil indigo carmine with contrast → 24-Fr resectoscope, finger in rectum → resect the veru until dye effluxes. Spare bladder neck, sphincter and rectum. Catheter overnight.
- **Results:** volume rises in most; sperm appear in about half. Cold-knife incision alone re-obstructs.

### Complications

- **Urine reflux** into the ducts, **epididymitis**, retrograde ejaculation, rectal injury, re-obstruction.

### Alternatives

- **Seminal vesiculoscopy** (4–6 Fr via utricle): avoids reflux; restenosis unknown. Sperm retrieval with ICSI (intracytoplasmic sperm injection).

**Diagram to draw:** sagittal prostate with midline cyst compressing the ducts at the veru.

> **Key points to remember:**
> - Low volume (<1 mL), acidic, fructose-negative azoospermia with palpable vasa.
> - TRUS: vesicle >1.5 cm; aspiration ≥3 sperm per high-power field confirms.
> - TURED: sperm return in about half; reflux and epididymitis are its price.

## SAQ 10 Tadalafil

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 3 Paper 3, Q3b

### Pharmacology

- Selective **PDE5 (phosphodiesterase type 5) inhibitor**; FDA (Food and Drug Administration) approved 2003.
- **Mechanism:** sexual stimulation releases NO (nitric oxide), raising cGMP (cyclic guanosine monophosphate), which relaxes cavernosal smooth muscle. PDE5 breaks down cGMP; tadalafil blocks this. It **augments, not initiates,** erection.
- Structurally distinct from sildenafil; cross-reacts with **PDE11**, little with PDE6 (retina).

| Property | Tadalafil (vs sildenafil) |
|---|---|
| Time to peak | 2 h (0.8 h) |
| Half-life | **17.5 h** (3–5 h) |
| Fatty food | No effect (reduces absorption) |
| Window | Widest (up to 36 h) |

### Uses and doses

1. **ED (erectile dysfunction):** 10–20 mg on demand, or 2.5–5 mg once daily. Intercourse success about 70%; 40–50% in diabetes and after prostatectomy.
2. **BPH (benign prostatic hyperplasia) with LUTS (lower urinary tract symptoms), with or without ED: 5 mg daily**, the only PDE5 inhibitor licensed. IPSS (International Prostate Symptom Score) falls about 6 points (placebo 3.6) **without Qmax (peak flow rate) change**; mechanism is NO–cGMP relaxation, perfusion and reduced afferent activity, not de-obstruction. Equal to tamsulosin for IPSS. EAU (European Association of Urology) 2022 and AUA (American Urological Association) 2021: option for moderate–severe LUTS.
3. **Also know:** fixed finasteride–tadalafil (FDA 2021); penile rehabilitation (benefit unproven); prevention of stuttering priapism (PDE5I paradox); pulmonary arterial hypertension.

### Adverse effects

- Headache, dyspepsia, **back pain and myalgia** (characteristic), flushing, nasal congestion. Visual effects rare.
- NAION (non-arteritic anterior ischaemic optic neuropathy): no proven link; stop if sudden visual loss.

### Contraindications and interactions

- **Nitrates in any form: absolute.** After tadalafil, no nitrates for **48 h** (24 h for sildenafil).
- Caution: α-blockers (start lowest dose of the added drug), aortic stenosis, hypotension. CYP3A4 inhibitors (ketoconazole, ritonavir) raise levels.
- Not studied: MI (myocardial infarction) or stroke within 6 months, unstable angina, BP <90/50 or >170/100 mmHg, retinitis pigmentosa.

**Diagram to draw:** NO–cGMP–PDE5 pathway in cavernosal smooth muscle.

> **Key points to remember:**
> - Half-life 17.5 h; food-independent; back pain.
> - 5 mg daily for LUTS: IPSS improves, Qmax does not.
> - No nitrates for 48 h after a dose.

## SAQ 11 Retrograde ejaculation

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 3 Paper 3, Q3f

### Definition

- RE (retrograde ejaculation): semen passes backwards into the bladder at orgasm because the **bladder neck fails to close**; orgasm is preserved, ejaculate absent or low volume.

### Physiology

| Phase | Event | Control |
|---|---|---|
| Emission | Vas, seminal vesicles, prostate deliver semen to posterior urethra | Sympathetic T10–L2 |
| Ejection | **Bladder neck closes**, external sphincter relaxes, bulbospongiosus contracts | Somatic S2–S4 + sympathetic reflex |
| Orgasm | Cerebral perception | Cortical |

### Causes

- **Anatomical: TURP (transurethral resection of prostate)** 42–100%, TUIP (incision) 5–45%; most other resections and enucleations; previous bladder neck reconstruction. Ejaculation-sparing techniques, PUL (prostatic urethral lift) and Aquablation preserve ejaculation.
- **Neurogenic: diabetic autonomic neuropathy** (commonest medical), RPLND (retroperitoneal lymph node dissection; less with nerve sparing), colorectal surgery, sympathectomy, aortic surgery, spinal cord injury, multiple sclerosis.
- **Drugs:** α-blockers (silodosin, tamsulosin), antipsychotics, antidepressants, thiazides.
- **Congenital:** Müllerian duct cyst, Wolffian duct anomaly, prune belly syndrome.

### Evaluation

- Low-volume or absent ejaculate with orgasm; infertility.
- **Post-ejaculatory urinalysis:** void, ejaculate, void again; centrifuge. **>5–10 sperm per HPF (high-power field)** confirms RE; significant if urinary sperm approach or exceed

antegrade sperm.
- Sperm and fructose in urine separate RE from **failure of emission**; TRUS (transrectal ultrasound) if EDO (ejaculatory duct obstruction) suspected.

### Treatment

- Reassure if fertility not desired; stop offending drug.
- **Sympathomimetics** (close bladder neck): **pseudoephedrine** 60–120 mg (120 mg 6-hourly the day before and twice on the day); midodrine, ephedrine, phenylpropanolamine. Side effects: dry mouth, hypertension.
- **Imipramine** 25 mg twice daily; antihistamines (brompheniramine) add to sympathomimetics.
- **Bladder neck reconstruction:** poor results; collagen injection experimental.
- **Sperm retrieval:** alkalinise urine (sodium bicarbonate), void or catheterise after orgasm, wash sperm in buffer → IUI (intrauterine insemination), IVF (in vitro fertilisation) or ICSI (intracytoplasmic sperm injection).
- **Prevention:** nerve-sparing RPLND, ejaculation-preserving procedures for BPH (benign prostatic hyperplasia).

**Diagram to draw:** bladder neck open at ejaculation with semen entering the bladder.

> **Key points to remember:**
> - TURP and diabetic neuropathy are the commonest causes.
> - Post-ejaculatory urine: sperm and fructose confirm RE.
> - Pseudoephedrine first; else urinary sperm retrieval for assisted reproduction.

## SAQ 12 CBAVD

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 4 Paper 3, Q3f

### Definition

- **CBAVD (congenital bilateral absence of the vas deferens):** both vasa absent, causing **obstructive azoospermia** with normal spermatogenesis.
- Often a **genital form of CF (cystic fibrosis)**: nearly all men with CF lack both vasa.

### Genetics

- **CFTR (cystic fibrosis transmembrane conductance regulator)**: chloride channel gene; more than 1600 mutations.
- Severe mutation on both alleles: CF. **Mild mutation on one or both alleles: CBAVD.** Commonest severe mutation **ΔF508**; the **5T variant** (intron 8) is a common mild allele.
- A minority have other mesonephric duct gene defects instead, with more **renal agenesis**.

### Clinical features

- **Vasa not palpable** (Meacham: a presumed vas slipping away three times is absent).
- Caput epididymis present and distended; body and tail often absent.
- Normal testis size, FSH (follicle-stimulating hormone) and testosterone.

- Seminal vesicles absent or hypoplastic: **semen volume <1.0 mL, acidic, fructose-negative**, azoospermic.
- Sinus or chest symptoms may reveal mild CF.

### Evaluation

- Semen analysis; FSH; **TRUS (transrectal ultrasound)** for seminal vesicles.
- **Renal ultrasound** (renal agenesis in about 1 in 10).
- **CFTR testing of the man and his female partner** (carrier rate about 1 in 20 in northern Europeans). Panels cover 25–40 mutations, so a negative panel leaves residual risk.
- Testis biopsy unnecessary when FSH and testis size are normal.

### Management

- **Genetic counselling**: if the partner carries a mutation, offspring may have CF; PGT (preimplantation genetic testing) or donor sperm can be offered.
- Reconstruction is impossible, so **surgical sperm retrieval + ICSI (intracytoplasmic sperm injection)**:

| Technique | Points |
|---|---|
| **MESA (microsurgical epididymal sperm aspiration)** | Best: many motile sperm; cryopreserve several vials |
| PESA (percutaneous epididymal aspiration) | Local anaesthetic; fewer sperm |
| TESA/TESE (testicular aspiration/extraction) | If epididymal retrieval fails |
| - ICSI results are good; one study reported more miscarriage and stillbirth. | |
| - **Also know:** CUAVD (unilateral absence) is a Wolffian duct defect with ipsilateral renal agenesis. | |

**Diagram to draw:** absent vas and seminal vesicle with intact testis and caput epididymis.

> **Key points to remember:**
> - CBAVD = mild CFTR mutation; test both partners.
> - Low-volume, acidic, fructose-negative azoospermia; normal FSH.
> - Renal ultrasound; MESA + ICSI.

## SAQ 13 Sertoli cell only syndrome

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 8 Paper 3, Q3a (non-obstructive azoospermia also answered in 2025-Winter Paper 2, Q3a)

### Definition

- SCOS (Sertoli cell only syndrome), **germ cell aplasia** (Del Castillo syndrome): seminiferous tubules lined only by Sertoli cells, with **complete loss of spermatogonia**. Leydig cells usually normal.
- The worst histological pattern of NOA (non-obstructive azoospermia).

## Types

| Type | Feature |
|---|---|
| Pure (complete, congenital) | All tubules lack germ cells |
| Mixed (focal, acquired) | Some tubules keep foci of spermatogenesis, the basis of sperm retrieval |

## Aetiology

- **Developmental:** primordial germ cells fail to migrate into the testis.
- **Genetic: complete AZFa (azoospermia factor a) deletion** gives pure SCOS; **AZFc** may give SCOS with foci; Klinefelter (47,XXY).
- **Acquired:** cryptorchidism, chemotherapy or radiation, mumps orchitis, torsion; idiopathic.

## Clinical features

- Normal virilisation; **azoospermia** with normal ejaculate volume; small soft testes.

## Evaluation

- Semen: azoospermia on two centrifuged samples.
- **FSH (follicle-stimulating hormone) high**, **inhibin B low** (Sertoli marker); LH (luteinising hormone) normal or high; testosterone normal or low.
- **FSH >7.6 IU/L and testis long axis ≤4.6 cm** suggest NOA.
- **Karyotype and Y microdeletion** before surgery.
- **Histology** confirms, usually at sperm retrieval.

| Pattern | Meaning | Micro-TESE retrieval |
|---|---|---|
| Hypospermatogenesis | All stages, fewer | Best |
| Maturation arrest | Stops at one stage | About 40% |
| **SCOS** | No germ cells | **About 30%** |

## Management

- **Counsel** on genetic findings.
- **Complete AZFa or AZFb deletion:** retrieval essentially nil; do not operate; donor sperm or adoption.
- **Micro-TESE (microdissection testicular sperm extraction):** at ×20–25, **larger, opaque, yellow tubules** (foci of spermatogenesis) are selected; SCOS tubules are thin, white and stringy. Live births are reported even in SCOS.
- Sperm used for **ICSI (intracytoplasmic sperm injection)**.
- Optimise first: stop exogenous testosterone; hormonal stimulation evidence weak.
- **Also know:** spermatogonial stem cell transplantation and in-vitro spermatogenesis are experimental.

**Diagram to draw:** tubule cross-section: normal layered epithelium versus Sertoli cells only.

> **Key points to remember:**
> - Complete loss of spermatogonia; high FSH, low inhibin B.
> - Complete AZFa deletion: pure SCOS, no retrieval.
> - Micro-TESE retrieval about 30%: worst histology, not hopeless.

# SAQ 14 Oxidative stress and male infertility

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 8 Paper 3, Q3b

## Definition

- **Oxidative stress (OS):** imbalance where ROS (reactive oxygen species) exceed the antioxidant capacity of semen. The term **MOSI (male oxidative stress infertility)** has been proposed.
- Low ROS levels are **physiological**: needed for capacitation, hyperactivation, acrosome reaction and sperm–oocyte fusion.

## Sources of ROS

| Source | Examples |
|---|---|
| **Leucocytes** (main) | Infection, prostatitis, epididymitis, leucocytospermia |
| **Immature sperm** | Excess residual cytoplasm |
| Varicocele | Heat and hypoxia |
| Lifestyle | **Smoking**, alcohol, obesity, heat |
| Others | Ageing, diabetes, pollutants, radiation |

## Why sperm are vulnerable

- Membrane rich in **polyunsaturated fatty acids** → lipid peroxidation.
- Little cytoplasm, so few antioxidant enzymes; limited DNA repair.
- **Seminal plasma antioxidants:** superoxide dismutase, catalase, glutathione peroxidase; taurine, vitamins C and E, zinc.

## Effects

- Lipid peroxidation → loss of **motility** and membrane fluidity.
- **Sperm DNA fragmentation** → poor embryo quality, miscarriage, lower ART (assisted reproductive technology) success.
- Mitochondrial damage and apoptosis.

## Evaluation

- Semen analysis: peroxidase-positive WBC (white blood cells) **>1 million/mL** = leucocytospermia.
- **Direct ROS:** chemiluminescence (luminol).
- **Indirect:** total antioxidant capacity, malondialdehyde (lipid peroxidation), ORP (oxidation-reduction potential).
- **DNA fragmentation tests:** TUNEL (terminal deoxynucleotidyl transferase dUTP nick-end labelling), SCSA (sperm chromatin structure assay), SCD (sperm chromatin dispersion), Comet.
- Use selectively (unexplained infertility, recurrent miscarriage, failed ART); not routine.

## Management

- **Lifestyle:** stop smoking, reduce weight and alcohol, avoid heat.

- **Treat cause:** antibiotics for infection; **varicocelectomy** for clinical varicocele (reduces DNA fragmentation).
- **Antioxidants** (vitamins C and E, CoQ10 [coenzyme Q10], zinc, selenium, carnitine): Cochrane review, low-quality evidence; **MOXI trial (2020)** showed no gain in semen parameters, DNA fragmentation or live birth. Avoid excess: **reductive stress** also harms.
- High DNA fragmentation with failed ICSI (intracytoplasmic sperm injection): **testicular sperm** has less fragmentation.

**Diagram to draw:** balance of ROS sources against seminal antioxidants, with effects on sperm.

> **Key points to remember:**
> - Leucocytes and immature sperm are main ROS sources.
> - Lipid peroxidation and DNA fragmentation are the key injuries.
> - Leucocytospermia >1 million/mL; antioxidant benefit unproven (MOXI).

## SAQ 15 Dhat syndrome

P3

**Type:** SAQ (10 marks)
**Source:** Practice Set 8 Paper 3, Q3c

### Definition

- **Dhat syndrome:** a **culture-bound syndrome** of South Asia. The man believes he is losing **semen ("dhat")** in urine or nocturnal emissions, and attributes weakness, fatigue and sexual problems to it.
- Term coined by **Wig** (India); listed in ICD-10 (International Classification of Diseases, 10th revision) and in the DSM-5 (Diagnostic and Statistical Manual, 5th edition) glossary of cultural concepts of distress.
- Related syndromes: **Shen-kui** (China), **Koro** (genital retraction).

### Cultural basis

- Traditional belief that semen is a **precious vital fluid** made from many drops of blood; its loss is felt to drain body and mind.

### Epidemiology

- Young men, often unmarried or newly married, rural, lower education; common in urology and sexual medicine clinics.

### Clinical features

- **Somatic:** fatigue, weakness, aches, poor appetite.
- **Psychological:** anxiety, low mood, guilt (about masturbation), poor concentration, sleep loss.
- **Sexual:** fear of ED (erectile dysfunction) or PE (premature ejaculation), worry about penis size and fertility.
- Whitish urine is usually phosphaturia, oxaluria or normal secretions.
- **Comorbidity: depression** and **anxiety** in many; somatisation.

### Evaluation

- Non-judgmental sexual history and belief system.
- Examination of genitalia; urine microscopy (phosphates, oxalates, sperm, pus cells).
- Exclude organic disease: **UTI (urinary tract infection), prostatitis, STI (sexually transmitted infection), diabetes**, true spermatorrhoea.
- Screen for depression and anxiety.

### Management

**1. Counselling (main)**

- **Sex education:** semen is continuously produced; nocturnal emission and masturbation are normal and harmless.
- Reassure without dismissing.

**2. Psychological therapy:** CBT (cognitive behavioural therapy) for misbeliefs and anxiety.

**3. Drugs**

- **SSRIs (selective serotonin reuptake inhibitors)** for depression, anxiety or associated PE; short anxiolytic course if needed.
- Treat genuine ED or PE.

**4. Avoid harm**

- Avoid unnecessary "tonics", antibiotics and repeated tests; they reinforce the belief.
- Refer to psychiatry when depression is significant.

### Prognosis

- Good with counselling; drop-out is common.

**Diagram to draw:** flowchart: complaint of semen loss → exclude organic cause → counsel → treat depression or anxiety.

> **Key points to remember:**
> - Culture-bound semen-loss anxiety of South Asia.
> - Depression and anxiety common; exclude UTI, prostatitis, STI.
> - Counselling and sex education first; SSRIs for comorbidity.

## SAQ 16 Male menopause (andropause, late-onset hypogonadism)

P4

**Asked as:**

- Male menopause. — *Summer 2017 Paper IV Q3e (SAQ)*
- Andropause — *Winter 2017 Paper IV Q3e (SAQ)*
- Male menopause — *Summer 2019 Paper IV Q3f (SAQ)*

**Type:** SAQ (10 marks)

*Read as: late-onset hypogonadism (testosterone deficiency of ageing men, "andropause"). Same topic as 2017-Summer Paper 4, Q3e and 2017-Winter Paper 4, Q3e; full answer given here.*

## Definition

- **Symptoms plus consistently low serum testosterone** in an ageing man.
- "Menopause" (or "Andropause") is a misnomer: the fall is gradual and partial. Preferred terms: PADAM (partial androgen deficiency of the ageing male), LOH (late-onset hypogonadism).

## Pathophysiology

- Testosterone falls slowly from middle age.
- Testosterone in the 70s is about one-third lower than in young men. (Another source: mean testosterone in the seventh decade is about **one-third lower** than in young men.)
- **Testis:** fewer Leydig cells, lower sperm output; FSH (follicle-stimulating hormone) rises about threefold.
- **Centre:** blunted feedback, irregular GnRH (gonadotrophin-releasing hormone) pulses.
- **SHBG (sex hormone-binding globulin) rises with age**, so free testosterone (bioavailable [free + albumin-bound] testosterone) falls more.
- Worse with obesity, diabetes and **opioids**.

## Clinical features

- **Sexual (most specific):** low libido, ED (erectile dysfunction), fewer morning erections.
- Fatigue, low muscle mass, osteoporosis, anaemia, low mood.
- **ADAM (Androgen Deficiency in the Aging Male)** questionnaire: sensitive, not specific.

## Diagnosis

- **Morning (7–11 am) total testosterone, twice (repeated).** AUA (American Urological Association): **<300 ng/dL** (≈10.4 nmol/L). EAU (European Association of Urology): about **12 nmol/L**; calculated free testosterone if borderline. Calculated bioavailable testosterone when SHBG is abnormal.
- **LH (luteinising hormone):** high in primary, low or normal in secondary hypogonadism. Check **prolactin** if LH is low (Low LH → **prolactin**); pituitary MRI (magnetic resonance imaging) if testosterone <150 ng/dL or prolactin high.
- Baseline: PSA (prostate-specific antigen), DRE (digital rectal examination), haematocrit.

## Treatment

- Lifestyle first: weight loss, exercise, treat comorbidities.
- **TRT (testosterone replacement therapy)** for symptoms + confirmed low level: gels, injections (enanthate every 2–3 weeks; undecanoate every 10–14 weeks), oral undecanoate. Stop if no benefit at 3–6 months (another source: at about 3 months).
- **Contraindications:**
- Prostate or breast cancer.
- **Wish for fertility** (TRT stops sperm production [suppresses spermatogenesis]; use hCG [human chorionic gonadotrophin] or clomiphene).
- Haematocrit >50% (raised haematocrit).
- Untreated severe sleep apnoea; recent MI (myocardial infarction) or stroke.

## Monitoring

- At 3–6 months, then yearly: testosterone, **haematocrit** (stop/hold if >54%), PSA.

### Safety evidence

- **TRAVERSE (2023):** 5,246 men aged 45–80 at high cardiovascular risk. TRT was **non-inferior for major cardiac events**; more atrial fibrillation and pulmonary embolism.
- **Testosterone Trials**: modest sexual benefit, **no cognitive benefit**.

**Diagram to draw:** hypothalamic–pituitary–testis axis with ageing defects.

> **Key points to remember:**
> - Symptoms + low morning testosterone twice (AUA <300 ng/dL; EAU 12 nmol/L).
> - SHBG rises with age; LH separates primary from secondary; prolactin if LH low.
> - TRT harms fertility; watch haematocrit; TRAVERSE: cardiac non-inferiority.

## SAQ 17 Klinefelter's syndrome.

P4

**Type:** SAQ (10 marks)
**Source:** 2018-Summer Paper 4, Q3b

### Definition and genetics

- **47,XXY**: commonest sex chromosome aneuploidy and **commonest genetic cause of male infertility**; 1 in 500–1000 male births.
- Non-mosaic in **80–90%**; the rest mosaic **46,XY/47,XXY** (milder, more sperm) or 48,XXXY variants. Extra X from non-disjunction (maternal age).
- Underdiagnosed: most never diagnosed; increasingly detected on NIPT (non-invasive prenatal testing).

### Pathophysiology

- Seminiferous tubules **hyalinise** progressively from puberty; germ cells are lost. Leydig cells fail, so testosterone falls. Puberty starts on time, then stalls.

### Clinical features

| Age | Features |
|---|---|
| Infant | Usually normal; micropenis, cryptorchidism, hypospadias |
| Adolescent/adult | **Small firm testes (<3.5 cm, 1–4 mL)**, gynaecomastia, tall stature with long legs (only ~30%), sparse hair, female fat distribution |
| Neurodevelopment | Language, learning and attention difficulties |

### Hormones and tests

- **High FSH (follicle-stimulating hormone) and LH (luteinising hormone), low or low-normal testosterone, high oestradiol** (raised oestradiol:testosterone ratio). Inhibin B low.
- Semen: **azoospermia** in most (~8% have ejaculated sperm).
- **Karyotype** confirms (AUA/ASRM, American Urological Association/American Society for Reproductive Medicine, guideline: karyotype all men with non-obstructive azoospermia or severe oligozoospermia).

### Management

- **Fertility: micro-TESE** (microdissection testicular sperm extraction) + ICSI (intracytoplasmic sperm injection). Sperm found in **~40–50%** (68% in one series); offspring are usually chromosomally normal.
- Spermatogenesis declines with age, so retrieve early (late adolescence or young adulthood). Some give clomiphene, anastrozole or hCG (human chorionic gonadotropin) beforehand.
- **TRT (testosterone replacement therapy):** for hypogonadism (bone, muscle, mood, libido). It **suppresses spermatogenesis**, so offer sperm retrieval or banking **before** starting.
- Gynaecomastia: reduction mammoplasty if persistent. Speech, educational and psychological support.

### Associated risks

- Osteoporosis, metabolic syndrome, diabetes, VTE (venous thromboembolism), autoimmune disease.
- **Mediastinal (extragonadal) germ cell tumours**; male breast cancer (~1% lifetime).

**Diagram to draw:** hypothalamic–pituitary–testis axis showing high FSH and LH, low testosterone.

> **Key points to remember:**
> - 47,XXY; non-mosaic 80–90%; 1 in 500–1000.
> - Small firm testes, high FSH and LH, azoospermia.
> - Micro-TESE ~50% success; retrieve sperm before testosterone.
> - Risks: mediastinal germ cell tumour, breast cancer, osteoporosis.

## SAQ 18 Semen analysis nomenclature.

P4

**Type:** SAQ (10 marks)
**Source:** 2018-Summer Paper 4, Q3e

### Collection (WHO, World Health Organization, 2021, 6th edition)

- Masturbation into a clean wide-mouthed non-toxic container (special condom if needed); complete sample.
- Abstinence **2–7 days** (evidence: 1–2 days best). Keep at 20–37 °C; analyse within **1 hour**, after liquefaction (~30 min).
- At least **two samples, 2–3 weeks apart**, because of variability.

### Lower reference limits (5th centile, fertile men)

| Parameter | Lower limit |
|---|---|
| Volume | **1.4 mL** |
| Concentration | **16 million/mL** |
| Total count | **39 million/ejaculate** |
| Total motility | **42%** |
| Progressive motility | **30%** |

| Parameter | Lower limit |
|---|---|
| Normal forms (strict) | **4%** |
| Vitality | **54%** |
| Peroxidase-positive leucocytes | <1 million/mL |

- These are reference values, not fertility cut-offs. The 6th edition grades motility as progressive, non-progressive or immotile.

## Nomenclature

| Term | Meaning |
|---|---|
| Normozoospermia | All parameters at or above reference |
| Oligozoospermia | Total count (or concentration) below reference |
| Asthenozoospermia | Progressive motility below reference |
| Teratozoospermia | Normal forms below reference |
| OAT (oligoasthenoteratozoospermia) | All three abnormal |
| Azoospermia | No sperm, **including centrifuged pellet** |
| Cryptozoospermia | Sperm absent in fresh smear, present only after centrifugation |
| Aspermia | No ejaculate (dry orgasm) |
| Hypospermia | Volume below reference |
| Necrozoospermia | Low vitality (many dead sperm) |
| Leucocytospermia | Peroxidase-positive white cells above 1 million/mL |
| Haemospermia | Blood in ejaculate |
| Globozoospermia | Round-headed sperm without acrosome |

## Notes

- **Low volume or aspermia:** post-ejaculatory urine for retrograde ejaculation; TRUS (transrectal ultrasound) for EDO (ejaculatory duct obstruction).
- **Vitality:** eosin–nigrosin, or **HOS (hypo-osmotic swelling)** test, which keeps sperm usable for ICSI (intracytoplasmic sperm injection).
- Only azoospermia is definitive; values overlap between fertile and infertile men.
- **Also know:** Kruger strict morphology; DNA fragmentation; antisperm antibodies (MAR, mixed antiglobulin reaction); semen pH no longer recommended.

**Diagram to draw:** normal sperm (head, midpiece, tail) beside a WHO reference box.

> **Key points to remember:**
> - WHO 2021: 1.4 mL, 16 million/mL, 39 million, 42%/30%, 4%, 54%.
> - Cryptozoospermia = sperm only in pellet; aspermia = no ejaculate.
> - Two samples 2–3 weeks apart; abstinence 2–7 days.

# Rapid Revision Sheet · Section 9

### Spermatogenesis, azoospermia and non-obstructive azoospermia (LAQ)

- Spermatogenesis about 64 days; Ap divide every 16 days; six stages.
- Intratesticular testosterone 50 × serum; exogenous testosterone causes azoospermia.
- FSH → Sertoli, LH → Leydig; tight junctions between Sertoli cells form the blood–testis barrier.
- Confirm azoospermia on two centrifuged samples.
- FSH ≤7.6 IU/L and testis >4.6 cm → obstructive; FSH >7.6 and testis ≤4.6 cm suggest NOA.
- Karyotype and Y microdeletion in all.
- AZFa/b complete deletion: do not operate (no retrieval).
- NOA: micro-TESE + ICSI; sperm in about half; SCO 30%.
- Maturation arrest: arrest at one stage; early (spermatocyte) or late (spermatid).
- FSH and testis size may be normal in maturation arrest: biopsy diagnoses.
- Micro-TESE retrieval in maturation arrest about 40%; none with complete AZFb deletion.

### Azoospermia: testicular biopsy and surgical sperm retrieval techniques (LAQ)

- Azoospermia: no sperm in centrifuged pellet on two samples.
- Biopsy indication: azoospermia, normal testes, vasa and FSH.
- Open biopsy is the gold standard; fix in Bouin's, never formalin; no forceps; touch prep.
- OA: normal FSH and testes; ICSI essential.
- OA: MESA (capillary action); MESA is best: many motile sperm, multiple vials.
- Move proximally; capillary action, never suction.
- TESA: 23 G needle, suction, local anaesthesia; for OA with normal spermatogenesis.
- TESA/TESE when epididymal sperm absent.
- NOA: micro-TESE best; finds sperm in about half of NOA; TESA yield poor.
- AZFa/AZFb deletion: 0% retrieval (no sperm), do not operate.
- All surgically retrieved (testicular) sperm need ICSI; cryopreserve extras.
- Haematoma commonest; epididymal biopsy most serious.

### Surgical management of ED and penile prostheses (types, biofilms) (LAQ)

- Prosthesis is the last line but gives 80–90% satisfaction.
- 3-piece inflatable gives the best rigidity and flaccidity.
- Malleable rods: nitinol (Tactra), silver (Genesis); almost never fail.
- Glucose <200 mg/dL, chlorhexidine–alcohol, coated device, no-touch technique.
- Biofilm: microbes in a matrix; antibiotics fail.
- Infection 1–3%; coated devices and no-touch technique; infected implant: remove all parts; Mulcahy salvage 84%.
- Satisfaction 80–90%; repairs 15% at 5 years.
- Arterial revascularisation only for young men with post-traumatic focal stenosis.

### Priapism: diagnosis and management (LAQ)

- Blood gas decides the type: ischaemic pO2 <30, pCO2 >60, pH <7.25.
- Aspiration plus phenylephrine first; distal shunt ± tunnelling before proximal.
- ED risk 0% ≤12 h, 100% ≥36 h; consider early prosthesis beyond 36 h.

- Beyond 36 h or refractory: early penile prosthesis (or analgesia and counselling).
- High flow: observe (about 60% resolve), then embolise; repeat embolisation rather than surgery.

**Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED (LAQ)**

- nNOS initiates, eNOS maintains; PDE5 ends (terminates) erection.
- Noradrenaline α1 is the main anti-erectile transmitter.
- Dopamine/oxytocin promote; serotonin/prolactin inhibit.
- Rigid phase: ischiocavernosus, suprasystolic pressure.
- Duplex: PSV <25 cm/s arterial; EDV >5 cm/s, RI <0.75 venous leak.
- Diabetic ED: check HbA1c, testosterone and cardiac risk.

**Discuss the role of urologist in reproductive endocrinology and male infertility practice. (LAQ)**

- FSH >7.6 IU/L + testis ≤4.6 cm = spermatogenic failure.
- Exogenous testosterone is a contraceptive: never give it for infertility.
- Hypogonadotropic hypogonadism: hCG ± FSH; most treatable cause.
- Varicocelectomy only for palpable varicocele with abnormal semen; micro-TESE for non-obstructive azoospermia.

**Male reproductive immunology and immunobead test (SAQ)**

- BTB = Sertoli tight junctions; plus active tolerance (galectin-1, TGF-β, androgens).
- ASA: vasectomy commonest; ≥50% motile sperm bound is significant; head-bound IgA matters most.
- Direct IBT = sperm-bound antibody; indirect = antibody in serum, seminal plasma or mucus.
- Treatment of choice: IVF/ICSI.
- Leukocytospermia >1 million/mL.

**Premature ejaculation (SAQ)**

- ISSM: about 1 min (lifelong), ≤3 min (acquired) plus poor control plus distress.
- Lifelong is neurobiological; acquired is usually secondary.
- Dapoxetine 30–60 mg on demand; paroxetine gives the greatest delay.

**Oligospermia and oligoasthenozoospermia (SAQ)**

- WHO 2021: concentration 16 million/mL, progressive motility 30%; severe <5 million/mL.
- Karyotype and Y-microdeletion if under 5 million/mL.
- Repair palpable varicoceles only; never give testosterone.

**Management of Peyronie's disease (SAQ)**

- Ask the "straight penis" question; duplex before surgery.
- CCH: IMPRESS, stable 30–90° curve, intact erections.
- Surgery: stable ≥12 months; plication <60–70°, graft complex curves, prosthesis if ED.

**Haematospermia (SAQ)**

- Usually benign; prostate biopsy is the commonest identifiable cause.
- Full work-up if ≥40, persistent or recurrent, or associated haematuria (or with haematuria).
- Exclude prostate cancer and genitourinary tuberculosis.

- TRUS: seminal vesicle >1.5 cm or ejaculatory duct >2.3 mm suggests obstruction.

**Avanafil (SAQ)**

- Peak 0.3–0.5 h; doses 50, 100, 200 mg.
- Highly PDE5-selective: no blue vision.
- Nitrates absolutely contraindicated. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**CRE & BEP (SAQ)**

- CRE: NDM common in India; ceftazidime–avibactam + aztreonam for metallo-β-lactamase.
- BEP ×3 good risk, ×4 intermediate/poor; EP ×4 if bleomycin avoided.
- Bleomycin lung: low $FiO_2$ and restricted fluids at surgery. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Genetic causes of male infertility. Enumerate and classify. (SAQ)**

- Klinefelter is the commonest genetic cause; micro-TESE finds sperm in about half.
- AZFa or AZFb complete deletion: do not attempt retrieval.
- CBAVD = genital form of cystic fibrosis; test both partners.

**Ejacultory duct obstruction (SAQ)**

- Low volume (<1 mL), acidic, fructose-negative azoospermia with palpable vasa.
- TRUS: vesicle >1.5 cm; aspiration ≥3 sperm per high-power field confirms.
- TURED: sperm return in about half; reflux and epididymitis are its price.

**Tadalafil (SAQ)**

- Half-life 17.5 h; food-independent; back pain.
- 5 mg daily for LUTS: IPSS improves, Qmax does not.
- No nitrates for 48 h after a dose.

**Retrograde ejaculation (SAQ)**

- TURP and diabetic neuropathy are the commonest causes.
- Post-ejaculatory urine: sperm and fructose confirm RE.
- Pseudoephedrine first; else urinary sperm retrieval for assisted reproduction.

**CBAVD (SAQ)**

- CBAVD = mild CFTR mutation; test both partners.
- Low-volume, acidic, fructose-negative azoospermia; normal FSH.
- Renal ultrasound; MESA + ICSI.

**Sertoli cell only syndrome (SAQ)**

- Complete loss of spermatogonia; high FSH, low inhibin B.
- Complete AZFa deletion: pure SCOS, no retrieval.
- Micro-TESE retrieval about 30%: worst histology, not hopeless.

**Oxidative stress and male infertility (SAQ)**

- Leucocytes and immature sperm are main ROS sources.
- Lipid peroxidation and DNA fragmentation are the key injuries.
- Leucocytospermia >1 million/mL; antioxidant benefit unproven (MOXI).

**Dhat syndrome (SAQ)**

- Culture-bound semen-loss anxiety of South Asia.
- Depression and anxiety common; exclude UTI, prostatitis, STI.
- Counselling and sex education first; SSRIs for comorbidity. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

**Male menopause (andropause, late-onset hypogonadism) (SAQ)**

- Symptoms + low morning testosterone twice (AUA <300 ng/dL; EAU 12 nmol/L).
- SHBG rises with age; LH separates primary from secondary; prolactin if LH low.
- TRT harms fertility; watch haematocrit; TRAVERSE: cardiac non-inferiority.

**Klinefelter's syndrome. (SAQ)**

- 47,XXY; non-mosaic 80–90%; 1 in 500–1000.
- Small firm testes, high FSH and LH, azoospermia.
- Micro-TESE ~50% success; retrieve sperm before testosterone.
- Risks: mediastinal germ cell tumour, breast cancer, osteoporosis.

**Semen analysis nomenclature. (SAQ)**

- WHO 2021: 1.4 mL, 16 million/mL, 39 million, 42%/30%, 4%, 54%.
- Cryptozoospermia = sperm only in pellet; aspermia = no ejaculate.
- Two samples 2–3 weeks apart; abstinence 2–7 days.